Opportunity, Security, and the Allied Industrial Base: A Quantitative Network Approach to Coalition Supply Chains for Energy Technologies

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Abstract

No allied economy can efficiently reproduce the whole supply chain of energy technologies, so protection against concentrated suppliers must come from allocating capabilities across trusted partners: managed interdependence rather than autarky. We build a framework to measure and design it. Energy-security and economic-opportunity pillars score 219 economies in up to 33 technology–stage cells. A Strategic Index reads security as need, and an endowment screen admits extraction stages to a member's priority only where it has a resource base. A 57-member coalition (37 network nodes) is derived from UN General Assembly ideal points, agreements and world supply shares. Bilateral indices rate partners as suppliers and markets, and a mixed-integer program allocates production in 22 upstream and midstream stages under a diversification penalty calibrated to observed export concentration. China is the largest exporter in 10 of the 18 traded stages. The calibrated design spreads each stage over seven to ten producers (median Herfindahl index 0.152) and, because it scores capability rather than current capacity, places two-thirds of the median stage's production outside today's three largest exporting nodes and 47 percent of upstream output in non-high-income members. Battery midstream is the most common national priority (27 members), far more than the nine producers the network design assigns it, and 21 of the 57 priorities lie within 0.01 of the runner-up. The framework turns rankings into an auditable division of labour.

JEL codes: C43, C61, F14, F52, L52, Q48.

Keywords: allied industrial base; energy security; economic opportunity; composite indicators; friend-shoring; geopolitical alignment; supply-chain network design; industrial policy.

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1 Introduction

Energy security used to be a matter of fuels: reserves, import routes and stocks of oil and gas. As solar panels, batteries, grids and power electronics absorb a growing share of energy investment, it is increasingly a matter of manufactured systems and of the minerals and components they are made from (Carey et al., 2025; International Energy Agency, 2024). A country can now be secure in fuels and exposed in transformers, battery materials or magnets. These manufacturing chains are concentrated. In the trade data used in this paper, China is the largest world exporter in ten of the 18 upstream and midstream technology–stage baskets that have a product definition, with shares between 20 and 43 percent, and the refining of rare earths and the manufacture of magnets and battery anode material are concentrated in China at shares close to or above 90 percent (International Energy Agency, 2025a, 2026b). Concentration is not in itself a market failure; it reflects scale economies, learning and geology. But it determines how far a disruption travels through production networks (Acemoglu et al., 2012; Carvalho et al., 2021; Baqaee and Farhi, 2019), and it turns trade into leverage. Hirschman (1945) made the point for bilateral trade dependence, and Farrell and Newman (2019) show that states with jurisdiction over the central nodes of global networks can observe, restrict or deny access to them.

The response so far has been national. Governments have returned to industrial policy on a scale not seen for decades (Juhász et al., 2024; Evenett et al., 2024), often with sourcing rules that tie subsidies to where minerals are processed and components made (Bown, 2023). National programmes are meeting economic and political limits. Several economies have targeted the same manufacturing segments without the demand or cost position to fill the plants, and announced capacity has been cancelled as prices fell (International Energy Agency, 2026b); subsidy competition reaches allies (Clausing and Wolfram, 2023); and tighter public finances make it implausible for every large economy to fund a complete green-industrial agenda (Ladislaw, 2025). Smaller economies cannot replicate the demand-led strategies of the largest ones at all (Albertone et al., 2025). Autarky is the expensive alternative. Simulations of broad relocalisation find large losses of trade and output without consistent gains in resilience, and diversification usually does more for resilience than reshoring (OECD, 2025b; Thakur-Weigold and Miroudot, 2024). Deep fragmentation into blocs is costly too, most of all for small economies (Aiyar et al., 2023; Javorcik et al., 2024).

These observations underpin the proposition of an Allied Industrial Base (Campbell and Doshi, 2025; New Energy Industrial Strategy Center, 2026). Matching China's scale across the electro-industrial supply chains is beyond any one allied economy. The premise is that allied economies jointly hold the markets, capital, resources and know-how that none holds alone, but deploy them through competing national strategies (Campbell and Doshi, 2025; New Energy Industrial Strategy Center, 2026). The working paper accordingly frames the policy goal as *managed interdependence*: each strategically important capability should be available at scale from some trusted member, with redundancy where a failure would be very costly, specialisation where scale economies and learning dominate, and open global sourcing where the risk is small (New Energy Industrial Strategy Center, 2026). Implementing it requires answers to questions that the literature mostly leaves open: which

nodes of which supply chains are severe enough to coordinate, who counts as a trusted partner, and which member should produce what (New Energy Industrial Strategy Center, 2026, Section 5).

This paper asks how a coalition of economies could allocate energy-technology supply chains among its members, technology by technology and stage by stage, and builds the measurements that such an allocation needs. The unit of analysis is a cell (i, t, s): an economy, one of eleven technologies (batteries, coal, electric grid, electric vehicles, gas, geothermal, green hydrogen, nuclear, oil, solar and wind) and one of three stages (upstream extraction, midstream processing and manufacturing, downstream deployment). Two pillars score the cells of 219 economies: Energy Security (ES), how secure an economy's supply is, and Economic Opportunity (EO), how well placed it is to compete as a producer. Both are built from rank-based S-curves of public data, among them the IEA, the Energy Institute, the U.S. Geological Survey, UN Comtrade and the Atlas of Economic Complexity (Section 3), and a cell enters a ranking only if the economy is sovereign and observed data carry at least half of its weight. A Strategic Index combines the pillars with a policy index into a within-country ranking of sectors in which security enters as *need*, favouring sectors where a country is exposed but capable; an endowment screen keeps an extraction stage out of a member's priority unless the member has the resource. The coalition is derived rather than chosen: from ideal points estimated from UN General Assembly votes (Bailey et al., 2017), membership of U.S. trade and security agreements, and shares of world supply. It has 31 core members on the U.S.-leaning side of the Assembly with GDP of at least USD 100 billion and 26 partners admitted through agreements or as major suppliers; with the European Union as one node, the network has 37 nodes. Bilateral indices rate each partner as a supplier and as a market, taking account of alignment and, in four manufacturing stages, of the Chinese ownership of the partner's foreign-invested plants. Finally, a mixed-integer program allocates production among the 37 nodes in each of 22 upstream and midstream stages. It rewards producers' opportunity and security and strong bilateral ties and penalises concentration, with the penalty weight calibrated so that designed concentration matches the concentration of the members' exports today. Figure 1 sets out the architecture.

The paper makes four contributions. The first is measurement at the level of the supply-chain node. Security and opportunity are scored jointly, on a common scale and with a common variable architecture, for each stage of each technology; the scores decompose exactly into category and variable contributions, and a rankability rule keeps cells that rest mostly on imputation out of every ranking. Existing energy-security indices describe an energy system or a fuel, and green-complexity measures describe products (Mealy and Teytelboym, 2022; Growth Lab at Harvard University, 2024); neither pairs exposure with capability in the same cell. The second turns the pillars into priorities while keeping the normative choices visible: the need reading of security, the endowment screen, the basis of the policy index and the margin of each member's priority over its runner-up are reported with the results. The third is a rule-based coalition and bilateral layer. Alignment is measured from Assembly votes rather than asserted, the coalition and its tiers follow from stated thresholds, and the bilateral indices carry both alignment and a first measure of the common-mode risk that arises when suppliers in different countries share Chinese parents. The fourth links the

indices to an explicit allocation through network design in the facility-location tradition. A concave diversification penalty, calibrated on observed export concentration, sets the rate at which the design trades capability against concentration, so that the design spreads production among nodes rather than ranking them, a first step toward the working paper's aim of optimising combinations (New Energy Industrial Strategy Center, 2026); because the penalty treats each node as an independent source, it does not yet register that producers sharing a bloc, a region or a parent company diversify less. The pipeline is open and configuration-driven, and the corrections made during an end-to-end audit of an earlier version are documented in Appendix D.

Five sets of results stand out. First, the coalition is stronger on security than on opportunity. A coalition member leads 16 of the 30 security pairs and is in the top three of all 30, but the best-placed member leads only 8 of the 33 opportunity pairs, against 22 for China. The two pillars are correlated within pairs (median Spearman 0.59), largely through shared components: on the 23 pairs in which both pillars keep components of their own, the median falls from 0.55 to 0.30 when only the non-shared components are used. About 15 percent of the cells rankable on both pillars combine above-median opportunity with below-median security, the region the Strategic Index favours. Second, members' priorities concentrate on manufacturing. The index scores 3,154 cells in 174 countries, and every one of the 57 members has an allied pick: 43 are midstream, and battery midstream alone is the pick of 27. The endowment screen changes 41 picks, mostly away from upstream stages in which a member has a need but no resource. The picks are fragile: the median margin over the runner-up is 0.015, 21 picks lie within 0.01, and only five survive if security is read as strength rather than need. Third, alignment shapes the partnership rankings. Viewed from the 31 core members, the United States and Germany are the strongest partners and core members hold 24 of the top 25 places; giving the alignment component zero weight would lift China from 44th to 3rd. Fourth, the calibrated network is diversified but does not replicate current trade. At the calibrated diversification weight, $\lambda = 0.78$, the median designed Herfindahl index (0.152, or 6.6 equivalent producers) matches the members' observed export concentration (0.156). Each stage has seven to ten producers, no share exceeds 0.325, and 29 of the 37 nodes produce somewhere. Non-high-income nodes carry 47 percent of upstream and 43 percent of midstream production, and in the median traded stage two-thirds of production goes to nodes other than the three that export most today. The calibration is knife-edge ($\lambda = 0.77$ misses the target by 0.0002), but the design is not: either neighbouring value reassigns at most about one percent of production in the mean stage. Fifth, rankings are more robust than allocations. Country rankings within pairs barely move under random category weights (median within-pair Spearman 0.997 for security and 0.988 for opportunity), and an empirical test of vertical complementarities finds no out-of-sample gain from rewarding a stage for capability at adjacent stages; removing the portfolio caps, by contrast, reassigns 18.5 percent of production in the mean stage.

The results speak to some open questions of the working paper (New Energy Industrial Strategy Center, 2026, Section 10) but not to others. The design answers the question of how much duplication is enough only conditionally, by anchoring designed concentration to the concentration of

the members' exports today; within that anchor, it gives non-high-income members 43 to 47 percent of production, so a diversified allied network need not be a rich-country club. The paper does not test the hypotheses that allied specialisation reaches a resilience target at lower public cost than parallel national programmes, that demand coordination matters more than subsidy coordination, or that fencing nodes dominates excluding sectors: these need cost, revenue and bankability data that the indices do not contain (Section 13).

The rest of the paper is organised as follows. Section 2 sets out the conceptual framework and notation and Section 3 the data. Section 4 presents the normalisation, imputation and rankability rules. Sections 5 and 6 construct the Energy Security and Economic Opportunity pillars, and Section 7 studies the joint space they define. Section 8 measures geopolitical alignment and derives the coalition, Section 9 builds the Policy and Strategic indices, Section 10 the bilateral indices, and Section 11 the allied network design. Section 12 reports robustness analyses, and Section 13 discusses the results, next steps and limitations.

1.1 Related literature

The Allied Industrial Base lies where several literatures meet that are usually read separately (New Energy Industrial Strategy Center, 2026). We organise them by the streams that bear on its design and then turn to the measurement literatures on which the indices build.

Industrial policy and capabilities. Rodrik (2004) recast industrial policy as a process of discovery and strategic collaboration rather than the selection of winners, and Juhász et al. (2024) review a new empirical literature that is more favourable to it while stressing context, measurement and governance; text-based and intervention-level data now make policy measurable (Juhász et al., 2022; Evenett et al., 2024). The green variant rests on learning, coordination and environmental externalities (Rodrik, 2014), and its current form ties support to location (Bown, 2023; Bistline et al., 2023). This literature studies the relation between a state and its firms. The coordination failures it identifies, such as uncertain demand, missing suppliers and first-mover costs, do not stop at borders, and the governance of industrial interdependence among several states remains thinly studied. Our Policy index records the presence and density of policy by sector; it does not measure effectiveness.

Supply-chain resilience and economic security. Production-network models show that shocks to central suppliers need not average out (Acemoglu et al., 2012; Carvalho and Tahbaz-Salehi, 2019; Baqaee and Farhi, 2019), that disruptions propagate along supply chains (Carvalho et al., 2021), and that private sourcing decisions can leave networks fragile (Elliott et al., 2022). Grossman et al. (2023) analyse the choice between diversification and reshoring, and Baldwin and Freeman (2022) separate exposure from resilience. Policy work defines a critical dependency as the combination of high disruption risk, high importance and limited substitutability (OECD, 2025a), and finds diversification generally more effective than relocalisation (OECD, 2025b; Thakur-Weigold and

Miroudot, 2024). These results motivate scoring by stage, separating exposure (ES) from capability (EO), and imposing redundancy in the network design. Our indices do not model shock propagation, and the bilateral indices capture direct exposure only, not the indirect dependence that Freund et al. (2024) document.

Weaponized interdependence and allied scale. Farrell and Newman (2019) show how the hubs of global networks confer coercive power, which shifts attention from national market shares to network position, a modern form of the concern of Hirschman (1945). Campbell and Doshi (2025) argue that only a coalition can match China's productive scale and that alliances must pool capacity, not only political support. Friend-shoring (Yellen, 2022) and the evidence on fragmentation give this argument empirical content: trade and investment between geopolitical blocs have weakened relative to flows within them (Gopinath et al., 2025), U.S. sourcing has moved away from China toward economies that are themselves integrated with Chinese supply chains (Alfaro and Chor, 2023; Freund et al., 2024), and confining trade to aligned partners carries welfare costs (Góes and Bekkers, 2022; Javorcik et al., 2024; Aiyar et al., 2023).

Regional industrial policy and historical coordination. Albertone et al. (2025) show, for ASEAN, that cross-border industrial strategy must manage competition as well as cooperation, and that coordination without integration needs mechanisms that reveal synergies and monitor commitments. Historical cases point the same way. The Combined Boards of 1942–45 pooled the raw materials and production of the United States, the United Kingdom and Canada and allocated scarce output against total requirements (U.S. Department of State, 1942; U.S. War Production Board, 1944). Airbus combined national specialisation with shared demand and public risk-taking (European Commission, 2022; International Monetary Fund, 2024). The European Coal and Steel Community placed strategic sectors under common rules (European Union, 1951). Each worked around a concrete objective, clear roles and benefits reciprocal enough to sustain it.

Systemic bankability. Spokas et al. (2025) show that infrastructure technologies stall when any part of the surrounding system (offtake, permitting, enabling infrastructure, insurance, supply chains, delivery capacity) is missing, so that a designated capability is not a capability until it is bankable. Our indices do not measure bankability, which is the main reason the network design is a benchmark rather than a plan.

Competing schools. The policy debate divides into three schools (New Energy Industrial Strategy Center, 2026). The first defends multilateral openness and stresses the costs of fragmentation (Aiyar et al., 2023; OECD, 2025b). The second, closest to the Allied Industrial Base, accepts selective de-risking within managed interdependence. The third proposes a reciprocal bloc that conditions market access on balanced obligations and the exclusion of China (Cass, 2025). Our coalition rule belongs with the second: it admits nonaligned and even China-leaning economies through agreements or supply shares, and it fences off no sector as a whole.

Measurement. Four measurement literatures supply the building blocks. Energy-security indicators organise availability, dependence, diversity and affordability (Kruyt et al., 2009; Sovacool and Mukherjee, 2011; Ang et al., 2015; Jewell, 2011) and define security as the low vulnerability of vital systems (Cherp and Jewell, 2014; Winzer, 2012); mineral criticality combines supply risk with vulnerability (Graedel et al., 2012; Nassar et al., 2020; Kowalski and Legendre, 2023). Most such indices describe an energy system or a fuel, whereas ES applies their logic to each stage of each technology. Economic complexity treats exports as revealing capabilities (Balassa, 1965; Hidalgo et al., 2007; Hausmann and Klinger, 2006; Hidalgo and Hausmann, 2009; Hausmann et al., 2014), with relatedness predicting future specialisation (Hausmann et al., 2022) and applications to green products (Mealy and Teytelboym, 2022; Growth Lab at Harvard University, 2024); EO combines these signals with demand, cost, readiness and investment. The composite-indicator literature disciplines normalisation, weighting and uncertainty analysis (Nardo et al., 2008; Saisana et al., 2005; Paruolo et al., 2013), and our S-curve is the probability-weighting function of Karmarkar (1978). Alignment is measured with UN-vote ideal points (Bailey et al., 2017; Voeten, 2009), in the tradition of foreign-policy similarity measures (Signorino and Ritter, 1999) and with the bloc definitions of Gopinath et al. (2025). The network layer belongs to capacitated facility location and supply-chain network design (Geoffrion and Graves, 1974; Owen and Daskin, 1998; Melo et al., 2009; Daskin, 2013), with disruption-aware sourcing (Tomlin, 2006; Snyder et al., 2016) and the portfolio logic of diversification (Markowitz, 1952). It departs from that tradition by maximising an index-based score rather than minimising cost.

The gap. Read together, these literatures diagnose the problem better than they allocate a response (New Energy Industrial Strategy Center, 2026, Section 5): they establish why capabilities and concentrated dependencies matter, why allied scale can offset national limits and why autarky is costly, but not who should produce what, or how much duplication suffices. Nor do they offer a common measurement architecture that connects sector priorities, partner capabilities and performance. The indices of Carey et al. (2025) were a first step. They score eleven technologies across commodities, manufacturing and deployment on energy-security risk and economic opportunity, read the joint space as four strategic postures, and rate partners as safer, stronger or more prosperous with weights 0.4, 0.2 and 0.4. This paper takes them from rankings to network design. It rebuilds the pillars at the level of the cell with a rankability rule, derives the coalition from observed alignment and supply, rates every directed partnership, and adds an optimisation layer that turns the scores into a division of labour whose trade-offs are explicit and whose calibration is disclosed.

2 Conceptual framework

This section places the paper within the analytical architecture proposed for an Allied Industrial Base (New Energy Industrial Strategy Center, 2026), states which parts of that architecture the paper measures and which it leaves aside, and then defines the objects used in the rest of the paper.

Figure 1 summarises the construction and Table 3 collects the notation.

2.1 The allied industrial base as a design problem

The working paper defines an Allied Industrial Base by its purpose: trusted economies coordinate strategically important capabilities so that, together, they reach a resilience and scale that none could reach efficiently alone (New Energy Industrial Strategy Center, 2026). The object is the network: a capability should exist at sufficient scale somewhere among trusted partners, not in every member. The unit of policy is the *node*, a step such as mineral refining or component manufacture, because one supply chain can warrant different treatment at different steps. And the problem is one of allocation: which capabilities should be domestic, duplicated, specialised among partners or left to global markets.

The working paper orders the decision in three steps. *Severity* asks how damaging a disruption would be and how hard substitution or recovery would be; *coordination*, whether the problem needs international action rather than private diversification or national policy; and *specialization*, where the capability should reside and what should be duplicated. The evidence is organised in five analytical layers: node criticality and severity, country capability and opportunity, partner complementarity and reciprocity, systemic bankability, and network robustness under stress. The object is a combination of capabilities, not a ranking: a partner that ranks low in isolation can be indispensable if it supplies what the network lacks.

Table 1 maps the layers onto this paper. We measure each economy's relative exposure within a node by the need term $1 - \mathrm{ES}$ (rank-based, so it does not compare severity across nodes) and, for extraction, apply the endowment screen; capability by EO, with the Policy index recording where governments already concentrate their effort; complementarity, alignment and part of reciprocity by the coalition and the bilateral indices; and robustness only in part, by the diversification penalty and multi-sourcing constraints of the network design. Systemic bankability (Spokas et al., 2025), common-mode risk (beyond a first measure of the Chinese ownership of suppliers' foreign-invested plants), dynamic option value, reciprocal obligations, resilience floors and stress tests are not modelled.

Two simplifications matter. First, our cell, a technology–stage pair, is coarser than the working paper's node: Batteries Midstream, for example, groups mineral refining, electrode materials and cell manufacture. It is the finest level at which both pillars can be scored for most economies from public data. Second, the coordination step is assumed rather than tested: the 22 traded Upstream and Midstream stages are all treated as candidates, while Downstream deployment, largely non-traded, enters the Strategic Index but not the network design.

2.2 Cells, pillars and rankability

The unit of analysis is the cell (i, t, s): an economy $i \in \mathcal{I}$, a technology $t \in \mathcal{T}$ and a stage $s \in \mathcal{S}$, where Upstream is the extraction of fuels and minerals, Midstream their processing and the manufacture of components and equipment, and Downstream deployment and end use. $\mathcal{T}$ holds the eleven headline

Table 1: The analytical layers of an Allied Industrial Base and their counterparts in this paper

Layer	Measured here (section)	Not measured
1. Node criticality and severity	Country-relative exposure within each technology–stage pair, $1 - \mathrm{ES}_{its}$ (import dependence, supplier concentration, reserves; 5); endowment screen on extraction stages (9)	Severity compared across nodes; disruption horizons, inventories, qualification and build times, end-use criticality
2. Country capability and opportunity	EO_{its} by cell (6); Policy index and Strategic Index (9)	Plant-level cost curves, pipeline capacity, workforce, tacit know-how
3. Complementarity and reciprocity	Coalition rule from UN-vote alignment, agreements and supply shares (8); Friendshore, Partnership Opportunity and Development Potential (10); network edges (11)	Reciprocal obligations, export-restriction risk, market-access commitments
4. Systemic bankability	—	Offtake, price risk, permitting, enabling infrastructure, insurance, delivery capacity
5. Network robustness	Concave penalty on the Herfindahl index of designed shares, share bounds and multi-sourcing constraints, calibrated on observed concentration (11); Chinese-parent exposure in four Midstream cells (10)	Stress tests, common-mode failure through shared inputs, logistics or equipment, stockpiles, recovery time, dynamic option value

Notes: Layers and their content follow New Energy Industrial Strategy Center (2026); numbers in parentheses refer to sections of this paper.

technologies, fossil fuels included, so an economy has at most 33 cells, and $\mathcal{I}$ holds 219 economies after harmonisation, 195 sovereign states and 24 territories (Section 3).

Energy Security ES_{its} measures how secure i's supply of technology t at stage s is, higher values being more secure. It combines exposure (import dependence, the concentration of supplying markets) with resilience (reserves, production, consumption, access and investment). Economic Opportunity EO_{its} measures how well placed i is to capture value as a producer, from comparative advantage, export feasibility, demand, cost, technological readiness and investment. Security describes an economy as a user and opportunity as a producer, so the two can diverge: a fuel importer with a strong manufacturing base can be insecure Upstream and well placed Midstream. Shared components make part of their association mechanical (Section 7).

Both pillars are built in the same way (Section 4). Each raw variable x_{itsv}, oriented so that higher is better, is replaced by the S-curve of its percent rank within a comparison set g, usually the economies observed in the pair (t, s),

$$\tilde{x}_{itsv} = \phi_\gamma(r_{itsv}), \qquad \phi_\gamma(r) = \frac{r^\gamma}{r^\gamma + (1-r)^\gamma}, \qquad r_{itsv} = \frac{\mathrm{rank}_g(x_{itsv}) - 1}{n_g - 1}, \qquad \gamma = 0.5, \tag{1}$$

the probability-weighting function of Karmarkar (1978), which stretches the tails of a distribution relative to its middle. A category averages the components observed, or filled by an explicit missing-data rule, for the cell,

$$c_{itsk} = \frac{1}{|\mathcal{V}^*_{itsk}|} \sum_{v \in \mathcal{V}^*_{itsk}} \hat{x}_{itsv}, \tag{2}$$

and the pillar is the weighted mean over the categories $\mathcal{K}^P_{its}$ observed for the cell,

$$P_{its} = \frac{\sum_{k \in \mathcal{K}^P_{its}} w^P_k \, c_{itsk}}{\sum_{k \in \mathcal{K}^P_{its}} w^P_k}, \qquad P \in \{\mathrm{ES}, \mathrm{EO}\}, \tag{3}$$

with ten categories and fixed weights per pillar (2 to 6 for ES, 2 to 4 for EO). An unobserved category drops out with its weight rather than counting as zero, and each pillar decomposes exactly into category and variable contributions. A cell enters a ranking, a top-N list or the Strategic Index only if it is *rankable*: the economy is sovereign and at most half of the cell's index weight is imputed (equation (22)). This holds for 4,163 of 6,232 scored ES cells and 4,301 of 7,131 EO cells.

2.3 Policy and the Strategic Index

The Policy index POL_{its} records where a government directs its policy. It is the weighted mean of an index of the measures in the IEA Policies and Measures database tagged to the cell (weight 3, required) and a component that interacts the country's Climate Action Tracker rating with the technology's emission intensity (weight 2, where the country is rated). It covers 215 economies, 65 on both components and 150 on the IEA component alone, and the basis matters (Section 9). Measures from the New Industrial Policy Observatory enter only as a robustness variant (Section 12). The Strategic Index ranks the sectors of each country by the case for building there,

$$\mathrm{SI}_{its} = 0.2\,\mathrm{POL}^*_{its} + 0.3\,\mathrm{EO}_{its} + 0.3\,(1 - \mathrm{ES}_{its}) + 0.1\,\lambda_s + 0.1\,g_t, \tag{4}$$

where POL^*_{its} is the S-curve of POL across i's own scored sectors, the pillars enter at their cross-country levels, λ_s is a stage weight (Midstream 0.75, Upstream 0.5, Downstream 0.25) and $g_t \in [0, 1]$ rises as lifecycle greenhouse-gas intensity falls. A cell is scored only if the country has a Policy index and both pillar cells are rankable; nothing is imputed.

Security enters as *need*, for a reason of marginal value. Let V_i be the value to i of supply security in a sector, increasing and concave, and ρ_i the probability that an effort to build capacity there succeeds, increasing in opportunity and in existing policy commitment. An effort that raises security by Δ is worth about

$$\rho_i(\mathrm{EO}_{its}, \mathrm{POL}_{its})\, V_i'(\mathrm{ES}_{its})\, \Delta, \tag{5}$$

which rises with opportunity and policy and falls with the security already attained. The index linearises this ranking, so sectors in which a country is exposed but capable rank first; reading security as strength would rank existing strengths first, which EO already describes. Equation (5) motivates the signs, not the additive form or the weights, which are normative (Decancq and Lugo, 2013); Section 12 reports the strength reading.

Each coalition member's *allied pick* is its highest-scoring Upstream or Midstream sector outside geothermal and green hydrogen. Read as need, ES gives an economy without a resource its largest need at the extraction stage, so an *endowment screen* admits an Upstream stage to the pick only if

the member has recorded reserves or production of the stage's resources. The screen is a selection rule, not part of the index: values are unchanged, and 669 of the 998 scored Upstream cells lack a resource base. The network design uses neither the index nor the picks.

2.4 Coalition, bilateral indices and network design

The coalition. The coalition $\mathcal{N}$ is derived by rule (Section 8). Each voting economy is placed on a U.S.–China axis $s_i \in [0,1]$ from its UN General Assembly ideal point (equation (50)), and quartiles of s_i define U.S.-leaning, nonaligned and China-leaning blocs. The 31 *core* members are the U.S.-leaning economies with GDP of at least USD 100 billion. The 26 *partners* are members of a U.S. free-trade agreement, the Indo-Pacific Economic Framework, the Minerals Security Partnership, the G7 or NATO above the same GDP floor, and major suppliers outside the China-leaning bloc without a GDP floor: at least 5 percent of world production, refining, manufacturing capacity or exports in an Upstream or Midstream stage, or at least 10 percent of world reserves of a headline mineral, fuel or uranium. With the six EU members below the GDP floor the coalition spans 63 economies; the network treats the EU-27 as one node with stage-weighted member inputs, which gives 37 nodes. The dyadic alignment A_{ij} (equation (51)), the probability that two economies fall on the same side of the divide, links the coalition to the bilateral layer.

Bilateral indices. For a reporter i, a partner $j \neq i$ and a pair (t,s), Friendshore FS_{ijts} rates j as a supplier to i from i's imports from j, i's need, j's opportunity, i's outward investment in j, the alignment A_{ij} and, in four manufacturing stages, the share of j's foreign-invested plants without Chinese parents, less a climate-policy penalty (equation (66)). Partnership Opportunity PO_{ijts} rates j as a market for i's exports, and Development Potential DP_{jts} scores partners that are not high-income economies. The Partnership Strength Index combines best-five summaries of the three with weights 0.4, 0.4 and 0.2, those of the partnership score of Carey et al. (2025); the headline averages the views of the 31 core members. Table 2 shows how security and opportunity enter each index.

Network design. For each of the 22 Upstream and Midstream stages, a mixed-integer program over the 37 nodes chooses producers ($y_p \in \{0,1\}$), supplier links ($\ell_{pc} \in \{0,1\}$) and flows $f_{pc} \geq 0$, the share of the network's unit demand that p supplies to c, with production shares $x_p = \sum_c f_{pc}$:

$$\max_{y,\ell,f} \; \sum_p \sum_c f_{pc}(a_p + 0.5\, W_{pc}) - \lambda \sum_p x_p^2 \quad \text{s.t.} \quad \sum_p f_{pc} = d_c, \quad 0.05\, y_p \leq x_p \leq 0.40\, y_p. \tag{6}$$

The producer score a_p weights the S-curves of EO and ES across the nodes by 5/7 and 2/7, without a policy term; the edge weight $W_{pc} = 0.5\,\mathrm{FS}_{cpts} + 0.5\,\mathrm{PO}_{pcts}$ reads each directed index from the end of the relationship it describes; and the demand weight d_c mixes the consumer's need and economic size. At least three nodes produce, and every consumer draws on at least two suppliers, each providing at least 10 percent of its demand. The penalty on the Herfindahl index of production

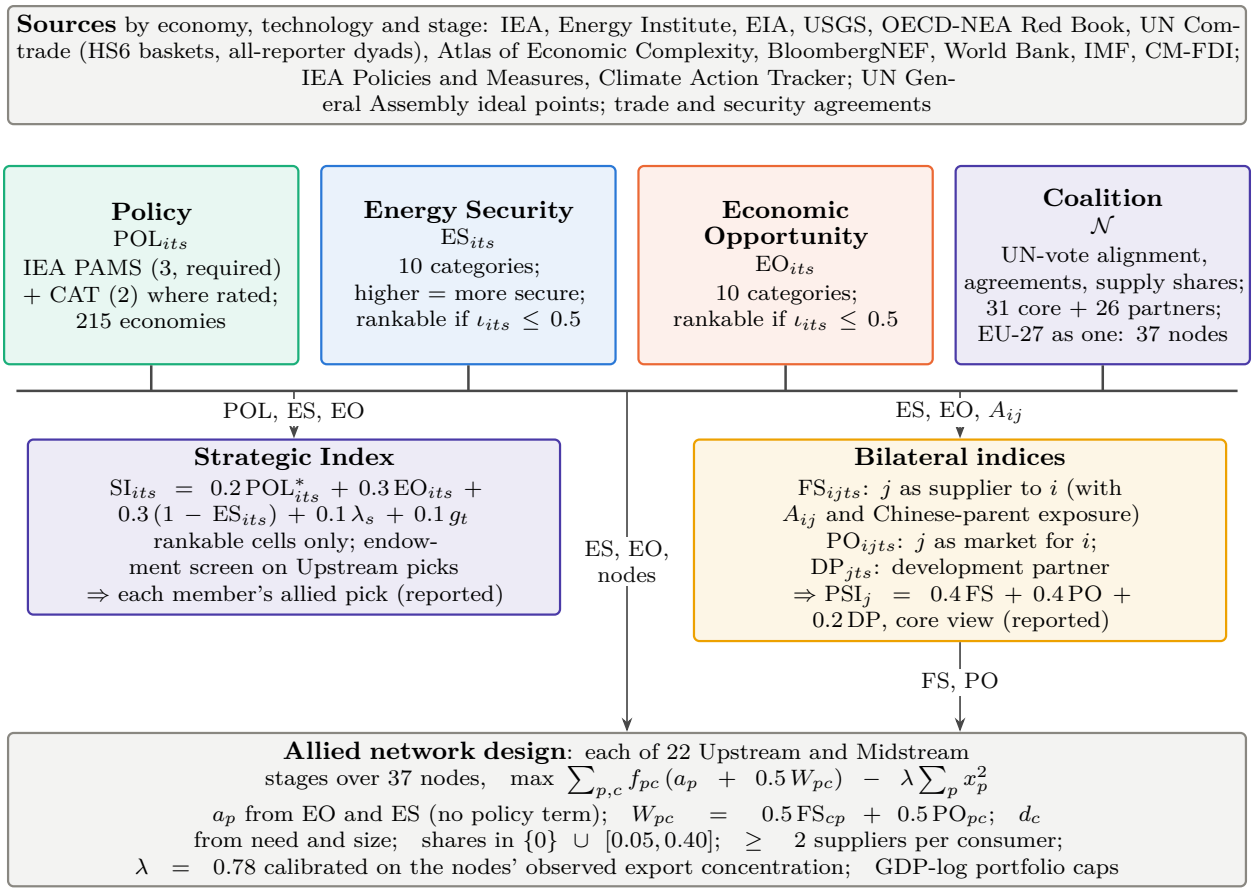


Figure 1: Architecture of the analysis. Sources feed four cell- or economy-level constructs: the Policy index, the two pillars, and the coalition, derived from alignment in the UN General Assembly, agreements and supply shares. The Strategic Index ranks each country's sectors and yields the allied picks; the bilateral indices rate partners as suppliers and markets and are summarised in the Partnership Strength Index. The network design allocates production among the coalition's nodes. In the terms of New Energy Industrial Strategy Center (2026), the pillars, the Policy index and the Strategic Index address relative exposure and capability, the coalition and the bilateral indices complementarity, and the network design specialisation and, in part, robustness. ι_{its} is the imputed share of a cell's index weight; the Strategic Index and the Partnership Strength Index are not inputs to the network design.

shares makes the program spread production among nodes rather than rank them, a first step toward optimising combinations. Because the penalty treats each node as an independent source, it does not register that producers sharing a bloc, a region or a parent company diversify less; Section 13 discusses a correlated-risk version. Its weight is calibrated: $\lambda = 0.78$ is the smallest value at which the median designed Herfindahl index across stages does not exceed the median observed export concentration of the nodes, a knife-edge choice that Section 11 discloses. An outer loop caps, by the logarithm of GDP, the number of stages in which a node is among the five largest producers.

Table 2: How security and opportunity enter each index

Index	Whose position	Energy security enters as	Economic opportunity enters as
Strategic Index SI_{its}	country i, deciding where to build	need of i: $1 - \mathrm{ES}_{its}$ (level), weight 0.3	capability of i: EO_{its} (level), weight 0.3
Friendshore FS_{ijts}	importer i, supplier j	need of the importer, ranked across i's own cells, weight 2 of 11	capability of the supplier, ranked across economies in (t, s), weight 2 of 11
Partnership Opportunity PO_{ijts}	exporter i, market j	need of the market, ranked across markets, weight 1 of 7	capability of the exporter, ranked across economies, weight 3 of 7
Development Potential DP_{jts}	partner j, not high income	need of the partner, weight 1 of 10	opportunity of the partner, weight 1.5 of 10
Network producer score a_p	node p as producer	security of the producer, S-curve across nodes, weight 2/7	capability of the producer, S-curve across nodes, weight 5/7
Network demand d_c	node c as consumer	need of the consumer, weight 0.5 (with size, 0.5)	—

Notes: Weights are nominal, before renormalisation over the components observed. Security enters as need wherever it describes the party that would gain from new supply, and as security only in the producer score of the network design, where a node's own security in a stage proxies its reliability as a supplier. Sections 9, 10 and 11 give the full definitions.

Table 3: Notation

Symbol	Meaning	Defined in
Sets and indices		
$\mathcal{I}$	the 219 economies after harmonisation	Sec. 3
$\mathcal{T}$, $\mathcal{S}$	eleven headline technologies; {Upstream, Midstream, Downstream}	Sec. 2.2
$\mathcal{N}$	the 57 coalition members (31 core, 26 partners); 37 network nodes	Sec. 8
i, j; t, s; k, v	economy (reporter), partner; technology, stage; category, variable	
p, c	network node as producer; as consumer	Sec. 11
(i, t, s), (i, j, t, s)	cell; directed dyad with reporter i and partner $j \neq i$	Secs. 2.2, 10
Normalisation and pillars		
x_{itsv}, r	raw variable; percent rank within comparison set g	Eq. (1)
ϕ_γ, γ	S-curve; curvature, $\gamma = 0.5$; $\phi_\gamma[\cdot]_g$ ranks within g	Eq. (1)
$\tilde{x}_{itsv}$, $\hat{x}_{itsv}$	theme index; after the missing-data rule	Eqs. (1), (2)
c_{itsk}, w_k^P	category score; weight of category k in pillar P	Eqs. (2), (3)
ES_{its}, EO_{its}	Energy Security and Economic Opportunity pillars, in $[0, 1]$	Eq. (3)
ι_{its}	imputed share of the cell's index weight; rankable if $\iota_{its} \leq 0.5$	Eq. (22)
Policy and Strategic Index		
POL_{its}, POL^*_{its}	Policy index; its S-curve across i's scored sectors	Eqs. (54), (56)
λ_s, g_t	stage weight; technology emissions weight (1 = cleanest)	Eqs. (4), (57)
SI_{its}	Strategic Index	Eq. (4)
Alignment and bilateral indices		
s_i, A_{ij}	position on the U.S.–China axis; dyadic alignment	Eqs. (50), (51)
FS_{ijts}, PO_{ijts}, DP_{jts}	Friendshore; Partnership Opportunity; Development Potential	Eqs. (66), (68), (69)
PSI_j	Partnership Strength Index	Eq. (72)
Network design		
a_p, W_{pc}, d_c	producer score; edge weight; demand weight ($\sum_c d_c = 1$)	Eq. (6)
y_p, ℓ_{pc}, f_{pc}, x_p	producer selection; supplier link; flow share; production share	Eq. (6)
λ, HHI	diversification weight; Herfindahl–Hirschman index	Eq. (6)

Notes: The stage weight λ_s carries a subscript and the diversification weight λ none; the axis position s_i carries an economy subscript and the stage s none. The production share x_p and the policy components x^v_{its} of Section 9 are distinct from the raw variable x_{itsv}. Later sections introduce further symbols where they are needed.

3 Data

The indices combine public statistical releases, licensed commercial datasets and a few project-authored tables. Appendix B lists every source with its publisher, coverage and release, Appendix A every variable built from it, and Online Appendix OA.1 the rule that selects a release, the harmonisation of economy names and the years each source covers.

3.1 Sources by pillar

Table 4 groups the sources by what they measure. Most report economies, but some report the world (commodity prices, demand projections, technology readiness) or a few regions (manufacturing capacity, relative equipment costs); such a variable shifts a pair's level without ranking economies within it. Coverage ranges from about two hundred economies for trade and energy statistics to a few dozen for levelised costs, a gap that the missing-data rules of Section 4.3 bridge.

Energy security. The Statistical Review of World Energy (Energy Institute, 2026), 2026 edition with data to 2025, supplies consumption per capita and its growth, the fuel balances behind energy imports, mineral production and the proved reserves of oil, gas and coal, which the Review holds at end-2020. Fossil production that the Review counts only in a regional residual is taken from the U.S. Energy Information Administration (U.S. Energy Information Administration, 2026), in the Review's units and paired with consumption of the same year, as an observation (Estonian coal, where the Review counts oil shale, stays missing). Critical-mineral reserves at end-2025 (cobalt, lithium, graphite, rare earths, copper, manganese, nickel, zinc and the platinum-group metals) come from the USGS Mineral Commodity Summaries 2026 (U.S. Geological Survey, 2026), with the erratum that puts Brazil's rare-earth reserves at 11 Mt; holders listed without a figure ("NA", "W", "s") count as zero in the technology roll-ups. Nuclear Upstream is scored on the OECD-NEA/IAEA Red Book (OECD Nuclear Energy Agency and International Atomic Energy Agency, 2026): identified recoverable resources below USD 260/kgU at 1 January 2025 (59 economies) and mine production in 2019–2024 (18 producers in 2024), with economies absent from the tables recorded as zero. The IEA Critical Minerals Dataset (International Energy Agency, 2026a) provides mining (Upstream) and refining (Midstream) supply by economy and the clean-technology demand that weights each mineral within a technology. Midstream capacity shares come from IEA Energy Technology Perspectives (International Energy Agency, 2024), taken over the world total including the source's "Other" row. Its 2035 announced-capacity shares are regional for Europe, so every EU member carries the EU's share (0.15 in Wind, 0.11 in Batteries, 0.24 in Green Hydrogen and 0.00 in Solar Midstream) in the foreign-dependency and market-share variables, which therefore do not separate the members. Commodity prices (IMF Primary Commodity Price System, International Monetary Fund, 2026b), clean-technology investment (Clean Investment Monitor, Rhodium Group and MIT Center for Energy and Environmental Policy Research, 2026) and three geospatial resource-potential datasets complete the pillar.

Table 4: Source families, levels of observation and what they feed

Family	Sources	Level	Feeds
Energy statistics	EI Statistical Review 2026; EIA International Energy Statistics (fuel production EI does not report)	Economy	ES: energy access, consumption, energy imports, reserves (fossil), production (minerals). EO: energy access, production
Reserves and uranium	USGS Mineral Commodity Summaries 2026; OECD-NEA/IAEA Red Book (Uranium 2026); EI proved reserves	Economy	ES: reserves, production (uranium). EO: production (uranium). Coalition supplier test
Trade	UN Comtrade all-reporter dyads (2020, 2024) and their World totals; Atlas of Economic Complexity (HS92); Comtrade critical-minerals extracts	Economy; dyad	ES: trade, minerals trade. EO: trade. Friendshore, Partnership Opportunity, network edges
Minerals and manufacturing	IEA Critical Minerals Dataset 2026; IEA ETP 2024 capacity shares; IEA EV regional data	Economy; 5–6 regions	ES and EO: foreign dependency, production; mineral demand weights
Demand and technology	IEA WEO 2025; IEA Global EV Outlook 2026; BNEF New Energy Outlook 2024 and Supply Chains 2025; BCG market sizing; IEA Clean Energy Technology Guide	World; region; United States	ES: consumption. EO: consumption, technology demand, technological readiness
Costs and prices	BNEF LCOE; IEA relative costs; ILOSTAT earnings; IMF lending rates, producer prices and commodity prices	Economy; region; world	ES: energy prices. EO: cost competitiveness, energy prices
Potential and investment	Global Solar and Wind Atlases; geothermal potentials; Clean Investment Monitor	Economy	ES: reserves (downstream). ES and EO: investment
Policy	IEA Policies and Measures; Climate Action Tracker; IPCC lifecycle emissions; GTA NIPO (robustness only)	Economy; technology	Policy index; technology weight of the SI
Bilateral and alignment	IMF Direct Investment Positions; CM-FDI projects; WDI; Doing Business; OECD CRS; UNGA ideal points; agreement memberships	Dyad; economy	PSI components; blocs, coalition, network demand and caps

Notes: ES = Energy Security; EO = Economic Opportunity; SI = Strategic Index; PSI = Partnership Strength Index. A source reported for a region or for the world is copied to the economies it covers by the theme that reads it. Appendix B gives publishers, coverage and releases; Appendix A the variables.

Economic opportunity. Export capability rests on the Growth Lab's reconciled HS92 data underlying the Atlas of Economic Complexity (The Growth Lab at Harvard University, 2019; Hausmann et al., 2014), newest year 2023, and on the 2024 Comtrade exports of Section 3.3. Demand comes from the IEA World Energy Outlook 2025 and Global EV Outlook 2026 (International Energy Agency, 2025b, 2026d), BloombergNEF's New Energy Outlook 2024 (BloombergNEF, 2024) and Supply Chains 2025 workbook, and a Boston Consulting Group sizing of the U.S. addressable market. Costs combine BloombergNEF levelised costs, IEA relative equipment costs, ILOSTAT earnings, IMF lending rates (placeholder zero readings dropped) and producer prices. The BloombergNEF and BCG inputs are licensed, which prevents a fully public replication of this pillar.

Technology readiness by stage. The IEA Clean Energy Technology Guide (International Energy Agency, 2026c) rates technology designs and components on an eleven-point readiness scale. Ordered rules assign each item to a technology and a stage or classify it as out of scope: 253 of the 634 items of the 2026 release map to a headline technology. A cell's readiness is computed from the items of its own stage; a stage without an item, 10 of the 33 cells, takes the technology-wide value. Readiness has no country dimension and is copied to every economy as an observation.

Policy. The Policy index combines the IEA Policies and Measures database (International Energy Agency, 2026e) (export of 22 January 2026), weight 3, and the Climate Action Tracker ratings (Climate Action Tracker, 2026) (39 countries and the EU, whose rating its unrated members take), weight 2. IEA PAMS is required and CAT added where it rates the country: 215 economies are scored, 65 on both and 150 on IEA PAMS alone (Section 9). The New Industrial Policy Observatory of Global Trade Alert (Global Trade Alert, 2026) enters only a robustness variant (Section 12). The technology weight of the Strategic Index uses the IPCC lifecycle-emission medians (Schlömer et al., 2014).

Bilateral, alignment and coalition inputs. Outward direct-investment positions come from the IMF Direct Investment Positions dataset. Positions with 16 conduit financial centres, the ten pass-through economies identified by Damgaard et al. (2024) and six offshore centres (Appendix A), are treated as unobserved on either side of the dyad. In battery, electric-vehicle, solar and wind manufacturing, the investment tie and the Chinese-parent exposure of Friendshore come instead from the Cleantech Manufacturing FDI dataset (Bandara and Net Zero Industrial Policy Lab, 2026), of whose 1,042 cross-border projects 905 qualify. Development potential uses the World Development Indicators (World Bank, 2026), the 2019 Doing Business scores (World Bank, 2020; World Bank Group, 2021) and OECD aid data. Geopolitical alignment uses the all-votes UN General Assembly ideal points of Bailey et al. (2017) (Dataverse V39, Voeten, 2009), averaged over sessions 74–78 (2019–2023), and coalition membership a project-authored list of U.S. free-trade agreements, IPEF, the Minerals Security Partnership, the G7 and NATO compiled from the official lists (Section 8).

3.2 The HS6 product crosswalk

Trade enters through baskets of six-digit Harmonized System (HS6) products, one per technology–stage pair, defined in one version-controlled table (`data/reference/energy_hs6_master.csv`) from which every trade and bilateral file is generated. With H_{ts} the codes in the basket of (t, s) and x_{ih} economy i's exports of product h, basket exports are

$$X_{its} = \sum_{h \in H_{ts}} x_{ih}, \tag{7}$$

and imports, world totals and bilateral flows are aggregated in the same way. Ores, concentrates, slag and scrap are Upstream, processed materials and components Midstream; Downstream activity is not a traded-goods stage.

The basket review. In October 2026 every code was tested for dominant use: it stays in a technology's basket only if that technology is a main use of the product. Ninety-two basket rows touching 69 codes failed (47 codes left every basket); each stays in the master with its reason. The crosswalk now holds 259 codes in 415 basket rows (298 in 490 before): 242 of the current nomenclature and 17 legacy codes that bridge the HS92 Atlas panels. The exclusions carried 92 percent of the pre-review coal Midstream basket's trade, 69 percent of gas Midstream and 89 percent of batteries Upstream, whose copper ores now belong to the grid (Table 5). Electric Grid Upstream held only electrical energy (HS 2716.00), which made net power exporters the grid's leading suppliers; it now holds seven copper and aluminium inputs, from ores and bauxite to cathodes and unwrought aluminium, so electricity is outside every basket. Switching and routing apparatus (HS 8517.62), 32 percent of the pre-review grid Midstream basket, is telecommunication equipment and left it.

Upstream baskets. Five Upstream baskets are new: Electric Vehicles (the battery-mineral ores and the rare earths of permanent magnets), Wind (rare earths), Nuclear (uranium ores and natural uranium compounds, the heading of most yellowcake, moved from Midstream), Solar (silicon metal) and Semiconductors (gallium, germanium and indium, carried for description only). The four that are scored trade USD 3.0 billion (wind) to 21.3 billion (electric vehicles) in 2024. With them, 18 of the 22 scored Upstream and Midstream pairs have a basket; only Geothermal and Green Hydrogen have none.

A many-to-many mapping. The baskets do not partition the product space: 105 codes (41 percent) belong to more than one basket, 63 to two, 33 to three and 9, the transformer, switchgear and cable headings, to four, since a power cable is an input to every technology that connects to the grid. Technology totals are therefore not additive, and the trade variables of technologies that share codes are mechanically correlated.

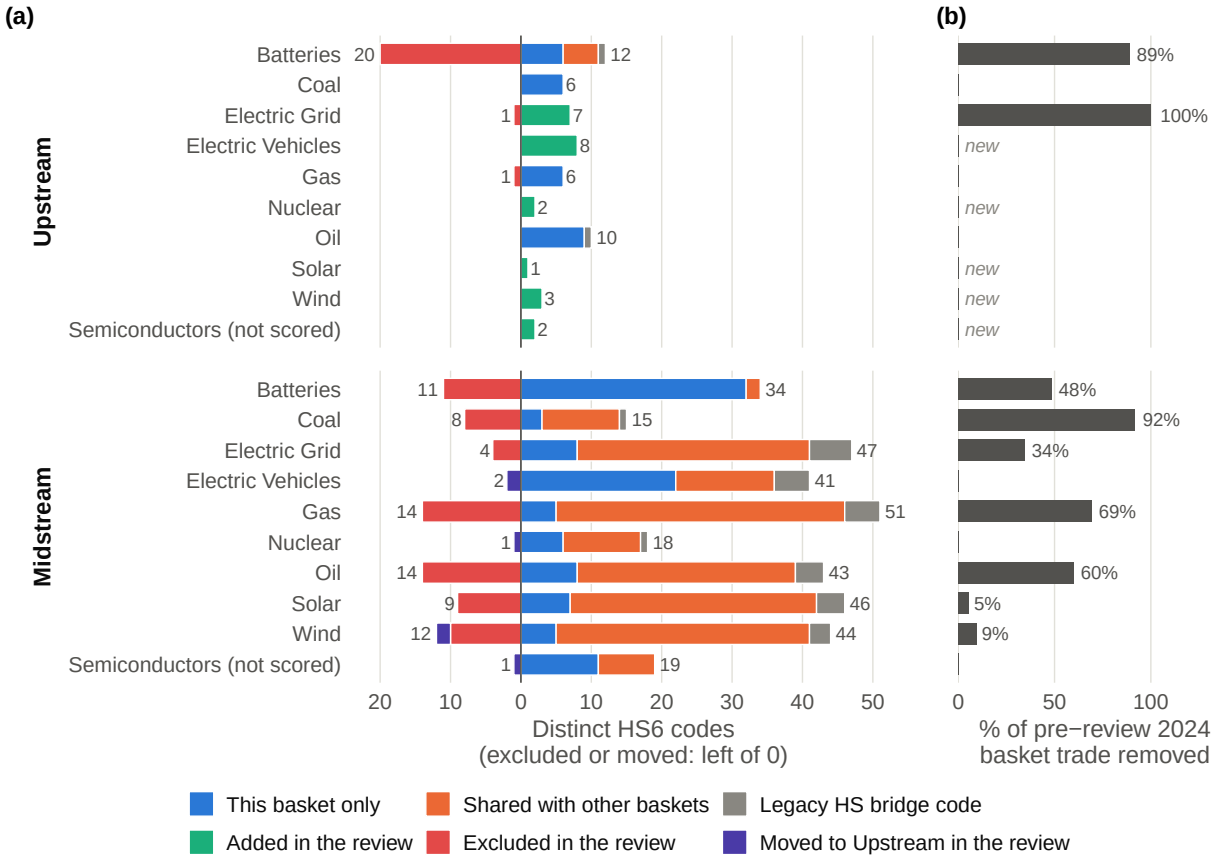


Figure 2: The HS6 baskets after the October 2026 review. Panel (a): distinct HS6 codes in each technology–stage basket, split into codes of that basket only, current codes shared with at least one other basket (the many-to-many part of the mapping), legacy HS bridge codes, and codes added by the review (codes moved from a technology's midstream basket count as added to its upstream basket); left of zero, the codes the review excluded from the basket and the codes it moved to the upstream stage. Panel (b): share of the pre-review basket's 2024 trade value carried by the excluded codes; "new" marks a basket the review created. Trade is the reconciled all-reporter Comtrade matrix (the importer's declaration, else the exporter's declaration $\times 1.06$), summed over exporter–importer pairs; a code in several baskets counts in each. Green hydrogen, geothermal and the downstream stage have no basket; semiconductors are carried in the master but not scored. Source: `data/reference/energy_hs6_master.csv`, `data/raw/comtrade_energy_dyads_all.csv`; `scripts/paper/fig_hs6_crosswalk.R`.

Table 5: HS6 baskets by technology and stage after the October 2026 basket review

		Codes in the basket				Review			Basket trade, 2024	
Technology	Stage	All	Legacy	Shared	Sub-sectors	Added	Moved	Excluded	USD bn	% removed
Batteries	Upstream	12	1	5	2	0	0	20	21.2	88.8
	Midstream	34	0	2	9	0	0	11	202.7	48.4
Coal	Upstream	6	0	0	2	0	0	0	192.5	0.0
	Midstream	15	1	12	3	0	0	8	20.5	91.9
Electric Grid	Upstream	7	0	0	3	7	0	1	321.0	100.0
	Midstream	47	6	36	12	0	0	4	449.6	34.3
Electric Vehicles	Upstream	8	0	8	8	8	0	0	21.3	new
	Midstream	41	5	17	7	0	2	0	1,003.2	0.0
Gas	Upstream	6	0	0	4	0	0	1	445.3	0.4
	Midstream	51	5	46	2	0	0	14	151.6	69.1
Nuclear	Upstream	2	0	0	2	2	0	0	11.4	new
	Midstream	18	1	12	3	0	1	0	24.6	0.0
Oil	Upstream	10	1	0	5	0	0	0	2,211.0	0.0
	Midstream	43	4	35	7	0	0	14	127.1	59.8
Solar	Upstream	1	0	0	1	1	0	0	3.8	new
	Midstream	46	4	38	9	0	0	9	711.5	5.5
Wind	Upstream	3	0	3	3	3	0	0	3.0	new
	Midstream	44	3	39	8	0	2	10	409.5	9.4
Semiconductors[a]	Upstream	2	1	0	1	2	0	0	1.1	new
	Midstream	19	0	8	8	0	1	0	1,348.1	0.0
Distinct codes		259	17	105	92	20	4	69		

Notes: Distinct HS6 codes in each technology–stage basket of `data/reference/energy_hs6_master.csv` after the review. "Legacy" codes are bridge codes of older HS revisions whose successors are all in the basket; "Shared" counts the codes that also belong to at least one other basket. "Excluded" counts the codes the review removed from the basket because the technology is not a dominant use of the product, and "Added" the codes it added (the copper and aluminium inputs that replace electrical energy, HS 2716.00, in electric-grid Upstream, and the Upstream baskets of electric vehicles, solar, wind, nuclear and semiconductors, which had none); "Moved" counts the codes it moved from a technology's Midstream to its Upstream basket (rare-earth compounds and metals, natural uranium compounds, unwrought gallium, germanium and indium), which count as added there, since a code sits in one stage of a technology; the rows stay in the master with the reason. Basket trade is the 2024 value in the reconciled all-reporter Comtrade matrix (the importer's declaration, else the exporter's mirror declaration × 1.06), summed over exporter–importer pairs; "% removed" is the share of the pre-review basket's value carried by the excluded codes (moved codes are not removals; "new" marks a basket the review created). A code in several baskets counts in each, so neither codes nor trade values add across rows. Green hydrogen, geothermal and the Downstream stage have no basket.

[a] Carried in the master; not a scored technology.

3.3 Bilateral trade and the trade themes

Bilateral flows come from an all-reporter pull of UN Comtrade (United Nations Statistics Division, 2026) for the endpoint years 2020 and 2024: 173 economies report in at least one year (168 in 2020; 142 for exports and 143 for imports in 2024), and 208 appear as reporter or partner. Comtrade records Taiwan as "Other Asia, n.e.s." (partner code S19), which is mapped to Taiwan's own code. Let $M^{\text{rep}}_{ji,h}$ be importer j's declared imports of product h from i and $X^{\text{rep}}_{ij,h}$ exporter i's declared exports to j. The flow in a cell is

$$F_{ij,ts} = \begin{cases} \sum_{h \in H_{ts}} M^{\text{rep}}_{ji,h} & \text{if } j \text{ reports imports in the cell that year,} \\ 1.06 \sum_{h \in H_{ts}} X^{\text{rep}}_{ij,h} & \text{otherwise, if } i \text{ reports exports to } j, \end{cases} \tag{8}$$

and unobserved otherwise; 1.06 is the IMF's conventional ratio of CIF import to FOB export values. Of the 701,214 exporter–importer–technology–stage cells observed in 2024, 22 percent are mirror values, but only 13 percent of the cells with positive trade and 4 percent of the value, because large traders report. Thirty importers that reported in 2020 did not in 2024, among them Russia, Iran, the United Arab Emirates and Viet Nam, and 65 of the 208 importers enter the 2024 matrix only through their partners' declarations, which omits their imports from exporters that do not report either.

The country-level trade themes read a file of exports and imports by reporter and code rebuilt from the World rows of the same pull: 142 exporting and 143 importing reporters in 2024 and every current-nomenclature basket code, including Qatar's exports of liquefied natural gas (USD 56.9 billion), which the earlier extract omitted. Hong Kong is left out of the Comtrade export measures (export size, revealed comparative advantage (Balassa, 1965) and export concentration) because its exports are mostly re-exports that the producers' own exports already count; it keeps its Atlas-based variables, and the bilateral matrix is not filtered. The export-concentration index is ranked across the cells of the eleven headline technologies only. Critical-minerals trade uses separate 2024 Comtrade extracts (130 import and 128 export reporters); the 2025 extracts, with three reporters, fall below the coverage floor of the release rule (Online Appendix OA.1).

3.4 Economies, coverage and rankability

Every economy name in every source is mapped to one canonical name per ISO3 code before anything is ranked, and world, regional, residual and historical aggregates are dropped. The headline pillars then cover 219 economies, each with its own ISO3 code, of which 195 are sovereign states (the 193 members of the United Nations, Kosovo and Taiwan) and 24 are territories, special administrative regions or the West Bank and Gaza; the same 219 economies (195 sovereign states and 24 territories) have an EO score, so the two pillars share one universe. Taiwan, which the World Bank's country table does not list, is carried as an economy throughout, including the trade themes and the bilateral matrix; it has an ES value in 28 headline cells and an EO value in 33. The pillars score 6,232 ES and 7,131 EO country–technology–stage cells in 30 and 33 headline pairs. In the alignment data

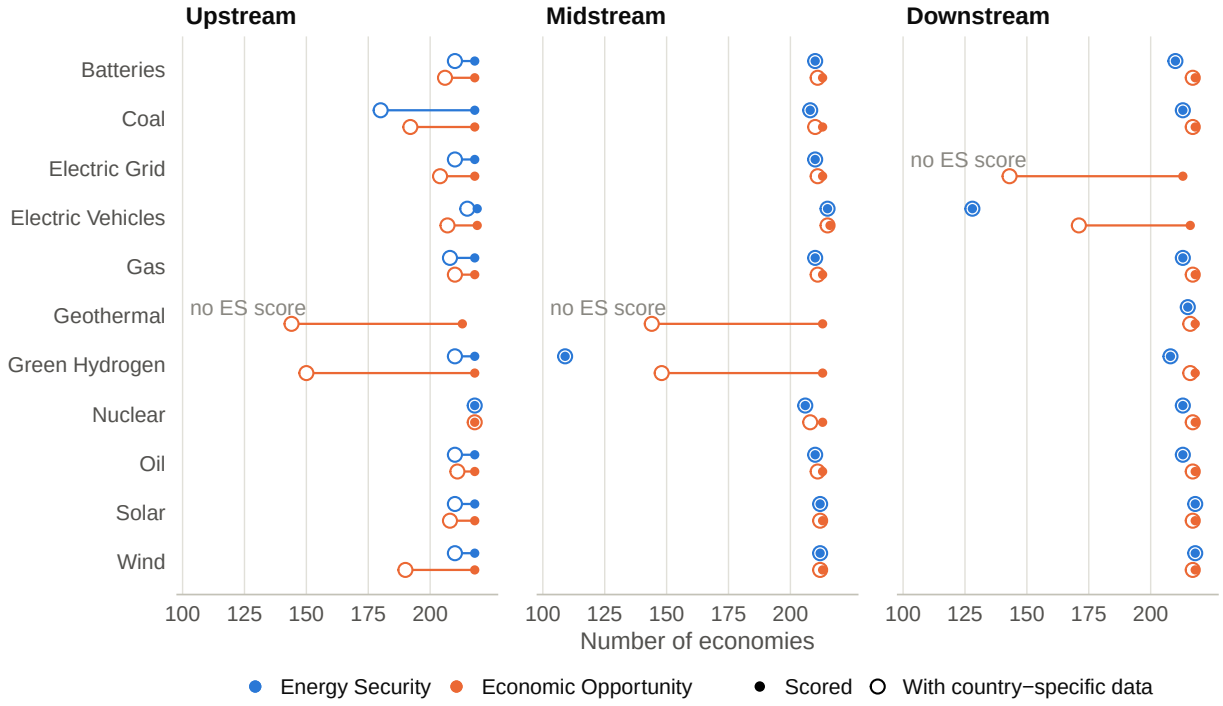


Figure 3: Economies scored and economies with country-specific data, by technology–stage pair. Filled dots: the number of economies with a pillar value in the pair. Hollow circles: the number of those economies for which at least one positively weighted, non-imputed component is country-specific, that is, takes at least two distinct observed values across the economies of the pair. World-level series copied to every economy, such as commodity-price volatility, world export concentration and technology readiness, do not count as country-specific. The segment between the two marks is the number of economies whose value rests only on imputed components and world-level series; where the two counts coincide the marks appear as a ringed dot. "No ES score" marks the pairs without an ES value. Every pair has status ok (no pair is constant or unscored). The axis starts at 100 economies. Sample: the eleven headline technologies. Source: variable contributions and pillar diagnostics in `data/processed/outputs/index_outputs.rds`; `scripts/paper/fig_method.R`.

the 193 economies that vote in the General Assembly have an ideal point; the other economies are treated as Section 8 describes.

The grids span almost the whole universe at every stage, but their width is set by the most widely distributed theme of each pair. Figure 3 separates economies scored from economies with country-specific data. In ES the two differ only upstream, where a commodity-price series copied to every economy sets the grid: by 4 economies in Electric Vehicles to 38 in Coal, and not at all in Nuclear once absence from the uranium registers is recorded as zero. In EO the gap is largest in Electric Grid downstream (143 of 213 economies with country-specific data). Territories rest more heavily on imputation (mean imputed weight 0.48 of a territory's ES cell against 0.29 for a sovereign state; 0.67 against 0.40 in EO).

Because coverage is uneven, every cell is scored but not every cell is ranked: a cell enters a ranking, a top-N list, the Strategic Index or the partner rankings only if the economy is sovereign and at most half of its index weight rests on imputed components (Section 4.7). The rule keeps

4,163 of the 6,232 ES cells and 4,301 of the 7,131 EO cells. Each component is taken at the latest year its source reports, so a cell can combine observations from 2023 to 2025 with projections to 2035 and, in ES reserves, end-2020 fossil stocks (Section 4.2).

4 Index construction: normalisation, imputation and aggregation

Construction has two layers. In the *theme layer* each source's builder orients its raw metrics (higher means more security or more opportunity) and maps them to the unit interval by a rank-based S-curve. In the *aggregation layer* the latest observation of each variable is selected, gaps are filled by explicit rules, and variables are averaged into categories and categories into pillars; nothing is re-ranked there. Online Appendix OA.2 gives the construction algorithm, the derivations and further diagnostics.

4.1 Percent ranks and the median S-curve

Let $x = (x_a)_{a \in A}$ be the values of an oriented raw variable over a comparison set A with n non-missing values. The percent rank and the normalised value, restating equation (1), are

$$r_a = \begin{cases} \dfrac{\operatorname{rank}(x_a) - 1}{n - 1}, & n \geq 2, \\ 0, & n = 1, \end{cases} \tag{9}$$

$$\phi_\gamma(r_a) = \frac{r_a^\gamma}{r_a^\gamma + (1 - r_a)^\gamma}, \qquad \gamma = 0.5, \tag{10}$$

with ties given the average of the ranks they occupy; a missing value stays missing. We write $\phi_\gamma[x]_g$ for the transform taken separately within each cell of a grouping g. The function is the one-parameter probability-weighting function of Karmarkar (1978) applied to ranks rather than probabilities, the unit-elevation member of the linear-in-log-odds family of Gonzalez and Wu (1999):

$$\operatorname{logit} \phi_\gamma(r) = \gamma \operatorname{logit} r. \tag{11}$$

The curve is median-centred ($\phi_\gamma(0) = 0$, $\phi_\gamma(1/2) = 1/2$, $\phi_\gamma(1) = 1$ for every γ), ordinal, hence insensitive to outliers (Conover and Iman, 1981), and symmetric, $\phi_\gamma(1 - r) = 1 - \phi_\gamma(r)$. With $\gamma < 1$ it is an inverse S: the bottom and top tenths of a comparison set occupy $[0, 0.25]$ and $[0.75, 1]$, while the middle half is compressed into an interval of width 0.27 (0.5 under the percent rank). The choice $\gamma = 0.5$ is a stated value judgement: moving into the lead of a distribution, or out of its bottom, is worth more than an equal move in rank within the middle, so dominance and absence, the distinctions most relevant to supply security, are set apart (Figure 4). Section 12 varies γ.

Zero anchor, comparison sets and cardinal inputs. Ranked with positive values, tied zeros would score mid-table (0.41 for $(0, 0, 0, 5)$), above an economy a register omits. Measures of activity

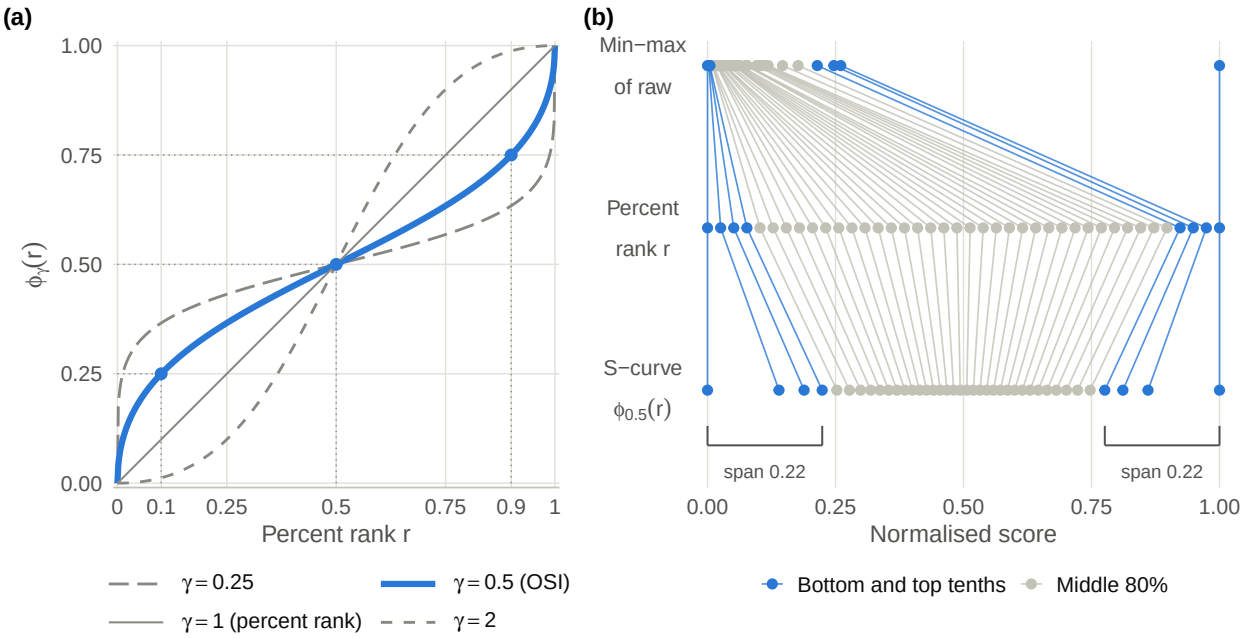


Figure 4: The median-centred S-curve. Panel (a): $\phi_\gamma(r) = r^\gamma/[r^\gamma + (1-r)^\gamma]$ against the percent rank r for $\gamma \in \{0.25, 0.5, 1, 2\}$. The highlighted curve is $\gamma = 0.5$, the value used for every component of the indices, and the marked points are $\phi_\gamma(0.1) = 0.25$, $\phi_\gamma(0.5) = 0.5$ and $\phi_\gamma(0.9) = 0.75$; $\gamma = 1$ is the percent rank itself. Panel (b): 40 draws of a right-skewed variable (lognormal, log-mean 0, log-standard deviation 1, fixed seed) under three normalisations, with each observation joined across the rows: a min-max rescaling of the raw value, the percent rank r of equation (9), and the S-curved index of equation (10) computed by the pipeline's own function. The four lowest and four highest observations are highlighted, and the brackets give the span the S-curve assigns to each group. No pipeline data are used. Source: `scripts/paper/fig_scurve.R`.

or endowment (investment, capacity, mineral supply and export shares, reserves and production roll-ups) therefore use an anchored form, with m positive values and rank^+ the average-tie rank among them:

$$\phi^0_\gamma[x](a) = \begin{cases} 0, & x_a = 0, \\ \phi_\gamma(\text{rank}^+(x_a)/m), & x_a > 0. \end{cases} \tag{12}$$

The comparison set is usually the economies observed in a pair; world-level series are ranked across technologies, pairs or minerals and copied to every economy. Index values are positions in a cross-section, not levels. Three EO inputs are cardinal transformations instead (Section 6).

4.2 Latest-year selection

For every economy, technology, stage, category, variable and theme the aggregation layer keeps the latest year present,

$$y^*_{itskvh} = \max\{y : \text{ a row } (i,t,s,k,v,h,y) \text{ exists}\}, \qquad x^h_{itskv} \equiv x^h_{itskv,y^*_{itskvh}}, \tag{13}$$

before imputation. It is a rule about the freshest information on each quantity, not a common reference year: 60 percent of the ES cells and 92 percent of the EO cells of the eleven headline technologies combine components from more than one year, with at most 3 distinct years in an ES cell and 6 in an EO cell (4 once the projection horizons after 2026 of the outlook series are set aside).

4.3 Missing data: observed-only imputation

Each pair's table is completed to the product of the economies in any of the pillar's rows for the pair and the component groups $g = (t,s,k,v,h)$ occurring there. The ES grids span 219, 215 and 219 economies at the three stages and the EO grids 219, 216 and 219. With O_g the economies observed in g and m_g the rule of the emitting theme (Table 6),

$$\hat{x}_{ig} = \begin{cases} x_{ig}, & i \in O_g, \\ 0, & i \notin O_g,\ |O_g| \geq 1,\ m_g = \text{zero}, \\ \bar{x}_g = \dfrac{1}{|O_g|}\displaystyle\sum_{j \in O_g} x_{jg}, & i \notin O_g,\ |O_g| \geq 1,\ m_g = \text{global average}, \\ \text{missing}, & \text{otherwise}. \end{cases} \tag{14}$$

Imputation uses observed values only: a fill is computed from the economies observed in the group, never from other fills, and a group observed for no economy stays missing. The zero rule, for registers of holders, producers and investors, places an unlisted economy at the bottom, level with an observed zero: absence from a complete register means none recorded, and leaving it missing would reward the absence of an endowment. The global-average rule, for attributes every economy

has in some degree, places it near 1/2, so that "not measured" is not read as "worst". Every fill is flagged and carried into the decompositions.

4.4 Category scores and available-case aggregation

A category is scored by a configured overall variable, the unweighted mean of its components present in the cell (a component reported by several themes is first averaged across them),

$$c_{kits} = \frac{1}{|\mathcal{V}^*_{kits}|} \sum_{v \in \mathcal{V}^*_{kits}} \tilde{x}_{itsv}, \qquad \mathcal{V}^*_{kits} = \{v \in \mathcal{V}_k : \ \tilde{x}_{itsv} \text{ not missing}\}, \tag{15}$$

which is c_{itsk} of equation (2) with the category index written first. With K_{its} the categories observed in the cell and w_k the nominal weights of Table 7, the pillar is the available-case weighted mean

$$\mathrm{ES}_{its} = \frac{\sum_{k \in K_{its}} w_k \, c_{kits}}{\sum_{k \in K_{its}} w_k} = \sum_{k \in K_{its}} \omega_{kits} \, c_{kits}, \tag{16}$$

$$\omega_{kits} = \begin{cases} w_k / \sum_{k' \in K_{its}} w_{k'}, & k \in K_{its}, \\ 0, & k \notin K_{its}, \end{cases} \tag{17}$$

and identically for EO_{its}. The rule is equivalent to imputing an unobserved category at the cell's own mean; zero-filling would instead penalise coverage rather than performance. The pillar is a weighted mean of S-curved scores, not itself a percentile.

Composition differs across pairs, not within them. Because every theme but the levelised-cost theme carries a rule, all scored economies share one category set in all of the 30 ES pairs and all of the 33 EO pairs (Online Appendix OA.2): within-pair comparisons are comparisons of one construct. Across pairs a cell has between 1 and 7 observed categories in ES (median 3; 21 percent of cells have a single category) and between 2 and 7 in EO (median 6). The effective weights therefore depart far from the nominal ones (Figure 5): ES Trade, nominally 0.095, carries 0.59 at the midstream stage and Consumption (0.119) 0.49 downstream, the gap between nominal and effective importance that Paruolo et al. (2013) stress. A category constant within a pair, such as Energy Prices or Technological Readiness, makes the pillar an affine function of the pillar without it,

$$\mathrm{ES}_{its} = \omega_{kts} \, \bar{c}_{kts} + (1 - \omega_{kts}) \, \mathrm{ES}^{(-k)}_{its}, \tag{18}$$

which shifts levels without reordering economies and matters only where levels are compared across pairs, as in the Strategic Index (Section 9). The weights are normative (Decancq and Lugo, 2013); Section 12 varies them.

Table 6: Missing-data rules by theme

Theme	Category	ES	EO	Rationale
Panel A: registers of holders, producers and investors (zero rule)				
`critical_minerals_production`	Production	zero	–	Unlisted economy has no recorded mine output
`critical_minerals_processing`	Foreign Dependency	zero	–	Unlisted economy has no recorded supply share
`critical_minerals_trade`	Minerals Trade	zero	–	No recorded exports of the mineral
`reserves`	Reserves	zero	–	Reserve tables (USGS, EI, Red Book) list every holder
`investment_momentum`	Investment	zero	zero	Investment and capacity records; no record means none recorded
`market_share_manufacturing`	Foreign Dependency	–	zero	No recorded manufacturing share
Panel B: attributes every economy has (global-average rule)				
`energy_access_consumption`	Energy Access	avg	avg	Coverage gap
`energy_consumption`	Energy Access, Consumption	avg	avg	Coverage gap
`solar_pv_potential`	Reserves	avg	avg[a]	Every territory has some resource potential
`wind_potential`	Reserves	avg	avg[a]	As above
`geothermal_potential`	Reserves	avg	avg[a]	As above
`import_dependence`	Energy Imports	avg	–	Coverage gap
`foreign_dependency`	Foreign Dependency	avg	–	Fills economies outside the theme's roster
`trade_concentration`	Trade	avg	–	Coverage gap
`export_feasibility`	Trade	–	avg	Coverage gap
`energy_prices`	Energy Prices	avg	avg	World series; the mean is the common value
`future_demand`	Technology Demand	–	avg	Market-level outlooks
`overcapacity_premium`	Technology Demand	–	avg	Market-level outlooks
`technological_readiness`	Technological Readiness	–	avg	Technology-level value
`cost_competitiveness`	Cost Competitiveness	–	avg	Coverage gap
`production_depth_momentum`	Production	–	avg	Coverage gap
Panel C: no rule (missing values stay missing)				
`lcoe_competitiveness`	Cost Competitiveness	–	none	Uncovered markets scored on the other cost components

Notes: "zero" assigns the bottom of the index scale and "avg" the mean of the observed index values within the technology×stage×category×variable×theme group, equation (14); either rule applies only in groups where at least one economy is observed. "–": the theme is not an input to the pillar. Source: `config/missing_data.yml`.
[a] Inert: the potential themes supply only the Reserves category, which EO does not score.

Table 7: Nominal category weights

Energy Security			Economic Opportunity		
Category	w_k	Share	Category	w_k	Share
Foreign Dependency	6	0.143	Technology Demand	4	0.133
Production	6	0.143	Trade	3	0.100
Consumption	5	0.119	Production	3	0.100
Energy Imports	4	0.095	Energy Prices	3	0.100
Reserves	4	0.095	Investment	3	0.100
Trade	4	0.095	Energy Access	3	0.100
Minerals Trade	4	0.095	Foreign Dependency	3	0.100
Energy Access	4	0.095	Cost Competitiveness	3	0.100
Investment	3	0.071	Consumption	3	0.100
Energy Prices	2	0.048	Technological Readiness	2	0.067
Total	42	1.000	Total	30	1.000

Notes: Share is $w_k / \sum_{k'} w_{k'}$, the weight a category would receive in a cell in which all ten are observed; no cell observes all ten. The weight applied in a cell is ω_{kits} of equation (17). Source: `config/weights.yml`.

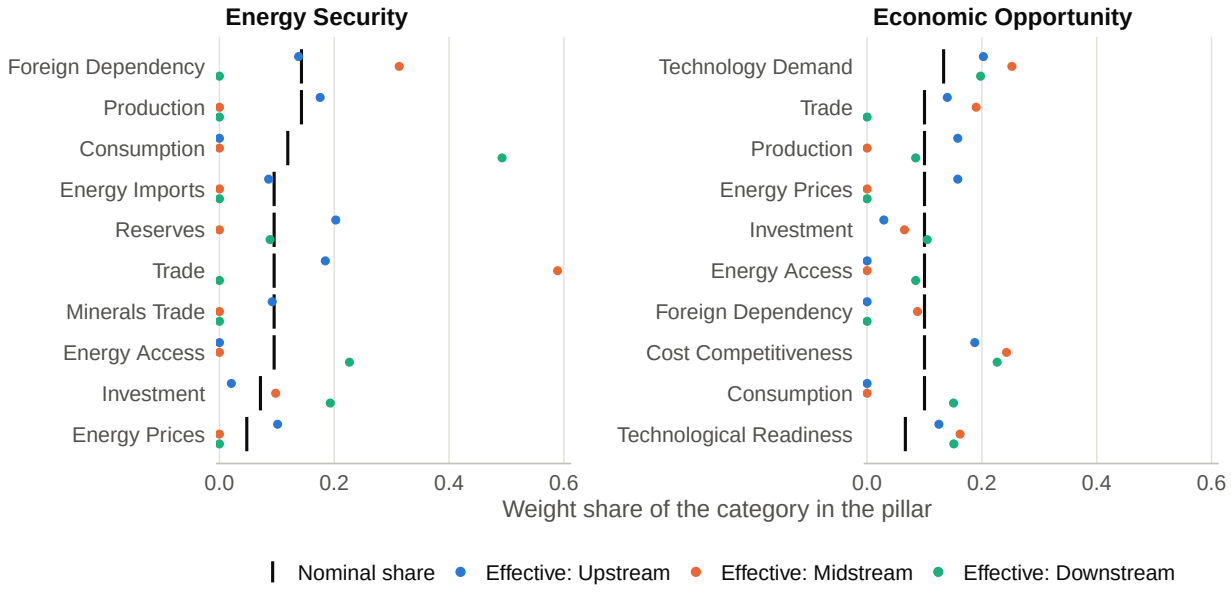


Figure 5: Nominal and effective category weights. Black ticks: nominal share $w_k / \sum_{k'} w_{k'}$ over the ten categories of the pillar (`config/weights.yml`; Table 7). Dots: mean effective weight ω_{kits} (equation (17)) over the scored cells of the eleven headline technologies at each stage, with $\omega_{kits} = 0$ where the category is not observed. A dot at zero marks a category that is never observed at that stage. The effective weights of a stage, like the nominal shares, sum to one over the categories. A dot to the right of the tick means that the available-case rule gives the category more weight than its nominal share. Cells: 2,181, 2,002 and 2,049 for ES and 2,394, 2,346 and 2,391 for EO (upstream, midstream, downstream). Source: category contributions in `data/processed/outputs/index_outputs.rds`; `scripts/paper/fig_method.R`.

4.5 Exact decompositions

With $C_{kits} = \omega_{kits} c_{kits}$ and $\lambda_{vkits} = 1/|\mathcal{V}^*_{kits}|$ for a present component (0 otherwise), every pillar value decomposes exactly into category and variable contributions,

$$\mathrm{ES}_{its} = \sum_{k \in \mathcal{K}^P} C_{kits}, \qquad \sum_{k \in \mathcal{K}^P} \omega_{kits} = 1, \tag{19}$$

$$\mathrm{ES}_{its} = \sum_{k \in \mathcal{K}^P} \sum_{v \in \mathcal{V}_k} \underbrace{\omega_{kits}\, \lambda_{vkits}\, \tilde{x}_{itsv}}_{\text{contribution of } v}, \qquad \sum_{k} \sum_{v} \omega_{kits}\, \lambda_{vkits} = 1. \tag{20}$$

The largest absolute difference between the index and the sum of its contributions is exactly zero for the category decomposition and 1.1×10^{-16} for the variable decomposition, in both pillars (7,794 ES and 7,661 EO cells, all technologies), so the attributions of Sections 5 and 6 are exact.

4.6 The imputed weight share

The share of a cell's weight carried by imputed components is

$$\iota_{its} = \sum_{k} \sum_{v} \omega_{kits}\, \lambda_{vkits}\, \mathbf{1}\big[\tilde{x}_{itsv} \text{ imputed}\big] \in [0, 1], \tag{21}$$

counting zero and average fills alike; a world-level series copied to every economy counts as observed. Imputed components carry on average 31 percent of the weight of an ES cell and 43 percent of an EO cell (medians 0.26 and 0.44; Figure 6). The ES burden is heaviest upstream (0.41, against 0.23 and 0.27), where Reserves, zero-filled for economies absent from the registers, is imputed at 0.84; the EO burden is flat across stages (0.41, 0.44, 0.43). No ES cell and 24 EO cells (the Isle of Man 14, Puerto Rico 8, Liechtenstein 2) rest entirely on imputation. All 30 ES and 33 EO headline pairs vary across economies.

4.7 The rankability rule

Every cell the sources reach is scored and enters the decompositions, but a cell enters a cross-country ranking, a top-N list or the Strategic Index only if it is rankable,

$$R_{its} = \mathbf{1}\big[i \in \mathcal{I}^{\mathrm{sov}}\big]\ \mathbf{1}\big[\iota_{its} \le \tau\big], \qquad \tau = 0.5, \tag{22}$$

where $\mathcal{I}^{\mathrm{sov}}$ is the set of 195 sovereign states of the Correlates of War state-system list (Correlates of War Project, 2024) (the 193 UN members, Kosovo and Taiwan). The threshold reads as "observed data carry at least half of the cell's weight"; values are never altered, and the Partnership Strength Index applies the same rule to partners (Section 10).

Of the 6,232 headline ES cells, 4,163 (67 percent) are rankable, 607 belonging to territories and 1,462 to sovereign states over the threshold; in EO, 4,301 of 7,131 (60 percent), with 718 and 2,112 excluded (Figure 7). The rule bites hardest in upstream ES (52 percent rankable, against 70 and

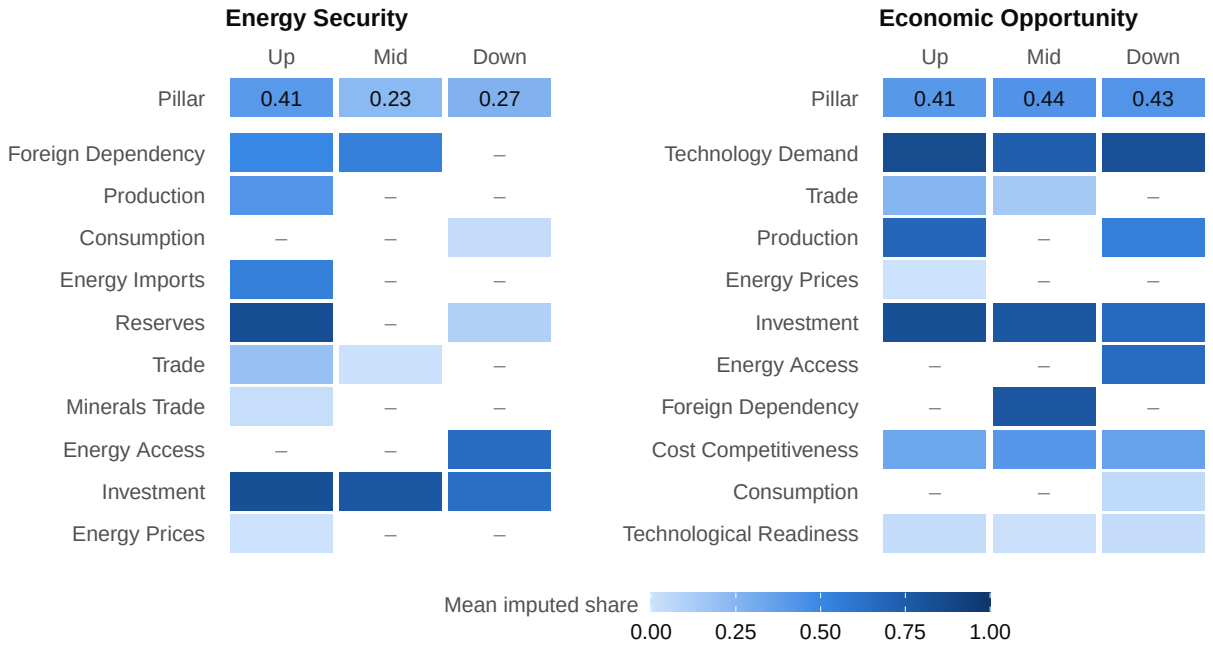


Figure 6: Imputation burden by category and stage. Category rows: the mean, over the headline cells in which the category is observed, of the share of the category score carried by imputed components, $\sum_v \lambda_{vkits} \mathbf{1}[\tilde{x}_{itsv}$ imputed]. Pillar row (with values): the mean imputed weight share ι_{its} of equation (21) over all scored cells of the stage, which weights the category rows by the effective weights. Dashes mark categories that are not observed at a stage. A component counts as imputed when a missing-data rule of Table 6 supplied it, zero fills included; world-level series copied to every economy count as observed. Online Appendix OA.2 tabulates the values. Source: variable contributions and pillar diagnostics in `data/processed/outputs/index_outputs.rds`; `scripts/paper/fig_method.R`.

80), where the registers cover a minority of economies, and in midstream and downstream EO (56 and 55 percent, against 70). Every sovereign state keeps a rankable ES cell, but 17 keep no rankable EO cell (Liechtenstein, Monaco and fifteen small island and low-income states). The rule is not level-neutral: since absence from a register scores at the bottom, excluded cells lean insecure. Of the 5,854 candidate Strategic Index cells, the 3,154 kept have a mean security need $1 - \text{ES}$ of 0.58 and EO of 0.49, against 0.68 and 0.44 for those removed. The rule states what the data can support rather than correcting the index; Section 12 reports other thresholds.

What the construction does not do. The stages of a technology are scored independently, although disruptions propagate along supply chains (Carvalho et al., 2021); dependence across stages enters only where a variable encodes it, as the mineral-supply terms of foreign dependency do. Section 12 reports the test of vertical complementarities behind this choice. Imputation assigns one value per uncovered economy, so it understates the uncertainty of imputed cells.

5 The Energy Security Index

The Energy Security pillar ES_{its} asks how far economy i can rely on the supply of technology t at supply-chain stage s: of the minerals it is built from, of the materials, components and equipment that embody it, and of the energy it delivers. Following Cherp and Jewell (2014), security is the low vulnerability of vital energy systems, scored on the dimensions the measurement literature shares: domestic resources, dependence on imports and on few suppliers, price stability and access (Kruyt et al., 2009; Sovacool and Mukherjee, 2011; Winzer, 2012; Ang et al., 2015); in the terms of Cherp and Jewell (2011), sovereignty, resilience and robustness. Because the "fuel" of a clean technology is a bundle of minerals and manufactured inputs, the pillar borrows the supply-risk logic of the criticality literature (Graedel et al., 2012; Nassar et al., 2020; Kowalski and Legendre, 2023; International Energy Agency, 2025a) and the concentration index of Hirschman (1945). A higher score means a more secure position; the Strategic Index (Section 9) uses the need $1 - \text{ES}_{its}$.

5.1 Construction

Ten categories (Table 8, weights summing to 42) are combined by the rules of Section 4 into the available-case weighted mean of equation (16). Registers of holders, producers, exporters or investors carry the zero rule and the zero-anchored S-curve of equation (12), the other themes the global-average rule (Table 6). The category bundle, and with it the effective weights of equation (17), differs across pairs but is common to all economies within one. Mineral-level terms are aggregated to a technology with demand weights from the IEA Critical Minerals Dataset (International Energy Agency, 2026a), Stated Policies scenario:

$$w^{y}_{mt} = \frac{D^{y}_{mt}}{\sum_{t'} D^{y}_{mt'}}, \qquad \tilde{w}^{y}_{mt}(M) = \frac{w^{y}_{mt}}{\sum_{m' \in M} w^{y}_{m't}}, \qquad m \in M, \tag{23}$$

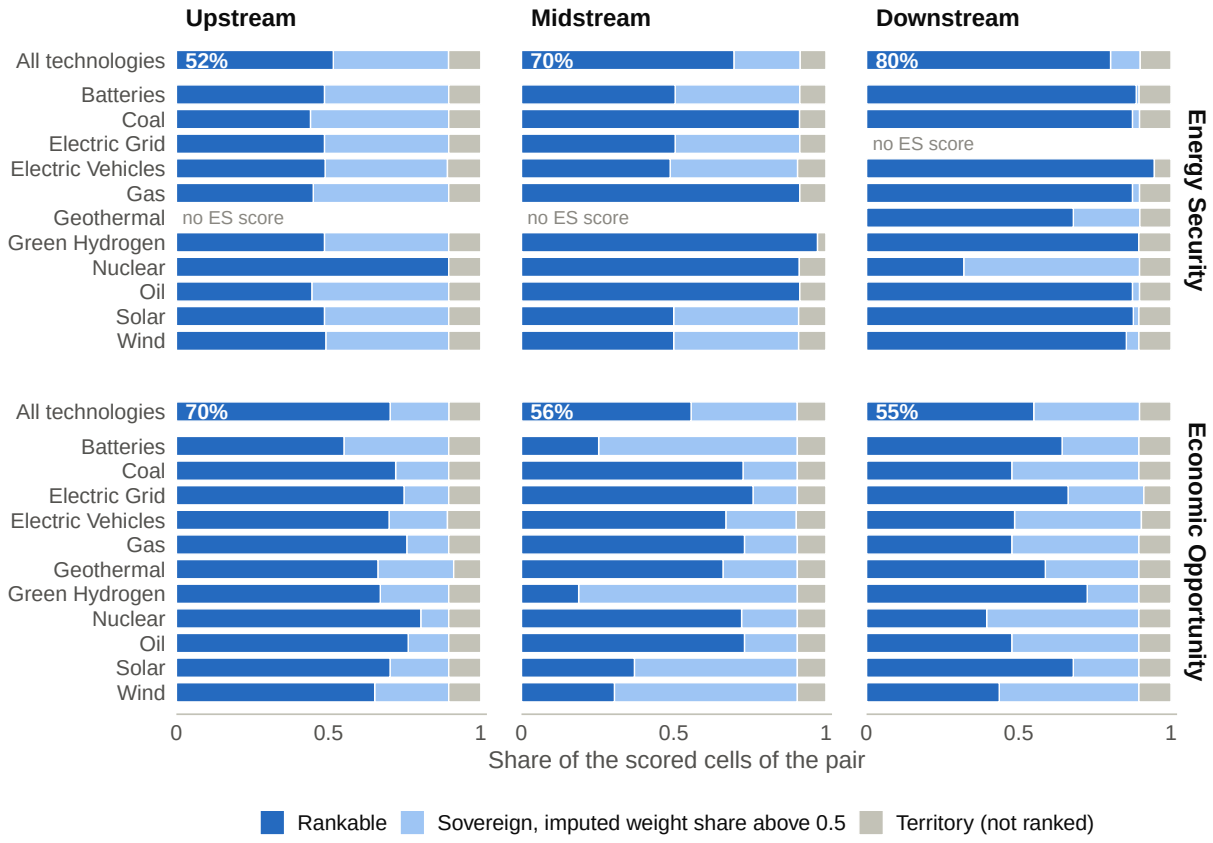


Figure 7: Rankability of the scored cells by technology–stage pair. Each bar divides the scored country–technology–stage cells of a pair into three exclusive groups: rankable cells (the economy is a sovereign state and the imputed weight share ι_{its} is at most $\tau = 0.5$); cells of sovereign states with $\iota_{its} > 0.5$; and cells of territories, special administrative regions and the West Bank and Gaza, which are scored but never ranked whatever their imputed share. The "All technologies" row pools the pairs of the stage, and its label gives the rankable share. Sovereignty follows the Correlates of War state-system list (195 states: the 193 members of the United Nations, Kosovo and Taiwan). "No ES score" marks the three pairs without an ES value. Sample: the eleven headline technologies, 6,232 ES and 7,131 EO cells; Online Appendix OA.2 gives the counts by stage. Source: `energy_security_index` and `economic_opportunity_index` in `data/processed/outputs/index_outputs.rds`; `config/rankability.yml`; `scripts/paper/fig_method.R`.

Table 8: Structure of the Energy Security pillar

Category (weight)	Stages: technologies	Components	Sources	Rule
Foreign Dependency (6)	Up, Mid: Bat, EV, Grid, H_2, Sol, Wind	own share and $1-$ HHI of mined (Up) or refined (Mid) mineral supply, 2025 and 2035; Mid: share of world manufacturing capacity (Bat, Sol, Wind), regional EV position (EV)	IEA Critical Minerals Dataset; IEA ETP; IEA Global EV Outlook	avg, zero
Energy Imports (4)	Up: Oil, Gas, Coal	production balance (EJ) and self-sufficiency	EI Statistical Review; EIA	avg
Reserves (4)	Up: all but Geo; Down: Sol, Wind, Geo	proved fossil reserves; demand-weighted mineral reserves; identified uranium resources; renewable resource potential	EI; USGS MCS 2026; Red Book; resource atlases	zero, avg
Trade (4)	Up, Mid: all but Geo, H_2	share of world basket exports; net basket imports/GDP; $1-$ world export HHI†	Atlas; UN Comtrade; WDI	avg
Minerals Trade (4)	Up: Bat, EV, Grid, H_2, Sol, Wind	demand-weighted share of world mineral exports; $1-$ HHI†	UN Comtrade	zero
Production (6)	Up: Bat, EV, Grid, H_2, Sol, Wind, Nuc	demand-weighted mine production; uranium mine output	EI; Red Book	zero
Energy Access (4)	Down: Coal, Oil, Gas, Sol, Wind, Nuc	energy supply per capita, level and growth	EI	avg
Consumption (5)	Down: all but EV, Grid	installed capacity per capita, level and growth to 2035	BNEF NEO 2024	avg
Energy Prices (2)	Up: all but Geo	commodity price volatility†	IMF PCPS	avg
Investment (3)	Up: Bat, EV; Mid: Bat, EV, Sol, Wind; Down: monitor technologies	investment, momentum, operating and pipeline capacity	Clean Investment Monitor	zero

Notes: Weights from `config/weights.yml`, renormalized over the observed categories of each cell (equation (16)). Up, Mid, Down: upstream, midstream, downstream; Bat batteries, EV electric vehicles, Grid electric grid, H_2 green hydrogen, Sol solar, Nuc nuclear, Geo geothermal. †Constant across economies within a pair. Rule: missing-data rule of the themes (avg: global average; zero). Variables in Appendix A.

where D^y_{mt} is the demand for mineral m from application t in the base year $y_0 = 2025$ or the horizon $y_1 = 2035$, and $\tilde{w}$ renormalizes over the set M of minerals a category observes: t's share of the mineral's clean-energy demand.

5.2 Categories

Foreign Dependency. The core component is the economy's position in the IEA's mined (upstream) and refined (midstream) supply of copper, cobalt, lithium, nickel, graphite and the magnet rare earths. With Q^y_{ims} the supply of mineral m by economy i at stage s,

$$\sigma^y_{ims} = \frac{Q^y_{ims}}{\sum_j Q^y_{jms}}, \qquad \mathrm{HHI}^y_{ms} = \sum_j \left(\sigma^y_{jms}\right)^2, \tag{24}$$

$$q_{imts} = \tfrac{1}{4}\Big(\phi^0_\gamma[\sigma^{y_0}]_s + \phi^0_\gamma[\sigma^{y_1}]_s + \phi_\gamma[1-\mathrm{HHI}^{y_0}]_s + \phi_\gamma[1-\mathrm{HHI}^{y_1}]_s\Big), \tag{25}$$

$$\mathrm{MS}_{its} = \frac{\sum_m w^{y_1}_{mt}\, q_{imts}}{\sum_m w^{y_1}_{mt}}. \tag{26}$$

A non-supplier scores zero on the shares but keeps the concentration terms. Midstream, the category adds the share of world manufacturing capacity including announced projects for batteries, solar and

wind (International Energy Agency, 2024), over the world total including the source's residual row, and four indices of the EV position of the economy's IEA region. The IEA reports the European Union as one region, so every EU member carries the EU's 2035 share (0.15 in wind, 0.11 in batteries, 0.00 in solar) and members are not ranked apart on it.

Energy Imports. For oil, gas and coal upstream, with production P_{if} and consumption C_{if} of fuel f in exajoules,

$$B_{if} = P_{if} - C_{if}, \qquad \beta_{if} = 100\,\frac{P_{if}}{C_{if}}, \qquad \mathrm{EI}_{if} = \tfrac{1}{2}\Big(\phi_\gamma[\beta]_f + \phi_\gamma[B]_f\Big), \tag{27}$$

the net balance and self-sufficiency, ranked within the fuel, from the Statistical Review for 2025 (Energy Institute, 2026). Where it lists no production for an economy and fuel, production is taken from the EIA's International Energy Statistics (U.S. Energy Information Administration, 2026), counted as observed and paired with consumption of the same year; Estonian coal, for which the Statistical Review counts oil shale, remains missing.

Reserves. Upstream, the category ranks proved oil, gas and coal reserves (end-2020, the last year the Energy Institute publishes them), reserves of nine critical minerals from the USGS Mineral Commodity Summaries 2026 (end-2025, with the erratum that puts Brazil's rare earths at 11 Mt) (U.S. Geological Survey, 2026), and identified recoverable uranium resources below USD 260/kgU from the OECD-NEA/IAEA Red Book (OECD Nuclear Energy Agency and International Atomic Energy Agency, 2026). With ρ_{im} the zero-anchored index of mineral m, zero for an unlisted holder,

$$R_{it} = \phi^0_\gamma\Big[\textstyle\sum_{m\in M_t} \tilde{w}^{y_0}_{mt}(M_t)\,\rho_{im}\Big]_t, \tag{28}$$

so a holder of one mineral does not score that mineral's rank for the whole technology. Holders the USGS lists without a figure (not available, withheld or small) count as zero. Downstream, the category scores solar, wind and geothermal resource potential.

Trade. The crosswalk of Section 3.2 assigns each pair a basket of HS six-digit products G_{ts}: nine midstream baskets and, upstream, the raw materials of batteries, electric vehicles (battery-mineral ores and magnet rare earths), wind (rare earths), solar (silicon metal), nuclear (uranium ores and natural uranium compounds), the fossil fuels and the grid, whose basket holds copper and aluminium inputs rather than electricity. The category is

$$c_{\mathrm{T},its} = \mathrm{mean}\Big\{\phi_\gamma[-\mathrm{HHI}_{ts}],\ \phi_\gamma\Big[\frac{\sum_{h\in G_{ts}} X_{ih}}{\sum_{h\in G_{ts}} W_h}\Big]_{ts},\ \phi_\gamma\Big[-\frac{\sum_{h\in G_{ts}}(M_{ih}-X_{ih})}{\mathrm{GDP}_i}\Big]_{ts}\Big\}: \tag{29}$$

the share of world basket exports and net basket imports relative to GDP, from the Atlas of Economic Complexity (The Growth Lab at Harvard University, 2019; Hausmann et al., 2014) and WDI (World Bank, 2026), and the concentration of world basket exports among UN Comtrade reporters in 2024

(United Nations Statistics Division, 2026), which excludes Hong Kong as a re-export hub and is ranked across the 18 basket cells of the headline technologies. That term takes one value per pair, so it moves the pair's level without reordering its economies.

Minerals Trade and Production. Minerals Trade averages the demand-weighted zero-anchored shares of world exports of the technology's minerals (for a non-reporter, partners' reported imports converted from CIF to FOB) with a pair-constant concentration term. Production ranks mine output like reserves,

$$\mathrm{P}_{it} = \phi^{0}_{\gamma}\Big[\textstyle\sum_{m\in M_t} \tilde{w}^{y_0}_{mt}(M_t)\,\pi_{im}\Big]_t, \tag{30}$$

and, for nuclear, uranium mine output from the Red Book. Fossil output enters only through Energy Imports.

Energy Access, Consumption, Prices and Investment. Downstream, Energy Access ranks energy supply per capita in 2025 and its growth since 2020, by fuel; Consumption ranks installed capacity per capita in 2024 and its projected growth to 2035 (BloombergNEF, 2024). Both read a deeper, expanding domestic system as more secure; treating high use as exposure would reverse the sign (Section 6.3). Energy Prices is $\phi_\gamma[-\bar{v}_t]$, with $\bar{v}_t$ the mean annualized volatility of monthly log changes in the technology's IMF commodity prices over 1, 5, 10 and 20 years (International Monetary Fund, 2026b), a world value. Investment ranks clean-technology investment, its three-year momentum, and operating and pipeline manufacturing capacity (Rhodium Group and MIT Center for Energy and Environmental Policy Research, 2026); upstream critical-minerals investment is split between batteries and electric vehicles by their IEA demand shares, 11.5 and 88.5 per cent.

5.3 Results

Coverage. The pillar scores 6,232 cells of the eleven headline technologies for 219 economies (195 sovereign, 24 territories) in 30 of the 33 pairs (Geothermal upstream and midstream and Electric Grid downstream have no category). No pair is constant and no cell fully imputed; the mean imputed weight share is 0.41, 0.23 and 0.27 by stage. All rankings and summaries use the 4,163 rankable cells (Section 4.6): 1,124 of 2,181 upstream, where the world price series places every economy in each pair while the mineral and fossil sources cover far fewer, 1,397 of 2,002 midstream and 1,642 of 2,049 downstream.

Distribution. The median rankable cell scores 0.28 upstream (interquartile range 0.19 to 0.37), 0.46 midstream (0.36 to 0.53) and 0.44 downstream (0.35 to 0.52), with a right-skewed upstream distribution (Figure 8): in the median upstream pair 71 per cent of rankable economies score zero on Reserves and 84 per cent on Production. Technology-stage fixed effects explain 44 per cent of the variance, country effects alone 12 per cent and both together 60 per cent (28 per cent of the within-pair variance), and the pair effects reflect the category bundles rather than the cell-constant

inputs. Rankings are meaningful within a pair; comparing one economy's levels across pairs, as the Strategic Index does, mixes content with construction.

Drivers. The exact category decomposition (equation (19), Figure 9) shows that upstream Trade supplies 31 per cent of the average score, Energy Prices 20, Minerals Trade 16 and Foreign Dependency 13, while Reserves and Production supply 7.2 and 3.1 per cent against mean effective weights of 20 and 19.5 per cent: for the typical economy the registers add nothing, for the few holders they decide the top. Midstream, Trade supplies 73 per cent and Foreign Dependency 26; downstream, Consumption 56 and Energy Access 24. The cell-constant inputs carry a mean effective weight of 0.21 upstream and 0.22 midstream but supply 36 and 26 per cent of the score. Reserves and Production correlate at a within-pair median of 0.84, so upstream the pillar counts mineral endowment twice.

Uranium. Nuclear upstream is scored on Energy Prices, Trade, Reserves and Production, with effective weights of 1/8, 1/4, 1/4 and 3/8 (Figure 10). Of the 195 rankable economies, 58 hold identified resources and 18 of them produce. The 105 economies with neither and without observed uranium trade share one value, 0.172; the other 32 without uranium lie between 0.145 and 0.201. Kazakhstan leads (0.84), ahead of Canada (0.74), Namibia and Australia (0.72); China is tenth (0.56) and the United States eleventh (0.53); 31 of the 57 coalition members hold resources. Read as a need, the common value becomes a uniform 0.83 that measures absence rather than a gap domestic supply could close. Because 86 of the 174 countries with a Strategic Index share it, the allied picks admit an upstream stage only where the country records reserves or production of its resources (Section 9.3), a pick-eligibility rule, not part of the index.

Leaders and the coalition. China leads 11 of the 30 pairs (Table 9): three upstream mineral pairs, five midstream pairs, through its shares of refined supply and manufacturing capacity (International Energy Agency, 2024, 2025a), and three downstream pairs; no other economy leads more than two. Upstream, leadership follows endowments (Saudi Arabia in oil, Qatar in gas, Kazakhstan in nuclear, Chile in the grid and solar); in Oil upstream China ranks 34th, the position of an importer (Figure 11). A coalition member is first in 16 pairs and in the top three of all 30. The 57 members have mean within-pair percentiles of 0.59, 0.72 and 0.62 by stage, against 0.47, 0.41 and 0.50 for other economies (Figure 12); the United States averages 0.95, China 0.92 (0.99 midstream), and Taiwan has the profile of a manufacturer without resources (0.75 midstream, 0.01 upstream). Canada is the one high-income member with a gap in the heatmap: its Downstream Consumption category is imputed at the global average in all eight cells it enters, because the BloombergNEF outlook behind it neither models Canada nor has a North American aggregate, so six of its nine scored Downstream cells are not rankable (the other economies with the whole category imputed are Taiwan, Bermuda and seven South Asian economies). The lead is unsurprising for a coalition of large or resource-rich economies; whether members' strengths complement one another is the subject of Section 11. Online Appendix OA.3 details the leaders, the tiers and the category correlations.

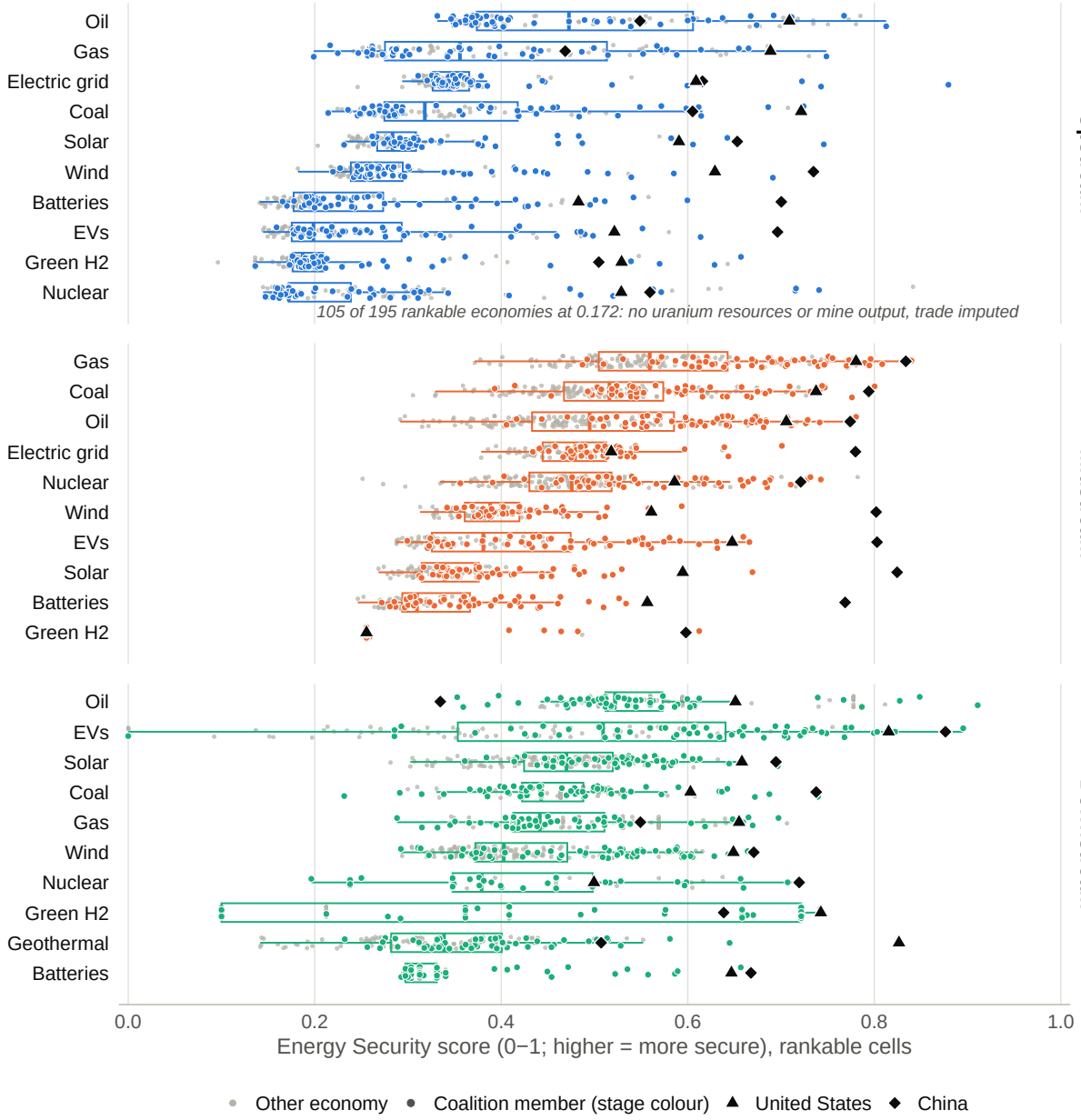


Figure 8: Energy Security by technology and stage, rankable cells (68 to 195 economies per pair; sovereign economy with at most half of the index weight imputed). Boxes span the interquartile range; gray dots are non-members, coloured dots coalition members in the stage colour; the triangle marks the United States and the diamond China. In Nuclear upstream, 105 of the 195 economies, which have neither uranium resources, production nor observed uranium trade, share the value 0.172 (Energy Prices plus an imputed Trade category). In Green Hydrogen midstream 90 per cent of economies share 0.26. Source: `energy_security_index` in `index_outputs.rds`, final run; `scripts/paper/fig_es.R`.

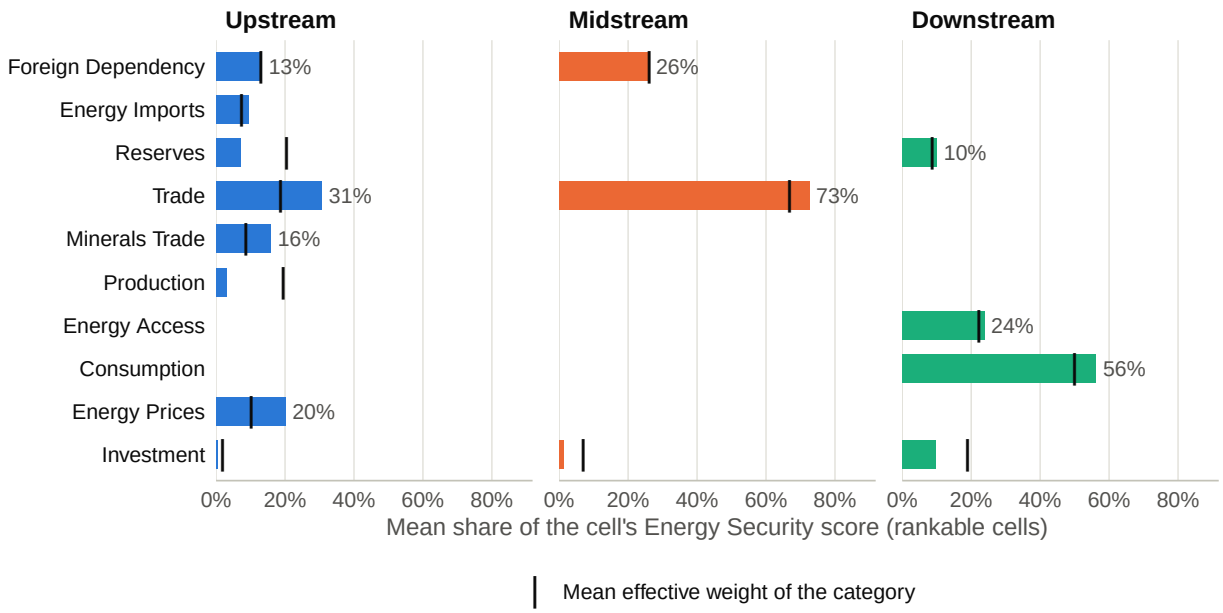


Figure 9: Average composition of Energy Security by stage, rankable cells. Bars: mean share of each category in the score, $\omega_{kits}c_{kits}/\text{ES}_{its}$ (equation (19)); ticks: the category's mean effective weight. A bar much shorter than its tick, as for Reserves and Production upstream, marks a category on which most economies score zero. The upstream Energy Prices and Trade shares include Nuclear upstream (15 and 28 per cent without it). Sample: 1,124 upstream, 1,397 midstream and 1,637 downstream cells. Source: `energy_security_outputs$category_contributions`; `scripts/paper/fig_es.R`.

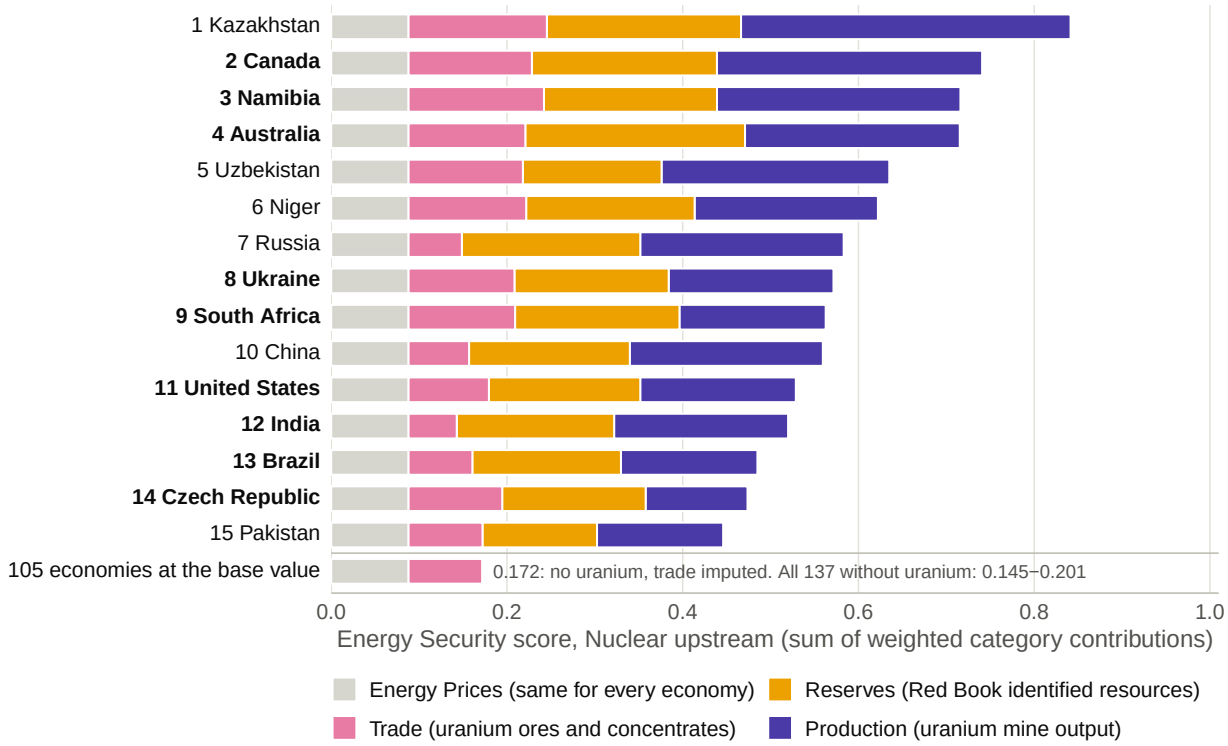


Figure 10: Composition of Energy Security in Nuclear upstream: the 15 highest-ranked of 195 rankable economies (coalition members in bold), split into Energy Prices (uranium price volatility, 0.088 for every economy), Trade (uranium ores, concentrates and natural uranium compounds), Reserves (identified resources below USD 260/kgU, OECD-NEA/IAEA Red Book) and Production (mine output). The bottom row stands for the 105 economies without uranium whose uranium trade is not observed (0.172). Source: `energy_security_index` and its category contributions; `scripts/paper/fig_es.R`.

Table 9: Leading economies in each technology-stage pair of the Energy Security pillar (rankable cells)

Technology	n	First	Second	Third	Best member (rank)	USA	CHN
Upstream							
Batteries	106	CHN 0.70	AUS 0.60	RUS 0.56	AUS (2)	9	1
Coal	96	AUS 0.72	USA 0.72	RUS 0.71	AUS (1)	2	8
Electric Grid	106	CHL 0.88	COD 0.74	PER 0.72	CHL (1)	5	4
Electric Vehicles	107	CHN 0.70	AUS 0.61	RUS 0.58	AUS (2)	5	1
Gas	98	QAT 0.75	RUS 0.73	USA 0.69	QAT (1)	3	33
Green Hydrogen	106	ZAF 0.66	RUS 0.64	IDN 0.63	ZAF (1)	7	8
Nuclear	195	KAZ 0.84	CAN 0.74	NAM 0.72	CAN (2)	11	10
Oil	97	SAU 0.81	VEN 0.79	IRQ 0.78	SAU (1)	12	34
Solar	106	CHL 0.75	CHN 0.65	PER 0.64	CHL (1)	5	2
Wind	107	CHN 0.73	AUS 0.69	USA 0.63	AUS (2)	3	1
Midstream							
Batteries	106	CHN 0.77	USA 0.56	CAN 0.53	USA (2)	2	1
Coal	190	POL 0.80	MOZ 0.80	CHN 0.79	POL (1)	10	3
Electric Grid	106	CHN 0.78	JPN 0.70	CHL 0.64	JPN (2)	22	1
Electric Vehicles	105	CHN 0.80	ESP 0.67	HUN 0.66	ESP (2)	4	1
Gas	192	KOR 0.84	CHN 0.83	ITA 0.81	KOR (1)	8	2
Green Hydrogen	106	IDN 0.61	CHN 0.60	RUS 0.49	IDN (1)	12^{95}	2
Nuclear	188	RUS 0.78	FRA 0.74	NLD 0.73	FRA (2)	30	5
Oil	192	KOR 0.78	CHN 0.77	GAB 0.74	KOR (1)	11	2
Solar	106	CHN 0.82	IND 0.67	USA 0.59	IND (2)	3	1
Wind	106	CHN 0.80	POL 0.59	USA 0.56	POL (2)	3	1
Downstream							
Batteries	186	CHN 0.67	DEU 0.66	USA 0.65	DEU (2)	3	1
Coal	186	IDN 0.74	CHN 0.74	JPN 0.69	IDN (1)	9	2
Electric Vehicles	121	TUR 0.90	CHN 0.88	VNM 0.82	TUR (1)	4	2
Gas	186	KWT 0.71	QAT 0.70	KOR 0.67	QAT (2)	6	24
Geothermal	146	USA 0.83	IDN 0.64	JPN 0.58	USA (1)	1	9
Green Hydrogen	186	USA 0.74	KOR 0.74	ALB 0.72^{49}	USA (1)	1	63
Nuclear	68	CHN 0.72	FRA 0.71	JPN 0.66	FRA (2)	17	1
Oil	186	QAT 0.91	OMN 0.85	ARE 0.83	QAT (1)	23	186
Solar	191	SAU 0.70	AUS 0.70	CHN 0.69	SAU (1)	4	3
Wind	186	CHN 0.67	CAN 0.66	USA 0.65	CAN (2)	3	1

Notes: ES_{its} for the eleven headline technologies, final run. Only rankable cells enter: a sovereign economy with at most half of the cell's index weight imputed. Economies are identified by ISO3 code and followed by their score; n is the number of rankable economies in the pair. "Best member" is the highest-ranked of the 57 coalition members, with its rank in parentheses; the USA and CHN columns give the ranks of the United States and China. A rank is one plus the number of rankable economies with a strictly higher score, so tied economies share a rank, and a superscript gives the number of economies that share it (the economy listed is the first of them alphabetically). Geothermal upstream, Geothermal midstream and Electric Grid downstream have no Energy Security score. Source: `energy_security_index` in `index_outputs.rds`; `scripts/paper/fig_es.R`.

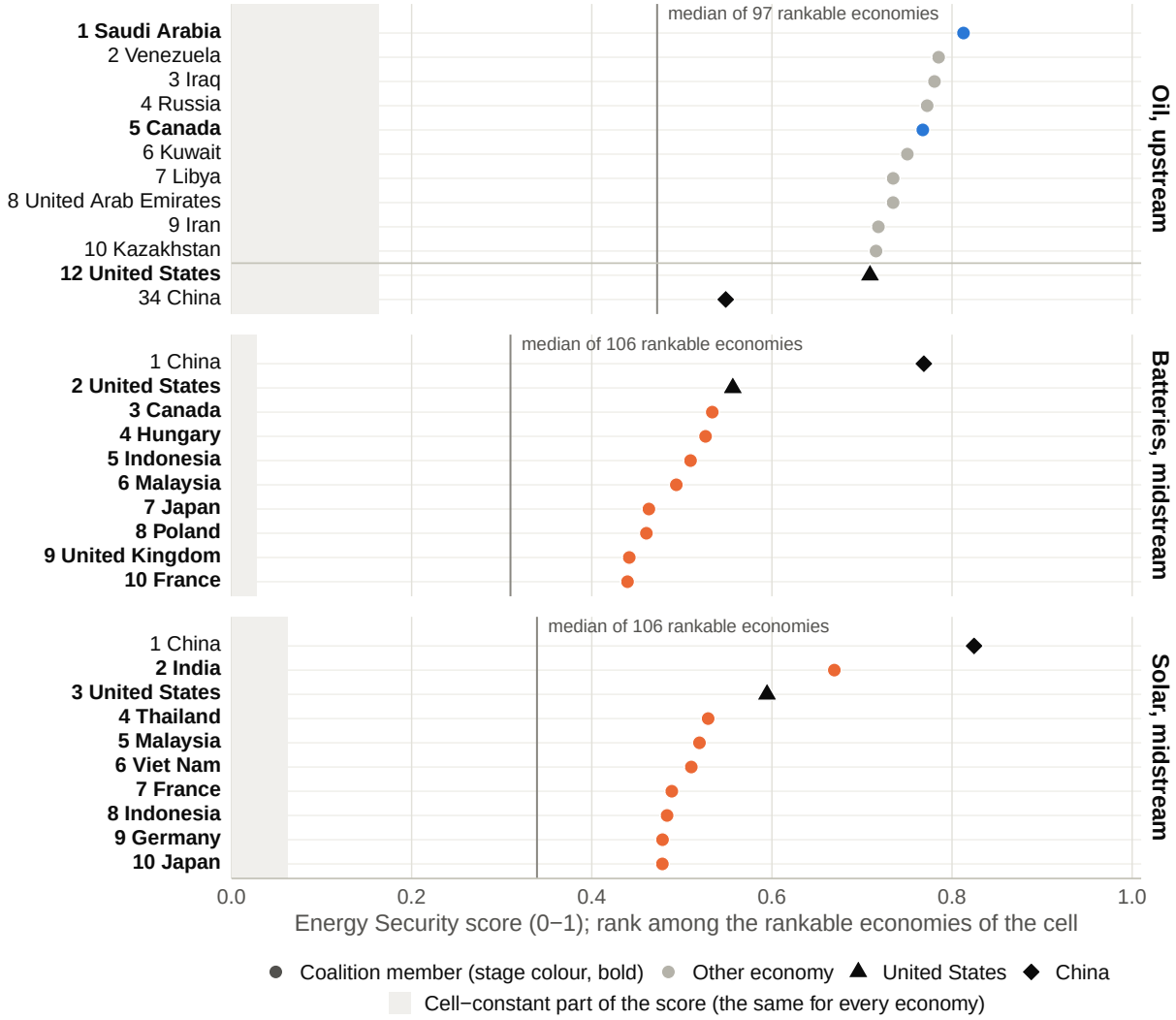


Figure 11: The ten highest-ranked economies in Oil upstream, Batteries midstream and Solar midstream (97, 106 and 106 rankable economies), with the United States (triangle) and China (diamond) appended when outside the top ten. Coalition members are in bold and the stage colour. The vertical line is the pair median; the shaded band is the cell-constant part of the score (0.16, 0.03 and 0.06). Source: `energy_security_index` and its variable contributions; `scripts/paper/fig_es.R`.

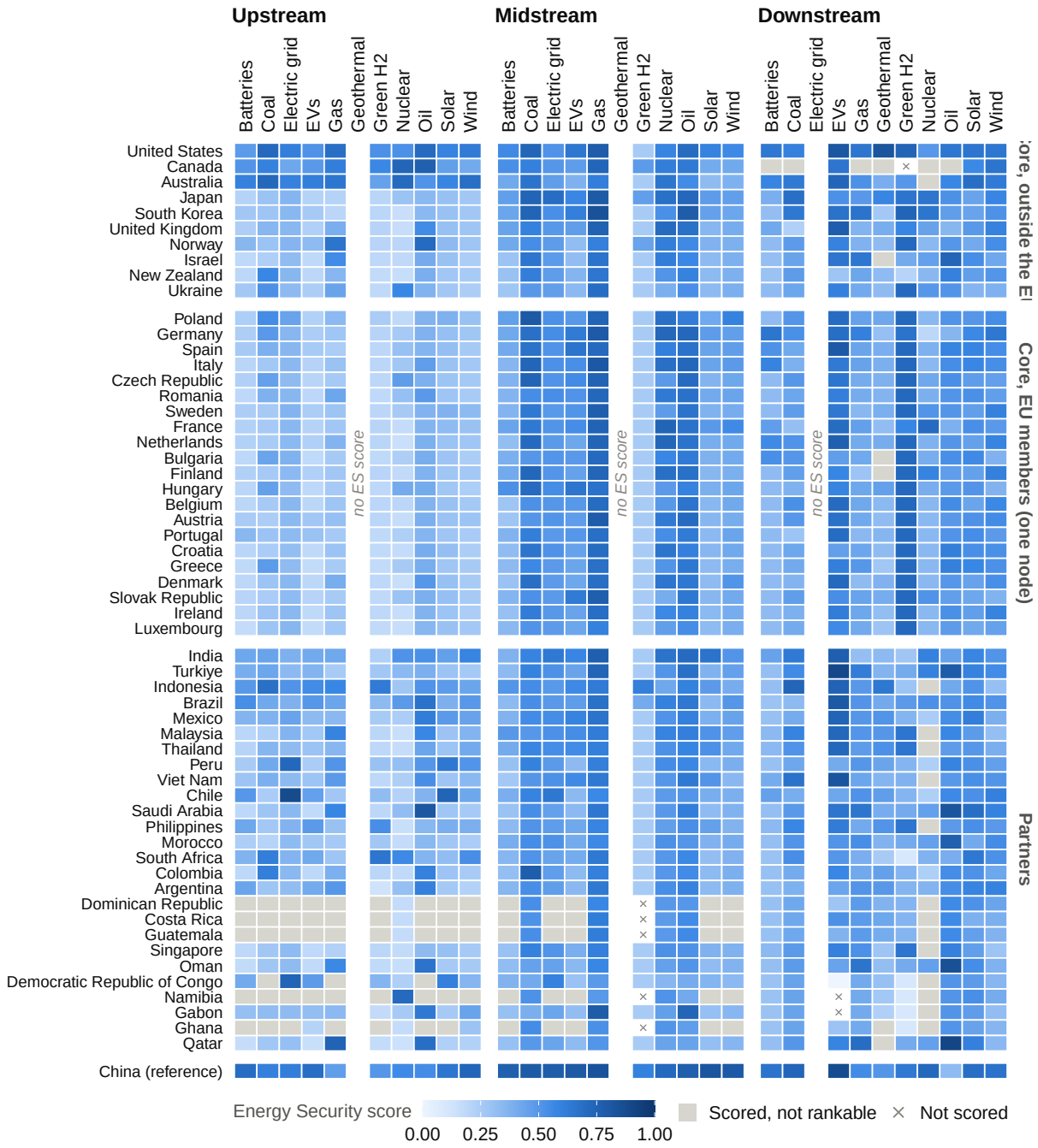


Figure 12: Energy Security of the 57 coalition members across the 33 sectors, grouped into the ten core members outside the EU, the 21 core EU members (part of the network's EU node) and the 26 partners, each ordered by mean within-pair percentile; China for reference. Colour gives ES_{its} for rankable cells; gray cells are scored but not rankable, crosses unscored, and the three sectors without an Energy Security score are blank. Canada's Downstream cells rest on an imputed Consumption category (its source, the BloombergNEF outlook, does not report Canada), so six of them are not rankable. Source: `energy_security_index` and `coalition_members`; `scripts/paper/fig_es.R`.

Caveats. Several pairs rank few distinct values: in the Strategic Index sample, Green Hydrogen midstream has 34 of 40 countries at one value, Nuclear upstream 86 of 174, Green Hydrogen downstream 45 of 158, Nuclear downstream 19 of 67 and Batteries downstream 38 of 138, and under the need reading each mass point becomes a uniformly high need (Section 9). Unquantified holders count as zero, and the downstream reading treats deeper domestic use of a fossil fuel as security.

6 The Economic Opportunity Index

The Economic Opportunity pillar EO_{its} asks how well placed economy i is to capture value from the expansion of technology t at stage s. Its premise comes from economic complexity: countries diversify into activities close, in capability space, to those they already perform competitively, and relatedness predicts new comparative advantage (Hidalgo et al., 2007; Hausmann and Klinger, 2006; Hausmann et al., 2014; Hidalgo and Hausmann, 2009; Hausmann et al., 2022; Hidalgo, 2021; Hausmann et al., 2007), including in low-carbon products (Mealy and Teytelboym, 2022; Growth Lab at Harvard University, 2024). We apply it at the supply-chain stage, where green industrial policy is now designed (Rodrik, 2014; Juhász et al., 2024), and add market size and growth, cost, technological maturity and committed capital. A higher score means more opportunity; EO_{its} is not the complement of ES_{its} (Section 6.3).

6.1 Construction

Ten categories (Table 10) are combined by the rules of Section 4 into the available-case weighted mean of equation (16). The weights sum to 30: eight categories carry 3, Technology Demand 4 and Technological Readiness 2. $\phi_\gamma[x]_g$ is the S-curve of equation (10) ($\gamma = 0.5$) within comparison set g, by default the pair (t, s). Technological Readiness and the input-cost component of Cost Competitiveness cover all 33 pairs. Technological Readiness, Energy Prices and three of the four Technology Demand components are constant within a pair: by equation (18) they shift a pair's level without reordering its economies. Appendix A lists every variable; Online Appendix OA.4 gives further detail.

6.2 Categories

Trade. Trade measures revealed and latent capability to export the goods of a stage, using the baskets G_{ts} of Section 3.2; the upstream baskets give every upstream pair except Geothermal and Green Hydrogen a Trade category. Besides the Atlas share of world basket exports (The Growth Lab at Harvard University, 2019), shared with ES, it ranks revealed comparative advantage (Balassa, 1965) from UN Comtrade (United Nations Statistics Division, 2026),

$$\mathrm{RCA}_{its} = \frac{E_{its}/T_i}{\sum_{j\in R} E_{jts} / \sum_{j\in R} T_j}, \tag{31}$$

Table 10: Structure of the Economic Opportunity pillar

Category	Weight (share)	Components (stage: technologies)	Sources	Higher score =	Missing rule
Trade	3 (0.10)	Share of world basket exports; Balassa RCA; basket export value; product-space feasibility; world export concentration† (Up, Mid: all but Geo, H_2)	Atlas HS92; UN Comtrade	larger, more specialised exports; closer capabilities; contestable market	avg
Production	3 (0.10)	Level, absolute change and a composite of fossil output (Up: Oil, Gas, Coal), uranium mine output (Up: Nuc), demand-weighted share of world mined mineral supply (Up: Bat, EV, Grid, H_2, Sol, Wind) or power generation (Down: Sol, Wind, Nuc, Coal, Gas, Oil)	EI Statistical Review; EIA; Red Book; IEA Critical Minerals Dataset	larger, faster-growing production	avg
Technology Demand	4 (0.13)	World supply in 2035 and its growth†; regional demand (12 economies) or the EV car market (Down: EV); US addressable market†; manufacturing overcapacity† (Mid: Bat, Sol, Wind, H_2)	IEA WEO 2025; BNEF NEO 2024; IEA GEVO 2026; BCG; BNEF	larger, faster-growing market; less overcapacity	avg
Technological Readiness	2 (0.07)	TRL Index of the items mapped to the technology and stage: bell-shaped level score and capped momentum† (all pairs)	IEA Clean Energy Technology Guide	readiness nearer TRL 6; faster maturation	avg
Energy Prices	3 (0.10)	Commodity price volatility† (Up: all but Geo)	IMF PCPS	lower volatility	avg
Investment	3 (0.10)	Investment; three-year momentum; operating and pipeline manufacturing capacity (Up: Bat, EV; Mid: Bat, EV, Sol, Wind; Down: Bat, EV, Geo, Nuc, Sol, Wind)	Clean Investment Monitor	more investment and capacity	zero
Energy Access	3 (0.10)	Energy supply per capita and its 2020–2025 growth (Down: Coal, Oil, Gas, Sol, Wind, Nuc)	EI Statistical Review	deeper, growing home market	avg
Foreign Dependency	3 (0.10)	Share of world manufacturing capacity incl. pipeline (Mid: Bat, Sol, Wind, H_2); regional EV production position (Mid: EV)	IEA ETP; IEA EV data	larger share of world manufacturing	zero
Cost Competitiveness	3 (0.10)	LCOE Index (Down: Bat, Coal, Gas, Sol, Wind); Input Cost Index of wages and capital costs (all pairs); relative equipment cost (Mid: Bat, Sol, Wind, H_2; Down: H_2)	BNEF; ILOSTAT; IMF; IEA	lower costs	none (LCOE); avg
Consumption	3 (0.10)	Installed capacity per capita and its growth to 2035 (Down: all but EV, Grid)	BNEF NEO 2024	deeper installed base	avg

Notes: Weights from `config/weights.yml`; shares $w_k/30$ apply where every category is observed (equation (16)). Abbreviations as in Table 8. †Constant across economies within a pair. Missing rules from `config/missing_data.yml` (Table 6); "avg" is the global-average rule.

with E_{its} basket and T_i total exports over the reporters R; the value of basket exports E; and latent capability, the mean Atlas distance $\delta_{i\eta}$ to the basket's HS four-digit headings $\mathcal{H}_{ts}$ (Hidalgo et al., 2007),

$$\text{feas}_{its} = 1 - \frac{1}{|\mathcal{H}_{ts}|} \sum_{\eta \in \mathcal{H}_{ts}} \delta_{i\eta}. \tag{32}$$

The fifth component is the world-market concentration term of the ES category, read as contestability. Then

$$c_{\text{T},its} = \text{mean}\Big\{\phi_\gamma[\text{ms}]_{ts},\ \phi_\gamma[\text{RCA}]_{ts},\ \phi_\gamma[E]_{ts},\ \phi_\gamma[\text{feas}]_{ts},\ \phi_\gamma[-\text{HHI}]\Big\}. \tag{33}$$

Hong Kong, a re-export hub, is excluded from the Comtrade measures (RCA, export value, concentration) but keeps its Atlas components, and the concentration term is ranked across the cells of the eleven headline technologies.

Production. Production measures the depth of existing output and its momentum. For oil, gas, coal and uranium upstream and electricity generation downstream, with output $P_{it,y}$ from the Statistical Review (Energy Institute, 2026) (filled from EIA where it lists none, Section 5.2) or the Red Book, and relative and absolute changes g^{rel}, g^{abs} over five years,

$$\text{size}_{its} = \phi_\gamma[P_{\cdot,2025}]_{ts}, \qquad \text{gabs}_{its} = \phi_\gamma[g^{\text{abs}}]_{ts}, \qquad O_{its} = \phi_\gamma\Big[\tfrac{1}{3}\big(\phi_\gamma[P_{\cdot,2025}] + \phi_\gamma[g^{\text{rel}}] + \phi_\gamma[g^{\text{abs}}]\big)\Big]_{ts}. \tag{34}$$

For the six mineral-intensive technologies upstream, mined supply S_{im} of each mineral in the IEA base year y_0 and in 2035 (International Energy Agency, 2026a) enters as a share of base-year world supply W_m, weighted by the technology's demand share $\tilde{w}_{mt}$:

$$\begin{aligned} s^y_{it} &= \sum_m \tilde{w}_{mt}\, \frac{S_{im,y}}{W_{m,y_0}}, \qquad \text{size}_{its} = \phi_\gamma[s^{y_0}]_{ts}, \\ \text{gabs}_{its} &= \phi_\gamma[s^{2035} - s^{y_0}]_{ts}, \qquad O_{its} = \phi_\gamma[2\,\text{size} + \text{gabs} + h_t]_{ts}, \end{aligned} \tag{35}$$

so growth is an absolute change in base-year world-share units and h_t is a concentration term common to the pair. In every cell $c_{\text{P},its} = \frac{1}{3}(\text{size} + \text{gabs} + O)$.

Technology Demand. Four sources each contribute one composite. Two energy outlooks weight growth twice as heavily as size:

$$\text{GD}_t = \phi_\gamma\Big[\tfrac{1}{3}\big(\phi_\gamma[D_{\cdot,2035}] + 2\,\phi_\gamma[D_{\cdot,2035} - D_{\cdot,2024}]\big)\Big], \tag{36}$$

$$\text{RD}_{it} = \phi_\gamma\Big[\tfrac{1}{3}\big(\phi_\gamma[D_{\cdot t,2035}]_t + 2\,\phi_\gamma\big[D_{\cdot t,2035}/D_{\cdot t,2024} - 1\big]_t\big)\Big]_t, \tag{37}$$

where $D_{f,y}$ is world supply of fuel f in the World Energy Outlook's Stated Policies Scenario (International Energy Agency, 2025b), ranked across seven fuels, and $D_{it,y}$ primary energy consumption in the twelve economies BloombergNEF models (BloombergNEF, 2024); downstream electric vehicles use the EV market composite of the Global EV Outlook (International Energy Agency, 2026d)

instead. The third source is BCG's US serviceable addressable market, one value per pair; the fourth is midstream overcapacity, with ρ_t the ratio of manufacturing capacity to forecast demand in BloombergNEF's supply-chain data,

$$\mathrm{OV}_t = 2^{-\rho_t}, \tag{38}$$

so that capacity equal to demand scores 1/2 and each further multiple halves the score (Figure 13c). Then

$$c_{\mathrm{TD},its} = \mathrm{mean}\left\{\mathrm{GD}_t,\ \mathrm{RD}_{it},\ \phi_\gamma[\mathrm{SAM}]_{ts},\ \mathrm{OV}_t\right\}. \tag{39}$$

Technological Readiness. Opportunity is taken to be greatest around the demonstration stage, where learning is fast and first-mover positions are open. The IEA Clean Energy Technology Guide (International Energy Agency, 2026c) grades technology designs by readiness level (TRL) for 2020–2025; its post-TRL-9 adoption stages are read as levels 9 to 11. Ordered rules map each item to a technology and a stage, and a stage without items takes the technology-wide value. For item n with level x_n and change Δ_n,

$$b(x) = \begin{cases} 0, & x \le 2 \text{ or } x \ge 11, \\ \cos\!\left(\frac{\pi}{2}\,\frac{6-x}{4}\right), & 2 < x \le 6, \\ \cos\!\left(\frac{\pi}{2}\,\frac{x-6}{5}\right), & 6 < x < 11, \end{cases} \qquad \psi(\Delta) = \left(\frac{\min\{\max\{\Delta, 0\}, 3\}}{3}\right)^{1/2}, \tag{40}$$

$$\mathrm{TRL}_{ts} = \frac{1}{|N_{ts}|} \sum_{n \in N_{ts}} \left[0.7\, b(x_n) + 0.3\, \psi(\Delta_n)\right], \tag{41}$$

a cardinal index constant within a pair but different across a technology's stages. Of the Guide's 634 items, 253 map to a headline technology; the index runs from 0.31 (Geothermal downstream) to 0.72 (Gas downstream).

Foreign Dependency: manufacturing position. In EO this category measures the production base from which a country can serve world demand. For batteries, solar, wind and green hydrogen (the IEA's electrolyser column) it is the S-curve of the share of world manufacturing capacity including announced projects (International Energy Agency, 2024), over the IEA world total including its residual row; for electric vehicles, four indices of the EV position of the economy's IEA region, whose five regions partition the world. It carries the zero rule. Every EU member takes the EU's regional 2035 share (0.15 in wind, 0.11 in batteries, 0.24 in green hydrogen, 0.00 in solar), so EU members are not ranked apart on it.

Cost Competitiveness. Three components, one per source. The levelised cost of electricity of a benchmark plant (BloombergNEF, 2024), today and in 2050,

$$\mathrm{LCOE}_{it} = \tfrac{1}{2}\Big(\phi_\gamma[-L_{\cdot,y_r}]_t + \phi_\gamma[-L_{\cdot,2050}]_t\Big), \tag{42}$$

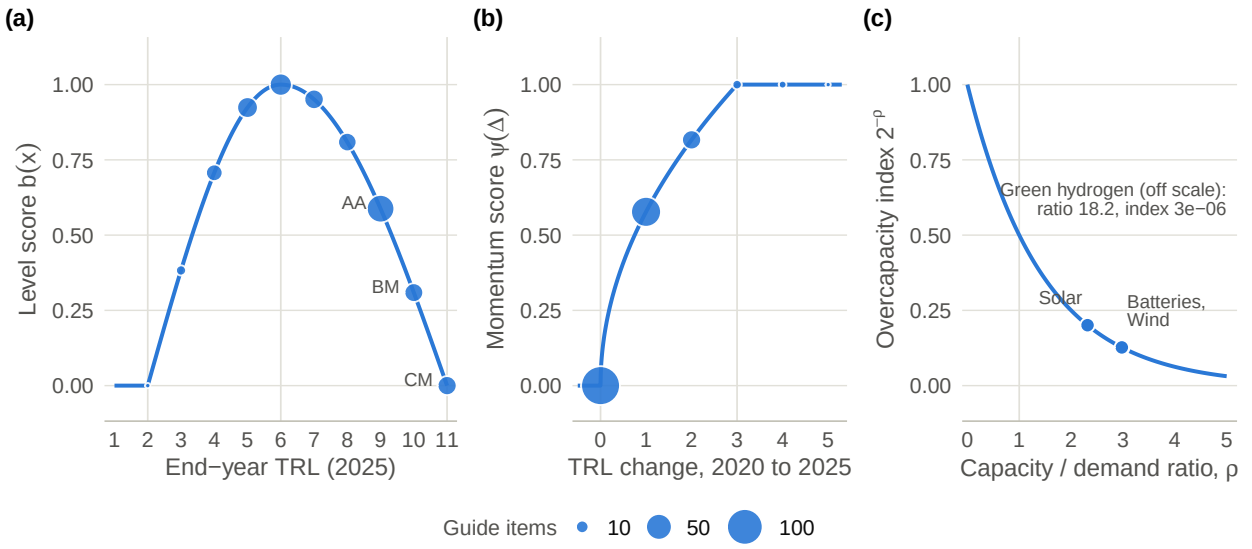


Figure 13: The three cardinal transformations of the Economic Opportunity pillar. (a) The level score $b(x)$ of equation (40) against the 2025 readiness level; circle area is proportional to the number of Guide items mapped to a headline technology at that level (253 items; TRL 9 to 11 are the Guide's adoption stages: awaiting adoption, building momentum, commercial in many markets). (b) The momentum score $\psi(\Delta)$ against the 2020–2025 change, capped at three steps. (c) The overcapacity index $2^{-\rho}$ of equation (38), with the four BloombergNEF technologies at their final-build values (green hydrogen, at $\rho = 18.2$, lies off the scale). Source: the Guide, `technological_readiness_tbl.rds`, `overcapacity_premium_tbl.rds`; `scripts/paper/fig_eo_transforms.R`.

is defined downstream for five technologies, without imputation. The Input Cost Index prices labour, $\ell_{is} = \phi_\gamma[\text{wage}]_s$ (ILOSTAT monthly earnings at market exchange rates in the activity that proxies the stage), and capital, combining the lending rate winsorised and rescaled to $[0, 1]$, R_i, with producer-price inflation,

$$\chi_i = 0.6\, R_i + 0.4\, \phi_\gamma[\text{infl}]_i, \tag{43}$$

weighted by the stage's factor shares (Table 27),

$$\text{ICI}_{its} = \frac{\alpha^L_{ts}}{\alpha^L_{ts} + \alpha^K_{ts}}\,(1 - \ell_{is}) + \frac{\alpha^K_{ts}}{\alpha^L_{ts} + \alpha^K_{ts}}\,(1 - \chi_i). \tag{44}$$

The third is the IEA's regional equipment cost relative to China, z_{rt}, and

$$c_{\text{CC},its} = \text{mean}\left\{\text{LCOE}_{it},\ \text{ICI}_{its},\ 1 - \phi_\gamma[z]_t\right\}. \tag{45}$$

Categories common to both pillars. Energy Prices, Investment, Energy Access and Consumption enter EO with the components and index values they have in ES (Section 5.2). Price volatility is read as investment risk for producers, investment and pipeline capacity as revealed investor commitment, and per-capita energy use and installed capacity as the home market.

Table 11: Raw signals shared by the Energy Security and Economic Opportunity pillars

Category (ES / EO weight)	Shared signal	Reading in ES (security)	Reading in EO (opportunity)	Same index values
Energy Access (4 / 3)	EI energy supply per capita, level and growth	availability at the point of use; equally the demand to be secured	depth and growth of the home market	yes
Consumption (5 / 3)	BNEF installed capacity per capita, level and growth	depth of the installed base	home market for equipment and services	yes
Investment (3 / 3)	CIM investment, momentum and capacity	future domestic supply	revealed investor commitment	yes
Energy Prices (2 / 3)	IMF commodity price volatility	price risk for users	investment risk for producers	yes (cell constant)
Trade (4 / 3)	Atlas export market share; world export concentration	a supplier rather than a dependent; few suppliers is a risk	revealed capability; a concentrated market is less contestable	two of three / five
Foreign Dependency (6 / 3)	IEA manufacturing-capacity share; IEA EV regional position	domestic capacity reduces dependence	production base to serve world demand	yes; different missing rules
Production (6 / 3)	mined output of the technology's minerals	domestic supply (demand-weighted, zero-anchored)	demand-weighted share of world supply and its projected change	no

Notes: Nominal category weights of `config/weights.yml` (ES out of 42, EO out of 30). "Same index values": the component enters both pillars as the same theme-level value; imputed values can differ where the missing-data rules or country grids differ.

6.3 Shared signals: exposure and capability

Table 11 lists the raw signals the pillars share, each with the same sign in both. For export share the readings agree; energy use and installed base also measure the demand to be secured; and a concentrated world market may signal an entrenched incumbent, as coded, or unmet demand for an alternative supplier. A shared component induces a positive correlation between the pillars unrelated to any economic link, while in the Strategic Index (equation (55); variable decomposition of equation (20))

$$\frac{\partial \,\mathrm{SI}_{its}}{\partial x_v} = 0.3\,(\omega^{\mathrm{EO}}_{k(v),its}\lambda^{\mathrm{EO}}_{v,its} - \omega^{\mathrm{ES}}_{k(v),its}\lambda^{\mathrm{ES}}_{v,its}), \tag{46}$$

where $\omega^{P}_{k,its}$ is the effective weight of the component's category in pillar P and $\lambda^{P}_{v,its}$ its weight within the category: a shared signal moves the index only as far as it weighs more in one pillar than in the other. The distinctive information lies in the non-shared components, which Section 7 separates.

6.4 Results

Coverage and rankability. The pillar scores 7,131 cells of the eleven headline technologies for 219 economies (195 sovereign, 24 territories) in all 33 pairs, none constant. Twenty-four cells are fully imputed (the Isle of Man 14, Puerto Rico 8, Liechtenstein 2); the mean imputed weight share is

0.43 (0.41, 0.44 and 0.43 by stage). The rankability rule admits 4,301 cells for 178 economies, from 40 in Green Hydrogen midstream to 175 in Nuclear upstream. The excluded cells would mostly have ranked low: they hold only three of the 330 top-ten places computed over all scored economies, and no first place.

Distribution and variance. The median rankable cell scores 0.500 upstream, 0.501 midstream and 0.488 downstream (interquartile ranges 0.434–0.542, 0.427–0.561 and 0.427–0.551; Figure 14). Differences between technologies dominate: pair medians range from 0.367 to 0.585 upstream and from 0.351 to 0.579 midstream, against a mean within-pair interquartile range of 0.037 and 0.082. The clean-technology midstream pairs sit lowest because they carry Foreign Dependency and Investment under the zero rule. Technology-stage fixed effects explain 57.9 per cent of the variance, country effects alone 17.7 per cent and both 77.2 per cent (45.8 per cent of the within-pair variance; ES: 44.4 and 27.8 per cent). Cross-pair comparisons of EO levels, as in the Strategic Index, compare mostly pair effects.

Drivers. In the level (Figure 15), Cost Competitiveness has the largest share at every stage (21.9, 29.2 and 27.5 per cent), followed by Technology Demand. In the within-pair variance, Technological Readiness and Energy Prices contribute nothing and Technology Demand 2 to 6 per cent; the categories that order economies are Cost Competitiveness (39.2, 37.6 and 21.7 per cent by stage), Trade and Production upstream, Trade and Foreign Dependency midstream, and Consumption and Investment downstream. Six technology-level constants carry 38.9 per cent of the average rankable cell's effective weight (46.3, 40.4 and 27.9 per cent by stage), imputed country inputs 17.5 per cent and observed country-specific information 43.6 per cent (Table 12): the pillar's cross-country content is thinnest upstream.

Leaders. China leads 22 of the 33 pairs, the United States five, Chad three, and Canada, Indonesia and Saudi Arabia one each (Table 13); emerging economies, mostly China, hold 26 first places and 46.4 per cent of the top-ten places. Chad's three pairs (Geothermal upstream and midstream, Electric Grid downstream) are those in which only Cost Competitiveness varies, and its leads rest on a recorded lending rate at the floor of the rescaled rate: artefacts, not evidence of opportunity. The median leader's margin is 0.043; China's in Green Hydrogen, Wind, Solar and Batteries midstream are 0.14 to 0.20. In Batteries midstream (Figure 17) China's lead over the United States comes from its manufacturing share and costs; in Solar midstream five Southeast Asian partners take five of the top ten places.

The coalition. Over their rankable cells the 31 core members have a mean within-pair percentile of 0.66, the 26 partners 0.49, other economies 0.44 and China 0.96 (Figure 16). The core is strongest midstream (0.73). The best-placed member is first in 8 pairs and in the top three of 30; in 14 pairs it is a partner, and Indonesia ranks second behind China in five upstream mineral pairs. Both ends of the member ranking are partly construction: four members (Türkiye, Brazil, Argentina, Ghana)

Table 12: Effective weight of the Economic Opportunity pillar by type of input

	Mean effective weight (per cent)			Constants' share
	Technology-level constants	Imputed, country-varying	Observed, country-varying	of the level (per cent)
Upstream	46.3	19.7	33.9	44.0
Midstream	40.4	14.6	45.0	40.2
Downstream	27.9	17.4	54.7	30.4
All rankable cells	38.9	17.5	43.6	38.7
All scored cells	37.7	30.1	32.3	39.5

Notes: The effective weight of variable v in cell (i, t, s) is the renormalised category weight times the variable's weight within its category, $\omega_{kits}\lambda_{vkits}$ (equation (20)); it sums to one in every cell. Technology-level constants are the variables whose value has zero range across all scored economies of the pair: the TRL Index (one value per technology and stage), the WEO global-demand composite, the BCG addressable market, the overcapacity index, the world-market concentration term of Trade and commodity-price volatility. Imputed components are those filled by a missing-data rule (global average or zero) for the economy in question. The first three columns sum to 100 within a row, up to rounding. The last column is the mean share of EO_{its} contributed by the constants. Rows by stage and "All rankable cells": means over the 1,681 upstream, 1,307 midstream and 1,313 downstream rankable cells of the eleven headline technologies (a sovereign economy with at most half of the cell's weight imputed); "All scored cells": means over all 7,131 scored cells. Source: `economic_opportunity_variable_contributions` in `data/processed/outputs/index_outputs.rds`; `scripts/paper/fig_eo.R`.

record nominal lending rates at the winsorisation ceiling, which depresses every pair, and the BNEF regional composite ranks some modelled economies below unmodelled ones (Online Appendix OA.4).

Table 13: Leading economies in each technology-stage pair of the Economic Opportunity pillar (rankable cells)

Technology	n	First	Second	Third	Best member (rank)	USA	CHN
Upstream							
Batteries	120	CHN 0.65	IDN 0.59	AUS 0.57	IDN (2)	7	1
Coal	157	IDN 0.52	CHN 0.51	IND 0.48	IDN (1)	7	2
Electric Grid	163	CHN 0.71	CHL 0.65	RUS 0.65	CHL (2)	27	1
Electric Vehicles	153	CHN 0.68	IDN 0.62	AUS 0.57	IDN (2)	4	1
Gas	165	USA 0.61	CHN 0.60	CAN 0.57	USA (1)	1	2
Geothermal	141	TCD 0.48	CMR 0.48	COG 0.48	PRT (4)	68	6
Green Hydrogen	146	CHN 0.62	IDN 0.62	ZMB 0.57	IDN (2)	62	1
Nuclear	175	CAN 0.65	CHN 0.62	KAZ 0.62	CAN (1)	6	2
Oil	166	USA 0.66	CHN 0.64	CAN 0.64	USA (1)	1	2
Solar	153	CHN 0.70	IDN 0.66	THA 0.61	IDN (2)	23	1
Wind	142	CHN 0.70	IDN 0.59	USA 0.58	IDN (2)	3	1
Midstream							
Batteries	54	CHN 0.77	USA 0.63	HUN 0.55	USA (2)	2	1
Coal	155	CHN 0.59	POL 0.57	IDN 0.56	POL (2)	14	1
Electric Grid	162	CHN 0.68	THA 0.64	POL 0.62	THA (2)	14	1
Electric Vehicles	145	CHN 0.74	ESP 0.64	USA 0.63	ESP (2)	3	1
Gas	156	CHN 0.71	IND 0.67	ITA 0.65	IND (2)	5	1
Geothermal	141	TCD 0.64	CMR 0.64	COG 0.64	PRT (4)	58	5
Green Hydrogen	40	CHN 0.71	DEU 0.50	USA 0.50	DEU (2)	3	1
Nuclear	154	CHN 0.65	FRA 0.65	IND 0.62	FRA (2)	21	1
Oil	156	CHN 0.72	AUT 0.67	ITA 0.66	AUT (2)	11	1
Solar	79	CHN 0.79	IND 0.64	USA 0.63	IND (2)	3	1
Wind	65	CHN 0.74	POL 0.54	IND 0.52	POL (2)	4	1
Downstream							
Batteries	140	CHN 0.75	USA 0.67	ITA 0.63	USA (2)	2	1
Coal	104	CHN 0.68	IDN 0.66	IND 0.66	IDN (2)	15	1
Electric Grid	141	TCD 0.68	CMR 0.68	TGO 0.67	THA (11)	109	20
Electric Vehicles	105	CHN 0.75	VNM 0.73	THA 0.70	VNM (2)	29	1
Gas	104	USA 0.72	SAU 0.71	CHN 0.68	USA (1)	1	3
Geothermal	128	USA 0.61	IDN 0.46	PRT 0.42	USA (1)	1	16
Green Hydrogen	158	USA 0.65	DEU 0.60	THA 0.60	USA (1)	1	4
Nuclear	86	CHN 0.72	FRA 0.67	JPN 0.64	FRA (2)	15	1
Oil	104	SAU 0.71	IRN 0.69	QAT 0.68	SAU (1)	8	98
Solar	148	CHN 0.75	USA 0.67	IND 0.67	USA (2)	2	1
Wind	95	CHN 0.70	USA 0.61	CAN 0.60	USA (2)	2	1

Notes: EO_{its} for the eleven headline technologies. Only rankable cells enter: a sovereign economy with at most half of the cell's index weight imputed (`config/rankability.yml`). Economies are identified by ISO3 code and followed by their score; n is the number of rankable economies in the pair (40 to 175). "Best member" is the highest-ranked of the 57 coalition members (31 core, 26 partners), with its rank in parentheses; the USA and CHN columns give the ranks of the United States and China. A rank is one plus the number of rankable economies with a strictly higher score. All 33 pairs are scored and none is constant across economies. In geothermal upstream and midstream and electric-grid downstream, Cost Competitiveness is the only category that varies across economies. Codes: AUS Australia, AUT Austria, CAN Canada, CHL Chile, CMR Cameroon, COG Congo, DEU Germany, ESP Spain, FRA France, HUN Hungary, IDN Indonesia, IND India, IRN Iran, ITA Italy, JPN Japan, KAZ Kazakhstan, POL Poland, PRT Portugal, QAT Qatar, RUS Russia, SAU Saudi Arabia, TCD Chad, TGO Togo, THA Thailand, VNM Viet Nam, ZMB Zambia, USA United States, CHN China. Source: `economic_opportunity_index` in `data/processed/outputs/index_outputs.rds`; `scripts/paper/fig_eo.R`.

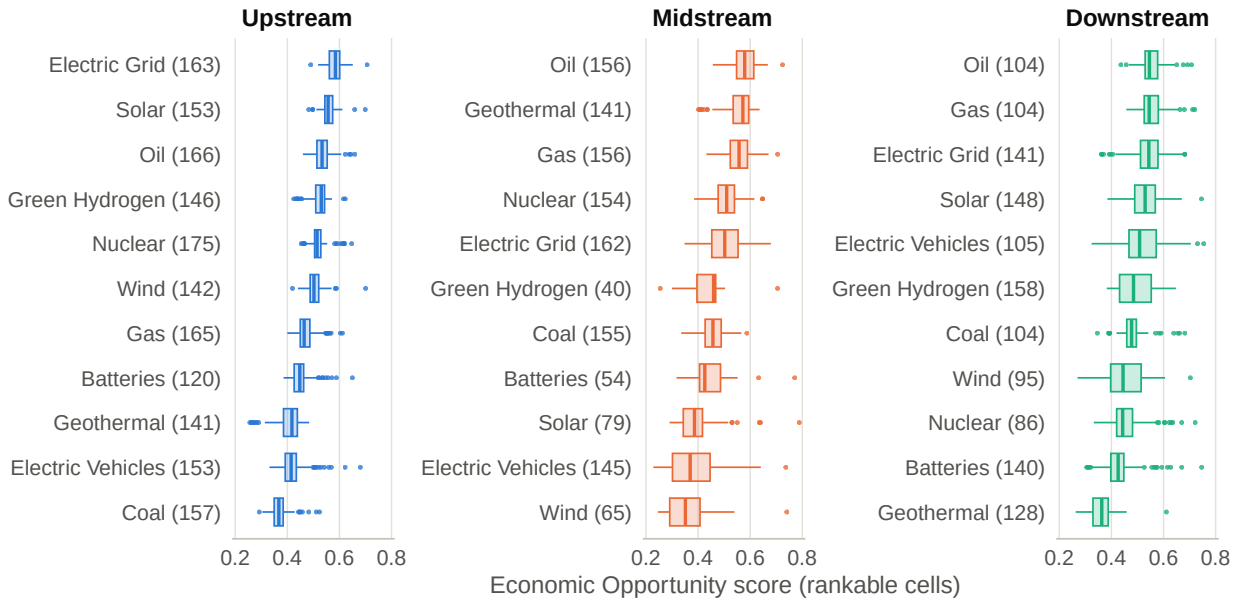


Figure 14: Economic Opportunity by technology and stage over the rankable economies of each pair (number in parentheses, 40 to 175; sovereign economy with at most half of the index weight imputed). Boxes show the median and interquartile range, whiskers 1.5 interquartile ranges. Sample: 4,301 rankable of 7,131 scored cells; all 33 pairs have status *ok*. Source: `economic_opportunity_index` in `index_outputs.rds`; `scripts/paper/fig_eo.R`.

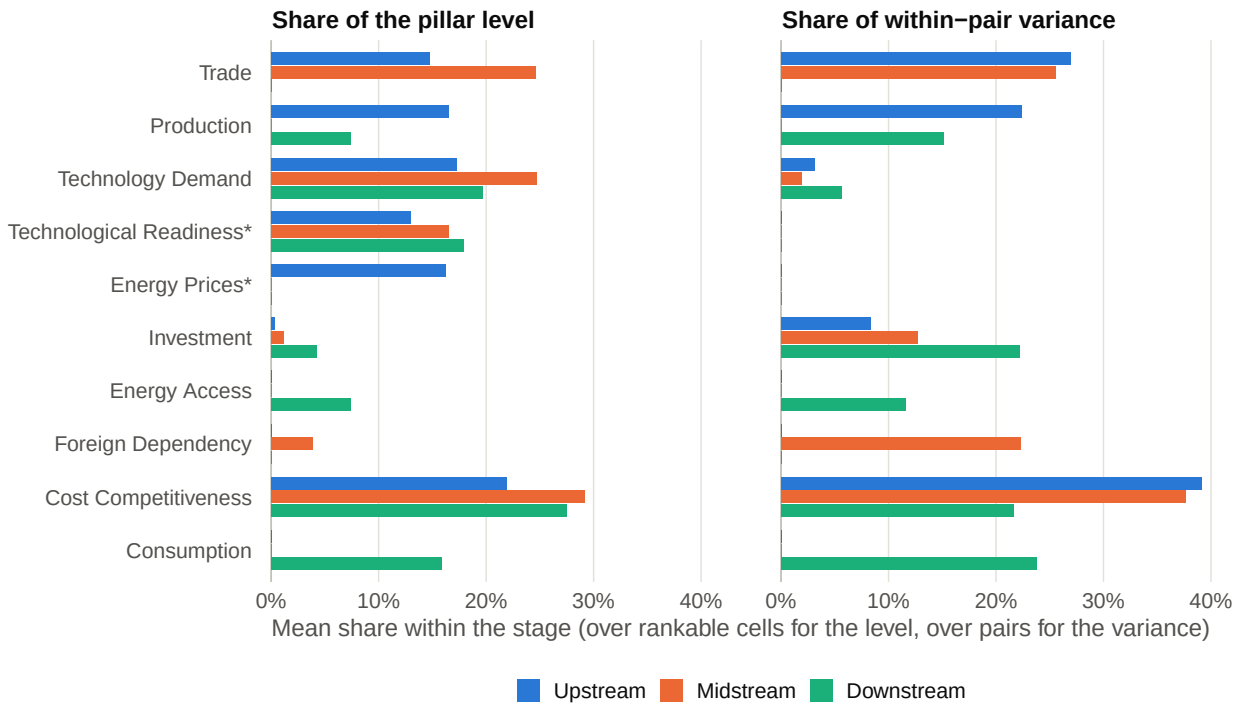


Figure 15: Composition of Economic Opportunity by stage. Left: mean share of each category in the level, $\omega_{kits}c_{kits}/\mathrm{EO}_{its}$ (equation (19)). Right: mean share in the within-pair variance, $\mathrm{cov}(\omega_{kits}c_{kits}, \mathrm{EO}_{its})/\mathrm{var}(\mathrm{EO}_{its})$. Starred categories (Technological Readiness, Energy Prices) are constant within every pair; Technology Demand is constant except for its BNEF regional and EV-market composites. Sample: 4,301 rankable cells. Source: category contributions in `index_outputs.rds`; `scripts/paper/fig_eo.R`.

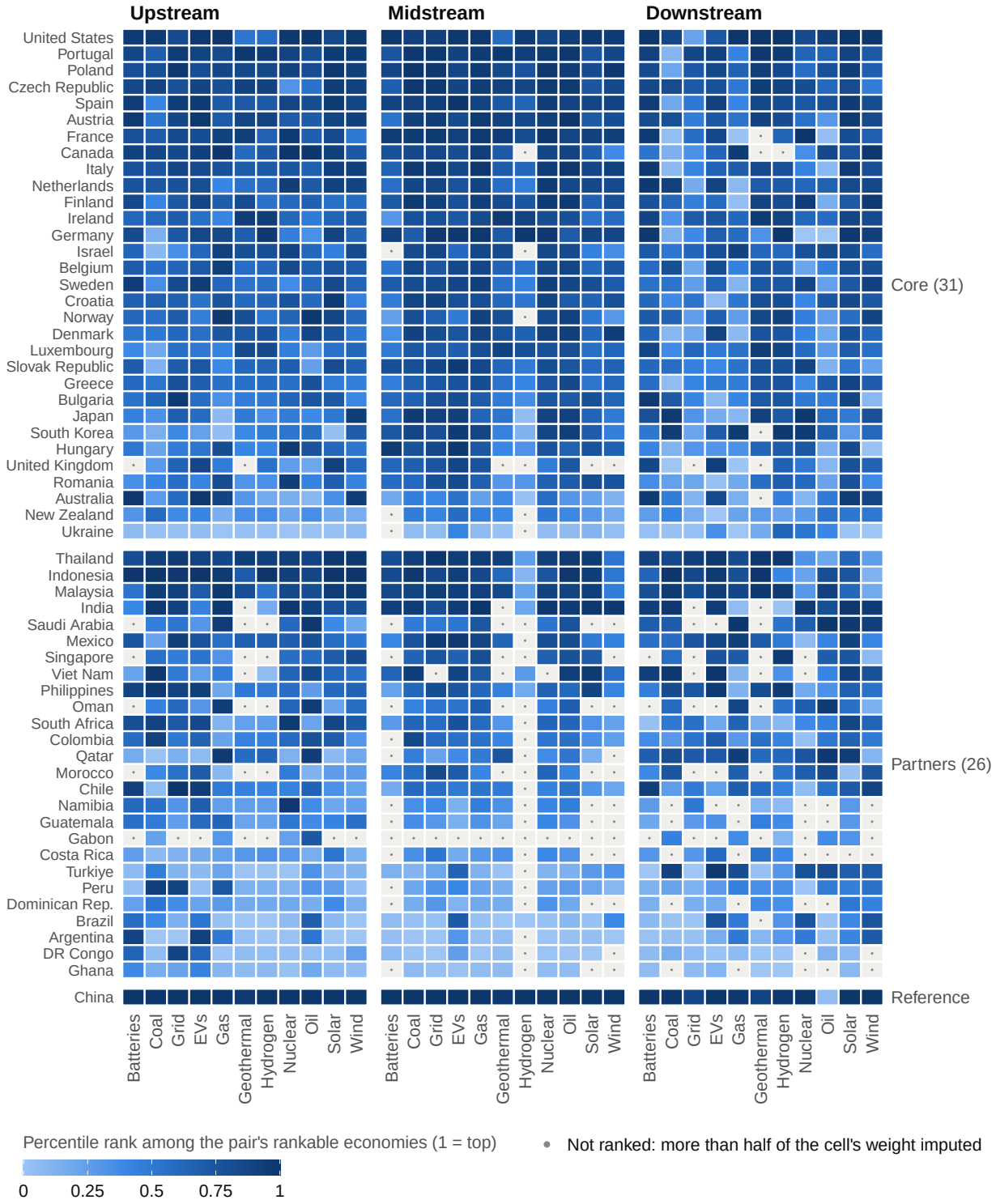


Figure 16: Within-pair percentile rank of EO_{its} (equation (9)) of the 31 core members, the 26 partners (Section 8) and China, by pair. Ranks rather than levels are shown because pair effects account for 58 per cent of the variance of the level. Gray cells with a dot are scored but not rankable (more than half of the weight imputed). Source: `economic_opportunity_index`, `coalition_members`; `scripts/paper/fig_eo.R`.

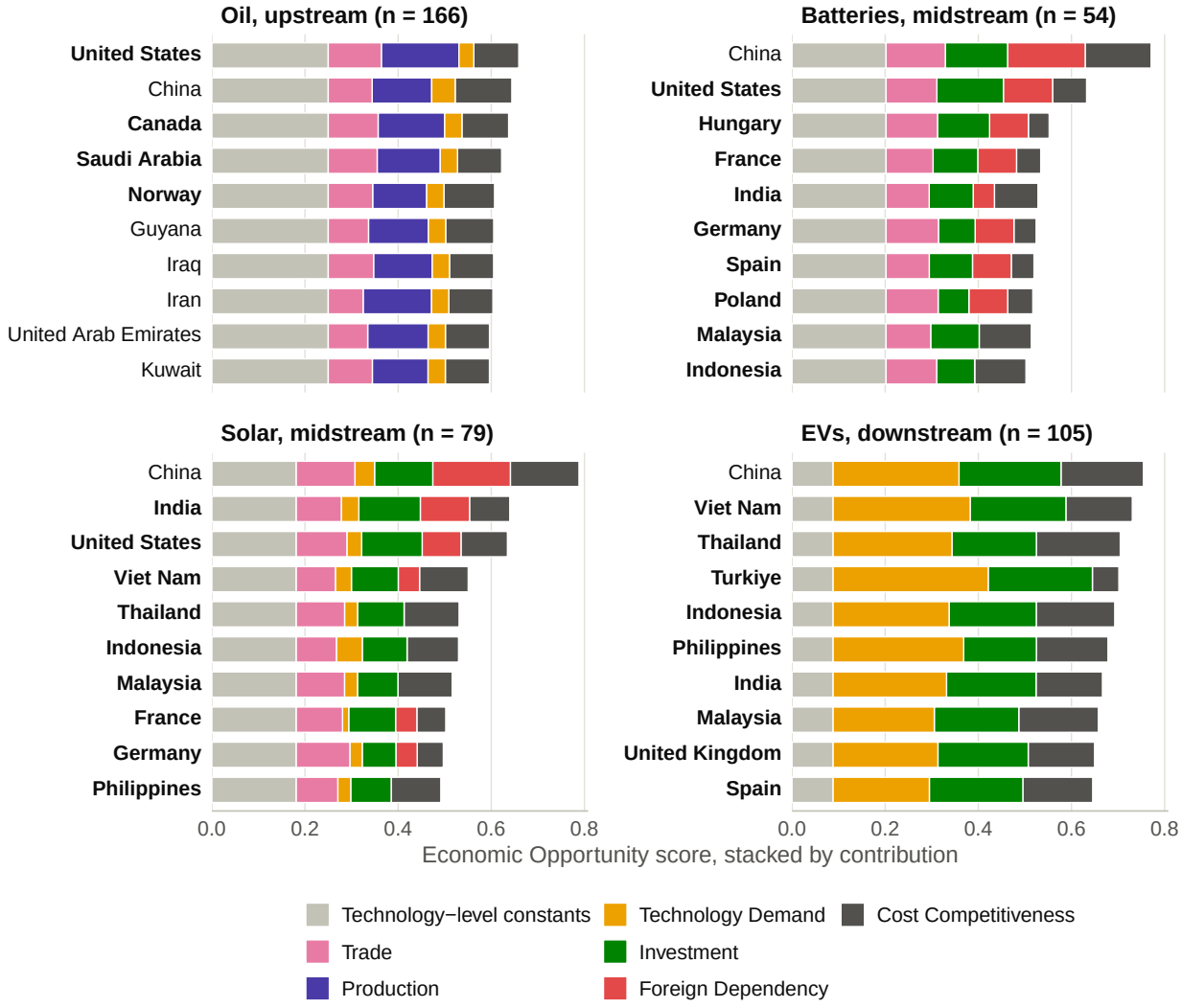


Figure 17: The ten highest-scoring rankable economies in Oil upstream ($n = 166$), Batteries midstream (54), Solar midstream (79) and EV downstream (105), stacked by the variable decomposition of equation (20); colour identifies the category. The gray segment sums the technology-level constants (0.250, 0.202, 0.181 and 0.088), the same for every economy in a pair. Coalition members in bold. Source: `economic_opportunity_index` and its variable contributions; `scripts/paper/fig_eo.R`.

7 The opportunity-security space

Each country-technology-stage cell is a point $(\mathrm{ES}_{its}, \mathrm{EO}_{its})$ of the unit square. The Strategic Index of Section 9 reads the two pillars together, and its pillar terms in equation (55) carry equal weights of opposite sign, so they reduce to the gap between opportunity and security:

$$0.3\,\mathrm{EO}_{its} + 0.3\,(1 - \mathrm{ES}_{its}) = 0.3 + 0.3\,\Delta_{its}, \qquad \Delta_{its} \equiv \mathrm{EO}_{its} - \mathrm{ES}_{its}. \tag{47}$$

The index favours cells in which opportunity exceeds security, and treats alike a cell that is secure and well placed and one that is neither. Because it ranks a country's own sectors, what matters is how Δ_{its} varies across them:

$$\mathrm{Var}_i(\Delta) = \mathrm{Var}_i(\mathrm{EO}) + \mathrm{Var}_i(\mathrm{ES}) - 2\,\mathrm{Cov}_i(\mathrm{EO}, \mathrm{ES}). \tag{48}$$

If the pillars rise together across a country's sectors, the gap is compressed and the policy term and sector priors move the ranking more than their nominal weights suggest (Paruolo et al., 2013; Becker et al., 2017).

Part of the association is built in. Four categories enter both pillars with identical values, and Trade and Foreign Dependency share components (Table 11). By the variable decomposition of equation (20), each pillar splits into the contribution of the shared components and of the rest, $P_{its} = A^P_{its} + B^P_{its}$, and for any set of cells

$$\mathrm{Cov}(\mathrm{ES}, \mathrm{EO}) = \mathrm{Cov}(A^{\mathrm{ES}}, A^{\mathrm{EO}}) + \mathrm{Cov}(A^{\mathrm{ES}}, B^{\mathrm{EO}}) + \mathrm{Cov}(B^{\mathrm{ES}}, A^{\mathrm{EO}}) + \mathrm{Cov}(B^{\mathrm{ES}}, B^{\mathrm{EO}}). \tag{49}$$

With fixed weights the first term is a positively weighted sum of the covariances of the shared components: it records construction, not an economic link. In the gap those components enter with coefficients $\omega^{\mathrm{EO}}\lambda^{\mathrm{EO}} - \omega^{\mathrm{ES}}\lambda^{\mathrm{ES}}$ and partly cancel (equation (46)), so the information on where the gap is large lies mainly in B^{ES} and B^{EO}.

We report within-pair rank correlations from the full pillars and from B^{ES} and B^{EO} alone, and classify cells by each pillar's position relative to its within-pair median. The four regions correspond to the four strategic postures of Carey et al. (2025), which read the same two-dimensional space, in the labelling of the dashboard built on these indices: opportunity without security (leapfrog: build capability), leaders (export promotion), laggards (friendshore: secure supply through partners, the case of Sections 10 and 11) and security without opportunity (alternative value creation). The correspondence is a reading aid, with the median of each pair as the cut. As New Energy Industrial Strategy Center (2026) stresses, security risk and economic opportunity are not synonyms, and the same exposure can warrant different postures; the results below measure how far the two pillars carry distinct information.

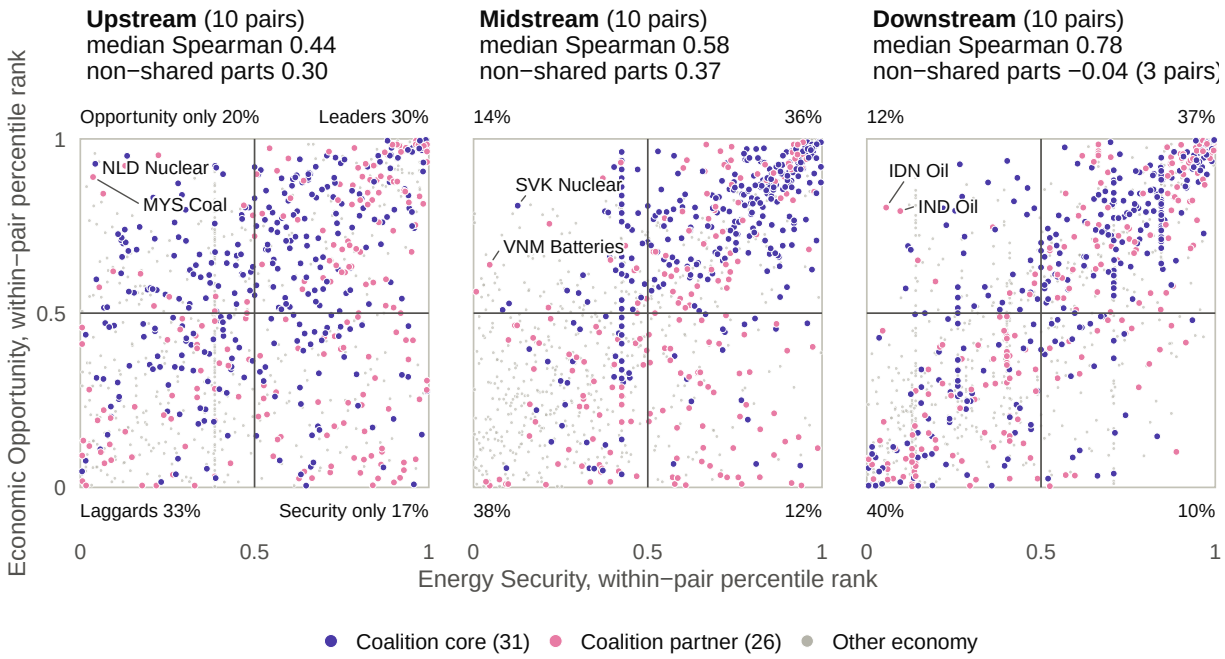


Figure 18: The opportunity-security space. Each point is a cell rankable on both pillars, at its within-pair percentile rank on ES_{its} (horizontal) and EO_{its} (vertical); panels are stages. Violet: the 31 core members; pink: the 26 partners; gray: other economies. Lines at 0.5 mark the within-pair medians and divide the regions (a cell at the median counts as low); corner percentages are the stage's shares. Panel headers give the median within-pair Spearman correlation from the full pillars and from the non-shared parts B^{ES}, B^{EO} of equation (49). The dense column left of the upstream security median is Nuclear upstream, where 87 of 175 economies without uranium share $\mathrm{ES}_{its} = 0.172$. Sample: 3,157 cells, 175 economies, 30 pairs. Source: `index_outputs.rds`; `scripts/paper/fig_joint.R`.

7.1 Results

Sample and co-movement. Both pillars score 6,232 cells; 3,157 cells for 175 economies in 30 pairs are rankable on both (607 territory cells and 2,468 sovereign cells over the imputation threshold are set aside), among them 1,500 cells of the 57 coalition members. The median within-pair Spearman correlation of ES_{its} and EO_{its} is 0.59 (interquartile range 0.45 to 0.77), from −0.21 in Green Hydrogen midstream to 0.94 in Electric Vehicles midstream (Table 14). It rises along the chain: 0.44 upstream, 0.58 midstream and 0.78 downstream (Figure 18), a gradient that also holds within most technologies.

How much is mechanical. Shared components carry 61 per cent of the effective weight of ES_{its} (25, 62 and 91 per cent by stage) but 28 per cent of that of EO_{its}; in seven downstream pairs every security component also enters opportunity. On the 23 pairs where both pillars keep components of their own, the median correlation falls from 0.55 to 0.30 when only non-shared components are used. The shared term accounts for 61 per cent of the pooled within-pair covariance (7, 42 and 87 per cent by stage). What remains runs mainly through trade: the largest cross-pillar category

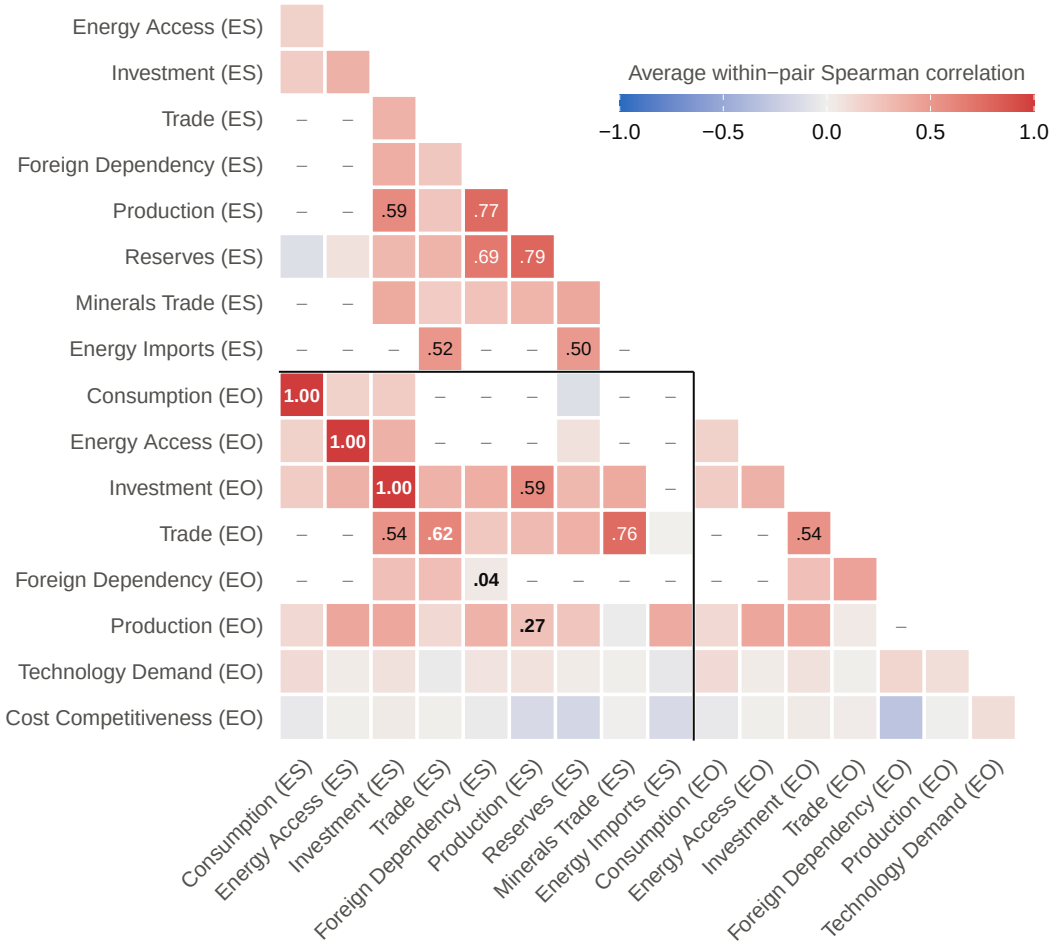


Figure 19: Average within-pair Spearman correlation between the categories of the two pillars (pairs with at least 20 rankable economies in which both categories vary). The black lines separate the cross-pillar block (bottom left) from the within-pillar blocks; bold numbers mark categories entering both pillars under the same name. Energy Prices and Technological Readiness, constant within every pair, are omitted. Sample: as Figure 18. Source: category scores in `index_outputs.rds`; `scripts/paper/fig_joint.R`.

correlation is between security's Minerals Trade and opportunity's Trade (0.76), while Technology Demand and Cost Competitiveness are nearly orthogonal to every security category (Figure 19). Net of shared inputs, security and opportunity are close to unrelated across economies except where exports reveal both.

Regions. Pooling all pairs, 15.1 per cent of cells have opportunity without security, 34.6 per cent are leaders, 13.2 per cent have security without opportunity and 37.1 per cent are laggards (12.6, 37.7, 15.3 and 34.4 per cent if ties count as high). The shares largely restate the rank correlation: a Gaussian copula with each pair's correlation implies an off-diagonal share of 14.6 per cent, against 14.2 observed. Opportunity without security is most common upstream (19.5 per cent), where the pillars are least related. The largest pair shares, Green Hydrogen midstream (45 per cent) and Nuclear upstream (28), are tie artefacts (Figure 20); without them the pooled share is 14.0 per cent.

Table 14: Energy Security and Economic Opportunity within technology-stage pairs, by stage

	Upstream	Midstream	Downstream	All
Technology-stage pairs	10	10	10	30
Country cells	999	1,048	1,110	3,157
Spearman correlation of ES_{its} *and* EO_{its} *within pairs*				
Median across pairs	0.44	0.58	0.78	0.59
Lowest pair	0.03	−0.21	0.40	−0.21
Highest pair	0.63	0.94	0.92	0.94
Pooled within-pair ranks	0.39	0.60	0.71	0.57
Components shared by the two pillars				
Effective weight in ES	0.25	0.62	0.91	0.61
Effective weight in EO	0.26	0.20	0.37	0.28
Pairs with both non-shared pillars defined	10	10	3	23
Median correlation, non-shared pillars	0.30	0.37	−0.04	0.30
Median correlation, full pillars, same pairs	0.44	0.58	0.58	0.55
Share of level covariance, shared term	0.07	0.42	0.87	0.61
Share of level covariance, non-shared term	0.58	0.20	0.00	0.16
Share of cells by region of the space (percent)				
Opportunity without security	19.5	13.8	12.4	15.1
Leaders (both high)	30.2	36.0	37.2	34.6
Security without opportunity	17.2	12.5	10.2	13.2
Laggards (both low)	33.0	37.7	40.2	37.1

Notes: Eleven headline technologies; cells rankable on both pillars only (a sovereign economy with at most half of the pillar's weight imputed), which leaves 3,157 of the 6,232 cells that both pillars score, for 175 economies. A technology-stage pair enters when both pillars score it and neither pillar's diagnostics flag it as constant or unscored, which leaves 30 of the 33 pairs: Energy Security does not score downstream grid or upstream and midstream geothermal. Correlations are Spearman correlations across economies within a pair; the pooled row is the Pearson correlation of within-pair percentile ranks over all cells of the stage. A component is shared when the other pillar contains a component of the same category with identical observed values; its effective weight is its category weight times its within-category weight, averaged over cells. Non-shared pillars are the weighted means of the remaining components with the weights renormalised; they are undefined where every component of ES is shared, as in 7 downstream pairs, and the last two correlation rows use only the pairs in which both are defined. The covariance shares decompose the within-pair covariance of the pillar levels as in equation (49), pooled over pairs with weights $n-1$: the shared term is $\mathrm{Cov}(A^{\mathrm{ES}}, A^{\mathrm{EO}})$, the non-shared term $\mathrm{Cov}(B^{\mathrm{ES}}, B^{\mathrm{EO}})$, and the two cross terms make up the remainder. Regions are defined by the within-pair medians of the two pillars, a cell at the median counting as low. Source: `index_outputs.rds`; `scripts/paper/fig_joint.R`.

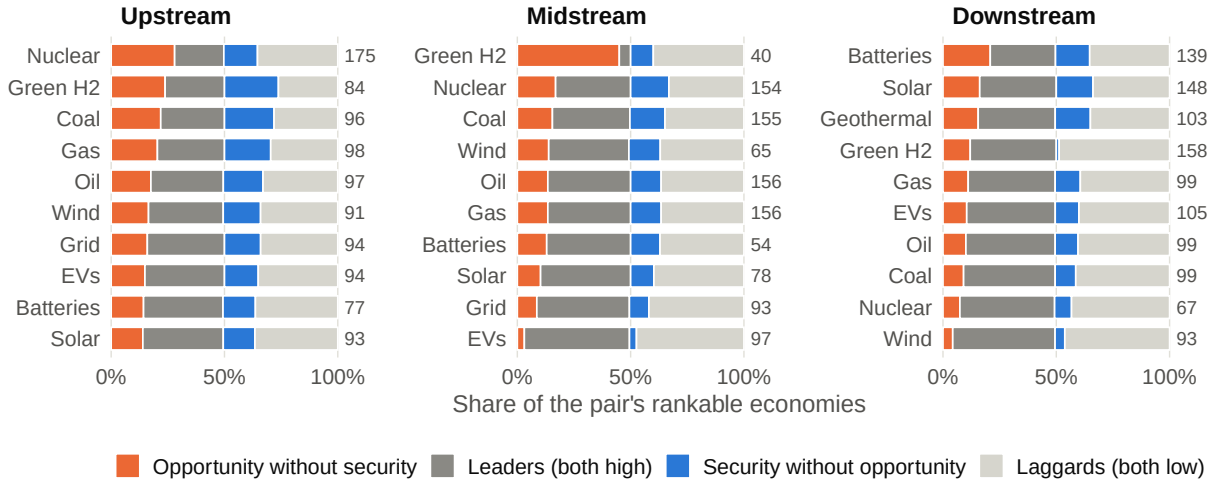


Figure 20: Shares of each pair's rankable economies in the four regions of Figure 18, by stage, ordered by the share with opportunity but not security (orange); blue marks security without opportunity, dark gray leaders, light gray laggards. The two longest orange segments are ties at the security median: in Green Hydrogen midstream 34 of 40 economies share the lowest ES_{its}, and in Nuclear upstream 87 of 175 share the median value. Source: as Figure 18.

The coalition. Of the members' cells, 49.3 per cent are leaders and 12.8 per cent have opportunity without security (core 55.1 and 16.4 per cent, partners 40.4 and 7.4; Figure 21), partly because scale raises both pillars; the United States is a leader in 29 of 30 pairs. Austria, Portugal and Luxembourg have the most leapfrog-type pairs (11 of 30 each); Brazil (19 of 29), Peru and Türkiye have the most security without opportunity.

Within countries and the Strategic Index. Across the sectors of each of the 167 countries with at least five Strategic Index cells, the correlation of EO_{its} and ES_{its} averages 0.46 (from -0.56 to 0.84), and the covariance term removes on average 36 per cent of $\mathrm{Var}_i(\mathrm{EO}) + \mathrm{Var}_i(\mathrm{ES})$ (42 per cent for members), leaving a standard deviation of Δ_{its} of 0.13. The index does favour the leapfrog region (mean 0.563 there, against 0.511 to 0.545 elsewhere), and 8 of the 57 allied picks lie in it (14.0 per cent, against 6.7 per cent of eligible sectors). Four of the nine Nuclear midstream picks come from small partners with four or five eligible sectors; their security ranks lie at or above a compressed median, so the picks reflect the technology weight of equation (55) (0.999 for nuclear against 0.42 for gas), not a security floor. Across a country's sectors the index correlates 0.63 with the level gap but only 0.19 with the gap in within-pair ranks: it reflects which pairs score high opportunity and low security for everyone at least as much as where the country stands. Online Appendix OA.5 gives further detail and caveats.

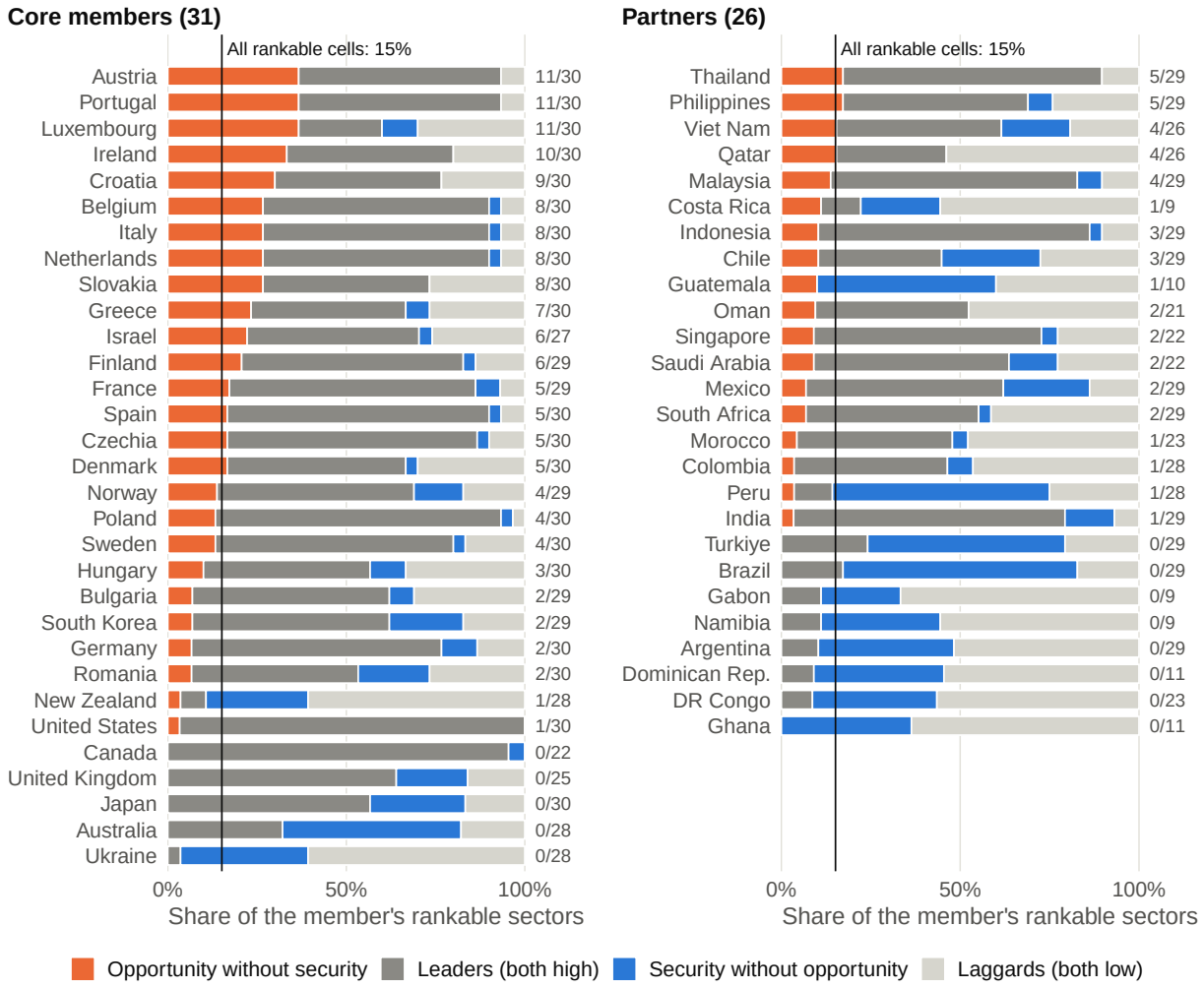


Figure 21: Each coalition member's rankable pairs divided among the four regions: the 31 core members (left) and 26 partners (right), ordered by the share with opportunity but not security; the fraction gives that count over the member's rankable pairs (9 to 30). The vertical line is the all-cell share (15 per cent). Nuclear upstream and Green Hydrogen midstream account for 25 of the 192 member pairs in orange. Source: as Figure 18.

8 Geopolitical alignment and the allied coalition

Three parts of the analysis need a coalition: the allied picks of the Strategic Index (Section 9), the viewpoint from which the Partnership Strength Index rates partners (Section 10) and the nodes of the network design (Section 11). We derive it by rule from two observable attributes, an economy's alignment in the United Nations General Assembly and its weight in world supply, rather than fixing it by hand.

8.1 Ideal points and the U.S.–China axis

Alignment is measured with the ideal points of Bailey et al. (2017), which place every voting state on one dimension of revealed foreign-policy preference, bridged so that positions are comparable across sessions. We use the all-votes series of the July 2026 release (Dataverse V39; Voeten, 2009) and average each economy's ideal point θ_i over sessions 74 to 78 (2019–2023), which covers the 193 economies that vote in their own right. Each is placed on the segment between the two anchors,

$$s_i \;=\; \min\Big\{1,\; \max\Big\{0,\; \frac{\theta_i - \theta_{\mathrm{CHN}}}{\theta_{\mathrm{USA}} - \theta_{\mathrm{CHN}}}\Big\}\Big\}, \tag{50}$$

so that $s_i = 0$ is China's position and $s_i = 1$ that of the United States ($\theta_{\mathrm{USA}} = 2.34$, $\theta_{\mathrm{CHN}} = -1.02$). Seven economies vote further from the United States than China does and are clamped to zero (Cuba, Iran, Nicaragua, North Korea, Syria, Venezuela and Zimbabwe). The axis records revealed preferences relative to two anchors, not alliance or any bilateral relationship.

Following Gopinath et al. (2025), blocs are quartiles of the axis: the U.S.-leaning bloc is the quarter of voting economies closest to the United States ($s_i \geq 0.511$, 49 economies), the China-leaning bloc the quarter closest to China ($s_i \leq 0.142$, 49), and the 95 in between are nonaligned. Membership is hard: the bloc indicator p_i is 1 for U.S.-leaning, $\frac{1}{2}$ for nonaligned and 0 for China-leaning economies. Because the United States often votes in small minorities, the distribution is skewed toward China (median $s = 0.22$), and a midpoint rule would call 142 of 193 economies China-leaning; quartiles keep that bloc to the quarter of the Assembly closest to China. After the cuts are set, 44 territories inherit their sovereign's bloc and Taiwan, Palestine and Kosovo are neutral ($p_i = \frac{1}{2}$); these assignments serve only the bilateral indices, and an economy without its own vote cannot be a core member.

For a pair of economies the dyadic alignment is

$$A_{ij} \;=\; p_i\, p_j + (1 - p_i)(1 - p_j), \tag{51}$$

which would be the probability that i and j fall on the same side of the divide if p_i were a probability; with hard blocs it is 1 for the same bloc, 0 for opposite blocs and $\frac{1}{2}$ whenever either is nonaligned. It is a Friendshore component with weight 2 of 11 (Section 10.2) and through Friendshore enters the edge term of the network design. Figure 22 shows the axis by coalition tier.

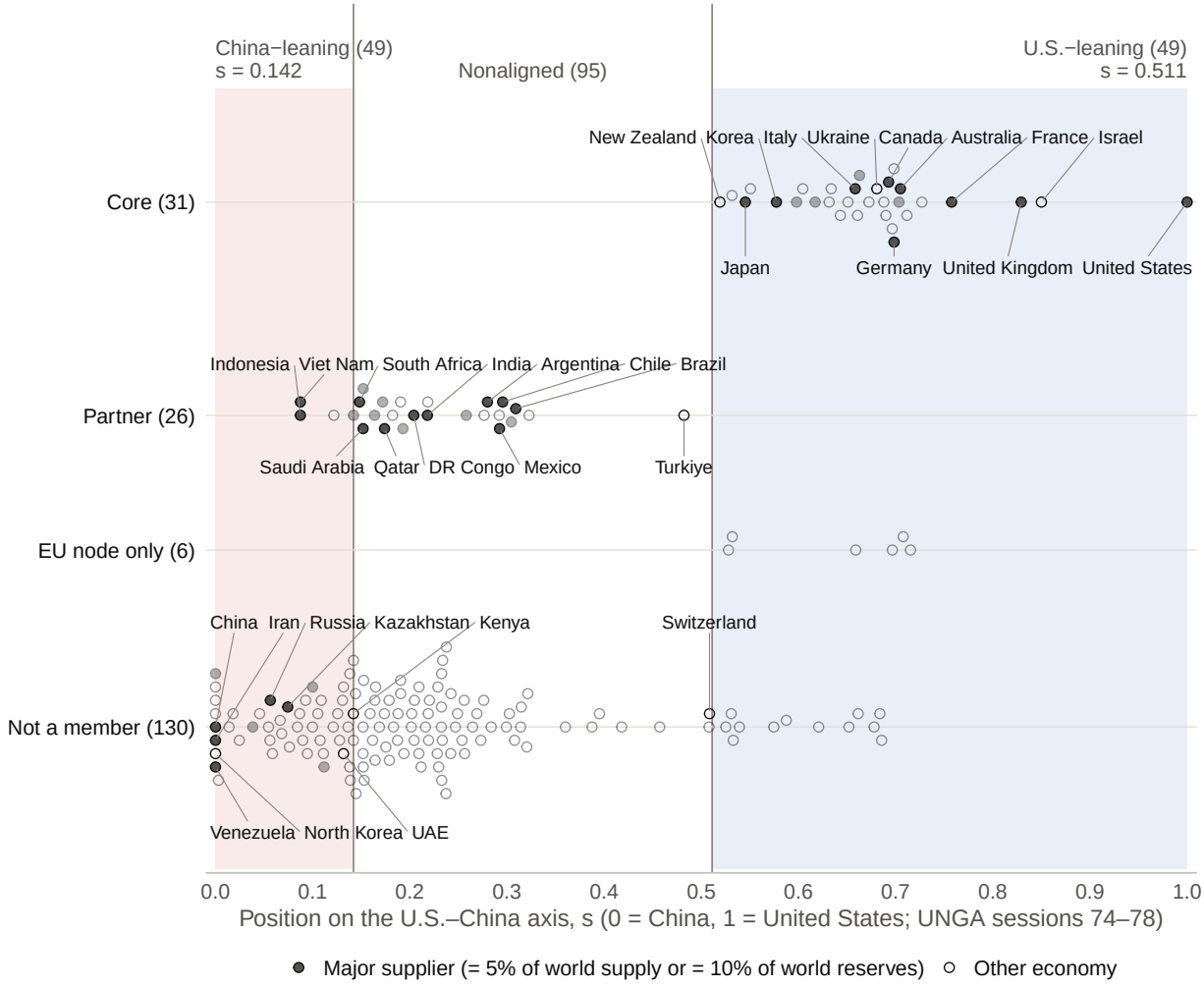


Figure 22: Positions on the U.S.–China axis and coalition tiers. Each point is one of the 193 economies with their own votes in the General Assembly, at s_i of equation (50): the mean all-votes ideal point of Bailey et al. (2017) over sessions 74–78 (2019–2023), projected on the segment from China ($s = 0$) to the United States ($s = 1$) and clamped to $[0, 1]$. Vertical lines are the quartile cuts ($s \leq 0.142$ China-leaning, $s \geq 0.511$ U.S.-leaning; 49, 95 and 49 economies). Rows are the 31 core members, the 26 partners, the six EU members that enter only through the EU node and the 130 other voting economies; vertical position within a row carries no information. Filled markers are major suppliers (at least 5 percent of world supply in an Upstream or Midstream stage, or at least 10 percent of world reserves of a headline mineral, fuel or uranium). G20 economies and selected other cases are labelled. Source: `geo_ideal_points_tbl.rds`, `coalition_tbl.rds`, `major_supplier_shares_tbl.rds`.

8.2 A coalition in two tiers

The *core* of the coalition $\mathcal{N}$ is the U.S.-leaning bloc restricted to economies with their own UN votes and a GDP of at least USD 100 billion (2025; World Bank, 2026): 31 economies. An economy outside the core is a *partner* if it qualifies by either of two routes. The agreement route admits members of a U.S. free trade agreement, the Indo-Pacific Economic Framework (IPEF), the Minerals Security Partnership (MSP), the G7 or NATO with GDP of at least USD 100 billion, whatever their bloc. The supplier route admits, without a GDP floor, any economy that is not China-leaning and is a major supplier of an Upstream or Midstream stage of a headline technology: it holds at least 5 percent of world supply in one measure of the stage (exports of the stage's HS6 basket, fuel or uranium production, mineral mining or refining, or manufacturing capacity), or at least 10 percent of world reserves of a resource the stage extracts, read from the reserves tables of the Energy Security pillar (U.S. Geological Survey, 2026; Energy Institute, 2026; OECD Nuclear Energy Agency and International Atomic Energy Agency, 2026). The reserves bar is twice the supply bar because a stock is not yet supply; holders that USGS does not quantify count as zero. Online Appendix OA.6 lists the source and year of each measure.

The rule yields 26 partners (Table 15, Figure 23): ten qualify through agreements only, nine as suppliers only and seven by both routes. Twenty-three are nonaligned and three China-leaning: Indonesia and Viet Nam, members of IPEF, and Oman, a U.S. free-trade partner. The 57 members account for 73.4 percent of world GDP (core 57.7, partners 15.7); China accounts for 16.6 percent.

Forty economies pass the supplier test, and every one that is not China-leaning is a member. The nine China-leaning major suppliers are kept out: China, Russia, Iran, Kazakhstan, Algeria, Uzbekistan, Venezuela, Myanmar and Zimbabwe, with largest qualifying shares from 94.4 percent of world graphite refining (China) and 45.1 percent of exports of the uranium basket (Kazakhstan) to 5.1 percent of gas exports (Algeria). Eleven members hold at least 10 percent of a world reserve (Indonesia, China-leaning, cannot use the supplier route), and for none of the other ten is the reserves leg decisive. Brazil, in none of the five agreements, holds 23.9 percent of world graphite reserves, 16.7 percent of manganese, 14.6 percent of rare earths and 11.4 percent of nickel, and also supplies 18.4 percent of world exports of silicon metal (the Solar Upstream basket). Among all economies the reserves leg is decisive only for Venezuela, which is kept out.

Two economies a reader might expect are not members. Switzerland is nonaligned at $s = 0.508$, three thousandths below the U.S.-leaning cut, belongs to none of the agreements, and its largest supply share is 1.1 percent; its former supplier route rested on electricity exports, no longer part of the Electric Grid basket (Section 3). Taiwan has no UN vote and is therefore neutral, appears in no agreement list and has no World Bank GDP; its largest share is 1.5 percent (Solar Midstream exports), and its semiconductor strength lies outside the energy baskets. Both remain partners in the bilateral indices but are not network nodes.

The coalition is long in mining and short in processing: its economies mine 81.9 percent of the world's cobalt and 60.0 percent of its lithium but refine 17.4 and 24.6 percent, and 5.5 percent of graphite, against China's 75.6, 73.7 and 94.4 percent (International Energy Agency, 2026a); Online

Table 15: Composition of the allied coalition

Group	n	Economies
Core: U.S.-leaning bloc, GDP $\geq$ USD 100bn	31	United States, Germany, Japan, United Kingdom, France, Italy, Canada, Spain, Korea, Australia, Netherlands, Poland, Belgium, Ireland, Sweden, Israel, Austria, Norway, Denmark, Romania, Czechia, Portugal, Finland, Greece, New Zealand, Hungary, Ukraine, Slovakia, Bulgaria, Croatia, Luxembourg
Partners, agreement route only	10	Türkiye (NATO), Indonesia (IPEF), Singapore (IPEF, U.S. FTA), Thailand (IPEF), Viet Nam (IPEF), Morocco (U.S. FTA), Dominican Rep. (U.S. FTA), Guatemala (U.S. FTA), Oman (U.S. FTA), Costa Rica (U.S. FTA)
Partners, supplier route only	9	Brazil (Solar Up 18% [+3]), Saudi Arabia (Oil Up 13%), Argentina (Batteries Up 6% [+1]), South Africa (Hydrogen Up 52% [+3]), Qatar (Gas Up 14%), Ghana (Batteries Up 10% [+2]), DR Congo (Batteries Up 66% [+9]), Gabon (Batteries Up 25% [+2]), Namibia (Nuclear Up 12%)
Partners, both routes	7	India (IPEF, MSP), Mexico (U.S. FTA), Philippines (IPEF), Malaysia (IPEF), Colombia (U.S. FTA), Chile (U.S. FTA), Peru (U.S. FTA)
EU members in the EU node only	6	Lithuania, Slovenia, Latvia, Estonia, Cyprus, Malta
Major suppliers kept out (China-leaning)	9	China (Batteries Mid 94% [+16]), Russia (Hydrogen Up 29% [+6]), Iran (Gas Up 6% [+1]), Kazakhstan (Nuclear Up 45%), Algeria (Gas Up 5%), Uzbekistan (Nuclear Up 8%), Venezuela (Oil reserves 18%), Myanmar (Wind Up 31% [+1]), Zimbabwe (Batteries Up 9% [+2])
Reserves holders between 5% and 10%, not members	6	Madagascar (Graphite 8.7%), Mozambique (Graphite 8.1%), Kuwait (Oil 5.9%), Tanzania (Graphite 5.8%), Niger (Uranium 5.7%), New Caledonia (Nickel 5.1%)
Added by the MSP Forum sensitivity	3	Kazakhstan, Uzbekistan, Ecuador
Hand-picked list of the previous draft: not members	2	UAE, Kenya
Hand-picked list of the previous draft: members	21	11 in the core, 10 partners

Notes: Economies are ordered by GDP within a group. Core: the U.S.-leaning quartile bloc of the all-votes ideal points of Bailey et al. (2017) over UNGA sessions 74–78 (2019–2023), $s \geq 0.511$, with the economy's own UN votes and latest World Bank GDP of at least USD 100bn. Partners (not core): a member of a U.S. free trade agreement, IPEF, the Minerals Security Partnership (MSP), the G7 or NATO with GDP of at least USD 100bn (qualifying agreements in parentheses), or, without a GDP floor, an economy that is not China-leaning and holds at least 5% of world supply in one measure of an Upstream or Midstream stage of a headline technology (exports of the stage's HS6 basket, fuel or uranium production, mineral mining or refining, or manufacturing capacity) or at least 10% of world reserves of a headline mineral, fuel or uranium. For the supplier route, the parentheses give the largest qualifying share: the technology–stage cell of a supply measure, or the resource when only reserves qualify, and in brackets the number of further qualifying cells. The six EU members below the GDP floor are part of the EU network node only. The MSP Forum sensitivity admits Forum members by the agreement route. The previous draft's 23 members were fixed by hand.

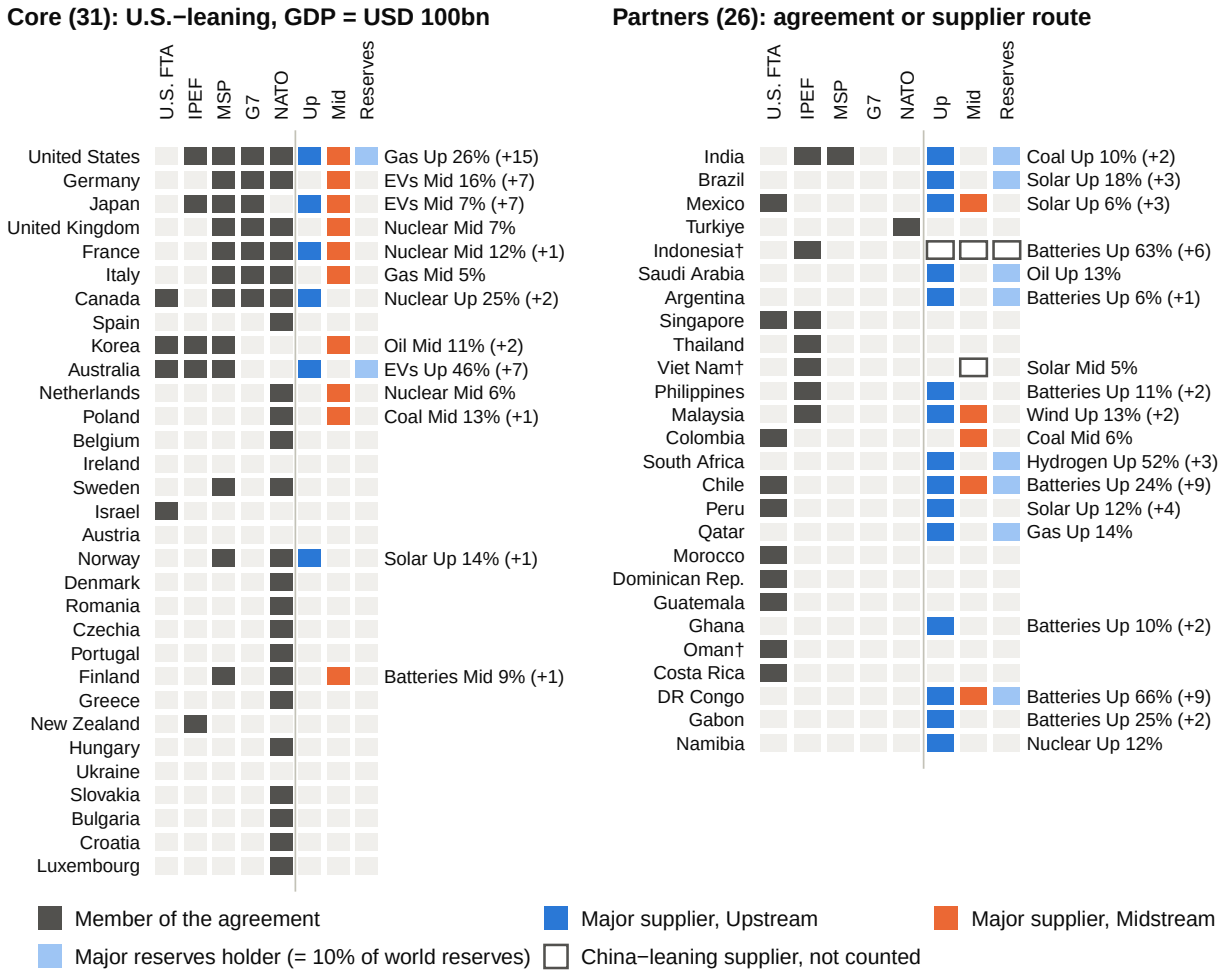


Figure 23: Routes into the coalition. Rows are the 31 core members (left) and the 26 partners (right), ordered by GDP. Gray tiles mark membership of the five agreements of the agreement route (U.S. free trade agreement, IPEF, MSP, G7, NATO). Coloured tiles mark the supplier test: at least 5 percent of world supply in an Upstream (blue) or Midstream (orange) stage, or at least 10 percent of world reserves (light blue). Outlined tiles (†) mark economies that pass the test but are China-leaning and so cannot use the supplier route; Indonesia and Viet Nam enter through IPEF and Oman through its U.S. free trade agreement. The core requires neither route. The text gives the largest qualifying share, with the number of further qualifying cells in parentheses. Source: `coalition_tbl.rds`, `major_supplier_shares_tbl.rds`, `economic_security_agreements.csv`.

Appendix OA.6 gives the supply cover by stage.

8.3 Network nodes and the European Union

The EU-27 enters the network design as a single node, because the Union conducts a common commercial policy and its members' supply chains are integrated in a single market. Twenty-one EU members are core members in their own right; the six below the GDP floor (Cyprus, Estonia, Latvia, Lithuania, Malta and Slovenia, 1.6 percent of EU GDP) belong to the network only through the node. The network therefore has 37 nodes representing 63 economies: 11 core nodes (the United States, the EU, Japan, the United Kingdom, Canada, Korea, Australia, Israel, Norway, New Zealand and Ukraine) and 26 partner nodes. The node's demand enters the design through ranks and is therefore compressed relative to the EU's share of coalition GDP (Section 11).

Members' inputs are aggregated into the node with stage-specific weights (Figure 24): production for Coal, Gas and Oil Upstream, exports of the stage's HS6 basket for 15 cells and GDP for the four Geothermal and Green Hydrogen cells, which have no basket. The weights are concentrated: Germany carries the largest weight in 11 of the 22 cells, and France 97.9 percent of the EU's uranium-basket exports although it mines no uranium, trade that reflects conversion and trading rather than mine output.

8.4 Alignment within the coalition

Because the blocs are hard, alignment within the coalition takes a simple form (Figure 25). Of the 666 pairs of network nodes, 8.7 percent have $A_{ij} = 1$ (pairs of core nodes and pairs among the three China-leaning partners), 86.3 percent have $A_{ij} = \frac{1}{2}$ because one of the two is a nonaligned partner, and 5.0 percent have $A_{ij} = 0$: each core node paired with Indonesia, Viet Nam or Oman. The alignment component of Friendshore thus separates core from partner suppliers and penalises the three China-leaning partners as suppliers to the core, but leaves the ordering of the 23 nonaligned partners to the other components.

8.5 Sensitivity and comparison with the hand-picked list

Two alternative vote series test the axis. With session 78 alone the Spearman correlation of s with the headline is 0.966, 89.1 percent of economies keep their bloc and the coalition keeps 57 members, with three changes at the margin. With the final-passage series over sessions 76–80 (2021–2025) the correlation is 0.965 and the core is unchanged, but the China-leaning cut falls to 0.055: Russia, Kazakhstan and Myanmar leave the China-leaning quartile and would enter as major suppliers, for 60 members. The core is robust to the vote series; the exclusion of China-leaning suppliers is the fragile element of the rule. Admitting the MSP Forum as a further agreement adds Kazakhstan, Uzbekistan and Ecuador.

The derived coalition replaces a list of 23 economies fixed by hand in an earlier version (Appendix D). Twenty-one remain; the United Arab Emirates and Kenya, both China-leaning, in no

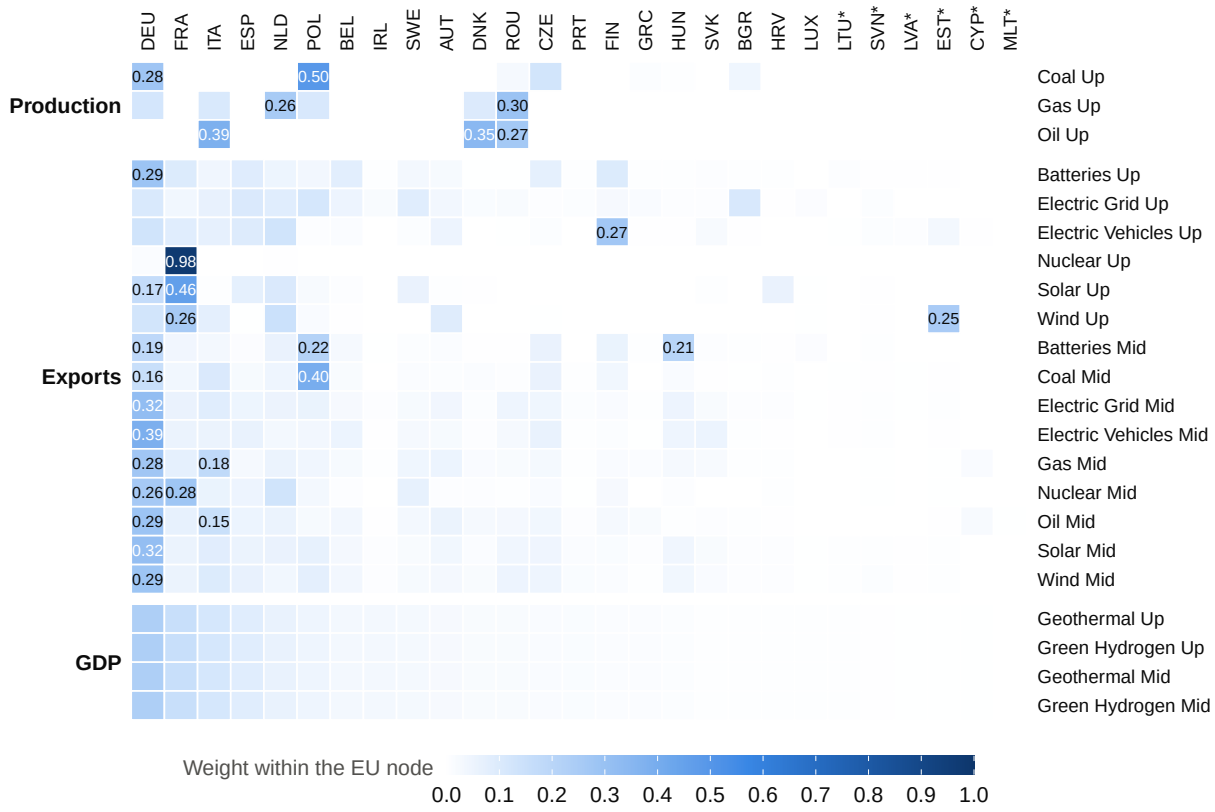


Figure 24: Member weights within the EU network node. Columns are the 27 EU members, ordered by GDP; an asterisk marks the six members below the GDP floor, which enter the network only through the node. Rows are the 22 Upstream and Midstream cells, grouped by the basis of the weights: members' production (Energy Institute; 3 cells), members' shares of world exports of the stage's HS6 basket (15 cells) or, for the four stages without a basket, GDP shares (Geothermal and Green Hydrogen, Upstream and Midstream). Weights in a row sum to one; weights of at least 0.15 under the production and export bases are printed. Median Herfindahl index of the weights: 0.34 (production), 0.139 (exports), 0.117 (GDP). Source: `eu_stage_weights_tbl.rds`.

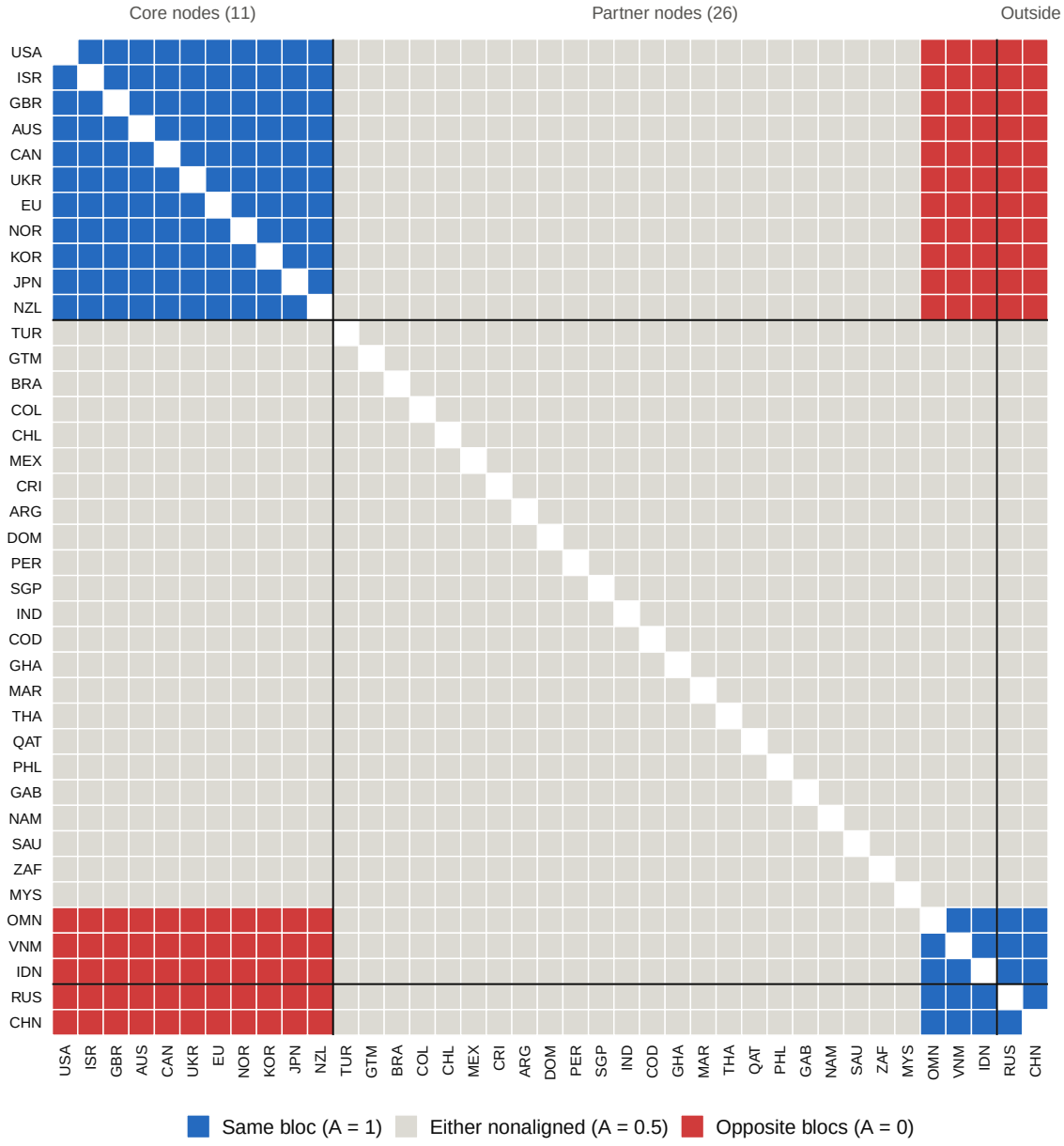


Figure 25: Dyadic alignment among the network nodes. Each cell is A_{ij} of equation (51): 1 for the same bloc, 0 for opposite blocs and $\frac{1}{2}$ when either economy is nonaligned. The 37 network nodes, 11 core and 26 partner, are followed by Russia and China for reference, ordered by tier and, within a tier, by s (for the EU node, the GDP-weighted mean of its members, 0.677; all 27 members are U.S.-leaning). Source: `geo_alignment_tbl.rds`, `allied_network_nodes.csv`.

agreement and without a qualifying share, leave; 36 economies join.

The rule and a functional definition of "allied". The working paper on the Allied Industrial Base (New Energy Industrial Strategy Center, 2026) argues that "allied" should be a functional rather than a treaty category, with partners assessed on reliability and contractual performance, economic complementarity, strategic alignment, burden sharing, technology-security requirements and contribution to system resilience, and with membership varying across sectoral compacts. The derived coalition implements part of that programme: the axis measures strategic alignment, the supplier route contribution to world supply and reserves, and the agreement route economic-security ties with the United States that are already institutionalised. It does not measure reliability, the rule of law, burden sharing or technology-security requirements, and it defines one coalition for all 22 stages. The supplier route gives the boundary a sectoral margin (nine partners qualify only as suppliers of particular stages), but a coalition defined stage by stage, admitting a resource holder to a minerals compact without admitting it to one on sensitive grid equipment, is left to future work.

9 The Policy index and the Strategic Index

This section constructs a Policy index POL_{its} of recorded policy activity in each cell and a Strategic Index SI_{its} that combines it with the two pillars and two fixed sector weights into a ranking of each country's sectors, from which one allied pick is drawn for each coalition member. Both have normative weights (Nardo et al., 2008; Decancq and Lugo, 2013); we state each and report the choices to which the rankings are sensitive.

9.1 The Policy index

No measure of industrial policy captures whether a policy is well designed (Juhász et al., 2024, 2022; Evenett et al., 2024). The Policy index measures the intensity and direction of state attention to a cell, which indicates where capacity is likely to be built and sustained (Rodrik, 2004, 2014), not whether that attention is warranted. It has two components.

IEA Policies and Measures. The IEA database records national, subnational and international energy and climate policies with their status, jurisdiction, type and text (International Energy Agency, 2026e). Each policy h is assigned to technologies and stages by keyword rules and given a strength σ_h, the product of type, status and jurisdiction weights and of discounts for cross-cutting or text-only assignments. The component is the S-curve of the log summed strength, ranked over all country–technology–stage cells jointly so that it keeps differences in policy density across countries and sectors,

$$x_{its}^{\mathrm{IEA}} = \phi_\gamma\Big[\log\Big(1+\textstyle\sum_{h\in\mathcal{H}_{its}}\sigma_h\Big)\Big]_{\mathrm{all}}. \tag{52}$$

Online Appendix OA.7 gives the tagging rules and weights.

Climate Action Tracker. The CAT component interacts a country's overall climate-policy rating, mapped to $\kappa_i \in \{0.25, 0.5, 0.75, 1\}$ (critically insufficient to almost sufficient; Climate Action Tracker, 2026), with the technology weight g_t defined below,

$$x_{its}^{\mathrm{CAT}} = \kappa_i \, g_t + (1 - \kappa_i)(1 - g_t) \qquad \text{for every stage } s. \tag{53}$$

Its slope in g_t is $2\kappa_i - 1$: a rating of "insufficient" or better tilts a country's cells toward clean technologies, "highly insufficient" leaves them flat and "critically insufficient" tilts them toward fossil ones (Figure 26, panel a). EU members without a rating of their own take the EU's.

Composite and eligibility. The Policy index is the weighted mean of the components observed in the cell,

$$\mathrm{POL}_{its} = \frac{\sum_{v \in \mathcal{V}_{its}} \omega_v \, x_{its}^{v}}{\sum_{v \in \mathcal{V}_{its}} \omega_v}, \qquad (\omega_{\mathrm{IEA}}, \omega_{\mathrm{CAT}}) = (3, 2). \tag{54}$$

The IEA component is required and CAT is added where the Tracker rates the country. All 215 economies in the IEA database receive an index: 65 on both components and 150 on the IEA component alone, with the basis recorded. Requiring both components, the rule until October 2026, is a robustness variant.

Why NIPO and the dual-use score are excluded. Earlier versions of the composite also carried the New Industrial Policy Observatory of Global Trade Alert (NIPO; Global Trade Alert, 2026; Evenett et al., 2024), with weight 4, and a dual-use score, with weight 1. The dual-use score is a project-authored judgement of each sector's defence relevance, identical for every country: it measures no country and would act as a further sector prior read as policy. NIPO is the most direct measure of industrial policy, but its allocation through product codes reaches only Upstream and Midstream cells, which changed the effective weights by stage, and the component available when the composite was fixed covered 23 economies. It returns as a robustness variant built from the October 2026 run (77 jurisdictions; Online Appendix OA.7).

9.2 The Strategic Index

For a cell in the sample, the index is

$$\mathrm{SI}_{its} = 0.2\,\mathrm{POL}^{*}_{its} + 0.3\,\mathrm{EO}_{its} + 0.3\,(1 - \mathrm{ES}_{its}) + 0.1\,\lambda_s + 0.1\,g_t, \tag{55}$$

where the policy term is the within-country S-curve of the Policy index over the country's in-sample sectors,

$$\mathrm{POL}^{*}_{its} = \phi_\gamma\big[\mathrm{POL}\big]_i \qquad \text{over } \big\{(t', s') : \ (i, t', s') \text{ in the sample}\big\}, \tag{56}$$

and EO_{its} and ES_{its} enter at their cross-country levels. Every term lies in $[0, 1]$ and the weighted terms add up to the index exactly.

Policy, need and scaling. The policy term rewards the sectors on which a government already concentrates its attention, where programmes and administrative capacity make further effort more likely to succeed (Rodrik, 2004; Juhász et al., 2024). It is ranked within the country because the volume of recorded policy depends on governments' transparency and compilers' attention; the pillars keep their cross-country levels, so that a sector in which a country leads the world counts for more than one in which it is merely its own best. Security enters as need, $1 - \mathrm{ES}_{its}$: new capacity is most valuable where it reduces an exposure rather than duplicating a strength (Section 2); the strength reading is reported below. Nominal weights are not effective weights (Paruolo et al., 2013; Becker et al., 2017), and the index ranks sectors within a country rather than comparing levels across countries.

Fixed sector weights. The stage weights are $\lambda_{\text{Midstream}} = 0.75$, $\lambda_{\text{Upstream}} = 0.5$ and $\lambda_{\text{Downstream}} = 0.25$: clean-energy supply chains are most concentrated in processing and manufacturing (International Energy Agency, 2021, 2024), extraction is bound to geology and deployment is largely non-traded. The technology weight g_t is a min-max index of lifecycle greenhouse-gas intensity. With c_r the median intensity ($\mathrm{gCO_2e/kWh}$) of generation technology r and R_t the rows of technology t,

$$g_t = \frac{1}{|R_t|} \sum_{r \in R_t} \frac{\max_{r'} c_{r'} - c_r}{\max_{r'} c_{r'} - \min_{r'} c_{r'}} \in [0, 1], \tag{57}$$

from the AR5 lifecycle medians (Schlömer et al., 2014) and, for oil-fired generation, the SRREN median of 840 $\mathrm{gCO_2e/kWh}$ (Moomaw et al., 2011); technologies without a row take proxies (Online Appendix OA.7). The weight runs from 0 for oil, 0.02 for coal and 0.42 for gas to 0.97 for solar and 0.999 for nuclear and wind. The two weights add a sector prior of 0.025 (Oil Downstream) to 0.175 (Wind Midstream) to every country's index (Figure 26, panel b): a stated preference for clean, midstream activity, not a measurement, which decides the ranking where a country's measured terms are flat. The same g_t scales the climate penalty of the bilateral indices (Section 10).

Implicit sector priors. Because the pillars enter at their cross-country levels, two further sector-level terms act as priors in each country's ranking, in addition to the stage and technology weights. Inputs that are the same for every economy in a pair (world demand, market size and overcapacity, TRL, price volatility and the world-concentration terms of Trade) shift that pair's pillar levels for all countries; on the scored cells they contribute on average 0.18 to EO_{its} and 0.07 to ES_{its}, and their net contribution to the index varies across sectors from -0.065 to 0.012. And stage-level differences in zero-inflation (in the median Upstream pair 71 percent of rankable economies score zero on Reserves; Section 5) lower Upstream security and so raise Upstream need for every country. The first term can be measured: across the 57 allied picks, the cell-constant inputs account for a median of 45 percent of the margin over the runner-up, favour the pick in 36 cases and exceed the whole margin in 18. Removing them from the pillar levels keeps 29 of the 57 picks, and entering opportunity and need as within-pair ranks keeps 27 (Section 12).

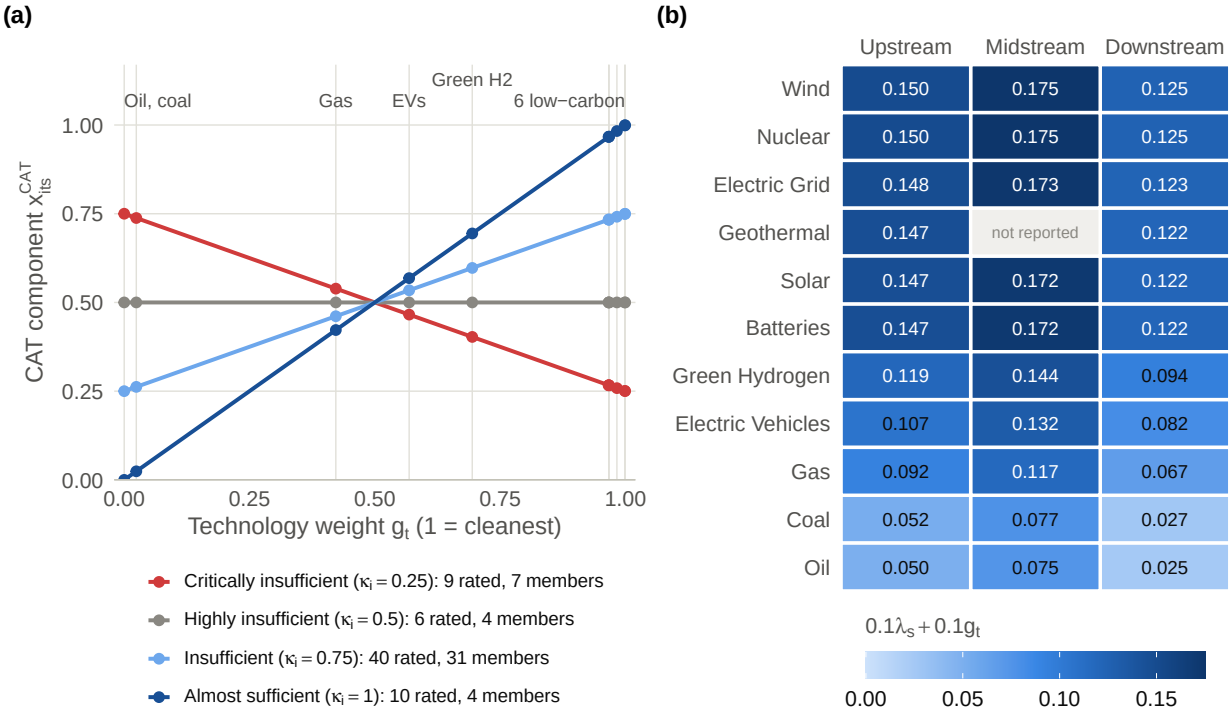


Figure 26: The country-invariant inputs of the Strategic Index. Panel (a): the Climate Action Tracker component of the Policy index, $x_{its}^{\mathrm{CAT}} = \kappa_i g_t + (1-\kappa_i)(1-g_t)$, against the technology weight g_t (IPCC life-cycle emissions, 1 = cleanest) for the four CAT ratings, with the eleven technologies marked on each line; the slope is $2\kappa_i - 1$. The legend counts the countries CAT rates in each category in the final build (65 in all, 26 of them EU members through the EU's rating) and the coalition members among them (46 of 57: 31 insufficient, 7 critically insufficient, 4 highly insufficient, 4 almost sufficient). Panel (b): the sector prior $0.1\lambda_s + 0.1g_t$ that the stage weights λ_s (Upstream 0.5, Midstream 0.75, Downstream 0.25) and the technology weights add to the Strategic Index of every country; Geothermal Midstream is not a reported sector. Source: the pipeline's functions and `data/raw/ipcc_ghg_intensity.csv`, with the CAT ratings of `policy_component_tbl` in `index_outputs.rds`; `scripts/paper/fig_si_fixed_inputs.R`.

9.3 Sample rules and the allied pick

Nothing is imputed in the Strategic Index. A cell is scored only if the country has a Policy index and both of its pillar cells are rankable under the rule of Section 4 (the economy is sovereign and at most half of each pillar cell's index weight is imputed); an unscored cell stays in the output with its reason recorded. Geothermal Midstream, which has no Energy Security pillar, is excluded by rule; a check that removes any sector whose pillar is constant across the scored countries finds none in the final build.

Allied pick. For each coalition member i, let $\mathcal{A}_i$ be its scored Upstream and Midstream sectors outside Geothermal and Green Hydrogen, less the Upstream sectors barred by the endowment screen below. The allied pick is

$$(t, s)_i^* = \arg\max_{(t,s)\in\mathcal{A}_i} \ \mathrm{SI}_{its}. \tag{58}$$

Only the traded stages, which coalition supply chains can organise across borders, are eligible; Geothermal and Green Hydrogen have no HS6 basket and Midstream Energy Security pillars that are constant or concentrated at one value. The pick summarises the index for each member, EU members included; the network design does not use it.

Endowment screen. With security read as need, an economy without a resource sits at the floor of the extraction stage's Energy Security pillar and has maximal need there. The screen makes an Upstream stage eligible for a member's pick only if the member has recorded reserves or production of the stage's resources (technology-level Reserves or Production greater than zero). It is a rule of the pick, not of the index: values are unchanged and screened cells are flagged.

9.4 Results

Coverage and basis of the Policy index. All 57 coalition members have a Policy index, 46 on both components and 11 on the IEA component alone: Israel and Ukraine in the core and nine partners (the Democratic Republic of the Congo, the Dominican Republic, Gabon, Ghana, Guatemala, Malaysia, Namibia, Oman and Qatar). The two components measure different things (Figure 27): over the 65 economies scored on both, their Spearman correlation is -0.05 across cells and -0.02 within a country at the median, and the composite follows the Tracker component within a country (median 0.79) more than the IEA one (0.48). Ranking these countries' cells on the IEA component alone gives a median within-country Spearman of 0.48 with the composite over their policy cells and 0.44 over their Strategic Index cells: the basis matters for a country's internal ranking.

Sample and distribution. The Strategic Index covers 3,154 cells of 174 countries in 30 sectors (Electric Grid Downstream and Geothermal Upstream have no scored cell). Of the 6,476 cells of policy-eligible countries, 2,700 fail the rankability rule on one or both pillars and 622 lack an Energy

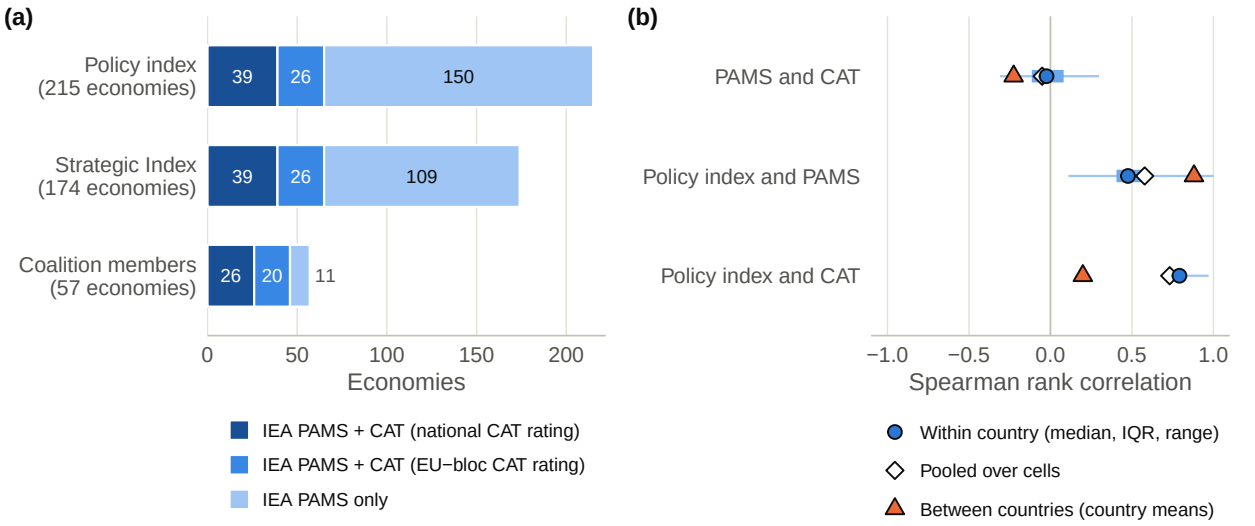


Figure 27: The basis of the Policy index. Panel (a): number of economies by the components of their Policy index, for all economies with an index (215), the economies with a Strategic Index cell (174) and the coalition members (57). "IEA PAMS + CAT" combines the IEA Policies and Measures database (weight 3) with the Climate Action Tracker rating (weight 2), national or, for EU members, the Union's; "IEA PAMS only" marks economies the Tracker does not rate. Panel (b): Spearman correlations among the two components and the composite over the 33 cells of the 65 economies scored on both: median, interquartile range and range of the within-country correlations (59 economies whose Tracker component varies across cells), pooled over cells and across country means. Source: `policy_component_tbl`, `policy_index` in `index_outputs.rds`.

Security cell. Every coalition member is scored, with a median of 29 sectors. The index (mean 0.535, standard deviation 0.082) is organised by sector far more than by country: sector fixed effects explain 72.6 percent of its variance and country fixed effects 4.4 percent (Figure 28). The highest cells are Nuclear Upstream cells of small economies without uranium, where Energy Security sits at a floor shared by 86 of the 174 countries and need is maximal: Nuclear Upstream is the top sector of 62 countries, 55 of them without recorded uranium. This construction effect of the need reading motivates the endowment screen. Green Hydrogen Midstream (34 of 40 countries at one value), Green Hydrogen Downstream, Nuclear Downstream and Batteries Downstream also have floors; no allied pick lies in a floor cell.

Decomposition. The five terms account for 18.8 (policy), 27.6 (opportunity), 32.4 (need), 9.3 (stage) and 11.9 percent (technology) of the index level, but very different shares of its within-country variance: on average 40.7 percent for the technology weight, 34.1 percent for need and 32.0 percent for policy, against −1.4 percent for opportunity and −5.4 percent for the stage weight. Opportunity moves against need (median within-country Spearman of EO and $1 - \text{ES}$ −0.47, negative in 164 of 167 countries; −0.49 and negative for all 57 coalition members), because a country's strongest opportunities tend to be the sectors in which it is already secure; the two offset, and the ranking is set by the technology weight, need and the policy rank (Figure 29).

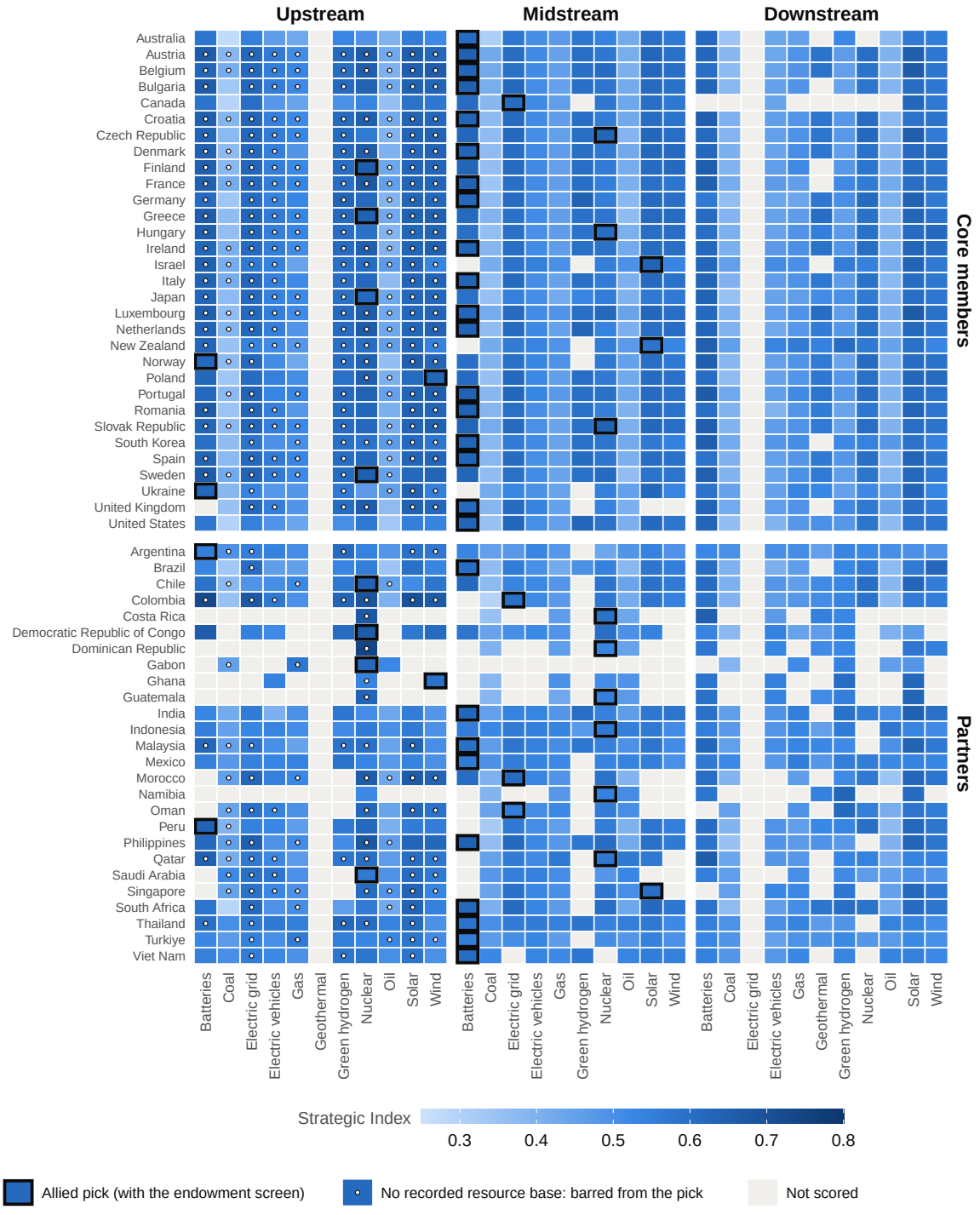


Figure 28: The Strategic Index across the coalition. Rows are the 57 members (31 core, then 26 partners), columns the 33 technology–stage cells grouped by stage; colour is SI_{its} (equation (55)). Grey cells are not scored (a pillar cell not rankable or missing, or Geothermal Midstream, excluded by rule). A dot marks an Upstream cell in which the member has no recorded reserves or production of the stage's resources: the endowment screen bars it from the pick, but its value is unchanged. The outlined cell in each row is the member's allied pick (equation (58)). Source: `strategic_index` in `index_outputs.rds`; `allied_strategic.csv`.

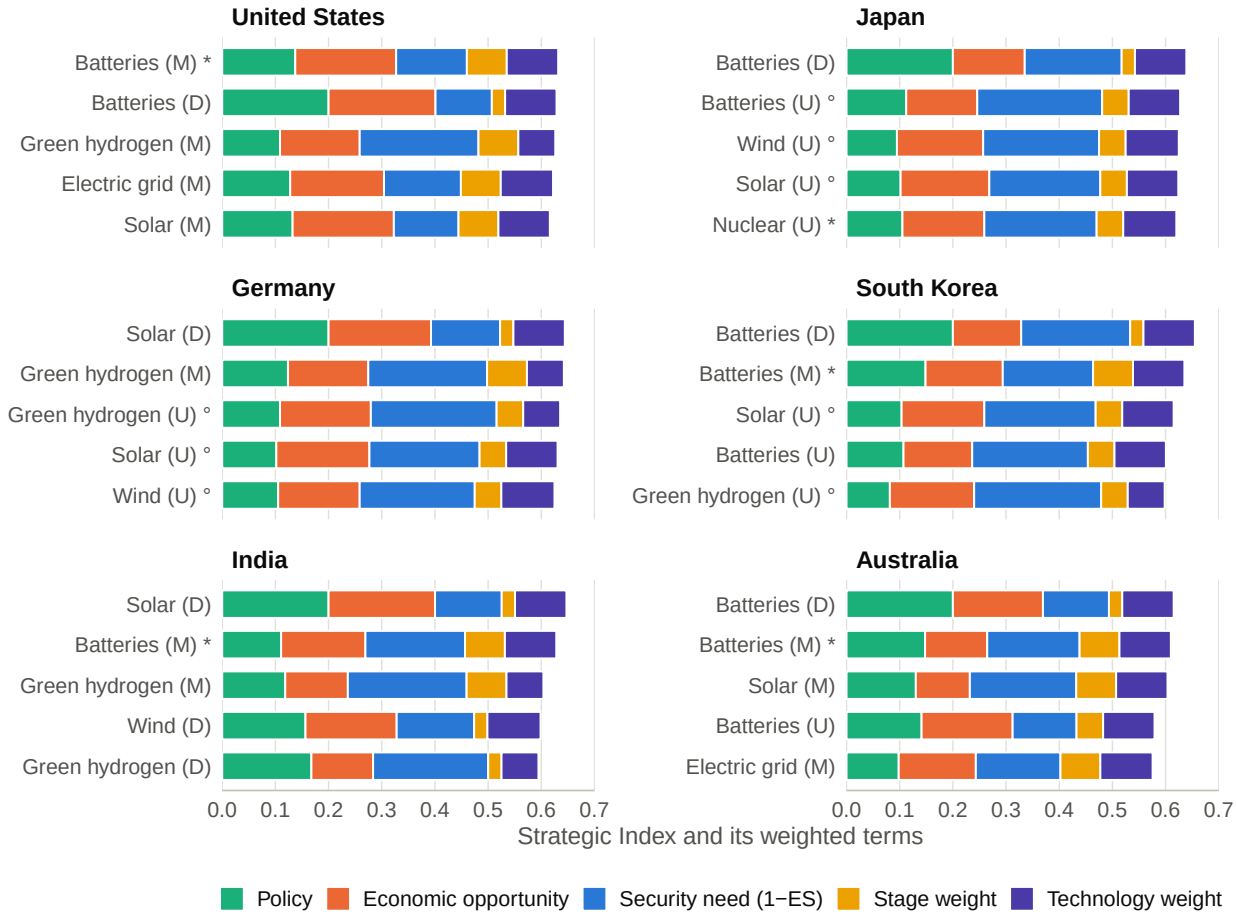


Figure 29: Decomposition of the Strategic Index for the five highest-ranked sectors of six coalition members. Each bar stacks the weighted terms of equation (55): $0.2\,\mathrm{POL}^{*}_{its}$, $0.3\,\mathrm{EO}_{its}$, $0.3\,(1-\mathrm{ES}_{its})$, $0.1\,\lambda_s$ and $0.1\,g_t$; bar length is SI_{its}. (U), (M) and (D) mark the stage; an asterisk marks the member's allied pick and a circle an Upstream sector barred by the endowment screen. Source: `strategic_index` in `index_outputs.rds`.

Allied picks. Every member has a pick, and none is tied (Table 16). Batteries Midstream is the pick of 27 members (among them the United States, Germany, Korea, India and Australia), Nuclear Midstream of nine, Nuclear Upstream of eight, Batteries Upstream and Electric Grid Midstream of four each, Solar Midstream of three and Wind Upstream of two (Ghana and Poland): 43 Midstream and 14 Upstream picks. The picks are sectors a member needs and supports with policy rather than sectors in which it already competes: within a member's eligible sectors the pick ranks first on policy for 22 members and on need for 23, but on opportunity for none. The eight Nuclear Upstream picks (Chile, the Democratic Republic of the Congo, Finland, Gabon, Greece, Japan, Saudi Arabia and Sweden) all hold identified uranium resources, but small ones, 0.015 to 0.79 percent of world identified resources below USD 260/kgU, and none has recorded mine production in 2019–2024. These picks are still need-driven: a negligible resource ranks near the floor of the security pillar, so the screen removes the no-resource artefact but not its small-resource counterpart. Requiring recorded production rather than any reserves removes all eight Nuclear Upstream picks (seven members move to another sector and Gabon is left without an eligible one) and the two Wind Upstream picks of Ghana and Poland.

The picks are fragile. The median margin over the runner-up is 0.015; it is below 0.005 for 10 members (17.5 percent) and below 0.01 for 21 (36.8 percent), against a within-country standard deviation of 0.080, so the table reports the runner-up beside each pick. The distribution also moved during the final data revisions: in a run of 8 October, when 45 members had a pick, Electric Grid Midstream was the pick of 17 and Batteries Midstream of 11, before completing the energy-trade file filled missing codes of the grid basket.

The Nuclear Midstream picks of small partners are not an Energy Security floor effect. Costa Rica, the Dominican Republic and Guatemala each have four eligible sectors, the Midstream stages of nuclear, oil, gas and coal (Namibia these and Nuclear Upstream). The technology weight ($g_t = 0.999$ for nuclear against 0.42 for gas, 0.02 for coal and 0 for oil) puts Nuclear Midstream's sector prior 0.058, 0.097 and 0.100 ahead of the three fuels, more than any other term separates pick from runner-up, while Nuclear Midstream Energy Security is compressed around its median (interquartile range 0.095, against 0.12–0.15 in the fossil Midstream sectors), so need adds only 0.020–0.042.

Picks and the network design. The pick is a national priority read from need and policy, not an allied assignment. Aggregated, the picks reproduce the duplication that an Allied Industrial Base is meant to avoid: 27 members would prioritise Batteries Midstream, which the network design of Section 11 gives to nine producer nodes. Of the 27, 20 hold a designed role in that stage, 14 of them only through the EU node, and seven (Australia, Brazil, Mexico, the Philippines, South Africa, Türkiye and Viet Nam) do not. Across all members, 28 of the 57 hold a designed share in their pick stage (the column "Design share" of Table 16), and eight (Costa Rica, the Dominican Republic, Gabon, Ghana, Guatemala, New Zealand, Türkiye and Ukraine) produce nothing in the design.

Table 16: Allied strategic selection: pick, runner-up, margin and robustness of the pick

Member	Pick	SI	Runner-up	Margin	Design share	Pick without the screen	Same pick under (1)	Same pick under (2)
Core members (31 of 31 with a pick)								
Australia	Batteries (M)	0.610	Solar (M)	0.0063	–	same	–	✓
Austria	Batteries (M)	0.647	Solar (M)	0.0187	0.105	Nuclear (U)	–	✓
Belgium	Batteries (M)	0.634	Solar (M)	0.0142	0.105	Wind (U)	–	✓
Bulgaria	Batteries (M)	0.651	Nuclear (U)	0.0101	0.105	Batteries (U)	–	–
Canada	Electric grid (M)	0.615	Batteries (M)	0.0052	0.050	same	–	–
Croatia	Batteries (M)	0.638	Electric grid (M)	0.0235	0.105	Nuclear (U)	–	✓
Czech Republic	Nuclear (M)	0.632	Solar (M)	0.0071	0.205	Wind (U)	–	✓
Denmark	Batteries (M)	0.636	Solar (M)	0.0109	0.105	Nuclear (U)	–	✓
Finland	Nuclear (U)	0.658	Batteries (M)	0.0245	–	same	–	✓
France	Batteries (M)	0.642	Electric grid (M)	0.0421	0.105	Nuclear (U)	–	✓
Germany	Batteries (M)	0.623	Nuclear (U)	0.0073	0.105	Solar (U)	–	–
Greece	Nuclear (U)	0.643	Batteries (M)	0.0252	–	Batteries (U)	–	✓
Hungary	Nuclear (M)	0.606	Wind (M)	0.0015	0.205	Batteries (U)	–	–
Ireland	Batteries (M)	0.636	Solar (M)	0.0244	0.105	Nuclear (U)	–	✓
Israel†	Solar (M)	0.636	Electric grid (M)	0.0487	–	Batteries (U)	–	✓
Italy	Batteries (M)	0.639	Nuclear (U)	0.0181	0.105	Batteries (U)	–	–
Japan	Nuclear (U)	0.620	Batteries (M)	0.0231	–	Batteries (U)	–	✓
Luxembourg	Batteries (M)	0.628	Solar (M)	0.0005	0.105	Nuclear (U)	–	✓
Netherlands	Batteries (M)	0.635	Electric grid (M)	0.0242	0.105	Nuclear (U)	–	✓
New Zealand	Solar (M)	0.587	Electric grid (M)	0.0269	–	Batteries (U)	–	✓
Norway	Batteries (U)	0.618	Batteries (M)	0.0148	–	Solar (U)	–	✓
Poland	Wind (U)	0.633	Batteries (U)	0.0089	–	Nuclear (U)	–	✓
Portugal	Batteries (M)	0.644	Nuclear (U)	0.0007	0.105	Wind (U)	–	–
Romania	Batteries (M)	0.643	Nuclear (U)	0.0195	0.105	Batteries (U)	–	–
Slovak Republic	Nuclear (M)	0.644	Batteries (M)	0.0054	0.205	Wind (U)	–	–
South Korea	Batteries (M)	0.635	Batteries (U)	0.0348	0.055	same	–	✓
Spain	Batteries (M)	0.636	Nuclear (U)	0.0200	0.105	same	–	–
Sweden	Nuclear (U)	0.649	Wind (U)	0.0145	–	Batteries (U)	–	✓
Ukraine†	Batteries (U)	0.624	Solar (M)	0.0018	–	Solar (U)	–	✓
United Kingdom	Batteries (M)	0.617	Electric grid (M)	0.0042	0.050	Nuclear (U)	–	✓
United States	Batteries (M)	0.632	Electric grid (M)	0.0096	0.295	same	–	✓
Partners (26 of 26 with a pick)								
Argentina	Batteries (U)	0.551	Solar (M)	0.0114	0.085	Solar (U)	–	–
Brazil	Batteries (M)	0.606	Solar (U)	0.0088	–	Electric grid (U)	–	–
Chile	Nuclear (U)	0.645	Batteries (M)	0.0251	–	same	–	✓
Colombia	Electric grid (M)	0.597	Solar (M)	0.0195	–	Batteries (U)	✓	✓
Costa Rica	Nuclear (M)	0.592	Gas (M)	0.1285	–	Nuclear (U)	✓	✓
Democratic Republic of Congo†	Nuclear (U)	0.664	Batteries (U)	0.0053	–	same	–	✓
Dominican Republic†	Nuclear (M)	0.532	Gas (M)	0.0733	–	Nuclear (U)	✓	–
Gabon†	Nuclear (U)	0.616	Oil (U)	0.0921	–	same	–	–
Ghana†	Wind (U)	0.589	Electric vehicles (U)	0.0469	–	same	–	✓
Guatemala†	Nuclear (M)	0.546	Oil (M)	0.0768	–	Nuclear (U)	✓	–
India	Batteries (M)	0.628	Wind (M)	0.0437	0.135	same	–	✓
Indonesia	Nuclear (M)	0.568	Solar (M)	0.0007	–	same	–	–
Malaysia†	Batteries (M)	0.596	Solar (M)	0.0109	0.135	Solar (U)	–	✓
Mexico	Batteries (M)	0.564	Electric grid (M)	0.0064	–	same	–	–
Morocco	Electric grid (M)	0.609	Batteries (M)	0.0014	0.050	Solar (U)	✓	–
Namibia†	Nuclear (M)	0.542	Nuclear (U)	0.0223	–	same	–	–
Oman†	Electric grid (M)	0.556	Nuclear (M)	0.0011	–	Solar (U)	–	–
Peru	Batteries (U)	0.639	Nuclear (U)	0.0189	–	same	–	✓
Philippines	Batteries (M)	0.648	Solar (U)	0.0190	–	Nuclear (U)	–	–
Qatar†	Nuclear (M)	0.573	Oil (M)	0.0031	–	Batteries (U)	–	–
Saudi Arabia	Nuclear (U)	0.567	Electric grid (M)	0.0156	0.050	Solar (U)	–	–
Singapore	Solar (M)	0.601	Electric grid (M)	0.0084	0.050	Solar (U)	–	–
South Africa	Batteries (M)	0.620	Solar (M)	0.0019	–	Solar (U)	–	✓
Thailand	Batteries (M)	0.587	Solar (M)	0.0182	0.061	Solar (U)	–	–
Turkiye	Batteries (M)	0.556	Solar (M)	0.0124	–	Solar (U)	–	–

Continued on the next page

Table 16: continued

Member	Pick	SI	Runner-up	Margin	Design share	Pick without the screen	Same pick under (1)	Same pick under (2)
Viet Nam	Batteries (M)	0.606	Nuclear (U)	0.0273	–	same	–	–
Members with the same pick (of 57)						16	5	31

Notes: For each coalition member with a Strategic Index, the pick is the Upstream or Midstream sector outside Geothermal and Green Hydrogen with the highest index (equation (58)), after the endowment screen bars Upstream sectors in which the member has no recorded reserves or production; the runner-up is the second-highest such sector and the margin the difference between their indices; (U) Upstream, (M) Midstream. No member has a tie. Design share is the share of the stage's production that the network design (Section 11, $\lambda = 0.78$) assigns to the member's node, for EU members the EU node (– if none). "Pick without the screen" is the pick when the endowment screen is switched off ("same" if unchanged); the index values are the same in both cases. Columns (1) and (2) mark whether the pick, with the screen, is unchanged when the index is recomputed on the same cells (1) with security strength ES in place of need $1 - \text{ES}$ and (2) without the stage and technology weights. All 57 members have a pick. † Policy index on IEA PAMS alone (no CAT rating; 11 members); the other members' Policy index combines IEA PAMS and CAT. Source: `index_outputs.rds` and `allied_strategic.csv` of the final run, which the script reproduces; `scripts/paper/fig_strategic.R`.

The endowment screen. Of the 998 scored Upstream cells in the screened technologies, 669 have no recorded resource base (306 of the coalition's 509). The screen changes 41 of the 57 picks (Figure 30): without it 47 picks would be Upstream (Nuclear 17, Batteries 12, Solar 12, Wind 5 and Electric Grid 1), many in members without the resource, such as Nuclear Upstream for France and the United Kingdom. The index forgone is small for most members (median 0.026; largest for the Dominican Republic, 0.216). An economy cannot become an allied supplier of a resource for which it has no recorded reserves or production.

The security sign and other normative choices. Reading security as strength, ES in place of $1 - \text{ES}$, gives an index that agrees only weakly with the headline (Figure 31): the median within-member Spearman correlation is 0.36, only 5 of the 57 picks survive, and picks move to Nuclear Midstream (28), Gas Midstream (10) and Electric Grid Midstream (7). The sign is the most consequential normative choice: the need reading asks where an economy should build capability to reduce its exposure, the strength reading where it can build on capability it has. The other choices keep median within-member correlations of 0.88–0.96 but change many picks. Of the 57 members, 31 keep their pick without the stage and technology priors, 30 without the policy term, 20 when opportunity and need are ranked within the country, and 34 when the October 2026 NIPO component is added back (27 of the 50 members with NIPO data); policy from IEA PAMS alone keeps 45 (34 of the 46 members scored on both components). The headline pick stays in the variant's top three for 86 to 96 percent of members. Requiring both policy components removes the 11 members scored on the IEA component alone and leaves the other cells unchanged. A member's sector ordering is thus robust to these choices, but its single pick is not (Online Appendix OA.7; Section 12).

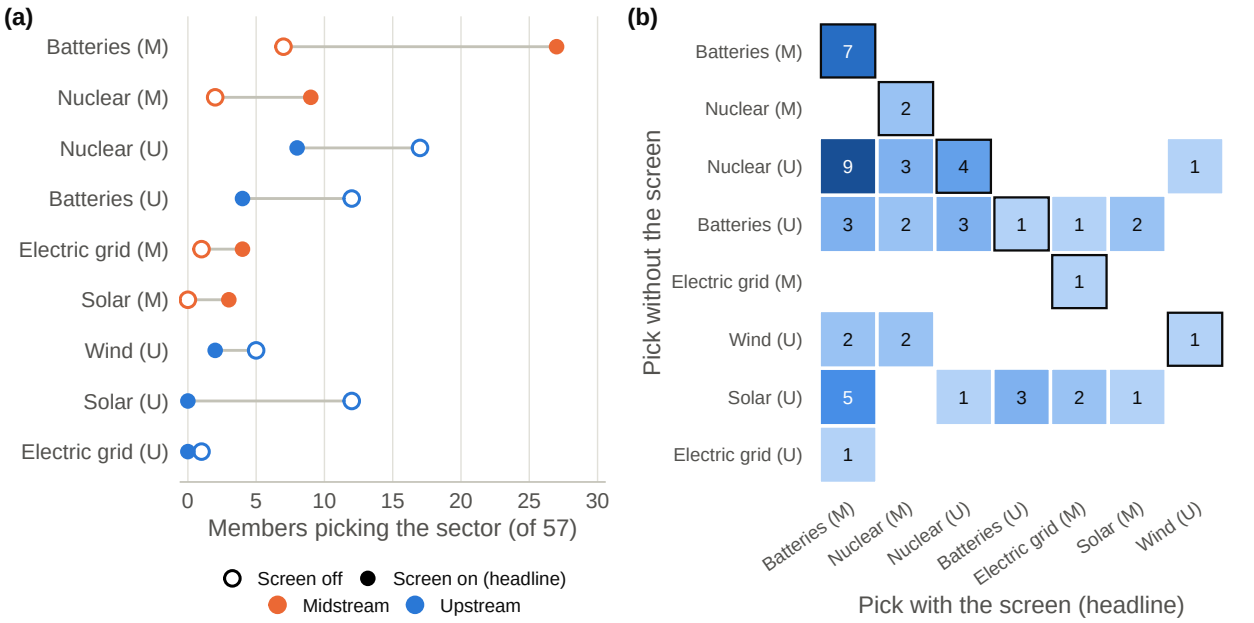


Figure 30: Allied picks with and without the endowment screen. Panel (a): number of the 57 members whose pick is each sector, with the screen (filled, the headline) and without it (open); colour is the stage. Panel (b): transition between the two selections; each tile counts the members whose pick without the screen is the row and with it the column; outlined diagonal tiles are unchanged picks (16 members). Index values are the same in both selections; the screened selection equals `allied_strategic.csv`. Source: `strategic_index` in `index_outputs.rds`; `select_allied_strategic_sectors()` with the screen on and off.

10 Bilateral partnership indices

The choice of partner trades the security of trusted suppliers against the efficiency cost of narrowing the supplier set (Yellen, 2022; Aiyar et al., 2023; Gopinath et al., 2025; Javorcik et al., 2024; Grossman et al., 2023), and sourcing is already being reallocated across partners (Alfaro and Chor, 2023; Freund et al., 2024). Its unit is the pair of countries. This section builds three bilateral pillars and a composite: the *Friendshore* index FS rates partner j as a supplier of (t, s) to economy i; the *Partnership Opportunity* index PO rates j as a market for i's exports; the *Development Potential* index DP rates j alone as a development partner; and the *Partnership Strength Index* PSI combines them (Table 17). The network design (Section 11) uses the dyad-level FS and PO and the partner-level DP, not the PSI.

10.1 Dyads, trade and shared building blocks

A dyad (i, j, t, s) pairs reporter i and partner $j \neq i$, among 219 economies, in a headline technology t and stage s. In FS_{ijts} the reporter is the importer and j the candidate supplier; in PO_{ijts} the reporter is the exporter and j the candidate market. Both rate j from i's side, so a supply edge from producer p to consumer c is read as FS_{cpts} and PO_{pcts} (Section 11). $\phi_\gamma[x]_g$ is the S-curve of Section 4 within each cell of grouping g (ts: all dyads of a cell; its: reporter i's partners in the cell;

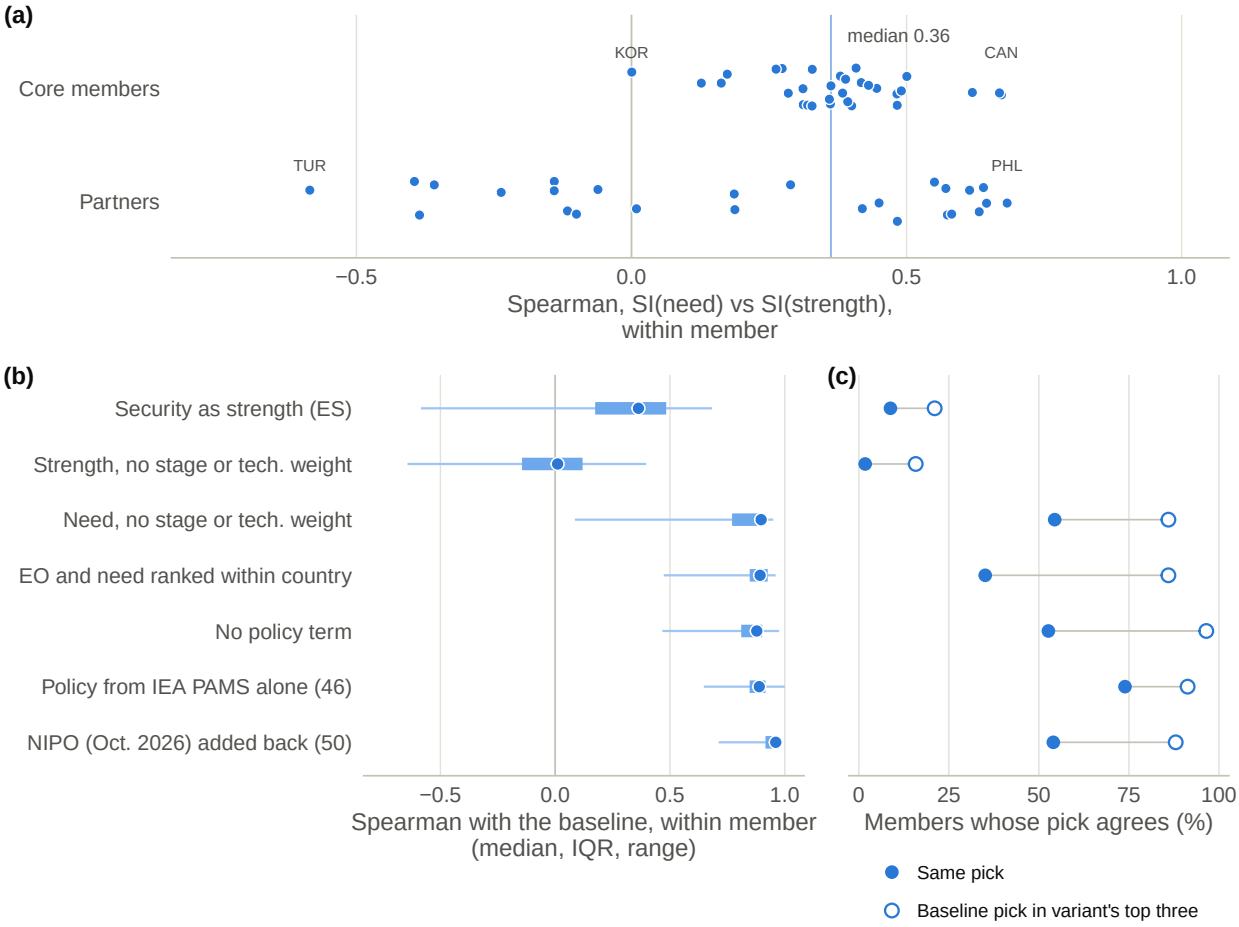


Figure 31: Sensitivity of the Strategic Index to its normative choices, coalition members. Panel (a): within-member Spearman correlation between the headline index (security as need, 1 − ES) and the index with security as strength (ES), all other terms unchanged; one point per member, by tier, median marked. Panel (b): within-member Spearman correlation of each variant with the headline (median, interquartile range and range): strength; strength without the stage and technology weights; need without those weights; opportunity and need ranked within the country; no policy term; policy from the IEA component alone (46 members scored on both); the October 2026 NIPO component added back (50 members with NIPO data). Panel (c): share of members whose pick is unchanged (filled) and whose headline pick is among the variant's top three (open); the IEA-alone and NIPO rows are shares of the 46 and 50 members whose index the variant changes, and over all 57 members 45 and 34 picks are unchanged (Table 22). All variants keep the endowment screen and the headline cells. Source: `index_outputs.rds`; the NIPO component of the October 2026 run.

Table 17: The bilateral pillars: units, direction, components and weights

	Friendshore (FS)	Partnership Opportunity (PO)	Development Potential (DP)
Unit	dyad (i, j, t, s)	dyad (i, j, t, s)	partner (j, t, s)
Reporter i	importer, sourcing (t, s)	exporter of (t, s)	none
Partner j	candidate supplier	candidate export market	non-high-income economy
Components (weight)	τ^M_{ijts} imports from j (2); ν_{its} reporter need (2); e_{jts} partner EO rank (2); o_{ijts} outward investment in j (2); A_{ij} UNGA alignment (2); $1-\chi_{jt}$ Chinese-parent exposure (1; four Midstream cells); reporter EO rank (0)	τ^X_{ijts} exports to j (3); ε_{its} reporter EO rank (3); η_{jts} partner ES need (1)	eight partner components, weights $1, 1, 1, 1, 1.5, 1.5, 2, 1$ (Table 18)
Climate penalty	$0.10\,(1-g_t)(\mathrm{CPI}_i + \mathrm{CPI}_j)$	$0.05\,(1-g_t)\,\mathrm{CPI}_j$	none
Final transform	$\phi_\gamma[\cdot]_{ts}$ over dyads, after flooring at 0	$\phi_\gamma[\cdot]_{ts}$ over dyads, after flooring at 0	$\phi_\gamma[\cdot]_{ts}$ over partners
Summary to PSI	best five (t, s) per dyad; mean over the core reporters	best five (t, s) per dyad; mean over the core reporters	best five (t, s) per partner
PSI weight	0.4	0.4	0.2

Notes: Weights from `config/weights.yml`; within each pillar they are renormalized over the components observed (equation (59)). The FS penalty coefficient (0.10 per side) is fixed in code, the PO coefficient configured. g_t is the technology emission weight of equation (57), CPI the Climate Action Tracker rating. In Batteries, Electric Vehicles, Solar and Wind Midstream o_{ijts} is the CM-FDI project tie and χ_{jt} is defined; elsewhere o_{ij} is the IMF position and χ is absent.

i: all units of reporter i). Missing components are dropped and the weights renormalized,

$$\mathcal{A}_{\boldsymbol{w}}(x_1, \ldots, x_C) = \frac{\sum_c w_c\, x_c\, \mathbf{1}\{x_c \text{ observed}\}}{\sum_c w_c\, \mathbf{1}\{x_c \text{ observed}\}}, \tag{59}$$

so a missing component is never scored as zero.

Bilateral trade. Trade comes from the all-reporter Comtrade dyads for 2020 and 2024 (United Nations Statistics Division, 2026): the importer's own declaration where it reports the cell, otherwise the exporter's mirror declaration times the CIF/FOB factor 1.06; a reporter that declares no trade with a partner records a zero, and non-reporting economies enter only through their partners' declarations. With H_{ts} the HS6 basket of the cell (Section 3), the cell flow is

$$M^y_{ijts} = \sum_{h \in H_{ts}} m^y_{ijh}, \tag{60}$$

a product in several baskets counting in each. Among dyads that trade, the 2024 level L and the growth since 2020 G are ranked within the cell and combined 2:1, then re-ranked among the reporter's own suppliers,

$$\tilde{\tau}^M_{ijts} = \mathcal{A}_{(2,1)}\big(\phi_\gamma[L]_{ts},\, \phi_\gamma[G]_{ts}\big), \qquad \tau^M_{ijts} = \phi_\gamma\big[\tilde{\tau}^M\big]_{its}, \tag{61}$$

so that τ^{M}_{ijts} measures j's place among i's sources. Recorded zeros form one tie block below every trading partner; a dyad with no observed 2024 flow takes the cell mean; the 15 cells without a basket carry no trade information. The export component τ^{X}_{ijts} is built the same way from i's exports.

Climate penalty. Each economy carries a Climate Action Tracker rating CPI_i (Climate Action Tracker, 2026) (0.25 to 1; EU members take the EU's, unrated economies the median, 0.75). The penalty is proportional to $(1-g_t)\,\mathrm{CPI}$, with g_t the technology weight of equation (57). Because g_t is constant within a cell, it re-orders dyads within carbon-intensive technologies by the parties' climate ambition (Clausing and Wolfram, 2023) rather than lowering those technologies against clean ones.

10.2 The Friendshore index

Friendshore asks how good a supplier j is for i in (t,s). Besides the import component it has the reporter's need, ranked among its own cells, and the partner's capability, ranked across countries,

$$\nu_{its} = 1 - \phi_\gamma[\mathrm{ES}]_i(i,t,s), \qquad e_{jts} = \phi_\gamma[\mathrm{EO}]_{ts}(j,t,s), \tag{62}$$

so that ν identifies which technologies i should friendshore, not which countries most need to. The dyadic alignment A_{ij} of equation (51) enters as it is, without ranking.

Outward investment. An outward position reveals an ownership or financing link that gives the investor a stake in the partner's capacity. Outside four cells, o_{ij} uses the reporter's outward direct-investment debt positions in the IMF Direct Investment Positions (International Monetary Fund, 2026a), averaged over 2020–2024 ($\bar{D}_{ij}$); positions involving 16 conduit financial centres, dominated by pass-through capital (Damgaard et al., 2024), are unobserved on both sides. With Y the 2023 GDP,

$$\tilde{o}_{ij} = \begin{cases} \mathcal{A}_{(1,1,1)}\big(\phi_\gamma[\bar{D}],\ \phi_\gamma[\bar{D}/Y_j],\ \phi_\gamma[(\bar{D}/Y_i)/Y_j]\big) & \text{if } \bar{D}_{ij} > 0, \\ 0 & \text{if } \bar{D}_{ij} \le 0, \end{cases} \qquad o_{ij} = \phi_\gamma[\tilde{o}]_i. \tag{63}$$

An absent record is missing, not zero. In Batteries, Electric Vehicles, Solar and Wind Midstream the tie o_{ijts} comes instead from the CM-FDI database of cross-border clean-technology manufacturing projects (Bandara and Net Zero Industrial Policy Lab, 2026): the projects of parents headquartered in i located in j, ranked on attributed counts and disclosed capital within the cell and then within the reporter.

Chinese-parent exposure. For supplier j and technology t in the same four cells, χ_{jt} is the share of j's foreign-invested manufacturing owned by parents headquartered in China, Hong Kong or Macao: capital-weighted where capital is disclosed and by count otherwise, half of a joint venture attributed to the listed parent, defined only where j hosts at least two projects, and set to one for

China, Hong Kong and Macao. It enters as $1 - \chi_{jt}$, so a supplier whose foreign-invested capacity belongs to Chinese parents scores lower.

Composite and index. With the reporter's own EO rank carried at weight zero,

$$\tilde{F}_{ijts} = \mathcal{A}_{\boldsymbol{w}^{\mathrm{FS}}}(\tau^{M}_{ijts},\, \nu_{its},\, e_{jts},\, o_{ijts},\, A_{ij},\, 1-\chi_{jt}), \qquad \boldsymbol{w}^{\mathrm{FS}} = (2,2,2,2,2,1), \tag{64}$$

$$\pi^{\mathrm{FS}}_{ijt} = 0.10\,(1-g_t)\,(\mathrm{CPI}_i + \mathrm{CPI}_j), \tag{65}$$

$$\mathrm{FS}_{ijts} = \phi_\gamma\big[\max(0,\ \tilde{F} - \pi^{\mathrm{FS}})\big]_{ts}. \tag{66}$$

The index rewards a match between the reporter's weaknesses and the partner's strengths, backed by revealed ties and political alignment.

10.3 The Partnership Opportunity index

A market is worth more to an exporter when the exporter is competitive in the technology, already sells there, and the market's insecurity signals demand for new supply. With the reporter's and the market's cross-country ranks,

$$\varepsilon_{its} = \phi_\gamma\big[\phi_\gamma[\mathrm{EO}]_{ts}(i,t,s)\big]_{ts}, \qquad \eta_{jts} = \phi_\gamma\big[1-\phi_\gamma[\mathrm{ES}]_{ts}(j,t,s)\big]_{ts}, \tag{67}$$

$$\tilde{P}_{ijts} = \mathcal{A}_{(3,3,1)}(\tau^{X}_{ijts},\, \varepsilon_{its},\, \eta_{jts}), \qquad \pi^{\mathrm{PO}}_{jt} = 0.05\,(1-g_t)\,\mathrm{CPI}_j, \qquad \mathrm{PO}_{ijts} = \phi_\gamma\big[\max(0,\ \tilde{P} - \pi^{\mathrm{PO}})\big]_{ts}. \tag{68}$$

Unlike the Friendshore need term, η_{jts} compares the market's insecurity with that of other markets. The penalty falls on the market's climate policy only.

10.4 The Development Potential index

Development Potential describes partner j alone: its enabling environment, energy-security need, economic opportunity, the official development assistance it receives in each stage, and its energy use (Table 18). It is defined for the 132 partners that the World Bank does not classify as high income,

$$\widetilde{D}_{jts} = \mathcal{A}_{\boldsymbol{w}^{\mathrm{DP}}}(d^{(1)}_{jts}, \ldots, d^{(8)}_{jts}), \qquad \boldsymbol{w}^{\mathrm{DP}} = (1,1,1,1,1.5,1.5,2,1), \qquad \mathrm{DP}_{jts} = \phi_\gamma\big[\widetilde{D}\big]_{ts}, \tag{69}$$

ranked across eligible partners within the cell. Only need, opportunity and stage aid vary across a partner's cells, so they decide which cells its best-five summary selects. Online Appendix OA.8 gives the 40-series WDI basket, the aid measure and the orientation choices.

Table 18: Components of the Development Potential index

Component	Varies by	Construction	Weight
$d^{(1)}$ development index	j	S-curve across economies of the mean of the observed S-curved WDI series of a 40-series basket (2022–2024 averages; 13 inverted)	1
$d^{(2)}$ business environment	j	S-curve of the Doing Business score (2019, last release)	1
$d^{(3)}$ investment environment	j	S-curve of $(d^{(1)} + d^{(2)})/2$	1
$d^{(4)}$ energy-security need	j, t, s	$1 - \mathrm{ES}_{jts}$	1
$d^{(5)}$ economic opportunity	j, t, s	EO_{jts}	1.5
$d^{(6)}$ stage aid	j, s	S-curve of ODA disbursements to the stage's sector (OECD CRS)	1.5
$d^{(7)}$ stage aid intensity	j, s	the same per unit of 2023 GDP	2
$d^{(8)}$ energy use	j	S-curve of energy use per capita	1

Notes: Weights from `config/weights.yml` (`partnership_development_components`). Stage aid: industry to Midstream, energy to Downstream, mineral resources and mining net of coal and upstream oil and gas to Upstream. A partner without a record of a component has it dropped, not scored as zero. Sources and the WDI basket: Online Appendix OA.8.

10.5 From dyads to partners: best-five summaries and the PSI

The PSI collapses technology and stage before reporters. For a dyad, the summary of a pillar is the mean of its five highest cells,

$$T_{ij}^{v} = \frac{1}{K_{ij}} \sum_{k=1}^{K_{ij}} v_{ij(k)}, \qquad K_{ij} = \min(5,\, n_{ij}), \tag{70}$$

and T_j^{DP} is the same over the partner's own cells. Partnerships in energy technologies are sector-specific, and an average over all cells would penalize a partner that is strong in a few technologies; the summary is therefore optimistic by design. With R the set of reporters whose view is taken, the category scores and the index are

$$C_j^k = \frac{1}{|R_j^k|} \sum_{i \in R_j^k} T_{ij}^k \;\; (k = \mathrm{FS}, \mathrm{PO};\; R_j^k \subseteq R), \qquad C_j^{\mathrm{DP}} = T_j^{\mathrm{DP}}, \tag{71}$$

$$\mathrm{PSI}_j = \frac{\sum_{k \in K_j} w_k\, C_j^k}{\sum_{k \in K_j} w_k}, \qquad (w_{\mathrm{FS}}, w_{\mathrm{PO}}, w_{\mathrm{DP}}) = (0.4,\, 0.4,\, 0.2), \tag{72}$$

where K_j is the set of categories observed for j (a high-income partner has no DP). The headline takes the viewpoint of the coalition's core: R is the 31 core members (Section 8), so that PSI_j is j's value to the economies that anchor the coalition; the full-coalition (57 reporters) and world views are alternatives. For one pair,

$$\mathrm{PSI}_{ij} = \mathcal{A}_{(0.4,0.4,0.2)}(T_{ij}^{\mathrm{FS}},\, T_{ij}^{\mathrm{PO}},\, T_j^{\mathrm{DP}}), \qquad i \neq j, \tag{73}$$

j's value to i. A partner is ranked only under the pillars' rankability rule: it is sovereign and at most half of its PSI weight rests on imputed or missing inputs. The PSI sees only direct links,

Table 19: Coverage of the bilateral partnership tables

	Friendshore	Opportunity	Development
Reporters	219	219	–
Partners	219	219	132
Reporter–partner pairs	47,742	47,742	–
Cells (i, j, t, s) or (j, t, s)	1,534,002	1,342,942	4,356
Technology–stage cells with an HS6 basket	18	18	–
Reporters with observed bilateral trade	208	–	–
Share of basket cells with observed trade	77.0%	77.5%	–
of which from the partner's mirror declaration	21.9%	21.8%	–
Reporters with IMF outward positions (other cells)	62	–	–
Share of cells with an investment component	18.1%	–	–
CM-FDI cells: share with a project tie	7.2%	–	–
CM-FDI cells: share with Chinese-parent exposure	12.4%	–	–
Share of cells with reporter need missing	12.5%	–	–
Partners with stage aid (Up/Mid/Down)	–	–	121/131/131
Partners in PSI_j (core view); pairs in PSI_{ij}	219 partners; 47,742 pairs		

Notes: Final rerun of 2026-10-12. Trade shares are over the dyad cells of the 18 technology–stage cells with an HS6 basket; observed trade is the importer's own declaration or, where the importer does not report, the exporter's mirror declaration times the CIF/FOB factor 1.06 (all-reporter Comtrade dyads, 2020 and 2024). The investment component is the IMF outward direct-investment position of the reporter in the partner (16 conduit financial centres excluded) except in the four cells (Batteries, Electric Vehicles, Solar and Wind Midstream) where it is the reporter's cross-border manufacturing projects in the partner recorded in CM-FDI. Development Potential covers the partners that are not high-income. Source: `partner_*_tbl.rds`, `partner_trade_coverage_tbl.rds`, `psi_tbl.rds`, `psi_dyad_tbl.rds`; `scripts/paper/fig_bilateral.R`.

although a replacement supplier may itself depend on the supplier it replaces (Freund et al., 2024; Carvalho et al., 2021), and it contains no prices or trade costs.

10.6 Results

Coverage and the trade matrix. Friendshore and Opportunity rate all 47,742 ordered pairs of the 219 economies, in 33 and 30 cells (Table 19). Over the 18 basket cells, 77.0 percent of the Friendshore trade components are observed (17.4 percent positive flows, 59.7 percent recorded zeros); mirror declarations supply 21.9 percent of the observed values, and without them 39.9 rather than 23.0 percent would be imputed. Dropping the mirror fill leaves the headline ranking almost unchanged (Spearman 0.997, the same top fifteen). IMF positions come from 62 reporters, 23 of them core members, covering 9,171 pairs; with the conduit centres excluded, none of the ten largest outward counterparts of the United States or Japan is a financial centre. CM-FDI contributes 905 of its 1,042 projects, 234 investor–destination pairs from 29 investing economies. In the headline view 192 of the 219 partners are ranked: 24 are not sovereign, and Liechtenstein, Monaco and Kosovo exceed the imputed-share threshold (Online Appendix OA.8).

The coalition Friendshore matrix. Among the core members supplier identity matters more than importer identity (Figure 32; standard deviation of suppliers' mean ratings 0.052 and 0.051 in Batteries and Solar Midstream, of importers' 0.036 and 0.033), and the matrix is asymmetric by

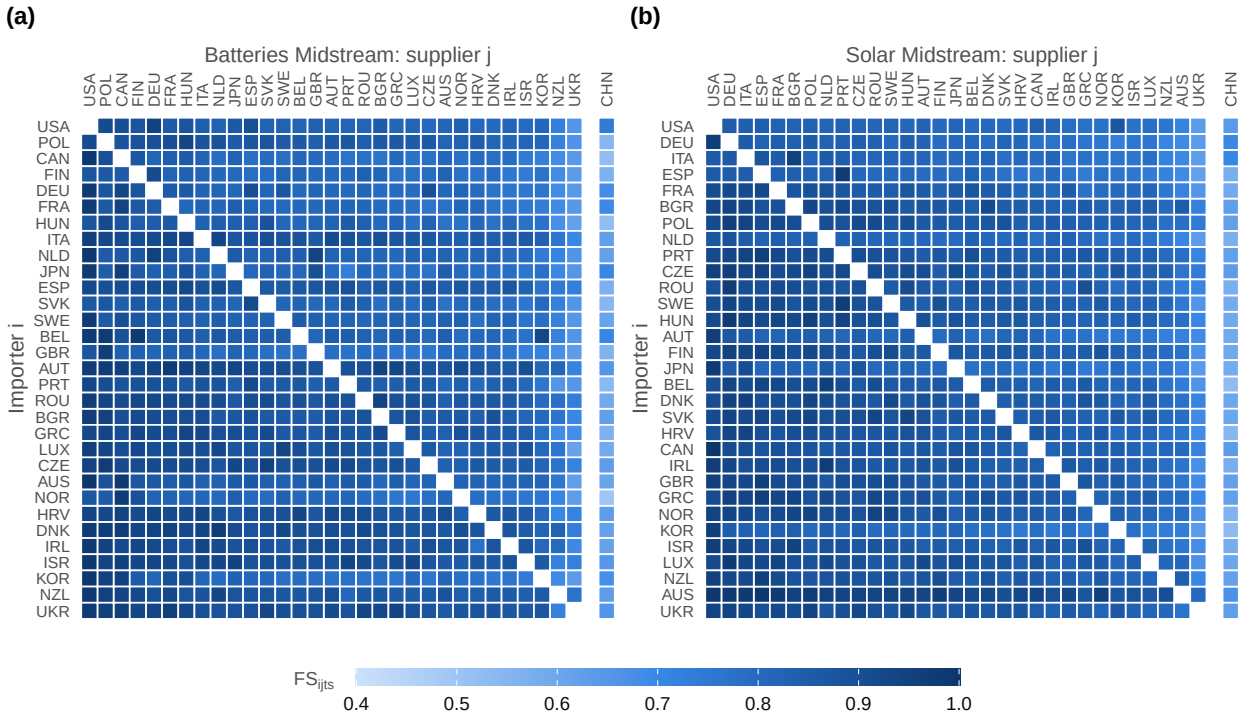


Figure 32: Friendshore index among the core members, (a) Batteries Midstream and (b) Solar Midstream. Rows are importers i, columns suppliers j; colour is FS_{ijts} (equation (66)), the within-cell S-curve of the weighted mean of the components less the climate penalty. Rows and columns are the 31 core coalition members ordered by their mean rating as suppliers; the last column is China as a supplier to each core member. The diagonal is empty. Every dyad shown has an observed import component (importer's declaration or mirror). Source: `partner_friendshore_tbl.rds`.

construction, since a capable, secure economy rates its partners low on need and is itself rated high on capability (Spearman of FS_{ijts} with FS_{jits} -0.40 and -0.41). China is rated 0.613 and 0.605 by the core, last of the 32 suppliers, although its trade and capability components are the highest; without alignment its rating would be 0.825 and 0.818.

What makes a top supplier. The leading suppliers of the United States (the United Kingdom, Germany, Canada) and of Japan (the United States, Portugal, Austria) are all U.S.-leaning except Thailand, seventh for Japan (Figure 33). Alignment is the largest contribution to the top ten's composites (29.1 and 29.0 percent), followed by partner capability (23.0 and 24.9), imports (18.7 and 20.1) and need (18.7 and 16.9); investment contributes 9.9 and 6.9 percent. China ranks 70th among the suppliers of the United States and 75th among those of Japan, and would rank 10th and 8th without alignment. Because absent investment records drop out rather than count as zero, and the summary keeps a dyad's five best cells, small aligned economies with a few strong cells rank high: Estonia and Lithuania are the ninth and tenth suppliers of the United States, and neither they nor the Netherlands, Ireland (both conduit centres, whose positions are excluded) and Slovakia have an investment record in their five best cells. Scoring an absent record at the lowest rank instead replaces all five in the United States' top ten, with Spain, Japan, Poland, Korea and Belgium, but

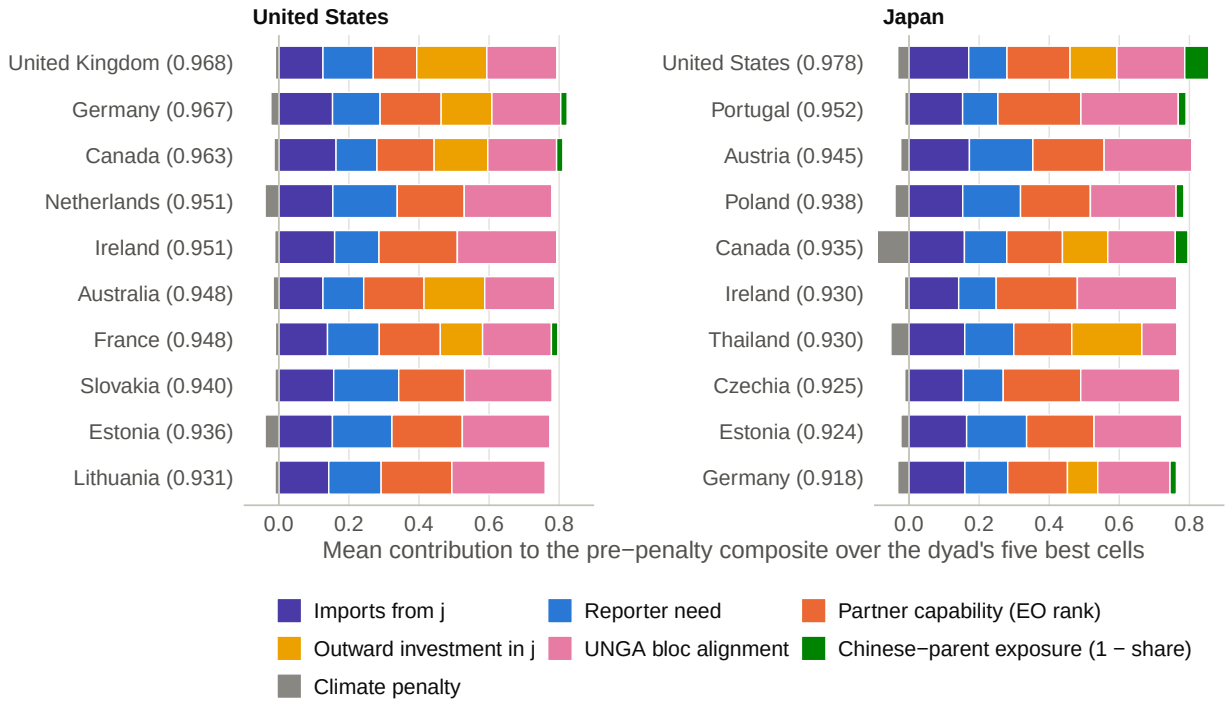


Figure 33: Composition of the Friendshore composite for the ten highest-ranked suppliers of the United States (left) and Japan (right). Each bar is the mean, over the five technology–stage cells behind the dyad's best-five summary T_{ij}^{FS} (in parentheses), of the weighted component contributions $w_c x_c / \sum_{c' \text{ obs}} w_{c'}$, which sum to the pre-penalty composite; the climate penalty is the negative segment. Components and weights: imports from j (2), reporter need (2), partner capability (2), outward investment in j (2; IMF positions, or CM-FDI projects in Batteries, Electric Vehicles, Solar and Wind Midstream), UNGA bloc alignment (2) and Chinese-parent exposure scored as 1−share (1, defined only in those four cells); the zero-weight reporter capability term is omitted. A missing segment means the component is unobserved for that dyad and dropped from the mean. Suppliers are ranked among partners that are sovereign and whose Friendshore dyad has an imputed weight share of at most 0.5 (158 for the United States, 170 for Japan). Source: `partner_friendshore_tbl.rds`, `partner_friendshore_top5_tbl.rds`; penalty recomputed from the technology emission index and the Climate Action Tracker ratings.

barely moves the headline index (Section 12.4).

The alignment component. Alignment carries weight 2 of 11 (18.2 percent) but 24.1 percent of the pre-penalty composite on average: for a core reporter it contributes 0.231 for a U.S.-leaning partner, 0.115 for a nonaligned one and zero for a China-leaning one. Setting its weight to zero changes the headline more than any other single choice examined: the Spearman correlation with the headline PSI_j is 0.869 (0.716 for C_j^{FS}), the median partner moves 17 places, and China would rise from 44th to 3rd, India from 39th to 16th, Indonesia from 62nd to 15th and Russia from 105th to 47th. The headline ranking is thus in large part a statement about political alignment with the United States; a purely economic reading of supplier strength should use the variant without it.

Chinese-parent exposure. Of the 905 CM-FDI projects, 406 have a parent headquartered in China, Hong Kong or Macao. The share is defined for 24 to 28 suppliers per covered cell, 47 economies in all, 31 of them coalition members (Figure 34); over the 94 supplier–technology cells outside China its median is 0.50, highest in solar (median 0.75) and lowest in wind (0). Some partners on which the coalition relies are heavily exposed, among them Türkiye, Thailand, Malaysia and Viet Nam, and in the core Korea (batteries 0.91) and Hungary (electric vehicles 0.64). The component implements one channel, parent-company ownership, of the common-mode upgrade proposed in the working paper on the Allied Industrial Base (New Energy Industrial Strategy Center, 2026): plants in two supplier countries owned by the same parent do not diversify a buyer's supply. With modest weight and narrow coverage its effect on the headline is small (within-cell Spearman of FS with and without it 0.99; no PSI_j rank moves by more than four places) but concentrated where it should be: the change in core members' ratings correlates at -0.93 with the share, and China loses 0.085–0.095 per cell. The component also raises the ratings of suppliers whose few foreign-invested plants have non-Chinese parents, whoever those parents are: with it, the core's mean rating of Russia over its covered cell rises from 0.32 to 0.40, and Argentina's from 0.42 to 0.50 (Figure 34b). It measures the absence of Chinese ownership, not the presence of trusted ownership. It sees ownership of foreign-invested plants only, not dependence on Chinese equipment, precursors or licences.

The partner-level PSI. In the headline view (Figure 35) the United States (0.936), Germany (0.934), the Netherlands, the United Kingdom, France, Italy, Spain, Belgium, Poland and Czechia (0.904) rank highest. Core members hold 24 of the top 25 places (median rank 16); the first non-member is Slovenia (25th), a member of the EU node. Partners rank lower (median 64.5), from Saudi Arabia (38th) and India (39th) to Gabon (141st). China ranks 44th, a strong market and a weak supplier in this view (21st on C_j^{PO}, 57th on C_j^{FS}), Taiwan 47th and Switzerland 51st. Opportunity varies little across partners (standard deviation 0.031, against 0.190 for C_j^{FS} and C_j^{DP}), so although nominally 40 percent of the index it accounts for 5.5 percent of the variance of PSI_j among the 130 partners with all three categories (Friendshore 63.8, Development 30.7 percent).

Views of the index. The core and full-coalition views agree closely (Spearman 0.981 over 192 ranked partners, median rank change 5, 13 of the top fifteen in common); the core and world views (188 ranked) less so (0.881, median change 17, ten in common) (Table 20). The views differ in how they place the large nonaligned and China-leaning economies: China is 44th, 9th and first in the core, full-coalition and world views, India 39th, 15th and 2nd, Indonesia 62nd, 32nd and 6th. The partners whose rating depends on who is asking are those whose alignment differs from that of the core.

Supplier and market views of the same edge. For coalition pairs, the importer's Friendshore rating of a supplier and the supplier's Opportunity rating of that importer as a market agree to a degree (Figure 36; Spearman 0.715 over 95,312 matched cells), partly through the shared capability

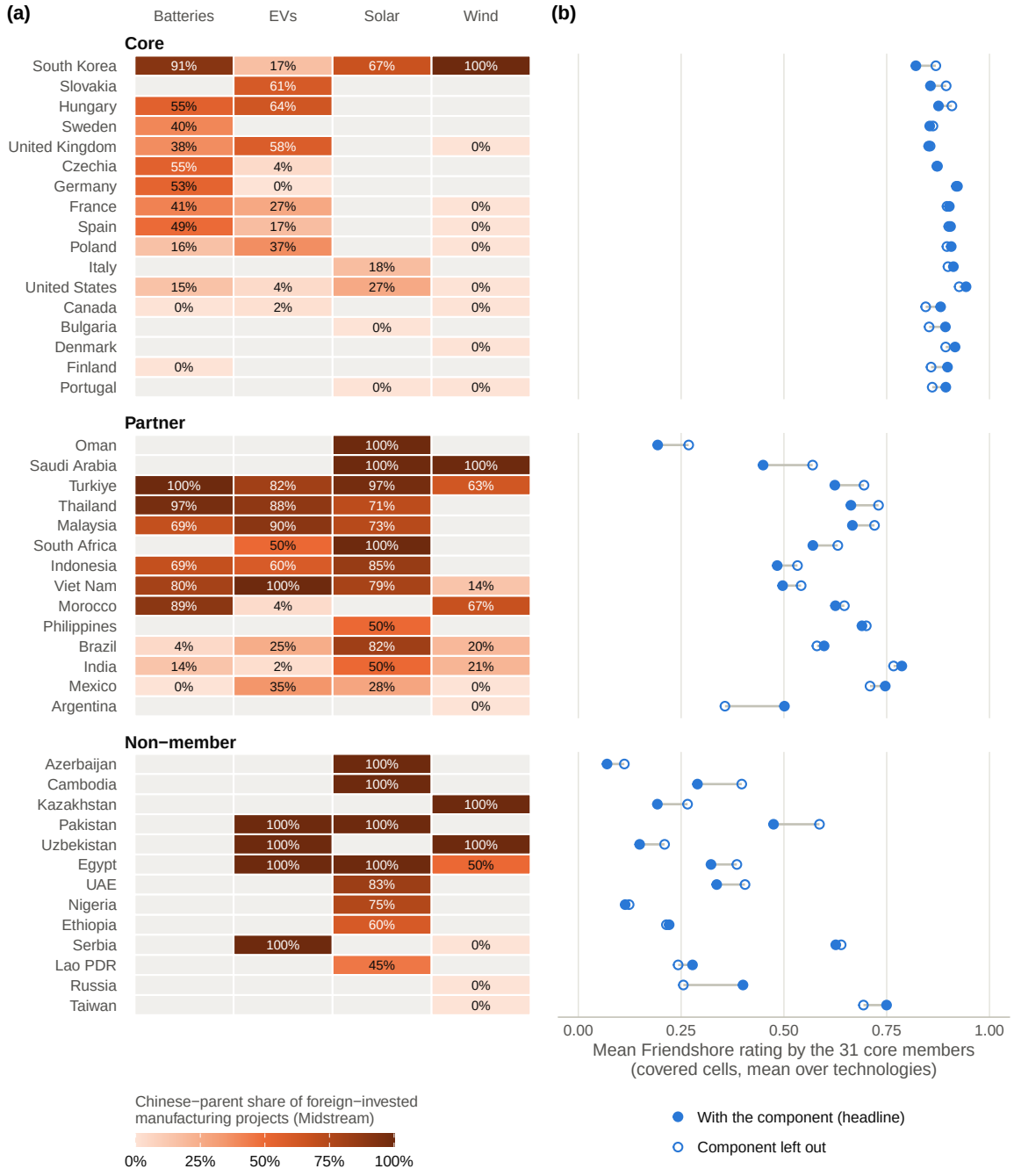


Figure 34: Chinese-parent ownership exposure of suppliers. Panel (a): for each economy hosting at least two foreign-invested manufacturing projects in a technology, the share of those projects owned by parents headquartered in China, Hong Kong or Macao (capital-weighted where capital is disclosed, by count otherwise; joint ventures attribute half to each parent), in the Midstream cells of batteries, electric vehicles, solar and wind; grey cells are undefined (fewer than two projects). Rows are grouped by coalition tier and ordered by the mean share; China, Hong Kong and Macao, set to one by construction, are omitted. Panel (b): the mean Friendshore rating of the economy by the 31 core members over its covered cells, with the component (filled) and with its weight set to zero and every later step re-run (open). Source: CM-FDI database, Net Zero Industrial Policy Lab, Johns Hopkins University (CC BY 4.0), 905 of 1,042 projects; `partner_cm_fdi_exposure_tbl.rds`, `partner_friendshore_tbl.rds`.

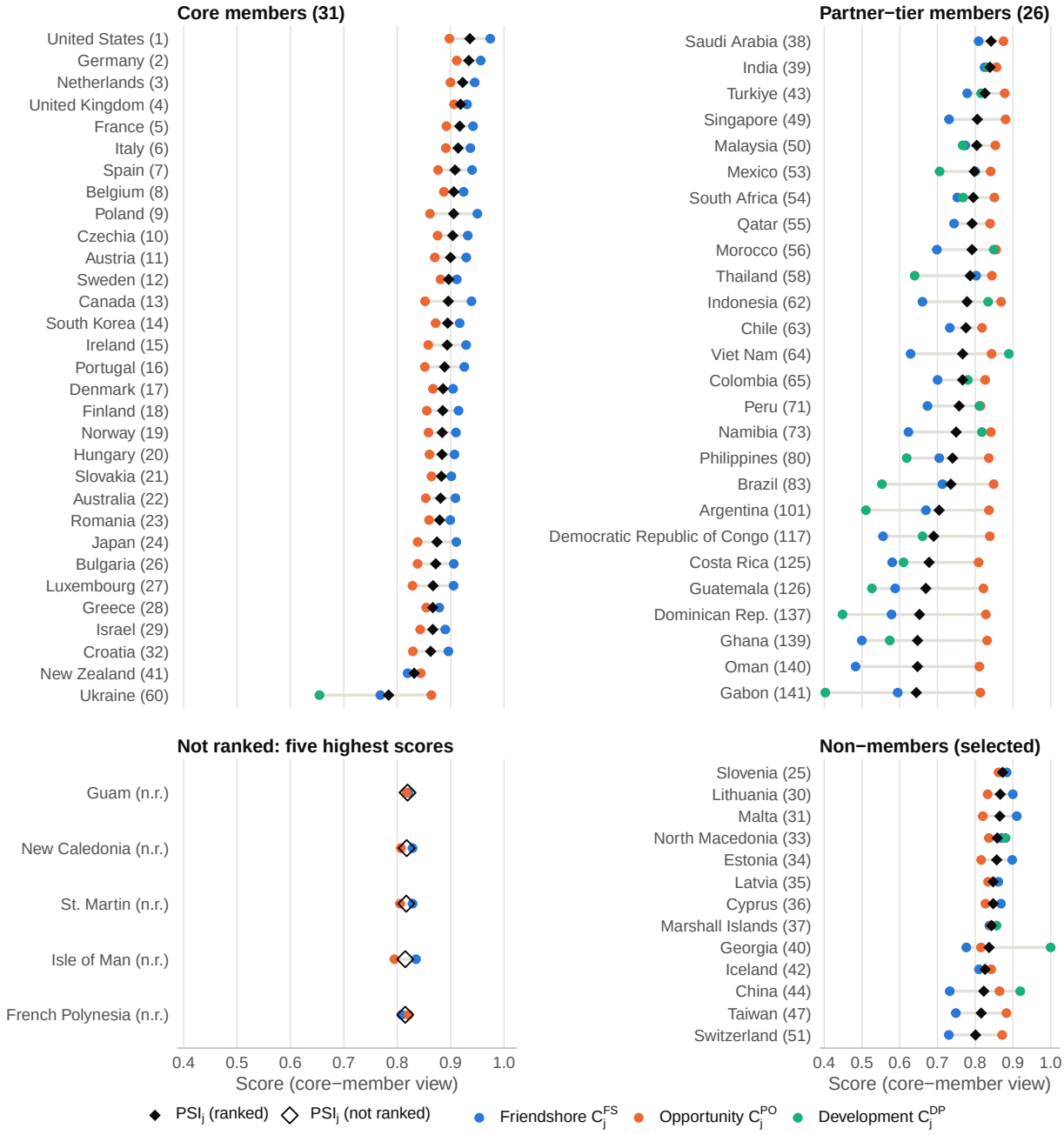


Figure 35: Partnership Strength Index in the headline view. Diamonds are PSI_j (equation (72)) with C_j^{FS} and C_j^{PO} averaged over the 31 core coalition members as reporters; dots are the three category scores, Friendshore C_j^{FS}, Partnership Opportunity C_j^{PO} and Development Potential C_j^{DP} (partners that are not high-income only). Panels: the 31 core members, the 26 partners, the ten highest-ranked non-members with China, Taiwan and Switzerland, and the five highest-scoring partners that are not ranked. Ranks in parentheses are among the 192 rankable partners (sovereign, imputed weight share at most 0.5); open diamonds and "n.r." mark partners without a rank. Source: `psi_tbl.rds`.

Table 20: The fifteen highest-ranked partners in the three views of the Partnership Strength Index

	Core members (31 reporters)		Full coalition (57)		World (218)	
Rank	Partner	PSI_j	Partner	PSI_j	Partner	PSI_j
1	United States	0.936	United States	0.893	China	0.813
2	Germany	0.934	Germany	0.879	India	0.787
3	Netherlands	0.922	France	0.868	France	0.783
4	United Kingdom	0.919	Netherlands	0.864	United States	0.782
5	France	0.917	Spain	0.861	Germany	0.780
6	Italy	0.914	United Kingdom	0.857	Indonesia	0.768
7	Spain	0.908	Italy	0.854	Netherlands	0.764
8	Belgium	0.906	South Korea	0.852	South Korea	0.761
9	Poland	0.906	China	0.850	Spain	0.760
10	Czechia	0.904	Belgium	0.848	United Kingdom	0.757
11	Austria	0.900	Canada	0.847	Italy	0.755
12	Sweden	0.896	Poland	0.840	Viet Nam	0.753
13	Canada	0.896	Czechia	0.838	Canada	0.750
14	South Korea	0.894	Austria	0.838	Austria	0.746
15	Ireland	0.893	India	0.836	Malaysia	0.746

Notes: PSI_j (equation (72)) with C_j^{FS} and C_j^{PO} averaged over the reporters of each view: the 31 core coalition members (the headline), the 57 core and partner-tier members, and every other reporter (218 per partner). Ranks among the rankable partners of each view (sovereign, imputed weight share at most 0.5): 192, 192 and 188 partners. Spearman correlations of PSI_j across the partners ranked in both views: core and full coalition 0.981 (192), core and world 0.881 (188), full coalition and world 0.948 (188). Source: `psi_tbl.rds`, `psi_full_coalition_tbl.rds`, `psi_world_tbl.rds`; `scripts/paper/fig_bilateral.R`.

term. Alignment, which enters only Friendshore, separates them: Friendshore exceeds Opportunity by 0.157 on average for same-bloc pairs and falls short by 0.24 for opposite blocs. The highest dyadic PSI_{ij} is Canada's rating of the United States (0.981).

Development potential. Among the 125 ranked partners, Georgia, Mongolia and Mauritius (1.00), Tuvalu and China (0.919) have the highest best-five summaries; the aid components, 35 percent of the nominal weight, provide 43.7 percent of the composite in their best cells. China's score is a reminder that the pillar rates absorptive capacity, not alignment.

11 Designing the Allied Industrial Base: a network optimization

The indices of Sections 5–10 score countries, sectors and pairs one at a time. The case for an Allied Industrial Base turns on a different question (New Energy Industrial Strategy Center, 2026): a top-ranked country may add little if its strengths duplicate the coalition's, and a lower-ranked one may complete a resilient chain, so the object is a country's marginal contribution to the system and the task is to optimise combinations rather than rankings. The same source sketches a portfolio objective, security, opportunity, development and durability gains net of cost premia, delay, duplication and residual coercion exposure, maximised under explicit constraints such as a resilience floor and minimum diversity; because the weights are political, it asks for a frontier of

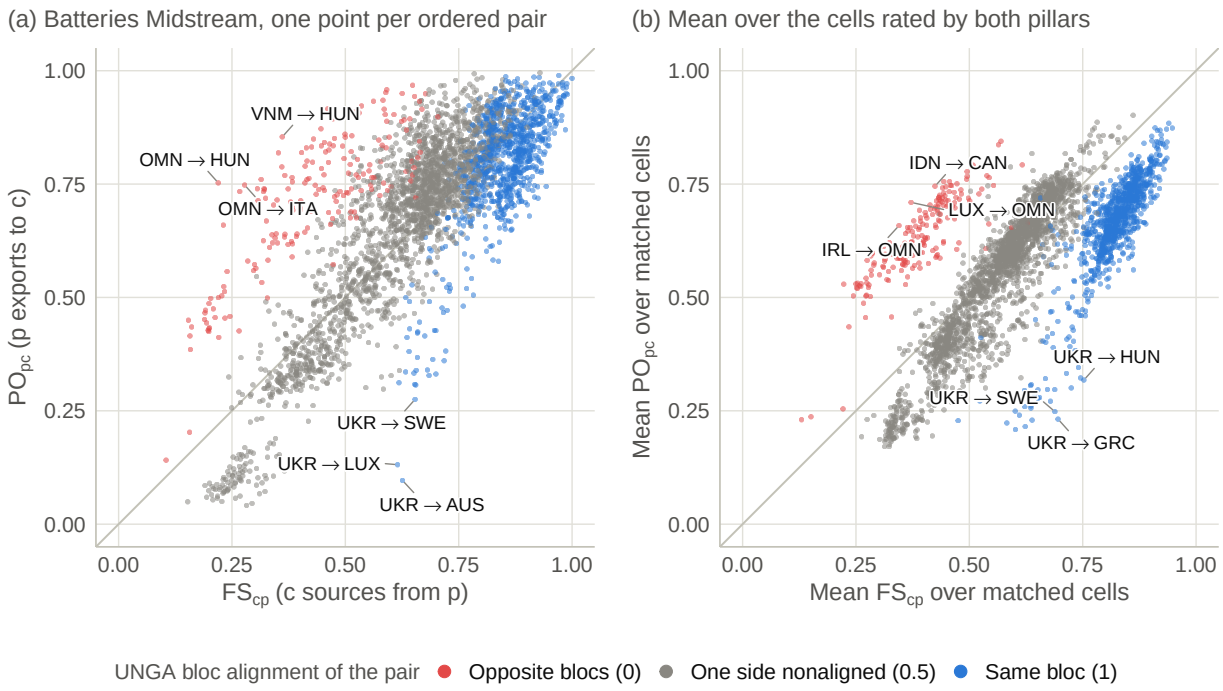


Figure 36: Supplier and market ratings of the same supply edge, coalition members. Horizontal: FS_{cp}, member c's Friendshore rating of member p as a supplier; vertical: PO_{pc}, p's Opportunity rating of c as a market. Panel (a): Batteries Midstream, one point per ordered pair of the 57 members (3,192 pairs). Panel (b): mean over the technology–stage cells rated by both pillars (28 to 30 per pair). Colour is the pair's dyadic UNGA alignment (1 same bloc, 0.5 when either is nonaligned, 0 opposite blocs); the line is the diagonal; the three pairs most favoured by each view are labelled. Source: `partner_friendshore_tbl.rds`, `partner_opportunity_tbl.rds`.

feasible choices rather than one correct weighting.

This section builds the part of that objective that the indices support. For each Upstream and Midstream stage of the eleven headline technologies, a mixed-integer program chooses which network nodes produce, what share each carries and which partners each supplies. It rewards fit, the capability of producers and the quality of their partnerships, and penalises concentration through a concave Herfindahl term, under share limits, a dual-sourcing floor and caps on the number of stages a node may lead; the penalty weight traces a frontier between fit and dispersion, and a rule picks one point on it. The program is a multi-sourcing variant of facility-location and network design (Geoffrion and Graves, 1974; Owen and Daskin, 1998; Melo et al., 2009; Daskin, 2013) with dual-sourcing constraints (Tomlin, 2006; Snyder et al., 2016). Its objective is a score, not a cost, and the design is normative: it describes what a planner with these preferences would choose, not where production will locate. Appendix C states the algorithms and Online Appendix OA.9 reports further results.

11.1 Nodes, stages and scores

The network consists of the 57 coalition members of Section 8 (31 core members and 26 partners) and the six EU members that belong to it only through the Union. The EU-27 is one node, so the 63 economies form 37 nodes; its inputs are aggregated from the members with the stage weights of Section 8, and edges between members are internal and dropped (Appendix C). Taiwan, Switzerland and the China-leaning suppliers outside the coalition are not nodes. A stage is a technology with $s \in \{\text{Upstream}, \text{Midstream}\}$, 22 in all; Downstream deployment serves each member's own market and is not part of the allied division of labour. All 37 nodes enter every stage, each as a potential producer p and as a consumer c.

Within each stage, EO and ES are re-ranked across the nodes with the S-curve of Section 4, $\tilde{v}_p = \phi_\gamma(r_p)$ with $r_p = (\text{rank}(v_p) - 1)/(n_v - 1)$ and $\gamma = 0.5$, so that a score measures a node's standing among its partners. The producer score is the weighted mean of the components observed,

$$a_p = \frac{\sum_{v \in \{\text{EO,ES}\}} \omega_v \, \mathbf{1}[\tilde{v}_p \text{ observed}] \, \tilde{v}_p}{\sum_{v \in \{\text{EO,ES}\}} \omega_v \, \mathbf{1}[\tilde{v}_p \text{ observed}]}, \qquad (\omega_{\text{EO}}, \omega_{\text{ES}}) = (0.5, 0.2), \tag{74}$$

that is, 5/7 and 2/7 when both are observed. Opportunity is the index closest to productive capability; security enters as reliability, since a secure node is a steadier supplier. The Policy index is not part of the score: its basis differs across members (IEA PAMS alone for 11 of them), and it enters the allied picks of Section 9 instead. In the two Geothermal stages no node has ES; in Green Hydrogen Midstream five small partners lack it.

Demand is an allocation key, not a forecast. With need $1 - \widetilde{\text{ES}}_c$ and size the S-curve of the rank of GDP (World Bank, 2026),

$$\tilde{d}_c = \begin{cases} \frac{1}{2}\,(1 - \widetilde{\text{ES}}_c) + \frac{1}{2}\,\text{size}_c & \text{if } \widetilde{\text{ES}}_c \text{ is observed,} \\ \text{size}_c & \text{otherwise,} \end{cases} \qquad d_c = \tilde{d}_c \Big/ \textstyle\sum_{c'} \tilde{d}_{c'}. \tag{75}$$

Because both terms are S-curved ranks, demand is rank-compressed: the EU node, with 24.6% of the nodes' GDP, has a median weight of 0.035 across stages, close to the United States' 0.033, and no node has more than 0.054 in any stage. Development potential (Section 10) has zero weight in the objective and serves only to rank build candidates (Online Appendix OA.9).

11.2 Partnership edges

The quality of supply from p to c combines the two directed partner indices of Section 10, each read from the end it describes: the consumer's friendshore rating of the producer and the producer's opportunity rating of the consumer's market,

$$W_{pc} = \tfrac{1}{2}\,\mathrm{FS}_{cpts} + \tfrac{1}{2}\,\mathrm{PO}_{pcts}, \qquad p \neq c. \tag{76}$$

Friendshore carries the consumer's view of the supplier, including UN-vote alignment and Chinese ownership of the supplier's foreign-invested manufacturing. A missing component counts as zero. Of the 29,304 stage-specific ordered pairs of distinct nodes, 90.3% carry both components and 9.7% friendshore only (every pair in the two Geothermal stages and the 180 pairs with a Green Hydrogen Midstream consumer without ES); none has neither. Self-supply is valued at the median edge of the stage,

$$W_{pp} = \mathrm{median}\{W_{qc} :\ q \neq c,\ W_{qc} \text{ observed}\}, \tag{77}$$

so domestic supply competes with a typical partnership, with neither home bias nor penalty.

11.3 The stage program

For stage (t, s) the decisions are the flows $f_{pc} \geq 0$, the share of the coalition's unit demand that p supplies to c; the selection indicators $y_p \in \{0, 1\}$; and the link indicators $\ell_{pc} \in \{0, 1\}$. Production shares are $x_p = \sum_c f_{pc}$, and the program is

$$\max_{f,\,y,\,\ell} \quad J = \sum_{p,c} f_{pc}\,(w_N\,a_p + w_E\,W_{pc}) + w_D \sum_p \delta_p\,y_p - \lambda \sum_p x_p^2 \tag{78a}$$

$$\text{s.t.} \quad \textstyle\sum_p f_{pc} = d_c \qquad \forall c, \tag{78b}$$

$$\underline{s}\,y_p \leq x_p \leq \bar{s}\,y_p, \qquad \textstyle\sum_p y_p \geq m_P \qquad \forall p, \tag{78c}$$

$$\varepsilon\,d_c\,\ell_{pc} \leq f_{pc} \leq d_c\,\ell_{pc}, \qquad \ell_{pc} \leq y_p, \qquad \textstyle\sum_p \ell_{pc} \geq m_S \qquad \forall p, c, \tag{78d}$$

$$y_p = 0 \qquad \forall p \in \mathcal{X}_{ts}, \tag{78e}$$

with $w_N = 1$, $w_E = 0.5$, $w_D = 0$, $\bar{s} = 0.40$, $\underline{s} = 0.05$, $m_P = 3$, $m_S = 2$, $\varepsilon = 0.10$ and $\lambda = 0.78$, where $\mathcal{X}_{ts}$ holds the nodes barred by the portfolio caps. Demand is met in full by network producers (the network is closed); no node carries more than 40% of a stage and a selected one at least 5%, a stand-in for minimum efficient scale; and every consumer has at least two suppliers, each carrying at least a tenth of its demand. These constraints bound the Herfindahl index of designed shares,

$\mathrm{HHI}^{\mathrm{D}}_{ts} = \sum_p x_p^2$ (Hirschman, 1945), for any λ:

$$\frac{1}{\min\{|\mathcal{E}_{ts}|, \lfloor 1/\underline{s} \rfloor\}} \;\le\; \mathrm{HHI}^{\mathrm{D}}_{ts} \;\le\; \lfloor 1/\bar{s} \rfloor\, \bar{s}^2 + \left(1 - \lfloor 1/\bar{s} \rfloor\, \bar{s}\right)^2 = 0.36, \tag{79}$$

where $\mathcal{E}_{ts}$ is the set of eligible producers.

The term $-\lambda\,\mathrm{HHI}^{\mathrm{D}}_{ts}$ is the diversification penalty, in score units per unit of concentration. Without it the objective is linear and the solution sits at corners, two producers at the cap and a third with the remainder. With it the marginal value of a share falls with its size, and at an interior optimum two producers' shares differ by their difference in marginal value divided by 2λ: at $\lambda = 0.78$ a score advantage of 0.0156 buys one percentage point of share. The penalty plays the role of the variance term in portfolio choice (Markowitz, 1952), without a model of disruption probabilities, and it is the part of the program that values a node by what it adds to a combination: once the leaders' marginal value has fallen, a lower-scoring node with good partnerships enters. If supplier outages were independent and equally likely, with probability π, the variance of the share of a stage lost would be $\pi(1-\pi)\mathrm{HHI}^{\mathrm{D}}_{ts}$, so λ reads as a price of that variance. The penalty is additively separable, however: a producer's marginal penalty depends only on its own share, and every node counts as an independent source, so two producers that share a bloc, a region or a Chinese parent diversify as much as two unrelated ones. The design is therefore a first step toward optimising combinations, not its full form (Section 13). The MILP carries the penalty through tangent cuts on a share grid of 0.01, in perspective form (error at most 2.5×10^{-5} per producer; Appendix C). Stages are built with `ompr` (Schumacher, 2023) and solved by GLPK (Makhorin, 2020) under a 300-second limit, with a greedy water-filling heuristic as fallback.

11.4 Portfolio caps

Solved independently, the stages would select the same strong nodes again and again, concentrating the base in a few economies and leaving many members without a stake. The design therefore caps the number of stages in which a node is a leading producer, rising with the logarithm of GDP:

$$\kappa_i = \min\Big\{\kappa_{\max},\, \max\Big\{\kappa_{\min},\; \mathrm{round}\Big(\kappa_{\min} + (\kappa_{\max} - \kappa_{\min})\,\frac{\eta_i - \min_j \eta_j}{\max_j \eta_j - \min_j \eta_j}\Big)\Big\}\Big\}, \tag{80}$$

where $\eta_i = \log(1 + \mathrm{GDP}_i)$, $\kappa_{\min} = 2$, $\kappa_{\max} = 10$ for the largest economy, and the EU's GDP summed over its members. A node's appearances are the stages in which it is among the five largest producers,

$$A_i = \big|\{(t,s) :\; x_{i,ts} > 0,\; \rho_{i,ts} \le K\}\big|, \qquad K = 5, \tag{81}$$

where $\rho_{i,ts}$ is a strict rank of the share (ties broken by producer score, then by code), so a stage contributes at most five appearances. After all stages are solved, each node with $A_i > \kappa_i$ is barred from the $A_i - \kappa_i$ stages in which its share times its score is smallest, those stages are re-solved, and the count is repeated until no node exceeds its cap: a heuristic coupling of the stages, not a joint optimum.

11.5 Calibrating the diversification weight

No price converts concentration into score, so λ is set by a rule with an observable anchor: the smallest value on a grid at which the median of $\mathrm{HHI}^{\mathrm{D}}_{ts}$ over the 22 stages is at most the median export Herfindahl index of the nodes themselves, $\mathrm{HHI}^{\mathrm{N}}_{ts}$, over the 18 stages with an HS6 basket (all but Geothermal and Green Hydrogen). $\mathrm{HHI}^{\mathrm{N}}_{ts}$ uses 2024 basket exports from the reconciled all-reporter Comtrade matrix (United Nations Statistics Division, 2026), with shares within the 37 nodes and the EU counting only its extra-EU exports. The target is 0.1556 (6.43 equivalent exporters). The design should be no more concentrated, in the median stage, than the coalition's own exports already are; a closed network of 37 nodes cannot match the dispersion of a world market, so world concentration would be the wrong anchor.

The grid has 36 points (0, 0.5, every 0.01 from 0.70 to 1.00, and 1.1, 1.2 and 1.5), each a full re-solve of the production configuration. The rule picks $\lambda = 0.78$, at which the median is 0.1520, or 6.58 effective producers (Figure 37; Appendix C tabulates the grid). The pick is knife-edge. At 0.77 the median is 0.1558, missing the target by 0.0002; the median is not monotone in λ (it rises in eight of the 35 grid steps, from 0.1485 at 0.79 to 0.1502 at 0.80, for instance), because interior shares move top-five membership and the portfolio loop then bars different nodes; and the pick moved with every revision of the inputs during the final data corrections. The rule, not the number, is therefore the specification, re-applied to every full rerun. The design it selects is less fragile than the number: at $\lambda = 0.77$ and 0.79 at most 1.1% of production moves in the mean stage (Section 12). The frontier prices diversification in score, not in resource costs: at the pick the summed node and edge terms are 5.8% below their value at $\lambda = 0$, at $\lambda = 1.5$ (median index 0.095) 10.6% below. This is the frontier of choices the conceptual objective asks for (New Energy Industrial Strategy Center, 2026).

11.6 Results

Solution. All 22 stages are solved by the MILP, none by the fallback, and the portfolio loop converges after seven passes; the 62 stage solves take 462 seconds, none more than 10.6.

Who produces what. Figure 38 and Table 21 show the designed division of labour: 201 producer roles, from seven producers (Nuclear Upstream) to ten per stage, median nine. $\mathrm{HHI}^{\mathrm{D}}_{ts}$ runs from 0.126 (Oil Midstream) to 0.209 (Nuclear Upstream); the largest share is 0.325 (Indonesia in Green Hydrogen Upstream, Canada in Nuclear Upstream), none reaches the cap, and 61 sit at the floor. Twenty-nine of the 37 nodes produce somewhere; eight never do (Turkey, New Zealand, Ukraine, the Dominican Republic, Guatemala, Ghana, Costa Rica and Gabon). Canada has the most roles (16), then the EU node (15; it leads no stage), the United States and Indonesia (13 each). How many roles a node takes follows its size through the caps (Spearman correlation 0.75 with the cap); which stages it takes follows its scores. Nodes outside the high-income group carry 47% of designed Upstream and 43% of Midstream production.

Three patterns arise from the construction rather than from observed capacity. Indonesia, China-leaning by its votes and a member through IPEF, leads six Upstream stages and Oil Midstream: the

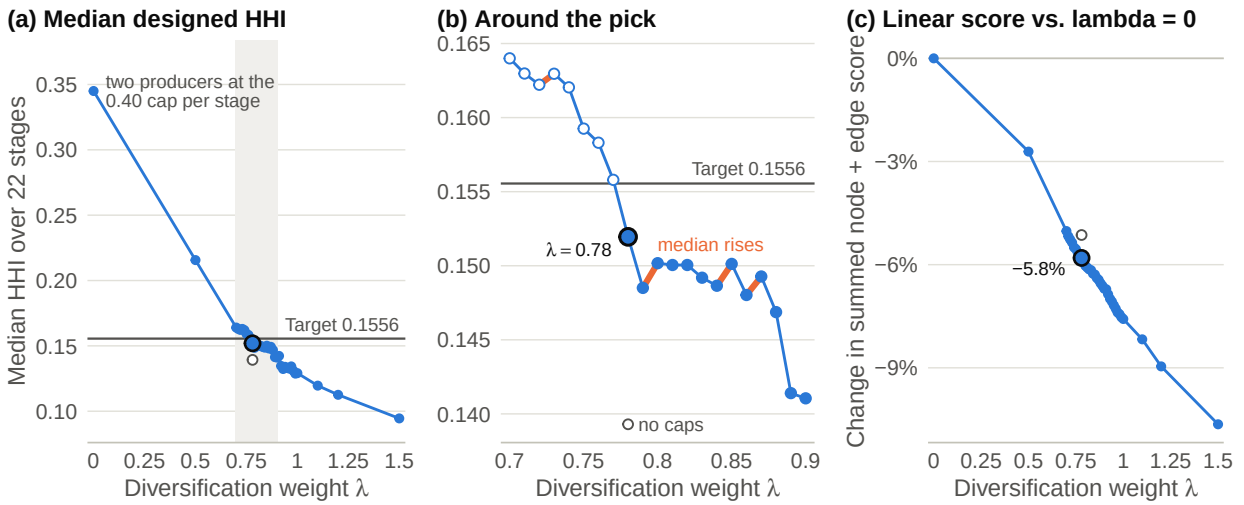


Figure 37: Calibration of the diversification weight λ. Panel (a): median over the 22 Upstream and Midstream stages of the Herfindahl index of designed production shares at each of the 36 grid values of λ (37 nodes, portfolio caps on, MILP with a 300-second stage limit); the horizontal line is the target, the median export Herfindahl index of the nodes over the 18 basket stages (0.1556); the large marker is the pick, $\lambda = 0.78$; the open circle the same design without portfolio caps. Panel (b): the neighbourhood $0.70 \leq \lambda \leq 0.90$; open markers lie above the target, filled markers meet it, and orange segments are steps at which the median rises although λ increases. Panel (c): the summed node and edge terms of the 22 stage objectives, without the Herfindahl penalty, in percent of their value at $\lambda = 0$. Source: calibration study on the inputs of the final rerun of 12 October 2026.

producer score has no alignment term, which reaches the design only through the edges. Thailand leads both Geothermal stages, where scores rest on EO alone and edges lack their opportunity half. And the design applies no endowment screen: by the screen of the allied picks (Section 9), 21 of the 90 designed roles in the ten screened Upstream stages are held by nodes without recorded reserves or production of the stage's resources, a median of 11% of stage production (none in Electric Vehicles, Gas and Oil Upstream; 36% in Solar Upstream). Their shares are supply they would have to source through trade or overseas equity.

Flows. Figure 39 shows three Midstream stages; India leads Solar and Wind Midstream (0.275 each) and the United States Batteries Midstream (0.295). The redundancy rules bind: 793 of the 811 consumer–stage pairs with positive demand draw on exactly two suppliers, and in every stage the weakest link carries exactly 10% of its consumer's demand. With no transport cost in the objective, routing follows the edge weights: the flow-weighted edge weight is 0.874 in the median stage against an unweighted mean of 0.58, and 15 to 28% of the mapped flows stay within a World Bank region.

Concentration and current exporters. Figure 40 sets the design against two benchmarks in the 18 basket stages. It meets the nodes' median export concentration by construction, but stage by stage the two are unrelated (Spearman correlation −0.07): the design spreads production where the nodes' exports are concentrated, as in Nuclear Upstream (Canada 54% of node exports), and

Table 21: Designed producers and production shares by technology and stage

Technology	Upstream	Midstream
Batteries	IDN .23, AUS .22, CAN .12, CHL .12, ARG .08, PHL .06, EU .06, GBR .05, ZAF .05	USA .29, IND .14, MYS .14, CAN .11, EU .10, THA .06, KOR .06, GBR .05, JPN .05
Coal	IDN .26, IND .15, VNM .12, COL .09, ZAF .07, PER .07, CAN .06, THA .06, EU .05, PHL .05	JPN .20, EU .20, COL .14, KOR .12, VNM .08, CAN .07, ISR .06, MEX .05, MYS .05
Electric Grid	CHL .31, MEX .14, COD .12, PER .12, PHL .07, AUS .05, GBR .05, JPN .05, NOR .05, ZAF .05	THA .22, MEX .15, JPN .13, MYS .11, EU .10, USA .08, CAN .05, IDN .05, IND .05, MAR .05
Electric Vehicles	IDN .24, AUS .21, USA .16, CHL .08, PHL .07, ARG .06, EU .05, THA .05, ZAF .05	USA .32, MEX .20, KOR .19, JPN .08, VNM .06, CAN .05, GBR .05, ZAF .05
Gas	USA .25, CAN .14, NOR .14, QAT .12, MYS .10, IDN .05, AUS .05, IND .05, OMN .05, SAU .05	IND .24, USA .18, EU .14, KOR .12, MEX .06, GBR .05, IDN .05, ISR .05, JPN .05, NOR .05
Geothermal	THA .26, MYS .15, NOR .13, ISR .10, EU .08, IDN .07, MEX .05, CAN .05, QAT .05, USA .05	THA .26, MYS .15, ISR .14, NOR .10, QAT .07, EU .07, CAN .06, IDN .05, MEX .05, USA .05
Green Hydrogen	IDN .33, THA .14, CAN .14, EU .12, MEX .09, ISR .08, GBR .05, PHL .05	USA .26, EU .21, VNM .14, NOR .08, PHL .07, SGP .07, KOR .06, ISR .05, JPN .05
Nuclear	CAN .33, NAM .22, IND .16, ZAF .14, ISR .05, SAU .05, THA .05	IND .24, EU .20, JPN .14, MYS .12, ISR .07, KOR .06, NOR .06, GBR .05, MEX .05
Oil	USA .23, SAU .17, CAN .17, NOR .12, OMN .08, QAT .08, IDN .05, IND .05, THA .05	IDN .20, IND .14, KOR .14, THA .13, VNM .09, JPN .08, EU .06, CAN .05, MYS .05, NOR .05
Solar	IDN .24, CAN .15, EU .12, GBR .12, NOR .10, ZAF .10, IND .07, SGP .06, MEX .05	IND .28, USA .20, VNM .14, THA .12, PHL .06, CAN .05, JPN .05, KOR .05, SGP .05
Wind	IDN .22, USA .22, AUS .20, JPN .08, ZAF .06, CAN .06, ISR .05, KOR .05, SGP .05	IND .28, USA .25, EU .15, JPN .09, MEX .06, BRA .06, KOR .05, VNM .05

Notes: Producers of each stage in the production design (37 network nodes, diversification weight $\lambda = 0.78$, portfolio caps on, every stage solved by the MILP), ordered by the designed production share $x_{p,ts}$, the share of the coalition's unit demand in the stage that the node supplies (ISO3 codes; EU = the EU-27 node; shares sum to one within a cell and are printed to two decimals). Source: `allied_network_specialization.csv`.

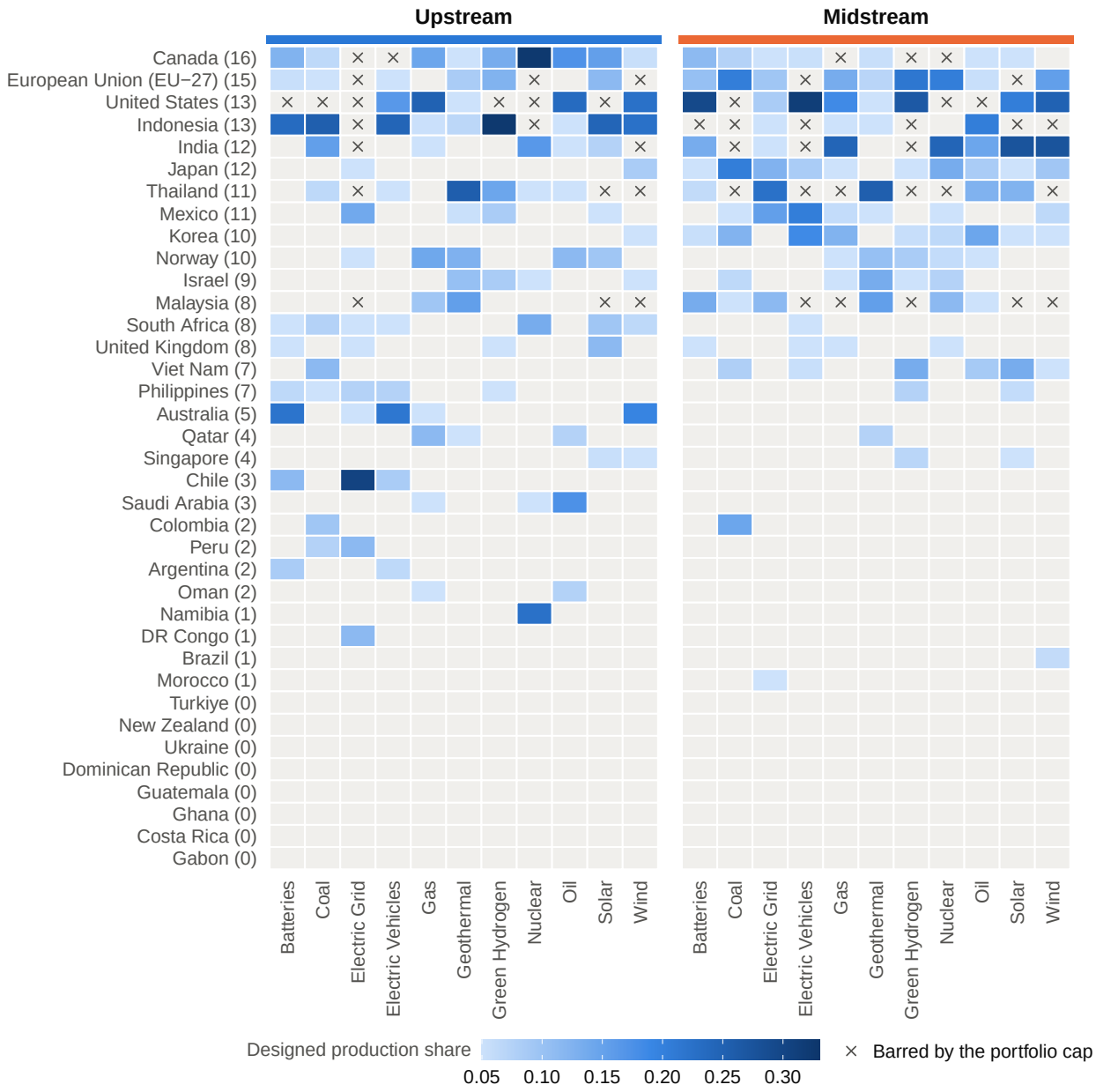


Figure 38: Designed division of labour across the allied network. Rows are the 37 network nodes (the EU-27 as one node), ordered by the number of stages in which they produce (in parentheses) and then by their summed share; columns are the eleven headline technologies within the Upstream and Midstream stages. Shading is the designed production share $x_{p,ts}$ from the floor 0.05 to the largest share, 0.325; light grey cells are stages in which the node does not produce, and a cross marks a stage from which the portfolio loop barred the node in its final pass. Production design: $\lambda = 0.78$, GDP-log portfolio caps, node policy term off, all 22 stages solved by the MILP. Source: `allied_network_specialization.csv` and `allied_network_diagnostics.csv`.

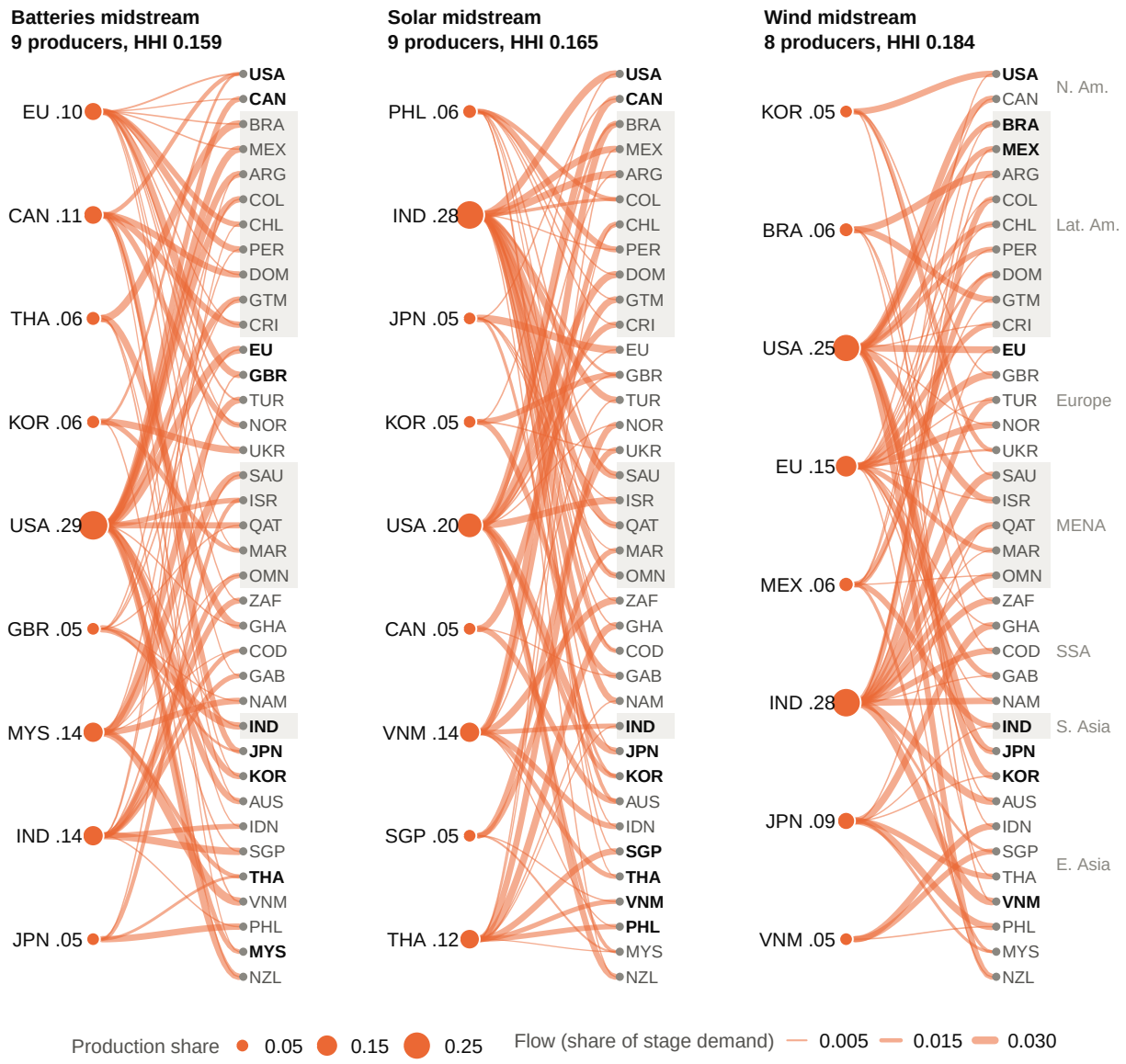


Figure 39: Designed flows in three Midstream stages. In each panel producers sit on the left, sized by their production share (printed next to the code; EU = the EU-27 node), and all 37 nodes sit on the right as consumers, in the same order in every panel: by World Bank region (shaded bands) and by GDP within region. A line is a flow f_{pc}, wider for larger flows; consumers that also produce are in bold. Producers are placed at the flow-weighted mean position of their consumers (the barycentre step of a two-layer layout computed with `igraph` (Csárdi and Nepusz, 2006)). Panel titles give the number of producers and $\mathrm{HHI}^{\mathrm{D}}_{ts}$. Production design ($\lambda = 0.78$, portfolio caps, MILP). Source: `allied_network_flows.csv` and `allied_network_specialization.csv`.

concentrates it where they are dispersed. It is less concentrated than world exports (median 0.101) in only five stages. What changes is the identity of suppliers: China is the largest world exporter in ten basket stages (20 to 43% of world exports), and the 37 nodes hold 74% of world exports in the median basket stage, so the closed design replaces by assumption 3 to 46% of current world supply, at a cost the model does not measure. Nor does it follow the ranking of current exporters: in the median basket stage 66% of designed production lies outside the three largest exporting nodes, and designed and observed export shares have a median Spearman correlation of 0.53. Economies that do not report to Comtrade enter these benchmarks only through their partners' mirror declarations.

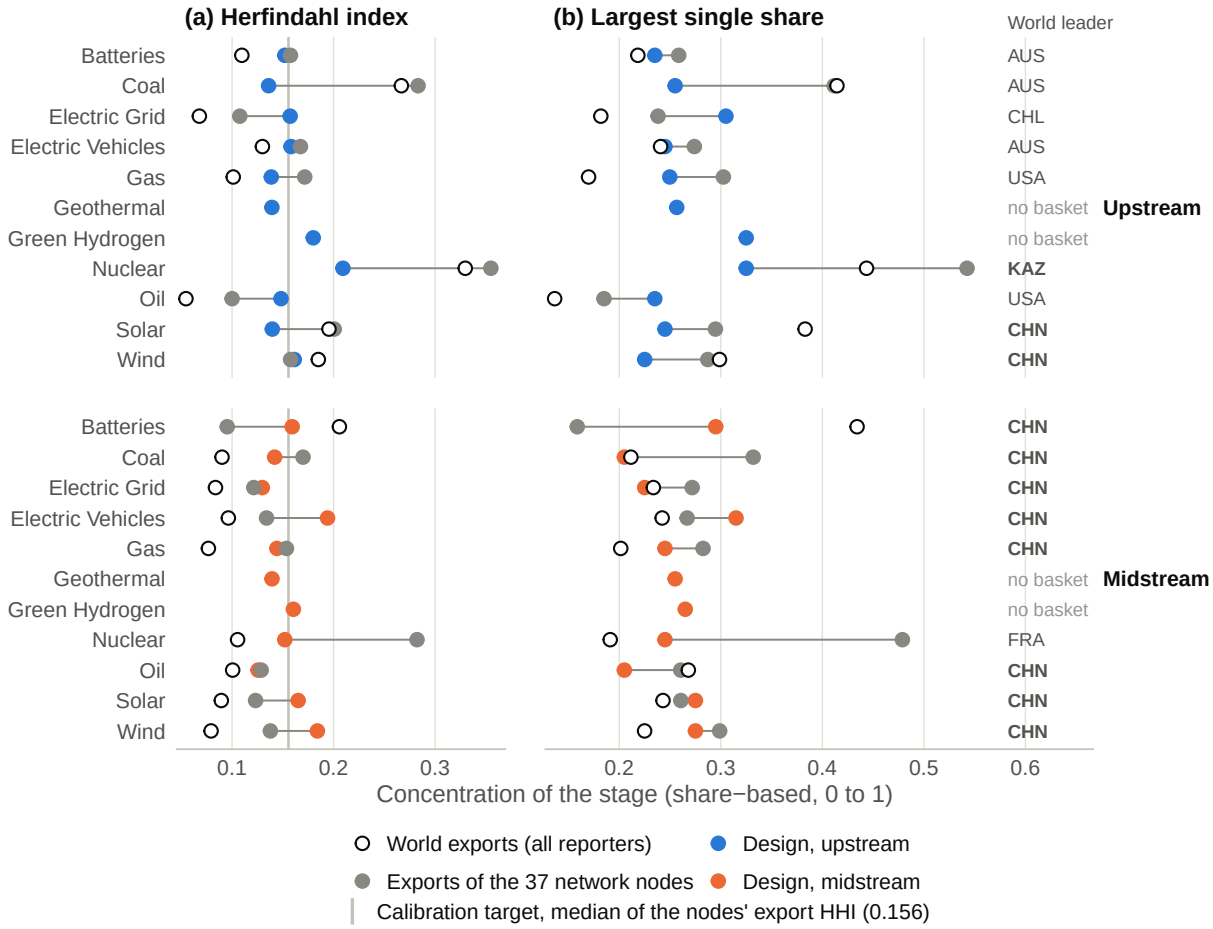


Figure 40: Designed versus observed concentration by technology and stage. Panel (a): Herfindahl index of designed production shares, $\mathrm{HHI}^{\mathrm{D}}_{ts}$ (filled circles coloured by stage); of the 2024 basket exports of the 37 nodes, $\mathrm{HHI}^{\mathrm{N}}_{ts}$ (grey; the EU-27 as one exporter counting only extra-EU exports; reconciled Comtrade matrix, importer report else the exporter's mirror $\times 1.06$); and of world exports across all reporters, $\mathrm{HHI}^{\mathrm{W}}_{ts}$ (open circles). The vertical line is the calibration target (0.156). Panel (b): the largest single share in each case; the codes at the right name the world's largest exporter of the stage's basket, in bold where it is outside the network. Geothermal and Green Hydrogen have no basket. Production design ($\lambda = 0.78$, caps on). Sources: `allied_network_diagnostics.csv`, `allied_network_observed_hhi.csv`, UN Comtrade (United Nations Statistics Division, 2026).

Portfolio caps. The caps range from two (Gabon, Namibia) to ten (the United States, the EU) and sum to 218, against 110 top-five appearances; eight nodes end at their cap, and the final pass bars 49 (node, stage) combinations in 17 stages (Figure 41). Without caps, the United States would be a top-five producer in 17 stages against a cap of ten. The caps pass appearances chiefly to Japan, Mexico, the Philippines, South Africa and Israel, cost 1.9% of the summed objective (at most 5.9%, in Coal Midstream) and raise the median $\mathrm{HHI}^{\mathrm{D}}_{ts}$ from 0.139 to 0.152, so the calibrated λ is specific to the capped configuration.

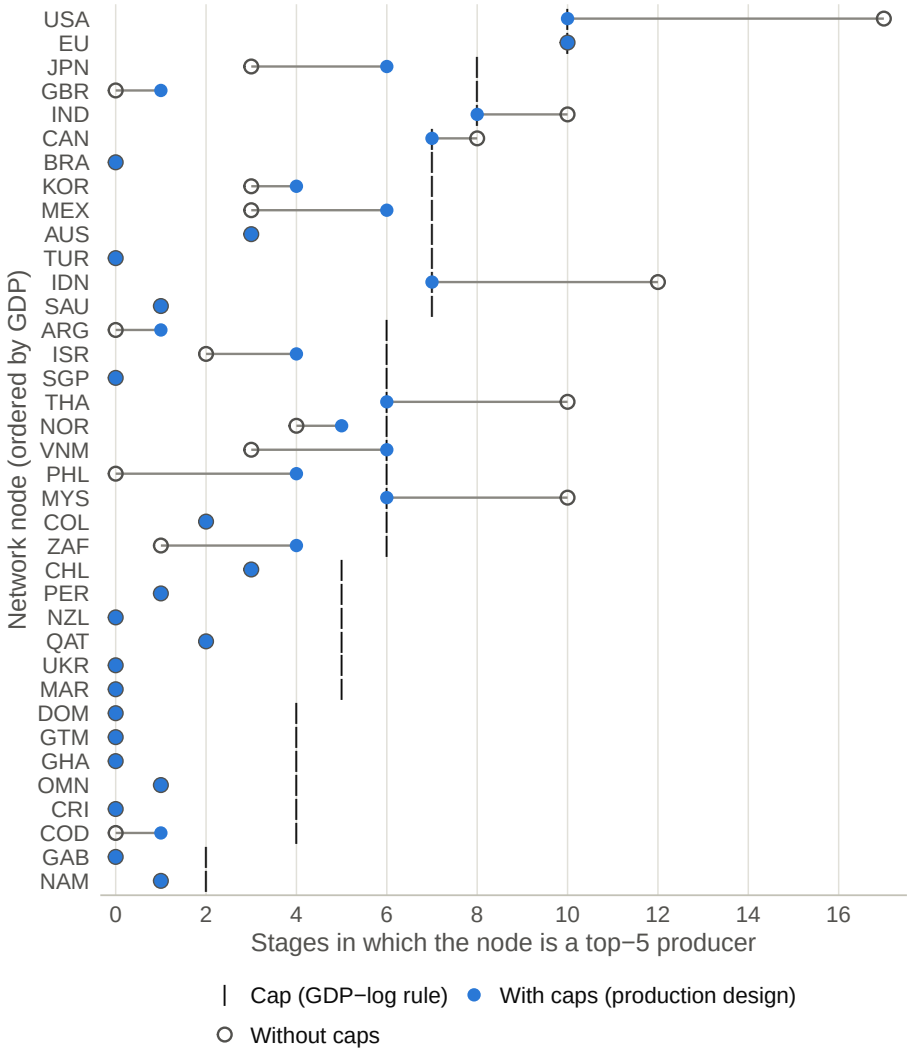


Figure 41: Portfolio caps and realised top-5 appearances. For each of the 37 nodes (EU = the EU-27 node), ordered by GDP, the vertical tick is the cap κ_i of (80), the filled circle the number of stages in which the node is a top-5 producer (81) in the production design, and the open circle the same count in the design solved without caps and otherwise unchanged ($\lambda = 0.78$). GDP: latest finite WDI value in current US dollars (World Bank, 2026), summed over the 27 members for the EU. Source: `allied_network_portfolio_caps.csv`, `allied_network_portfolio_counts.csv` and the no-caps run of the calibration study.

11.7 What the design does and does not show

Against the five analytical layers of the conceptual source (New Energy Industrial Strategy Center, 2026), the design joins the capability layer (node scores) to the complementarity layer (directed edges, with alignment and Chinese-parent exposure) under a minimal robustness layer (share caps, dual sourcing and the diversification penalty). It takes the severity layer as given, since all 22 stages are designed alike, and has no bankability layer; in the source's sequence from severity to coordination to specialization, it is the last step. Within these limits it illustrates, by construction rather than as a finding, the source's point that an allocation under a diversification constraint need not follow the ranking of current exporters; it does not show that the combinations are resilient, because the program models no shocks.

Four further limits apply. The objective is a score: no unit costs, capacities, ramp-up times or build costs enter. Stages are solved independently, so cell manufacturing can be placed where none of the upstream supply is, although shocks propagate along such links (Acemoglu et al., 2012; Carvalho et al., 2021). The network is closed: it shows how the coalition could divide its own requirement, not the cost of decoupling (Aiyar et al., 2023; Javorcik et al., 2024). And exposure is direct: dyadic edges do not register a partner's own dependence on non-members, although the economies that gained U.S. market share from China are themselves integrated into Chinese supply chains (Freund et al., 2024; Alfaro and Chor, 2023); the Chinese-parent component of friendshore captures one channel of such common-mode exposure, ownership, but not shared precursors, equipment or logistics.

12 Robustness and validation

We vary one element of the construction at a time, holding the others at the final configuration, and ask how far the results depend on choices rather than data (Saisana et al., 2005; Nardo et al., 2008; Saltelli et al., 2008). For the pillars the comparison that matters is the ranking of countries within a technology–stage pair, so the main statistic is the within-pair Spearman correlation with the baseline, summarised across pairs and computed over the baseline's rankable cells: 4,163 of 6,232 scored cells in the 30 headline ES pairs and 4,301 of 7,131 in the 33 EO pairs. Every variant is recomputed from stored inputs with the pipeline's own builders, which reproduce the published indices at the baseline settings. Online Appendix OA.10 gives the details and further tables.

12.1 The pillars

Within-pair rankings are almost insensitive to the category weights and to the curvature of the S-curve (Figure 42). Over 1,000 Dirichlet draws centred on the configured weights (coefficient of variation 0.3 for a category of average weight), the median within-pair correlation is 0.997 in ES and 0.988 in EO, and equal weights give 1.000. Re-curving every component to $\gamma = 0.25$, 1 (the plain percent rank) or 2 gives medians between 0.964 and 0.997 in ES and between 0.965 and 0.992 in EO, the lowest at $\gamma = 2$. Much of this stability is mechanical: all scored countries of a pair observe the same categories, so a category that is constant within the pair shifts every country alike

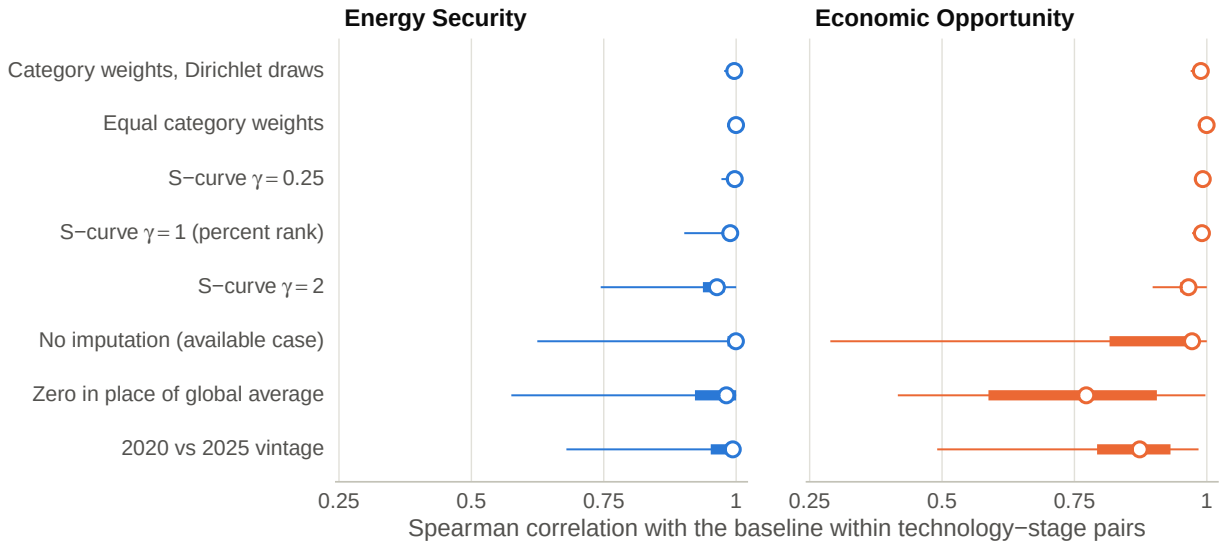


Figure 42: Stability of within-pair country rankings under alternative constructions of the pillars. Each row is one variant; for each technology–stage pair the Spearman rank correlation across countries between the baseline pillar and the variant is computed over the baseline's rankable cells, and the row summarises these correlations across pairs: open circle, median; thick bar, interquartile range; thin line, minimum to maximum. For the Dirichlet weight draws the per-pair value is the median over 1,000 draws. The S-curve rows rebuild both pillars from the theme tables with every component re-curved to $\gamma = 0.25$, 1 or 2 (baseline 0.5). "No imputation" removes every global-average rule, so uncovered countries drop the component; "Zero" fills them with 0. The last row compares the 2020 and 2025 vintages. Sample: 30 ES pairs (4,163 rankable cells) and 33 EO pairs (4,301). Source: `index_outputs.rds`, theme tables and vintages of the final run; `scripts/paper/fig_robustness.R`.

(equation (18)), and the median pair has only three varying categories in ES and four in EO. A high average can also hide wide rank bands for individual members (Online Appendix OA.10).

The imputation rule matters more. Without the global-average imputation ES is unchanged in the median pair (1.00) and EO nearly so (0.97), but the pooled EO correlation is 0.85 and the minimum 0.29; filling gaps with zeros lowers the EO median to 0.77. Outside the economies that the narrow sources cover, EO rankings depend on how gaps are filled, which is why the rankability rule withholds cells whose index rests mostly on imputed components. Its threshold is itself consequential: at $\tau = 0.3$ the Strategic Index shrinks from 3,154 cells for 174 countries to 1,577 cells for 107, and at $\tau = 0.7$ it grows to 5,156 cells for 194. The 2020 and 2025 vintages, which re-derive only the time-varying sources (about a third of the average cell's weight), give median within-pair correlations of 0.99 in ES and 0.87 in EO. Leaving out one category confirms that nominal weights and effective importance diverge (Paruolo et al., 2013; Becker et al., 2017): Technology Demand, the largest nominal EO weight, barely moves the median ranking, while Foreign Dependency, Cost Competitiveness and Trade carry most of it.

12.2 The Strategic Index and the allied picks

The ordering of a country's sectors is robust; its single top sector is not. Under 1,000 Dirichlet draws of the five weights of equation (55), the median within-country Spearman correlation is 0.96 (0.88 to 0.99), but a member keeps its allied pick in 73 per cent of the draws on average, 16 of the 57 members in at least 90 per cent and 10 in fewer than half (Figure 43). Retention follows the pick's margin over the runner-up (Spearman correlation 0.84), and the margins are small: a median of 0.015, below 0.005 for 10 members and below 0.01 for 21, smallest for Luxembourg (0.0005) and Indonesia (0.0007). Electric Grid and Nuclear Midstream picks rest on the finest margins (medians 0.003 and 0.007, against 0.014 for Batteries Midstream). The Nuclear Midstream picks of Costa Rica, the Dominican Republic, Guatemala and Namibia have wide margins for a reason other than capability, and not because of a floor in the security pillar: each may pick among only four sectors, the Midstream stages of nuclear, oil, gas and coal (Namibia also Nuclear Upstream), Midstream nuclear security is compressed around its median, and the technology greenhouse-gas weight favours nuclear (0.999, against 0.42 for gas, 0.02 for coal and 0 for oil); without the sector priors the Dominican Republic, Guatemala and Namibia move to Oil Midstream.

Table 22 reports discrete variants. Reading security as strength rather than need is the most consequential: the pooled Spearman correlation with the baseline is 0.33 and only 5 of the 57 picks survive. Without the sector priors 31 picks survive, with all terms ranked within country 20, without the policy term 30 and with equal weights 39. Switching the endowment screen off changes 41 picks, raising Upstream picks from 14 to 47: a country without uranium scores zero on the reserve and production terms, sits near the floor of the security pillar (86 of 174 countries share one ES value in Nuclear Upstream) and so shows a high need. The screen is a rule of eligibility, not part of the index, and without it the Upstream picks would largely reflect that floor. Tightening the screen to require recorded production, rather than any recorded reserves, removes all eight Nuclear Upstream picks, whose holders have small identified resources and no mine production, and the two Wind Upstream picks: nine members change pick and Gabon is left without an eligible sector. Two variants address the implicit sector priors of Section 9.2. Removing the cell-constant inputs from the pillar levels keeps the within-member ordering (median Spearman correlation 0.95) but only 29 of the 57 picks, and 23 picks then fall on Electric Grid Midstream; entering opportunity and need as within-pair ranks keeps 27 picks (median within-member correlation 0.80). The basis of the Policy index matters for its values more than for the picks: for the 65 countries with both components, the within-country correlation between policy on IEA PAMS alone and on the composite has a median of 0.48 (interquartile range 0.41 to 0.57), yet IEA PAMS alone keeps 45 picks, and requiring both components keeps the 46 CAT-rated members' picks exactly while removing the 11 scored on IEA PAMS alone. Adding back the October 2026 NIPO component (77 jurisdictions) keeps only 34 picks, shifting them from Batteries Midstream (27 to 15) towards Nuclear Upstream (8 to 14).

The picks also rest on the trade data. Setting the EIA-filled fossil production to zero changes no pick and removing the new Upstream trade baskets changes four, but dropping the Trade category keeps 29 of the 54 members that still have a pick, Trade on the Atlas basis 49, and the 2025 and

2020 vintages 48 and 44. The recorded history agrees: before the energy-trade file was completed, Electric Grid Midstream was the pick of 17 and Batteries Midstream of 11 of the 45 members then with a pick, against 4 and 27 of 57 in the final build. The allied pick is an informative but fragile summary, stable to the fill rules and basket additions of the final revision and sensitive to how trade is measured.

12.3 Vertical complementarities

The pillars score each stage separately, although shocks propagate along input–output links (Acemoglu et al., 2012; Carvalho et al., 2021) and capabilities in one activity ease entry into related ones (Hidalgo et al., 2007). A companion study tested whether a stage's EO should reward capability in the adjacent stage and its ES the security of its inputs. On the Atlas panel of 172 economies (1995–2023), with country–year and technology–year fixed effects, adjacent-stage capability is associated with a stage's market-share growth and entry into comparative advantage (Balassa, 1965) across countries, but the association is concentrated in fossil fuels and nuclear, close to zero for the clean-technology chains, zero within country–technology cells, and of no help out of sample. A sourcing-weighted input-security measure reflects mostly alignment, its post-2022 shift is driven by Russia, and it did not predict the import shortfalls of 2021–22, whereas the concentration of sourcing did. At the weights the evidence supports, an adjacent-stage EO term moves one pick and an input-security ES term moves ten, no more than a placebo with permuted scores (Online Appendix OA.10). We therefore keep the stages uncoupled and leave to future work plant-level co-location from the CM-FDI and BNEF project data, input shares weighted by input–output coefficients, a test of sourcing concentration on the 2023 export controls on graphite and gallium, and a co-location term in the network design.

12.4 The partnership index

The headline PSI averages the dyadic index over the 31 core members' viewpoints. The full-coalition view (57 reporters) gives nearly the same ranking (Spearman correlation 0.98 over 192 partners, 13 of the top 15 in common), although China moves from 44th to 9th; the world view (218 reporters) differs more (0.88, 10 of the top 15, median absolute rank change 17). The viewpoint is a choice of whose interests the index represents (Figure 44). Without the CM-FDI parts of friendshore, in the integration study on the inputs of 11 October 2026, the core PSI is practically unchanged (Spearman correlation 0.9998, largest rank change five places, same top 15).

Two further sensitivities concern Friendshore. Scoring an absent outward-investment record at the lowest rank rather than dropping it from the available-case mean leaves the core PSI almost unchanged (Spearman correlation 0.994, 14 of the top 15 in common, largest rank change 34 places, China from 44th to 37th), although it reorders the top suppliers of individual reporters (Section 10). The discretisation of alignment matters more. With continuous alignment, $p_i = s_i$ so that $A_{ij} = s_i s_j + (1 - s_i)(1 - s_j)$, the Spearman correlation with the headline is 0.904, 12 of the top 15 are shared, the median partner moves 13.5 places and China rises from 44th to 9th; with tercile

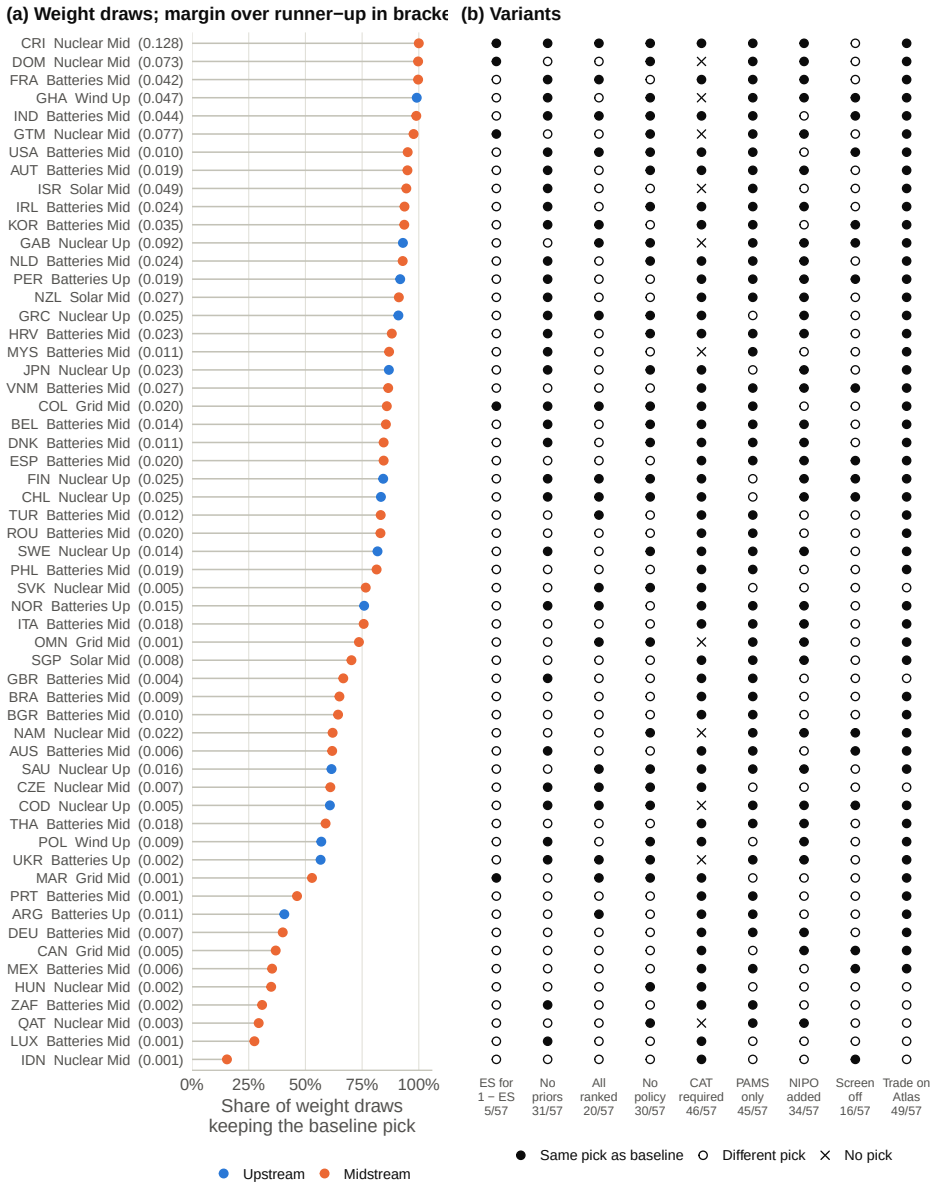


Figure 43: Robustness of the allied picks. Each row is a coalition member, its baseline pick (the eligible Upstream or Midstream sector with the highest Strategic Index) and, in brackets, the margin over its second eligible sector. Panel (a): share of 1,000 Dirichlet draws of the five Strategic Index weights, centred on $(0.2, 0.3, 0.3, 0.1, 0.1)$ with concentration 43.4, under which the pick is unchanged; colour gives the stage of the pick. Panel (b): filled circle, pick unchanged under the variant; open circle, changed; cross, no pick. Variants: security in place of need; no stage and technology priors; EO and 1 − ES S-curved within country; no policy term; both Policy components required; IEA PAMS only; the October 2026 NIPO component added back; the endowment screen off; Trade on the Atlas basis. Sample: 3,154 Strategic Index cells of 174 countries; 57 members, all with a baseline pick. Source: `index_outputs.rds`, `policy_component_tbl`, the October 2026 NIPO component; `scripts/paper/fig_robustness.R`.

Table 22: Allied picks by sector under the baseline and the variants of the Strategic Index

Panel A. Specification and Policy index									
Sector	Base-line	ES for 1 − ES	No priors	All ranked	No policy	Equal weights	CAT required	PAMS only	NIPO added
Nuclear Upstream	8	2	14	13	20	1	6	4	14
Batteries Upstream	4	1	3	4	2	2	3	4	6
Wind Upstream	2	0	2	0	3	1	1	1	3
Coal Upstream	0	0	1	0	0	0	0	0	0
Electric Grid Upstream	0	3	1	1	1	0	0	0	0
Electric Vehicles Upstream	0	0	1	0	0	0	0	0	0
Gas Upstream	0	1	0	0	0	0	0	0	0
Oil Upstream	0	2	3	0	0	0	0	0	0
Solar Upstream	0	0	2	4	1	0	0	0	2
Batteries Midstream	27	0	15	5	8	30	26	34	15
Nuclear Midstream	9	28	2	8	9	8	5	5	7
Electric Grid Midstream	4	7	1	15	13	4	3	3	5
Solar Midstream	3	2	2	1	0	9	2	6	4
Electric Vehicles Midstream	0	0	0	0	0	0	0	0	1
Gas Midstream	0	10	0	0	0	0	0	0	0
Oil Midstream	0	1	10	6	0	0	0	0	0
Wind Midstream	0	0	0	0	0	2	0	0	0
Members with a pick	57	57	57	57	57	57	46	57	57
Same pick as baseline	57	5	31	20	30	39	46	45	34

Panel B. Data: endowment screen, final data choices, trade and reserves inputs, vintage									
Sector	Base-line	Screen off	Zero for EIA	No new baskets	Trade on Atlas	No Trade	No Reserves	2025 vintage	2020 vintage
Nuclear Upstream	8	17	8	11	8	9	14	9	8
Batteries Upstream	4	12	4	4	4	1	1	4	5
Wind Upstream	2	5	2	1	2	1	2	2	1
Electric Grid Upstream	0	1	0	0	0	0	0	0	0
Solar Upstream	0	12	0	0	0	0	0	0	0
Batteries Midstream	27	7	27	25	26	13	24	25	23
Nuclear Midstream	9	2	9	9	4	0	8	4	4
Electric Grid Midstream	4	1	4	4	5	10	4	5	11
Solar Midstream	3	0	3	3	7	19	4	7	5
Electric Vehicles Midstream	0	0	0	0	0	1	0	0	0
Gas Midstream	0	0	0	0	1	0	0	1	0
Members with a pick	57	57	57	57	57	54	57	57	57
Same pick as baseline	57	16	57	53	49	29	50	48	44

Notes: Number of coalition members whose allied pick, the Upstream or Midstream sector (excluding Geothermal and Green Hydrogen, and Upstream cells without a recorded resource base) with the highest Strategic Index, falls in each sector. All 57 members have a pick in the baseline. Panel A: ES for 1 − ES, security in place of need; No priors, without the stage and technology terms; All ranked, EO and 1 − ES S-curved within country; No policy, without the policy term; Equal weights, 0.2 on each term; CAT required, both Policy components required (`OPSI_POLICY_REQUIRE_ALL=1`), which leaves 65 countries with a Policy index and 46 members with a pick; PAMS only, CAT dropped for every country; NIPO added, the October 2026 NIPO component (GTA NIPO as of 5 October 2026) re-enters the Policy index with weight 4 (IEA PAMS 3, CAT 2), IEA PAMS still required, for its 77 policy-eligible jurisdictions. Panel B: Screen off, every Upstream cell eligible, including the 669 of 998 scored Upstream cells without recorded reserves or production; Zero for EIA, the oil, gas and coal production that the Energy Institute does not report set to zero instead of the EIA value (the rule until 11 October 2026); No new baskets, the Trade category removed from Electric Vehicles, Wind, Nuclear and Solar Upstream, whose trade baskets were added on 11 October 2026; Trade on Atlas, the Trade category of both pillars replaced by its Atlas-based counterpart of the 2025 vintage; No Trade, No Reserves, the category dropped from the pillars that score it and the remaining categories renormalised; 2025 and 2020 vintage, both pillars taken from the vintage builds, in which only the time-varying sources differ. The Policy index is the headline one in Panel B. Source: `index_outputs.rds`, `data/processed/vintages`; `scripts/paper/fig_robustness.R`.

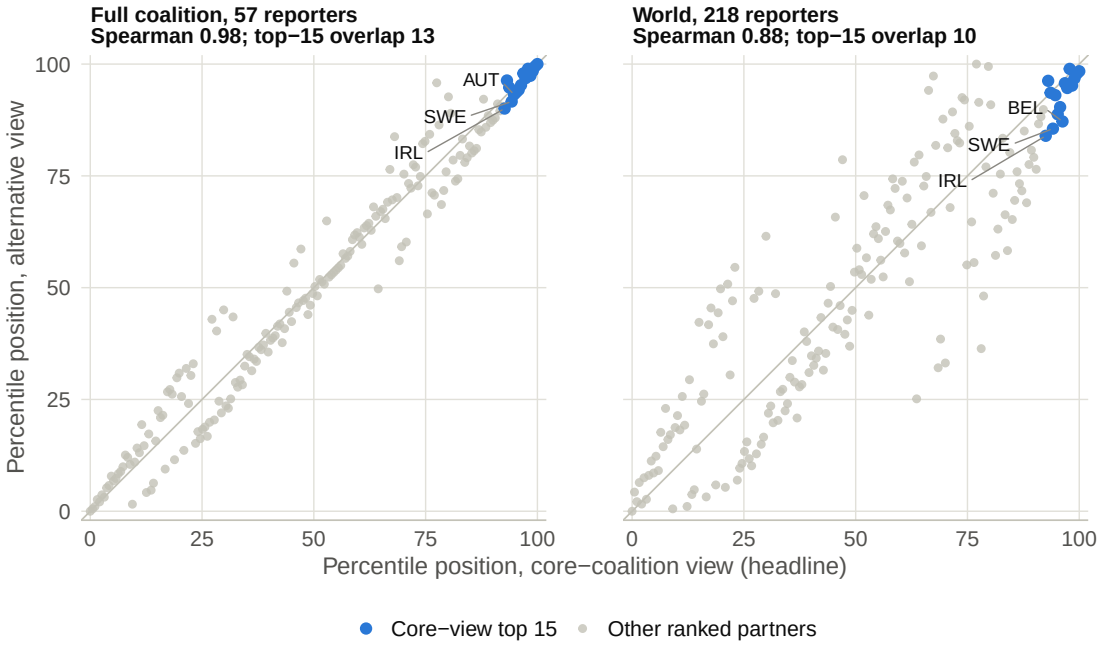


Figure 44: The partnership index from three viewpoints. Each point is a partner ranked in both views ($\tau = 0.5$), placed by its percentile position in the core-coalition view, the headline average over the 31 core members' ratings (horizontal), and in an alternative view (vertical): left, the full coalition of 57 reporters; right, the world of 218 reporters. Blue marks the top 15 of the core view; the labelled points are the three of them with the lowest position in the alternative view. Panel titles give the Spearman correlation with the core view and the overlap of the top 15. Source: `psi_tbl.rds`, `psi_full_coalition_tbl.rds`, `psi_world_tbl.rds`; `scripts/paper/fig_robustness.R`.

blocs (cuts at $s = 0.164$ and 0.305, 65 economies in each outer bloc) the correlation is 0.952, 14 of the top 15 are shared and China ranks 40th. The quartile cuts thus hold China down mainly through the step from a continuous to a three-valued alignment, less through where the cuts fall.

12.5 The network design

Figure 45 and Table 23 re-solve the design of Section 11 with one setting changed, on inputs assembled as in production (the re-run of the production configuration reproduces it to 5×10^{-16}). Each variant is compared with the production design stage by stage through the Jaccard overlap of the producer sets and the share of production reassigned, $\frac{1}{2}\sum_p |x^{\text{alt}}_{p,ts} - x_{p,ts}|$. The neighbouring values of λ leave the design almost unchanged: at 0.77, 21 of the 22 producer sets are identical and 0.5 per cent of production is reassigned in the mean stage; at 0.79, 19 and 1.1 per cent. The calibration is knife-edge in its number, not in what it selects. Demand matters little, since it enters the choice of producers only through the edges and redundancy constraints and is rank-compressed (size only: 3.4 per cent reassigned). The edge weight matters more ($w_E = 0$ and 1 reassign 8.4 and 10.8 per cent), and the portfolio caps most: without them 18.5 per cent of production is reassigned in the mean stage, 46.5 per cent in Electric Grid Upstream. Solved greedily with the caps, the

design is more concentrated (median index 0.160, above the calibration target) and scores 2.5 per cent lower. These tests concern the stability of the design to its specification, not its robustness to shocks, which the program does not model beyond the redundancy floor.

Table 23: Robustness of the network design to one setting at a time

	Against the production design				Design of the variant			
Variant	Jaccard	Min.	Identical stages	Re-assigned	Median HHI^D	Pro-ducers	Max. share	MILP stages
Production design	–	–	–	–	0.152	9	0.325	22
No portfolio caps	0.692	0.294	5	0.185	0.139	10	0.312	22
Greedy heuristic	0.836	0.600	2	0.086	0.160	9	0.346	0
Edge weight $w_E = 0$	0.840	0.538	6	0.084	0.146	9	0.295	22
Edge weight $w_E = 1$	0.825	0.455	7	0.108	0.157	9	0.355	22
Demand from need only	0.995	0.900	21	0.010	0.154	9	0.325	22
Demand from size only	0.941	0.600	15	0.034	0.159	9	0.325	22
$\lambda = 0.77$	0.995	0.889	21	0.005	0.156	9	0.325	22
$\lambda = 0.79$	0.980	0.750	19	0.011	0.149	9	0.325	22

Notes: Production design: $\lambda = 0.78$, GDP-log portfolio caps, edge weight $w_E = 0.5$, demand from the equal mix of need and size, MILP. Each other row re-solves the 22 Upstream and Midstream stage programs on the 37 network nodes with one setting of the production design changed. Jaccard: mean across stages of the Jaccard overlap $|A \cap B|/|A \cup B|$ of the stage's producer set with that of the production design; Min.: its smallest value; Identical stages: stages with the same producer set. Reassigned: mean across stages of the share of production assigned to different nodes, $\frac{1}{2}\sum_p |x^{\text{alt}}_{p,ts} - x_{p,ts}|$ (0 when the shares are unchanged, 1 when the producer sets are disjoint). Median HHI^D: median across stages of the Herfindahl index of designed shares. Producers: median number of producers per stage. Max. share: largest designed share in any stage. MILP stages: stages solved by the MILP (the greedy row solves all stages heuristically). Source: `scripts/paper/fig_network_robustness.R`.

12.6 What the analysis does and does not establish

Within the ranges explored, the within-pair rankings of ES are robust to weights, curvature and, over rankable cells, imputation, and those of EO to weights and curvature but not to imputation. The Strategic Index orders a country's sectors stably, while the allied pick, a single arg max, is robust only where its margin is large. There is no external validation: no outcome series of realised disruptions or later export growth exists against which to test the pillars, and stability shows that the rankings are determined by the data and the construction, not that they are right (Ravallion, 2012). The perturbations are local and one at a time, and several data choices are not varied: the exclusion of electricity from the grid basket and of Hong Kong's re-exports from the export measures, the treatment of non-reporting economies through mirror declarations, the rank compression of the EU's demand weight, the regional IEA 2035 manufacturing shares that every EU member inherits in the foreign-dependency and market-share variables, and the zero counted for unquantified USGS holdings. In the network design, the self-edge weight (the median observed edge), the dual-sourcing rule (at least two suppliers, each at least 10 percent of demand) and the share bound of 0.40 are not varied, and they shape the designed flows: self-supply is rare and most consumer–stage pairs draw on exactly two suppliers by construction.

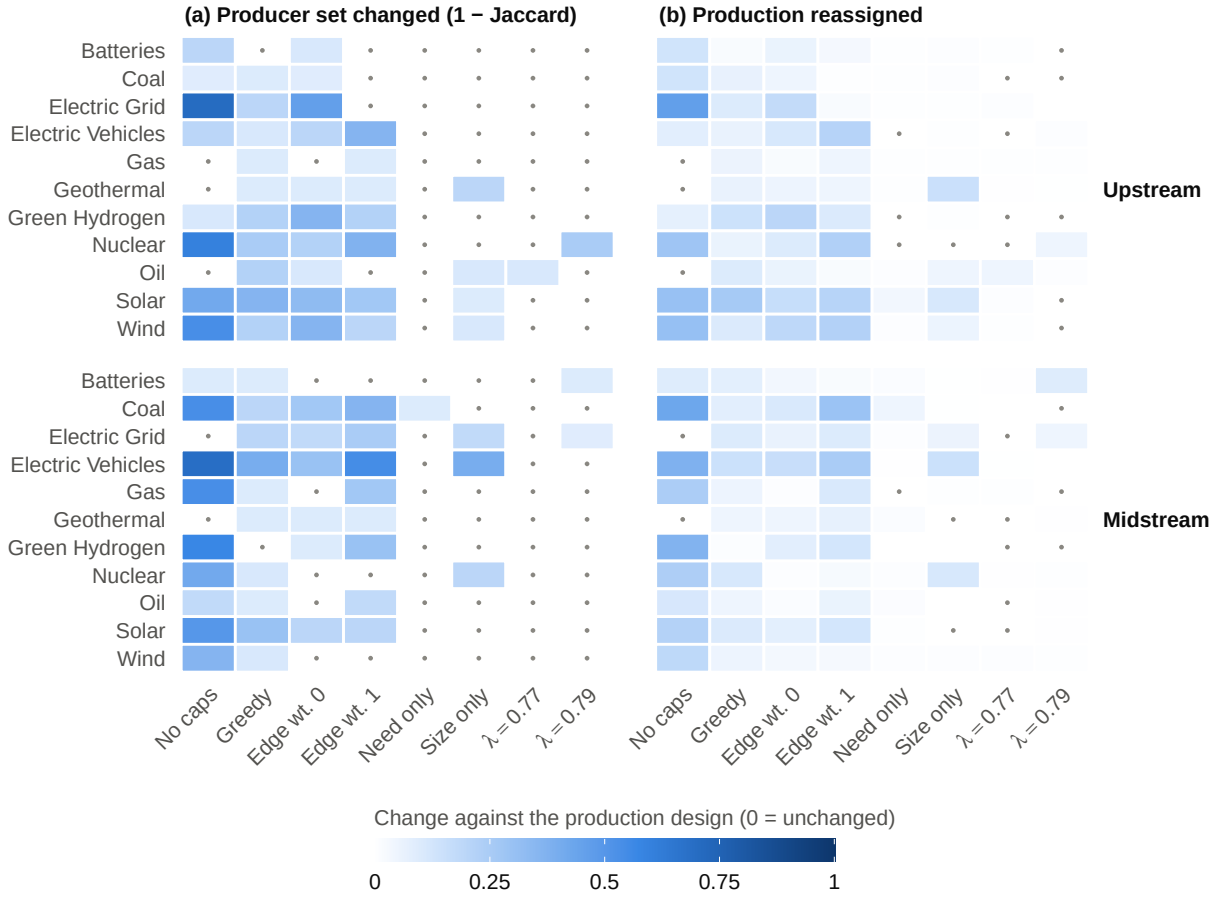


Figure 45: Robustness of the network design, by stage and variant. Each column re-solves the 22 Upstream and Midstream stage programs on the 37 nodes with one setting changed: no portfolio caps; the greedy heuristic instead of the MILP (caps on); edge weight $w_E = 0$ or 1 instead of 0.5; demand from need only or size only; and $\lambda = 0.77$ or 0.79 instead of 0.78. Panel (a): one minus the Jaccard overlap $|A \cap B|/|A \cup B|$ of the stage's producer sets in the variant and the production design. Panel (b): share of the stage's production assigned to different nodes, $\frac{1}{2}\sum_p |x^{\text{alt}}_{p,ts} - x_{p,ts}|$. Both are 0 when the stage is unchanged (dot) and 1 when the producer sets are disjoint. Source: `scripts/paper/fig_network_robustness.R`, on the inputs of the final rerun of 12 October 2026.

13 Discussion, next steps and conclusion

This section reads the results through the analytical design for an Allied Industrial Base of the New Energy Industrial Strategy working paper (New Energy Industrial Strategy Center, 2026), states what the paper does not establish, and sets out the extensions that design calls for.

13.1 What the results imply for an allied industrial base

Managed interdependence rather than autarky. The working paper defines the objective as managed interdependence at allied scale: strategically consequential capabilities should exist at sufficient scale somewhere in a trusted network, not in every member, because no single ally can reproduce China's scale across the electro-industrial stack (New Energy Industrial Strategy Center, 2026; Campbell and Doshi, 2025). China leads 22 of the 33 Economic Opportunity pairs and 11 of the 30 Energy Security pairs (Sections 5 and 6); a coalition member leads 16 security pairs but only 8 opportunity pairs. Within each member, opportunity and security need move in opposite directions: the median within-country Spearman correlation between EO and $1 - \text{ES}$ is -0.49 and negative for every member (Section 9). A member's strongest opportunities lie where it is already secure, and its needs lie where others are strong. The network design makes this arithmetic explicit (Section 11). It selects seven to ten producers per stage among 37 nodes, with a median designed Herfindahl index of 0.152 (6.6 equivalent producers), close by calibration to the nodes' own export concentration. Self-supply is positive in two stages only, a consequence of scoring a node's self-supply at the median edge weight and of the binding dual-sourcing rule (793 of the 811 consumer–stage pairs draw on exactly two partners) rather than an empirical finding; under these rules every node draws most of its requirement from other members.

The closed network is not the world market, however. In the median basket stage the 37 nodes account for 74 percent of world exports (54 to 97 percent), and China is the largest world exporter in 10 of the 18 basket stages, including eight of the nine Midstream ones, with shares of 20 to 43 percent. The design therefore replaces between 3 and 46 percent of observed world supply by assumption; whether the designated nodes could supply it, at what cost and how fast, lies outside the model. The evidence that broad relocalisation is costly and that diversification usually serves resilience better than reshoring (OECD, 2025b; Thakur-Weigold and Miroudot, 2024; Aiyar et al., 2023) applies to an allied wall as much as to a national one.

The node, not the sector, as the unit of treatment. The working paper argues that policy should be set by node, because one technology contains very different risks (New Energy Industrial Strategy Center, 2026). Our stages are coarser than its nodes, but they show the same heterogeneity. Technology and stage explain 44 percent of the variance of ES and 58 percent of EO, against 12 and 18 percent for countries alone, and concentration sits in different places along each chain. Midstream security is mostly a trade question (the Trade category supplies 73 percent of the Midstream score), and China dominates the Midstream baskets. Upstream concentration lies with resource holders

inside the network: Canada supplies 54 percent of the nodes' Nuclear Upstream exports, Australia 41 percent of Coal Upstream, and the EU 48 percent of Nuclear Midstream. Chinese-parent ownership of foreign-invested plants is high in solar (median share 0.75 across suppliers) and nil in wind (Section 10).

Node-level treatment departs from naive rankings in two places. First, the need reading of the Strategic Index rates a stage highest where a member lacks the resource. Without the endowment screen 47 of the 57 allied picks would be Upstream, 17 of them in Nuclear Upstream, many in members with no uranium. With the screen 43 picks are Midstream, and 41 of the 57 picks change (Figure 30). The eight Nuclear Upstream picks that remain all hold identified uranium resources, but small ones (0.015 to 0.79 percent of the world's) and none with recorded mine production in 2019–2024, so they are still need-driven: the screen removes the no-resource artefact but not its small-resource counterpart. The screen is a rule of pick eligibility and leaves every index value unchanged. Nor are the picks an allied assignment: 27 members would prioritise Batteries Midstream, which the network design gives to nine producer nodes, and only 28 of the 57 members hold a designed share in their pick stage. Second, the network does not reproduce the export ranking: the median Spearman correlation between designed production shares and observed export shares is 0.53, and in the median basket stage 66 percent of designed production goes to nodes outside the three largest exporting nodes. The design applies no endowment screen, so 21 of its 90 roles in the screened Upstream stages are held by nodes without a recorded resource base; their shares would have to be sourced through trade or overseas equity.

Partners in the designed network. The working paper asks an allied design to "optimize combinations, not rankings", valuing a partner for the capability it adds rather than for its stand-alone score (New Energy Industrial Strategy Center, 2026). As bilateral partners the 26 partner-tier members rank low: their median rank in the headline Partnership Strength Index is 64.5 of 192, against 16 for the core members, in part because UNGA alignment makes up on average 24 percent of the Friendshore composite. Yet, in a design that scores capability rather than current capacity, the partner tier carries 58 percent of designed Upstream production and 45 percent of Midstream production, and nodes outside the high-income group carry 47 and 43 percent. Eight nodes produce only Upstream, all resource economies (Australia, Chile, Saudi Arabia, Peru, Argentina, Oman, Namibia and the Democratic Republic of the Congo); eight others produce nothing and belong to the network as consumers. One caveat qualifies the reading: the producer score carries no alignment term, so Indonesia, which enters through IPEF and is China-leaning by its votes, leads six Upstream stages.

The three hypotheses. The working paper proposes three hypotheses for testing first (New Energy Industrial Strategy Center, 2026). The paper tests none of them; its results bear on each to a different degree. On the first, that allied specialisation reaches a resilience target at lower public cost than parallel national programmes, the design shows that seven to ten producers per stage reproduce the nodes' observed concentration and leave no consumer on a single supplier, but it

contains no costs, capacities or outlays; the 1.9 percent of the objective that the portfolio caps forgo is score, not money. On the second, that coordinating demand matters more than coordinating subsidies, the results are silent: the Policy index folds every policy type into one strength score, and network demand is an index of need and size, not a volume. On the third, that rules targeted at high-severity nodes deliver most of the benefit of broad decoupling at lower cost, the results document a precondition, heterogeneity across stages (the Strategic Index is 73 percent sector and 4 percent country variance), but not the benefit or cost of either rule, because the design excludes China from every stage, itself a sector-wide rule.

Normative choices and how readers can vary them. The weights of the Strategic Index, its need reading, the alignment weight in Friendshore, the diversification weight λ, the portfolio caps and the coalition thresholds are choices. As the working paper argues, such weights are political, and analysis should display the frontier of choices rather than one number (New Energy Industrial Strategy Center, 2026). Some choices matter a great deal (Section 12): reading the security term as strength rather than need keeps only 5 of the 57 picks; random draws of the five Strategic Index weights keep a member's pick in 73 percent of draws on average; dropping alignment from Friendshore moves China from 44th to 3rd in the Partnership Strength Index. The interactive explorer that accompanies the paper exposes these choices through four dials in its Strategic Insights module: *hawkishness* weights security (the need term, Friendshore, alignment and Chinese-parent exposure, the diversification weight), *bullishness* opportunity (the EO term, partnership opportunity, market size in network demand), *populism* policy signals and home manufacturing, and *greenness* climate performance, dropping coal, oil and gas at its highest settings. The paper's headline is one setting of these dials. The descriptive Sector Insights module shows the theme data (trade flows, reserves, production, investment including the CM-FDI projects, policy records) by technology and stage without index weights, so that a reader can inspect the evidence behind any score.

13.2 Limitations

No outcome validation. The indices are not validated against disruptions, investment or export growth. The one predictive test, in the study of vertical complementarities (Section 12.3), is a warning: sourcing-weighted input security did not predict the import shortfalls of 2021–22, whereas sourcing concentration did.

Data vintages and source conventions. Inputs come from different years (92 percent of opportunity cells mix years; fossil reserves date from end-2020, mineral reserves from end-2025), and the 2020 and 2025 vintages agree less for EO (median within-pair Spearman 0.87) than for ES (0.99). Holdings the USGS does not quantify (not available, withheld or small) count as zero in the technology roll-ups. The IEA's 2035 announced manufacturing-capacity shares are regional for Europe, so every EU member carries the EU's share (for example 0.15 in Wind Midstream). The BloombergNEF outlook behind Downstream Consumption does not report Canada, whose category is imputed in all eight cells it enters, leaving six of Canada's nine scored Downstream security cells unrankable.

Mirror trade. Economies that do not report to UN Comtrade enter the trade matrix only through their partners' declarations: 65 of 208 importers in 2024, among them two coalition members (Gabon and Viet Nam). Electricity is excluded from the grid basket, and Hong Kong's re-exports from the export measures.

A knife-edge λ. The calibrated value 0.78 meets the target by a small margin (0.77 misses it by 0.0002), the median designed concentration is not monotone in λ, and the pick moved with every input vintage. The design itself is stable: at 0.77, 21 of the 22 producer sets are unchanged.

Policy basis. The Policy index of 150 economies, 11 of them coalition members, rests on IEA PAMS alone. For the 65 economies with both components, ranking their cells on IEA PAMS alone rather than on the composite gives a median within-country Spearman correlation of only 0.48.

Thin and construction-driven picks. The pick is within 0.005 of its runner-up for 10 of the 57 members and within 0.01 for 21. The Nuclear Midstream picks of Costa Rica, the Dominican Republic, Guatemala and Namibia are not an Energy Security floor effect. Each may pick only four sectors, the Midstream stages of nuclear, oil, gas and coal (Namibia also Nuclear Upstream); Midstream nuclear ES is compressed around its median; and the technology weight (0.999 for nuclear against 0.42 for gas, 0.02 for coal and 0 for oil) decides the pick. The picks reflect a narrow choice set, not an opportunity in nuclear fuel.

Rank-compressed EU demand. The EU node holds 24.6 percent of the nodes' GDP but a median demand weight of 0.035, about that of the United States: network demand is a ranking, not a volume.

Coalition boundary. Taiwan (no UN vote, no listed agreement, at most 1.5 percent of any world stage) and Switzerland (whose only supplier route rested on electricity exports) are outside the coalition.

Layers not yet measured. Of the working paper's five analytical layers, the paper measures node severity only as country-relative exposure (concentration and dependence, ranked within each node), capability through EO, and partner complementarity through the bilateral indices and the network. It does not measure the fourth and fifth, bankability and robustness under stress (recovery time), nor common-mode risk beyond the ownership of foreign-invested plants or the dynamic option value of latent capacity.

13.3 Next steps and recommendations

The working paper orders the decision in three steps, severity, coordination and specialisation (New Energy Industrial Strategy Center, 2026). This paper offers a first, score-based version of the last step and partial measures for the first two. Table 24 lists ten extensions drawn from the working paper's implementation questions, analytical upgrades and research agenda, with the data and model change each requires. Five deserve emphasis.

First, a *complementarity matrix*. The network can already price a node's marginal contribution: re-solving the design without each node, or with each candidate added, measures what it adds to the network objective. A full solve of the 22 stages takes about 13 minutes at the headline setting, so

the matrix is affordable, and it would replace the stand-alone ranking of partners with the gap-filling value the working paper asks for.

Second, *common-mode risk and resilience floors*. Two suppliers in two countries do not diversify supply if they share a parent, a precursor or an equipment maker (New Energy Industrial Strategy Center, 2026; Farrell and Newman, 2019). The Chinese-parent component of Friendshore covers the first channel in four Midstream stages; the network should also cap, by constraint, the designed share resting on any one parent headquarters or precursor source. The redundancy rule of the design (at least two suppliers per consumer) is a first resilience floor. Floors tied to recovery time need volumes, capacities and inventories in place of unit demand, so that removing the largest producer for 6, 12 or 24 months yields unmet demand rather than a change in score. The two other upgrades need new inputs. *Dynamic option value*, the time and cost of ramping or requalifying capacity, calls for a multi-period design with build lead times, which the CM-FDI announcement and completion years could calibrate. *Bankability* needs the cost of capital, permitting times, offtake and delivery capacity by country and stage (Spokas et al., 2025), entered as a node-score component or a feasibility constraint.

Third, an *AIB Network Optimization Pilot* for the three chains the working paper proposes: rare earths and magnets, batteries and anodes, and solar wafers. Our baskets already isolate the magnet rare earths (Wind and Electric Vehicles Upstream), permanent magnets (Wind and Electric Vehicles Midstream), anode materials (Batteries Midstream) and polysilicon and wafers (Solar Midstream), but each chain must be split into sub-stages (separation, alloy, magnet; graphite, spherical graphite, anode; polysilicon, ingot and wafer, cell) with plant-level capacity and qualification times. The output would be a portfolio per chain: capacity by node, locations, minimum redundancy, priority projects and financing gaps.

Fourth, *differentiated Gates and Fences and sectoral compacts*. Combined with severity data on consequence and substitutability, the stage-level evidence assembled here (concentration, China's export share, Chinese-parent exposure, alignment) can sort stages into high, medium and low fences. The designed flows identify the partner sets for compacts with pooled demand and finance, and the build candidates (led by India, Indonesia and Viet Nam) show where development finance would back a designated role.

Fifth, a *system scorecard with off-ramps*. A recurring rerun already produces several of the working paper's system metrics: supplier counts, designed and observed concentration, the largest share and the trusted share of world supply. With inventory coverage, recovery time and the price premium added, it could trigger reviews automatically, for instance when the nodes' observed concentration in a stage rises above the calibration target, and retire support when it falls.

The hypotheses can then be tested. Forcing each node to supply its own demand gives the self-sufficiency counterfactual for the first, once costs enter. CM-FDI project histories can be set against demand instruments and supply subsidies for the second. Admitting China and other non-members as nodes, fenced stage by stage rather than excluded, would test the third, and varying the redundancy rule would show when another allied supplier stops reducing risk. The study of

vertical complementarities adds its own agenda (last row of the table).

Table 24: Next steps: extensions of the indices and the network design

Extension	Data needed	Model change
Complementarity matrix	None beyond the present inputs	Leave-one-node-out and add-one-candidate re-solves; marginal (or Shapley) value of each node in each stage
Common-mode risk	Plant ownership beyond foreign investment; precursor and equipment sourcing by plant	Constraint on the designed share sharing a parent headquarters or precursor source
Dynamic option value	Project lead times (CM-FDI announcement and completion years), pipeline capacity, technology readiness of substitutes	Multi-period design with capacity expansion and build lead times
Bankability and feasibility	Cost of capital, permitting times, offtake, insurance, delivery capacity (Spokas et al., 2025)	Bankability as a node-score component or a feasibility constraint
Resilience floors and recovery time	Demand volumes for 2030 and 2035, plant capacities, inventories	Volumetric flows; stress tests removing the largest producer for 6, 12 or 24 months
Network Optimization Pilot	Sub-stage HS6 codes, plant-level capacity, qualification times for magnets, anodes and wafers	Sub-stage nodes for three chains; portfolio by chain
Gates and Fences by node	Consequence and substitutability by stage; Chinese-parent exposure beyond four stages	Stage-specific access rules for non-members admitted as fenced nodes
Sectoral compacts	Offtake and procurement commitments; development and export finance	Pooled demand by compact; finance assigned to designed roles
System scorecard and off-ramps	Inventory coverage, price premium, capital mobilised	Annual rerun with review triggers on concentration and supplier counts
Vertical complementarities	Plant-level co-location; input–output coefficients; 2023 graphite and gallium controls	Co-location term in the network; tested before adoption

Notes: Extensions drawn from the implementation questions, analytical upgrades and research agenda of the NEIS working paper on the Allied Industrial Base (New Energy Industrial Strategy Center, 2026), matched to the data and model of this paper. "Re-solves" use the production configuration of Section 11 (37 nodes, $\lambda = 0.78$, portfolio caps).

13.4 Conclusion

This paper has built indices of energy security and economic opportunity for 219 economies, eleven technologies and three supply-chain stages; derived a coalition of 57 members from UN voting and supply shares; rated every ordered pair of economies as suppliers and markets; and designed an allied division of labour across 22 Upstream and Midstream stages. The results support the working paper's premise in a limited sense: no member is both secure and competitive across the chains, and a design that penalises concentration assigns roles very differently from a ranking of current exporters. Whether members' capabilities are complementary in the working paper's sense, and whether such a design is cheaper, more bankable or faster to recover, needs the complementarity matrix, costs, volumes and time. The indices and the network are best read as the analytical backbone of a recurring review, with the explorer showing how its answers change when the weights change.

References

Acemoglu, Daron, Vasco M. Carvalho, Asuman Ozdaglar, and Alireza Tahbaz-Salehi, "The Network Origins of Aggregate Fluctuations," *Econometrica*, 2012, *80* (5), 1977–2016.

Aiyar, Shekhar, Jiaqian Chen, Christian Ebeke, Roberto Garcia-Saltos, Tryggvi Gudmundsson, Anna Ilyina, Alvar Kangur, Tansaya Kunaratskul, Sergio Rodriguez, Michele Ruta, Tatjana Schulze, Gabriel Soderberg, and Juan P. Trevino, "Geoeconomic Fragmentation and the Future of Multilateralism," IMF Staff Discussion Note SDN/2023/001, International Monetary Fund, Washington, DC 2023.

Albertone, B., A. Lebdioui, and S. Y. Tham, "Regional Coordination of Green Industrial Policies: What Can We Learn from the ASEAN Experience?," TIDE Centre Working Paper 93, TIDE Centre, University of Oxford 2025.

Alfaro, Laura and Davin Chor, "Global Supply Chains: The Looming "Great Reallocation"," NBER Working Paper 31661, National Bureau of Economic Research 2023. Prepared for the 2023 Jackson Hole Economic Policy Symposium.

Ang, Beng Wah, W. L. Choong, and Tsan Sheng Ng, "Energy Security: Definitions, Dimensions and Indexes," *Renewable and Sustainable Energy Reviews*, 2015, *42*, 1077–1093.

Bailey, Michael A., Anton Strezhnev, and Erik Voeten, "Estimating Dynamic State Preferences from United Nations Voting Data," *Journal of Conflict Resolution*, 2017, *61* (2), 430–456.

Balassa, Bela, "Trade Liberalisation and "Revealed" Comparative Advantage," *The Manchester School*, 1965, *33* (2), 99–123.

Baldwin, Richard and Rebecca Freeman, "Risks and Global Supply Chains: What We Know and What We Need to Know," *Annual Review of Economics*, 2022, *14*, 153–180.

Bandara, P. and Net Zero Industrial Policy Lab, "Cleantech Manufacturing FDI Dataset (CM-FDI), Edition 2026.10.05," Johns Hopkins University 2026. Project-level data, 1,042 projects; downloaded 11 October 2026; licence CC BY 4.0.

Baqaee, David Rezza and Emmanuel Farhi, "The Macroeconomic Impact of Microeconomic Shocks: Beyond Hulten's Theorem," *Econometrica*, 2019, *87* (4), 1155–1203.

Becker, William, Michaela Saisana, Paolo Paruolo, and Ine Vandecasteele, "Weights and Importance in Composite Indicators: Closing the Gap," *Ecological Indicators*, 2017, *80*, 12–22.

Bistline, John E. T., Neil R. Mehrotra, and Catherine Wolfram, "Economic Implications of the Climate Provisions of the Inflation Reduction Act," *Brookings Papers on Economic Activity*, 2023, *2023* (1), 77–182.

BloombergNEF, "New Energy Outlook 2024," Technical Report, BloombergNEF, London 2024. Published May 2024; Economic Transition Scenario used. Proprietary data.

Bown, Chad P., "Industrial Policy for Electric Vehicle Supply Chains and the US-EU Fight over the Inflation Reduction Act," PIIE Working Paper 23-1, Peterson Institute for International Economics, Washington, DC 2023.

Campbell, Kurt M. and Rush Doshi, "Underestimating China: Why America Needs a New Strategy of Allied Scale to Offset Beijing's Enduring Advantages," *Foreign Affairs*, 2025. May/June 2025.

Carey, L., S. Ladislaw, B. Bartle, J. Ekberg, S. Mohanty, and E. Brooks, "Secure, Competitive, Global: The Playbook for United States Energy Statecraft in a New Technological Era," Report, RMI 2025.

Carvalho, Vasco M. and Alireza Tahbaz-Salehi, "Production Networks: A Primer," *Annual Review of Economics*, 2019, *11*, 635–663.

_ **, Makoto Nirei, Yukiko U. Saito, and Alireza Tahbaz-Salehi**, "Supply Chain Disruptions: Evidence from the Great East Japan Earthquake," *The Quarterly Journal of Economics*, 2021, *136* (2), 1255–1321.

Cass, Oren, "A Grand Strategy of Reciprocity: How to Build an Economic and Security Order That Works for America," *Foreign Affairs*, 2025. November/December 2025.

Cherp, Aleh and Jessica Jewell, "The Three Perspectives on Energy Security: Intellectual History, Disciplinary Roots and the Potential for Integration," *Current Opinion in Environmental Sustainability*, 2011, *3* (4), 202–212.

_ **and** _ , "The Concept of Energy Security: Beyond the Four As," *Energy Policy*, 2014, *75*, 415–421.

Clausing, Kimberly A. and Catherine Wolfram, "Carbon Border Adjustments, Climate Clubs, and Subsidy Races When Climate Policies Vary," *Journal of Economic Perspectives*, 2023, *37* (3), 137–162.

Climate Action Tracker, "Country Ratings," Climate Analytics and NewClimate Institute 2026. Accessed 2026; methodology at `https://climateactiontracker.org/methodology/cat-rating-methodology/`.

Conover, W. J. and Ronald L. Iman, "Rank Transformations as a Bridge between Parametric and Nonparametric Statistics," *The American Statistician*, 1981, *35* (3), 124–129.

Correlates of War Project, "State System Membership List, v2024," Correlates of War Project 2024. Accessed 7 October 2026.

Csárdi, Gábor and Tamás Nepusz, "The igraph Software Package for Complex Network Research," *InterJournal, Complex Systems*, 2006, p. 1695.

Damgaard, Jannick, Thomas Elkjaer, and Niels Johannesen, "What Is Real and What Is Not in the Global FDI Network?," *Journal of International Money and Finance*, 2024, *140*, 102971.

Daskin, Mark S., *Network and Discrete Location: Models, Algorithms, and Applications*, 2nd ed., Hoboken, NJ: John Wiley & Sons, 2013.

Decancq, Koen and María Ana Lugo, "Weights in Multidimensional Indices of Wellbeing: An Overview," *Econometric Reviews*, 2013, *32* (1), 7–34.

Elliott, Matthew, Benjamin Golub, and Matthew V. Leduc, "Supply Network Formation and Fragility," *American Economic Review*, 2022, *112* (8), 2701–2747.

Energy Institute, "Statistical Review of World Energy 2026," 75th edition, Energy Institute, in collaboration with Ember, Kearney and KPMG, London 2026.

European Commission, "Industrial Policy for the 21st Century: Lessons from the Past," European Economy Discussion Paper 157, European Commission 2022. Annex IV: Airbus.

European Union, "Treaty Establishing the European Coal and Steel Community," Signed in Paris, 18 April 1951 1951. With historical materials on the Schuman Declaration and the ECSC.

Evenett, Simon, Adam Jakubik, Fernando Martín, and Michele Ruta, "The Return of Industrial Policy in Data," IMF Working Paper WP/24/1, International Monetary Fund, Washington, DC 2024. Journal version: *The World Economy* 47(7): 2762–2788, doi:10.1111/twec.13608.

Farrell, Henry and Abraham L. Newman, "Weaponized Interdependence: How Global Economic Networks Shape State Coercion," *International Security*, 2019, *44* (1), 42–79.

Freund, Caroline, Aaditya Mattoo, Alen Mulabdic, and Michele Ruta, "Is US Trade Policy Reshaping Global Supply Chains?," *Journal of International Economics*, 2024, *152*, 104011.

Geoffrion, A. M. and G. W. Graves, "Multicommodity Distribution System Design by Benders Decomposition," *Management Science*, 1974, *20* (5), 822–844.

Global Trade Alert, "New Industrial Policy Observatory (NIPO) Database," St. Gallen Endowment for Prosperity through Trade 2026. July 2026 release used.

Góes, Carlos and Eddy Bekkers, "The Impact of Geopolitical Conflicts on Trade, Growth, and Innovation," WTO Staff Working Paper ERSD-2022-09, World Trade Organization, Geneva 2022.

Gonzalez, Richard and George Wu, "On the Shape of the Probability Weighting Function," *Cognitive Psychology*, 1999, *38* (1), 129–166.

Gopinath, Gita, Pierre-Olivier Gourinchas, Andrea F. Presbitero, and Petia Topalova, "Changing Global Linkages: A New Cold War?," *Journal of International Economics*, 2025, *153*, 104042.

Graedel, T. E., Rachel Barr, Chelsea Chandler, Thomas Chase, Joanne Choi, Lee Christoffersen, Elizabeth Friedlander, Claire Henly, Christine Jun, Nedal T. Nassar, Daniel Schechner, Simon Warren, Man yu Yang, and Charles Zhu, "Methodology of Metal Criticality Determination," *Environmental Science & Technology*, 2012, *46* (2), 1063–1070.

Grossman, Gene M., Elhanan Helpman, and Hugo Lhuillier, "Supply Chain Resilience: Should Policy Promote International Diversification or Reshoring?," *Journal of Political Economy*, 2023, *131* (12), 3462–3496.

Growth Lab at Harvard University, "Greenplexity: Green Growth Opportunities Tool," Interactive web tool 2024. Launched 20 November 2024; announcement at `https://growthlab.hks.harvard.edu/?p=9643`.

Hausmann, Ricardo and Bailey Klinger, "Structural Transformation and Patterns of Comparative Advantage in the Product Space," CID Working Paper 128, Center for International Development, Harvard University, Cambridge, MA 2006.

_ , César A. Hidalgo, Sebastián Bustos, Michele Coscia, Alexander Simoes, and Muhammed A. Yildirim, *The Atlas of Economic Complexity: Mapping Paths to Prosperity*, Cambridge, MA: MIT Press, 2014.

_ , Daniel P. Stock, and Muhammed A. Yıldırım, "Implied Comparative Advantage," *Research Policy*, 2022, *51* (8), 104143.

_ , Jason Hwang, and Dani Rodrik, "What You Export Matters," *Journal of Economic Growth*, 2007, *12* (1), 1–25.

Hidalgo, César A., "Economic Complexity Theory and Applications," *Nature Reviews Physics*, 2021, *3* (2), 92–113.

_ and Ricardo Hausmann, "The Building Blocks of Economic Complexity," *Proceedings of the National Academy of Sciences*, 2009, *106* (26), 10570–10575.

_ , Bailey Klinger, Albert-László Barabási, and Ricardo Hausmann, "The Product Space Conditions the Development of Nations," *Science*, 2007, *317* (5837), 482–487.

Hirschman, Albert O., *National Power and the Structure of Foreign Trade*, Berkeley, CA: University of California Press, 1945.

International Energy Agency, "The Role of Critical Minerals in Clean Energy Transitions," World Energy Outlook Special Report, IEA, Paris 2021.

_ , "Energy Technology Perspectives 2024," Technical Report, IEA, Paris 2024.

_ , "Global Critical Minerals Outlook 2025," Technical Report, IEA, Paris 2025.

_ , "World Energy Outlook 2025," Technical Report, IEA, Paris 2025. Annex A free dataset (World) used.

_ , "Critical Minerals Dataset," IEA data product 2026. 2026 edition (file `IEA Critical Minerals Dataset 2026.xlsx`).

_ , *Energy Technology Perspectives 2026, Part B: Clean Energy Technology Manufacturing and Trade*, Paris: IEA, 2026.

_ , "ETP Clean Energy Technology Guide," Interactive database of technology designs and components with technology readiness levels (TRL) 2026. Continuously updated; accessed September 2026.

_ , "Global EV Outlook 2026," Technical Report, IEA, Paris 2026.

_ , "Policies and Measures Database," IEA policy database 2026. Accessed 2026.

International Monetary Fund, "Industrial Policy in Europe: A Single Market Perspective," IMF Working Paper WP/2024/249, International Monetary Fund 2024.

_ , "Direct Investment Positions by Counterpart Economy (DIP, formerly CDIS)," IMF Data portal 2026. Outward positions by instrument and counterpart economy; accessed 2026.

_ , "Primary Commodity Price System (PCPS)," IMF Data portal 2026. Monthly/annual commodity prices; accessed 2026.

Javorcik, Beata, Lucas Kitzmüller, Helena Schweiger, and Muhammed A. Yıldırım, "Economic Costs of Friendshoring," *The World Economy*, 2024, *47* (7), 2871–2908.

Jewell, Jessica, "The IEA Model of Short-Term Energy Security (MOSES): Primary Energy Sources and Secondary Fuels," IEA Energy Papers 2011/17, OECD Publishing / International Energy Agency, Paris 2011.

Juhász, Réka, Nathan Lane, and Dani Rodrik, "The New Economics of Industrial Policy," *Annual Review of Economics*, 2024, *16*, 213–242.

_ , _ , Emily Oehlsen, and Verónica C. Pérez, "The Who, What, When, and How of Industrial Policy: A Text-Based Approach," Working Paper, SocArXiv 2022. Revised as "Measuring Industrial Policy: A Text-Based Approach," NBER Working Paper 33895 (2025), `https://www.nber.org/papers/w33895`.

Karmarkar, Uday S., "Subjectively Weighted Utility: A Descriptive Extension of the Expected Utility Model," *Organizational Behavior and Human Performance*, 1978, *21* (1), 61–72.

Kowalski, Przemyslaw and Clarisse Legendre, "Raw Materials Critical for the Green Transition: Production, International Trade and Export Restrictions," OECD Trade Policy Papers 269, OECD Publishing, Paris 2023.

Kruyt, Bert, Detlef P. van Vuuren, H. J. M. de Vries, and Heleen Groenenberg, "Indicators for Energy Security," *Energy Policy*, 2009, *37* (6), 2166–2181.

Ladislaw, S., "The Evolution of Green Industrial Policy," Paper, New Energy Industrial Strategy Center October 2025.

Makhorin, Andrew, "GLPK (GNU Linear Programming Kit)," Free Software Foundation 2020.

Markowitz, Harry, "Portfolio Selection," *The Journal of Finance*, 1952, *7* (1), 77–91.

Mealy, Penny and Alexander Teytelboym, "Economic Complexity and the Green Economy," *Research Policy*, 2022, *51* (8), 103948.

Melo, M. T., S. Nickel, and F. Saldanha da Gama, "Facility Location and Supply Chain Management – A Review," *European Journal of Operational Research*, 2009, *196* (2), 401–412.

Moomaw, William, Peter Burgherr, Garvin Heath, Manfred Lenzen, John Nyboer, and Aviel Verbruggen, "Annex II: Methodology," in O. Edenhofer et al., eds., *IPCC Special Report on Renewable Energy Sources and Climate Change Mitigation*, Cambridge and New York: Cambridge University Press, 2011.

Nardo, Michela, Michaela Saisana, Andrea Saltelli, Stefano Tarantola, Anders Hoffman, and Enrico Giovannini, *Handbook on Constructing Composite Indicators: Methodology and User Guide*, Paris: OECD Publishing, 2008. Joint OECD and European Commission Joint Research Centre publication.

Nassar, Nedal T., Jamie Brainard, Andrew Gulley, Ross Manley, Grecia Matos, Graham Lederer, Laurence R. Bird, David Pineault, Elisa Alonso, Joseph Gambogi, and Steven M. Fortier, "Evaluating the Mineral Commodity Supply Risk of the U.S. Manufacturing Sector," *Science Advances*, 2020, *6* (8), eaay8647.

New Energy Industrial Strategy Center, "Allied Industrial Base: Conceptual Overview, Literature Review and Analytical Design," Working Paper, New Energy Industrial Strategy (NEIS) Center October 2026.

OECD, *Economic Security in a Changing World*, Paris: OECD Publishing, 2025.

_ , *OECD Supply Chain Resilience Review*, Paris: OECD Publishing, 2025.

_ , "Creditor Reporting System (CRS)," OECD Data Explorer 2026. Aid disbursements by donor, recipient and sector; accessed 2026.

OECD Nuclear Energy Agency and International Atomic Energy Agency, *Uranium 2026: Resources, Production and Demand* number 7770. In 'NEA.', Paris: OECD Publishing, 2026.

Owen, Susan Hesse and Mark S. Daskin, "Strategic Facility Location: A Review," *European Journal of Operational Research*, 1998, *111* (3), 423–447.

Paruolo, Paolo, Michaela Saisana, and Andrea Saltelli, "Ratings and Rankings: Voodoo or Science?," *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, 2013, *176* (3), 609–634.

Ravallion, Martin, "Mashup Indices of Development," *The World Bank Research Observer*, 2012, *27* (1), 1–32.

Rhodium Group and MIT Center for Energy and Environmental Policy Research, "The Clean Investment Monitor," Quarterly database of U.S. clean-energy investment 2026. Accessed 2026.

Rodrik, Dani, "Industrial Policy for the Twenty-First Century," Working Paper, John F. Kennedy School of Government, Harvard University 2004.

_ , "Green Industrial Policy," *Oxford Review of Economic Policy*, 2014, *30* (3), 469–491.

Saisana, Michaela, Andrea Saltelli, and Stefano Tarantola, "Uncertainty and Sensitivity Analysis Techniques as Tools for the Quality Assessment of Composite Indicators," *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, 2005, *168* (2), 307–323.

Saltelli, Andrea, "Composite Indicators between Analysis and Advocacy," *Social Indicators Research*, 2007, *81* (1), 65–77.

_ **, Marco Ratto, Terry Andres, Francesca Campolongo, Jessica Cariboni, Debora Gatelli, Michaela Saisana, and Stefano Tarantola**, *Global Sensitivity Analysis: The Primer*, Chichester: John Wiley & Sons, 2008.

Schlömer, Steffen, Thomas Bruckner, Lew Fulton, Edgar Hertwich, Alan McKinnon, Daniel Perczyk, Joyashree Roy, Roberto Schaeffer, Ralph Sims, Pete Smith, and Ryan Wiser, "Annex III: Technology-Specific Cost and Performance Parameters," in O. Edenhofer et al., eds., *Climate Change 2014: Mitigation of Climate Change. Contribution of Working Group III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*, Cambridge and New York: Cambridge University Press, 2014, pp. 1329–1356.

Schumacher, Dirk, "ompr: Model and Solve Mixed Integer Linear Programs," R package version 1.0.4, CRAN 2023.

Signorino, Curtis S. and Jeffrey M. Ritter, "Tau-b or Not Tau-b: Measuring the Similarity of Foreign Policy Positions," *International Studies Quarterly*, 1999, *43* (1), 115–144.

Snyder, Lawrence V., Zümbül Atan, Peng Peng, Ying Rong, Amanda J. Schmitt, and Burcu Sinsoysal, "OR/MS Models for Supply Chain Disruptions: A Review," *IIE Transactions*, 2016, *48* (2), 89–109.

Sovacool, Benjamin K. and Ishani Mukherjee, "Conceptualizing and Measuring Energy Security: A Synthesized Approach," *Energy*, 2011, *36* (8), 5343–5355.

Spokas, K., D. Ulama, C. Greig, K. Waltzer, and S. Hobart, "Systemic Bankability Is the Key to Unlocking Energy Transition Speed and Scale," Report, Clean Air Task Force 2025.

Thakur-Weigold, B. and S. Miroudot, "Promoting Resilience and Preparedness in Supply Chains," OECD Trade Policy Papers 286, OECD Publishing, Paris 2024.

The Growth Lab at Harvard University, "International Trade Data (HS, 92)," Harvard Dataverse 2019. Continuously versioned (V18 as of April 2026); underlies the Atlas of Economic Complexity.

Tomlin, Brian, "On the Value of Mitigation and Contingency Strategies for Managing Supply Chain Disruption Risks," *Management Science*, 2006, *52* (5), 639–657.

United Nations Statistics Division, "UN Comtrade Database," Online database 2026. HS6 bilateral trade; accessed 2026.

U.S. Department of State, "Joint Memorandum of the President and the Prime Minister Creating a Combined Raw Materials Board," in "Foreign Relations of the United States, Conferences at Washington, 1941–1942, and Casablanca, 1943," Washington, DC: U.S. Government Printing Office, 1942.

U.S. Energy Information Administration, "International Energy Statistics," Bulk data file INTL.zip, U.S. Energy Information Administration, Washington, DC 2026. Fossil-fuel production and consumption by country; file dated 3 October 2026, accessed 11 October 2026; public domain.

U.S. Geological Survey, "Mineral Commodity Summaries 2026," Technical Report, U.S. Geological Survey, Reston, VA 2026. Data release "Commodity Salient U.S. and World Statistics", doi:10.5066/P1WKQ63T, with the version 1.3 erratum for Brazil's rare-earth reserves.

U.S. War Production Board, "War Production in 1944: Report of the Chairman of the War Production Board," Report, U.S. War Production Board, Washington, DC 1944.

Voeten, Erik, "United Nations General Assembly Voting Data," Harvard Dataverse 2009. Continuously versioned; Dataverse creator field lists E. Voeten; data page credits Voeten, Strezhnev and Bailey; current title "United Nations General Assembly Ideal Points" (V39, July 2026).

Winzer, Christian, "Conceptualizing Energy Security," *Energy Policy*, 2012, *46*, 36–48.

World Bank, *Doing Business 2020: Comparing Business Regulation in 190 Economies*, Washington, DC: World Bank, 2020.

_ , "World Development Indicators," World Bank DataBank 2026. Accessed 2026.

World Bank Group, "World Bank Group to Discontinue Doing Business Report," Statement September 2021. 16 September 2021.

Yellen, Janet L., "Remarks by Secretary of the Treasury Janet L. Yellen on Way Forward for the Global Economy," U.S. Department of the Treasury press release JY0714, remarks at the Atlantic Council, Washington, DC April 2022. 13 April 2022.

Appendix A Variable inventory

Tables 25 and 26 list every component of the overall variable that scores each category in `config/index_definition.yml`, for the Energy Security and Economic Opportunity pillars, as built in the final run. A final block in each table lists the configured components that no input theme of the pillar emits for a headline cell, with the reason. A category score is the mean of the listed components present in the cell (equation (15)). Index rows that a theme emits but no overall variable lists (the size and growth sub-indices of the demand composites, the level and momentum parts of the TRL Index, the two levelised-cost points, the sub-indices of the Input Cost Index) are carried for inspection and do not enter the pillars. Variables are named as in the processed theme tables and the variable-contribution tables, so every term of the decomposition of equation (20) can be traced here. Table 28 lists the inputs of the layers built on the pillars: the Policy index, the fixed terms of the Strategic Index, geopolitical alignment, the coalition rule and the partnership indices.

The columns give, for each component: its category, with the emitting theme in brackets; a one-line definition of the raw quantity, followed by the cells in which it is defined; the raw source; the transformation onto $[0, 1]$; the orientation, that is, the direction of the raw quantity that raises the index; and the missing-data rule of the emitting theme (Table 6): "zero", "avg" for the global average, or "none". $\phi_\gamma[x]_g$ is the S-curve of equation (10) with $\gamma = 0.5$ applied within the comparison set g, and ϕ^0_γ its zero-anchored variant (equation (12)); $g = (t, s)$ ranks countries within a technology-stage cell. Stages are abbreviated Up, Mid and Down, and technologies Bat (batteries), EV (electric vehicles), Grid (electric grid), H_2 (green hydrogen), Sol (solar), Wind, Nuc (nuclear), Geo (geothermal), Coal, Gas and Oil. The constructions are motivated in Sections 5 and 6; Appendix A.1 below collects the construction details of the Economic Opportunity components that the main text omits.

Table 25: Component variables of the Energy Security pillar

Variable	Category [theme]	Definition (cells)	Raw source	Transformation (S-curve group or other)	Orientation (index rises with)	Missing rule
`Mineral Supply`	Foreign Dependency [`foreign_dependency`; `critical_minerals_processing`]	Own share of world supply, and world supply concentration, of each of the technology's minerals, base year and 2035; mined at Up, refined at Mid (Up, Mid: Bat, EV, Grid, H_2, Sol, Wind)	IEA Critical Minerals Dataset: supply (base case), demand (Stated Policies)	Per mineral, mean of $\phi^0_\gamma[\sigma^{y_0}]$, $\phi^0_\gamma[\sigma^{y_1}]$, $\phi_\gamma[1-\mathrm{HHI}^{y_0}]$, $\phi_\gamma[1-\mathrm{HHI}^{y_1}]$ over all rows of the stage; 2035-demand-weighted mean over minerals (§5.2)	own share; lower concentration	avg; zero
`Market Share`	Foreign Dependency [`foreign_dependency`]	Share of world manufacturing capacity incl. announced projects (2035), world total incl. the source's "Other" row; China, Viet Nam, India, EU, US (Mid: Bat, Sol, Wind)	IEA manufacturing extract (ETP 2024)	$\phi_\gamma[\mu]$ across the five producers; every EU member carries the EU's regional share	capacity share	avg
`production`	Foreign Dependency [as above]	EV production of the country's region, 2024 (Mid: EV)	IEA EV regional data	ϕ_γ across country rows after copying regional values to members	production	avg
`sales`	Foreign Dependency [as above]	EV sales of the region, 2024 (Mid: EV)	IEA EV regional data	as above	sales	avg
`net_export`	Foreign Dependency [as above]	Net-export ratio of the region, $(P_R - S_R)/S_R$, 2024 (Mid: EV)	IEA EV regional data	as above	net exports	avg
`market_share`	Foreign Dependency [as above]	Region's share of world EV production, 2024 (Mid: EV)	IEA EV regional data	as above	world share	avg
`Production surplus/deficit`	Energy Imports [`import_dependence`]	Balance (production minus consumption, EJ) and self-sufficiency (100× production / consumption), 2025, production and consumption of the same year (Up: Oil, Gas, Coal)	EI Statistical Review; EIA International Energy Statistics for production EI does not report	Mean of $\phi_\gamma[B]_f$ and $\phi_\gamma[\beta]_f$ over the terms defined, within fuel (§5.2)	net production surplus	avg
`Oil Reserves`, `Gas Reserves`, `Coal Reserves`	Reserves [`reserves`]	Proved reserves, end-2020 (Up: Oil, Gas, Coal)	EI Statistical Review	ϕ_γ over reporting countries plus one zero anchor, anchor discarded	reserves	zero

Continued on next page

Table 25 (continued)

Variable	Category [theme]	Definition (cells)	Raw source	Transformation (S-curve group or other)	Orientation (index rises with)	Missing rule
`Batteries`, `Electric Vehicles`, `Electric Grid`, `Green Hydrogen`, `Solar`, `Wind Reserves`	Reserves [`reserves`]	Demand-weighted reserves of the technology's minerals among cobalt, lithium, graphite, rare earths, copper, manganese, nickel, zinc and PGMs, end-2025 (Up: Bat, EV, Grid, H_2, Sol, Wind)	USGS Mineral Commodity Summaries 2026 (v1.3 erratum: Brazil rare earths 11 Mt); IEA base-year demand	Per mineral ϕ_γ with a zero anchor; unlisted and unquantified (NA, W, s) holders = 0; weighted mean; ϕ^0_γ across countries (§5.2)	reserves	zero
`Nuclear Reserves`	Reserves [`reserves`]	Identified recoverable uranium resources (reasonably assured and inferred) below USD 260/kgU, 1 January 2025 (Up: Nuc)	OECD-NEA/IAEA Red Book, *Uranium 2026*, Table 1.2a	ϕ^0_γ across economies; absent = 0	resources	zero
`Overall Solar PV Potential Index`	Reserves [`solar_pv_potential`]	Total and mean photovoltaic output potential (Down: Sol)	Global Solar Atlas	$\phi_\gamma\left[\frac{1}{2}(\phi_\gamma[\mathrm{PV}^{\mathrm{tot}}] + \phi_\gamma[\mathrm{PV}^{\mathrm{mean}}])\right]$ across economies	potential	avg
`Overall Wind Potential Index`	Reserves [`wind_potential`]	Total and mean wind power density; share of area above the density threshold (Down: Wind)	Global Wind Atlas	Mean of the three ϕ_γ across economies	potential	avg
`Overall Geothermal Potential Index`	Reserves [`geothermal_potential`]	LCOE of a 500 MW and of a 5 GW enhanced-geothermal resource; total technical potential (Down: Geo)	Franzmann, Heinrichs and Stolten (2025), staged composite	Mean of three precomputed sub-indices, a missing one counted as 0	potential; lower cost	avg
`HHI`	Trade [`trade_concentration`]	Concentration of world exports of the cell's HS6 basket across reporters, Hong Kong excluded as a re-export hub (Up: Bat, Coal, EV, Gas, Grid, Nuc, Oil, Sol, Wind; Mid: all but Geo, H_2)	UN Comtrade 2024, World rows of the all-reporter pull	$\phi_\gamma[-\mathrm{HHI}]$ across the 18 basket cells of the headline technologies; one value per cell	lower concentration	avg
`market_share`	Trade [as above]	Share of world exports of the basket (same cells)	Atlas HS92 six-digit, newest year	$\phi_\gamma[\mathrm{ms}]_{ts}$ (§5.2)	export share	avg
`deficit_gdp`	Trade [as above]	Net basket imports relative to GDP (same cells)	Atlas HS92 six-digit; WDI GDP, current US dollars	$\phi_\gamma[-d]_{ts}$	smaller deficit	avg

Continued on next page

Table 25 (continued)

Variable	Category [theme]	Definition (cells)	Raw source	Transformation (S-curve group or other)	Orientation (index rises with)	Missing rule
`critmin_market_share`	Minerals Trade [`critical_minerals_trade`]	Share of world exports of each of the technology's minerals (Up: Bat, EV, Grid, H_2, Sol, Wind)	UN Comtrade critical-minerals extracts, 2024 (≥ 50 reporters; non-reporters from partners' imports); IEA base-year demand	ϕ^0_γ within mineral; base-year-demand-weighted mean over minerals with trade data (§5.2)	export share	zero
`critmin_hhi`	Minerals Trade [as above]	Concentration of world exports of each mineral (same cells)	as above	$\phi_\gamma[-\mathrm{HHI}^E]$ across minerals; weighted mean; same for every economy	lower concentration	zero
`Batteries`, `Electric Vehicles`, `Electric Grid`, `Green Hydrogen`, `Solar`, `Wind Production`	Production [`critical_minerals_production`]	Demand-weighted mine production of the technology's minerals among nine EI production minerals, 2025 (Up: Bat, EV, Grid, H_2, Sol, Wind)	EI Statistical Review mineral tables; IEA base-year demand	Per mineral ϕ_γ with a zero anchor, blank or unlisted = 0; weighted mean; ϕ^0_γ across countries (§5.2)	production	zero
`Nuclear Production`	Production [`critical_minerals_production`]	Uranium mine production, latest year (2024) (Up: Nuc)	OECD-NEA/IAEA Red Book, Table 1.18 (2019–2024), completed with observed zeros	ϕ^0_γ across economies	production	zero
`Energy consumption per capita`	Energy Access [`energy_access_consumption`; `energy_consumption`]	Total energy supply per capita, 2025, a missing fuel read as 0 (Down: Coal, Oil, Gas, Sol, Wind, Nuc)	EI Statistical Review	ϕ_γ within fuel	per-capita supply	avg
`Energy consumption per capita growth`	Energy Access [as above]	Relative change in per-capita supply, 2020–2025 (same cells)	EI Statistical Review	ϕ_γ within fuel; infinite growth from a zero base ranks highest	growth	avg
`installed_cap_index`	Consumption [`energy_consumption`]	Installed capacity per capita, 2024; H_2: final hydrogen use per capita (Down: all but EV, Grid)	BNEF New Energy Outlook 2024, Economic Transition Scenario	ϕ_γ within technology across 12 countries and 6 regions; regional value copied to members (§5.2)	installed base	avg

Continued on next page

Table 25 (continued)

Variable	Category [theme]	Definition (cells)	Raw source	Transformation (S-curve group or other)	Orientation (index rises with)	Missing rule
`elec_growth_index`	Consumption [as above]	Relative growth of the same quantity, 2024–2035 (same cells)	as above	as above	growth	avg
`Overall Price Volatility Index`	Energy Prices [`energy_prices`]	Mean annualised volatility of monthly log price changes of the technology's commodities, 1-, 5-, 10- and 20-year windows (Up: all but Geo)	IMF Primary Commodity Price System	$\phi_\gamma[-\bar{v}]$ across technologies; same for every economy (§5.2)	lower volatility	avg
`Annual Investment Index`	Investment [`investment_momentum`]	Actual investment in the latest year, constant 2024 US dollars (Up: Bat, EV; Mid: Bat, EV, Sol, Wind; Down: Bat, EV, Geo, Nuc, Sol, Wind)	Clean Investment Monitor	$\phi^0_\gamma[\ln(1+I)]$ within cell and year (§5.2)	investment	zero
`Investment Momentum Index`	Investment [as above]	$\ln(1+I_{\bar{\tau}}) - \ln(1+I_{\bar{\tau}-3})$ (same cells)	as above	ϕ_γ within cell, among countries investing in either year	investment growth	zero
`Operating Capacity Index`	Investment [as above]	Operating manufacturing capacity by product, 2025 snapshot (Up: Bat, EV; Mid: Bat, EV, Sol, Wind)	Clean Investment Monitor, capacity	ϕ^0_γ of the mean over products of $\phi^0_\gamma[\ln(1+K)]$	capacity	zero
`Pipeline Capacity Index`	Investment [as above]	Capacity under construction, announced, paused or not yet online, by product (same cells)	as above	as above, averaged over the four statuses and the products	capacity	zero
Configured components emitted for no headline cell of the ES pillar						
`Overall Market Share`, `Overall Production`	Foreign Dependency	Emitted only by the EO theme `market_share_manufacturing`.				
`Cobalt`, `Lithium`, `Graphite`, `Rare Earths`, `Copper`, `Manganese`, `Nickel`, `Zinc`, `PGMs` `Reserves` and ... `Production`	Reserves; Production	Per-mineral indices, emitted under the mineral as technology; they enter the pillar only through the demand-weighted technology roll-ups above.				

Continued on next page

Table 25 (continued)

Variable	Category [theme]	Definition (cells)	Raw source	Transformation (S-curve group or other)	Orientation (index rises with)	Missing rule
`Heat Pumps`, `Hydroelectric Power`, `Geothermal Reserves`	Reserves	Not produced: the IEA demand table assigns these technologies no demand for the reserve minerals.				
`Heat Pumps`, `Coal`, `Oil`, `Gas`, `Hydroelectric Power`, `Geothermal Production`	Production	Not produced by any ES theme; fossil output enters ES only through the balance of Energy Imports.				
`size`, `growth_abs`, `Overall Production`	Production	Emitted only by the EO theme `production_depth_momentum`.				
`Energy consumption per capita`, `Energy consumption per capita growth`	Consumption	Emitted under the Energy Access category, where they are scored.				

Table 26: Component variables of the Economic Opportunity pillar

Variable	Category [theme]	Definition (cells)	Raw source	Transformation (S-curve group or other)	Orientation (index rises with)	Missing rule
`market_share`	Trade [`export_feasibility`]	Share of world exports of the cell's HS6 basket (Up: Bat, Coal, EV, Gas, Grid, Nuc, Oil, Sol, Wind; Mid: all but Geo, H_2)	Atlas HS92 six-digit, newest year	$\phi_\gamma[\text{ms}]_{ts}$; identical to the ES component	export share	avg
`rca`	Trade [as above]	Balassa revealed comparative advantage in the basket (same cells)	UN Comtrade 2024: World rows of the all-reporter pull; total exports	$\phi_\gamma[\text{RCA}]_{ts}$, world over reporters in both files; Hong Kong excluded (eq. 31)	specialisation	avg
`export_size`	Trade [as above]	Value of basket exports to the world (same cells)	UN Comtrade 2024, as above	$\phi_\gamma[E]_{ts}$; a reporter without basket exports has $E = 0$; Hong Kong excluded	export value	avg
`feas`	Trade [as above]	One minus the mean Atlas distance to the basket's distinct HS4 headings (same cells)	Atlas HS92 four-digit, newest year with distances	$\phi_\gamma[\text{feas}]_{ts}$ (eq. 32)	proximity to existing capabilities	avg
`HHI`	Trade [as above]	Concentration of world basket exports across reporters (same cells)	UN Comtrade 2024	$\phi_\gamma[-\text{HHI}]$ across the 18 basket cells; identical to the ES component	lower concentration	avg
`size`	Production [`production_depth_momentum`]	Latest output: fossil production, 2025 (Up: Oil, Gas, Coal), uranium mine production, 2024 (Up: Nuc), power generation (Down: Sol, Wind, Nuc, Coal, Gas, Oil); demand-weighted share of world mined supply in the base year (Up: Bat, EV, Grid, H_2, Sol, Wind)	EI Statistical Review (EIA where EI has no row); Red Book; IEA Critical Minerals Dataset (mining rows)	Energy: $\phi_\gamma[P]_{ts}$; minerals: $\phi_\gamma[P^{y_0}]_t$ (eqs. 34, 35)	production	avg
`growth_abs`	Production [as above]	Energy and uranium: absolute change in output over five years; minerals: projected absolute change of the demand-weighted share, base year to 2035 (same cells)	as above	As for `size`	growth	avg
`Overall Production`	Production [as above]	Composite of level and growth (same cells)	as above	Energy: ϕ_γ of the mean of the level, relative-growth and absolute-growth S-curves; minerals: $\phi_\gamma[2\,\text{size} + \text{growth_abs} + h]$	production and growth	avg

Continued on next page

Table 26 (continued)

Variable	Category [theme]	Definition (cells)	Raw source	Transformation (S-curve group or other)	Orientation (index rises with)	Missing rule
`Overall Global Demand`	Technology Demand [`future_demand`]	World total energy supply in 2035 and its change from 2024, EJ; hydrogen: low-emissions production (Up, Mid, Down: Oil, Gas, Coal, Nuc, Sol, Wind, H_2)	IEA World Energy Outlook 2025, Annex A, Stated Policies	$\phi_\gamma\left[\frac{1}{3}(\phi_\gamma[D_{2035}] + 2\,\phi_\gamma[\Delta D])\right]$ across fuels; same for every economy (eq. 36)	demand and growth	avg
`Overall Demand` (BNEF)	Technology Demand [as above]	Primary energy consumption (hydrogen: final) in 2035 and its relative growth from 2024, in 12 modelled economies (same cells)	BNEF New Energy Outlook 2024, Economic Transition Scenario	$\phi_\gamma\left[\frac{1}{3}(\phi_\gamma[D_{2035}] + 2\,\phi_\gamma[\nu])\right]$ within fuel (eq. 37)	demand and growth	avg
`Overall Demand` (EV)	Technology Demand [as above]	Passenger-car EV stock, sales and sales share: 2025 level, 2022–2025 growth, 2035 level, 2025–2035 growth (Down: EV)	IEA Global EV Outlook 2026, Stated Policies	ϕ_γ of the mean of twelve S-curves, each within parameter (eq. 82)	market size and growth	avg
`Overall Addressable Market`	Technology Demand [as above]	US serviceable addressable market, 2020–2050, $bn; non-positive entries missing (26 cells: all stages of Bat, Coal, Gas, Geo, H_2, Nuc, Oil, Sol, Wind except Down Bat)	BCG	ϕ_γ[SAM] across cells; the US value is copied to other economies by the avg rule	market size	avg
`Overall Overcapacity`	Technology Demand [`overcapacity_premium`]	Mean ratio of manufacturing capacity to forecast demand over the supply-chain components (Mid: Bat, Sol, Wind, H_2)	BNEF Energy Transition Supply Chains 2025	$2^{-\rho_t}$, cardinal (eq. 38); copied to the trade theme's economies	less overcapacity	avg
`TRL Index`	Technological Readiness [`technological_readiness`]	Mean over the Guide's items mapped to the cell's technology and stage of $0.7\,b(x) + 0.3\,\psi(\Delta)$; technology-wide value where the stage has no item (10 of 33 cells) (all cells)	IEA ETP Clean Energy Technology Guide	Cardinal bell and capped concave momentum (eqs. 40, 41); copied to the trade theme's economies	readiness nearer TRL 6; faster maturation	avg
`Overall Price Volatility Index`	Energy Prices [`energy_prices`]	As in the ES pillar (Up: all but Geo)	IMF Primary Commodity Price System	$\phi_\gamma[-\bar{v}]$ across technologies; identical to ES	lower volatility	avg

Continued on next page

Table 26 (continued)

Variable	Category [theme]	Definition (cells)	Raw source	Transformation (S-curve group or other)	Orientation (index rises with)	Missing rule
`Annual Investment Index`, `Investment Momentum Index`, `Operating Capacity Index`, `Pipeline Capacity Index`	Investment [`investment_momentum`]	As in the ES pillar (same cells)	Clean Investment Monitor	As in the ES pillar; identical values	investment and capacity	zero
`Energy consumption per capita`, `Energy consumption per capita growth`	Energy Access [`energy_access_consumption`; `energy_consumption`]	As in the ES pillar (Down: Coal, Oil, Gas, Sol, Wind, Nuc)	EI Statistical Review	As in the ES pillar; identical values	per-capita supply and growth	avg
`Overall Market Share`	Foreign Dependency [`market_share_manufacturing`]	Share of world manufacturing capacity incl. announced projects (2035), world total incl. "Other", five producers; the electrolyser column mapped to H_2 (Mid: Bat, Sol, Wind, H_2)	IEA manufacturing extract (ETP 2024)	$\phi_\gamma[\mu]$ across the five producers; every EU member carries the EU's regional share	capacity share	zero
`production`, `sales`, `net_export`, `market_share`	Foreign Dependency [as above]	Regional EV production, sales, net-export ratio and world production share, 2024 (Mid: EV)	IEA EV regional data	ϕ_γ across country rows after copying regional values to members; identical to ES	production, sales, net exports, share	zero
`Overall Production`	Foreign Dependency [as above]	Row mean of the four EV indices (Mid: EV)	IEA EV regional data	Arithmetic mean, not re-ranked	as components	zero
`LCOE Index`	Cost Competitiveness [`lcoe_competitiveness`]	LCOE of a benchmark plant in the release's reference year and in 2050, Mid scenario (Down: Gas, Coal, Sol, Wind, Bat)	BNEF LCOE Data Viewer	$\frac{1}{2}(\phi_\gamma[-L_{yr}] + \phi_\gamma[-L_{2050}])$ within technology across BNEF markets (eq. 42)	lower cost	none
`Input Cost Index`	Cost Competitiveness [`cost_competitiveness`]	Labour and capital costs weighted by their cost shares in the technology and stage (all cells)	ILOSTAT earnings; IMF lending rates and producer prices	$w^L_{ts}(1-\phi_\gamma[\text{wage}]_s) + w^K_{ts}(1-\chi_i)$, $\chi_i = 0.6R_i + 0.4\phi_\gamma[\text{infl}]_i$, R_i winsorised min-max (eqs. 43, 44)	lower wages and cheaper capital	avg

Continued on next page

Table 26 (continued)

Variable	Category [theme]	Definition (cells)	Raw source	Transformation (S-curve group or other)	Orientation (index rises with)	Missing rule
`IEA Cost index`	Cost Competitiveness [as above]	Cost of producing the equipment relative to China (= 100), 2023, six regions (Mid: Bat, Sol, Wind, H_2; Down: H_2)	IEA relative-cost estimates	$1 - \phi_\gamma[z]$ across the six regions; members take their region's value	lower relative cost	avg
`installed_cap_index`, `elec_growth_index`	Consumption [`energy_consumption`]	As in the ES pillar (Down: all but EV, Grid)	BNEF New Energy Outlook 2024	As in the ES pillar; identical values	installed base and growth	avg
Configured components emitted for no headline cell of the EO pillar						
`Cobalt` ... `PGMs Production`; `Solar` ... `Geothermal Production` (22 names)	Production	Emitted only by the ES theme `critical_minerals_production`, which is not an EO input.				
`Mineral Supply`, `Market Share`	Foreign Dependency	Emitted only by the ES themes `foreign_dependency` and `critical_minerals_processing`.				
`Energy consumption per capita`, `Energy consumption per capita growth`	Consumption	Emitted under the Energy Access category, where they are scored.				

A.1 Construction details for the Economic Opportunity pillar

Trade. Feasibility uses the Atlas distances of the newest year for which they are published, and each HS four-digit heading counts once however many of its six-digit products the basket contains. A Comtrade reporter with no exports of the basket has $E_{its} = 0$ and ranks at the bottom on export size; an economy that does not report to Comtrade has no value on export size or RCA. Country-level components are ranked across the World Bank economies within the cell; Atlas codes outside that list are excluded before ranking but kept in the world denominators.

Production: mineral block. In equation (35), $S_{im,y}$ is mined supply of mineral m by economy i in the IEA supply table (International Energy Agency, 2026a), zero where the IEA does not name i for m; W_{m,y_0} is world mined supply in the base year, named producers plus the IEA's "Rest of world"; and $\tilde{w}_{mt}$ are the IEA base-year demand shares of technology t, renormalised over the minerals with supply data. The measure is therefore a demand-weighted share of world mined supply, comparable across minerals, and growth is its absolute change to 2035 in the same units. The concentration term $h_t = \phi_\gamma[-\sum_m \tilde{w}_{mt}\mathrm{HHI}_m]$ is common to every economy of the pair and does not affect the ranking. Economies the IEA names for no mineral take the global-average fill. Solar Upstream is in effect copper, since silicon and silver have no supply rows. The fuel and uranium cells are historical instead (equation (34)): oil, gas and coal from the Statistical Review, filled from the EIA International Energy Statistics where the Review has no row, and uranium mine production from the Red Book over 2019–2024.

Technology demand. The twelve BNEF economies are Australia, Brazil, China, France, Germany, India, Indonesia, Japan, South Korea, the United Kingdom, the United States and Viet Nam; BNEF's hydrogen is all hydrogen, not only the electrolytic hydrogen that the WEO's low-emissions measure is closer to. For downstream electric vehicles, for each parameter $p \in \{\text{EV stock}, \text{EV sales}, \text{EV sales share}\}$ of passenger cars in the economies the Global EV Outlook reports (International Energy Agency, 2026d), with $v_{ip,y}$ its value (historical for 2022 and 2025, Stated Policies projection for 2035), four S-curves are ranked across countries within the parameter: the 2025 level, growth over 2022–2025, the 2035 level and growth over 2025–2035. The composite is

$$\mathrm{EVD}_i = \phi_\gamma\Big[\,\mathrm{mean}_p\,\mathrm{mean}_{\ell\in\{1,\dots,4\}}\,\phi_{\gamma_{p\ell}}(v)\Big], \tag{82}$$

and carries the name of the BNEF composite (`Overall Demand`), which has no electric-vehicle fuel, so the two never meet in one cell. The BCG addressable market covers nine technologies at three stages; its one non-positive entry, downstream batteries, marks a cell BCG does not size and is set to missing. The overcapacity ratio averages the capacity-to-demand ratios of the components of each midstream supply chain (for solar, polysilicon, wafers, cells and modules; for green hydrogen, electrolyser stacks).

Technological readiness. Items of the Clean Energy Technology Guide are mapped to a technology and a stage, or classified as out of scope, by the ordered rules of `config/iea_clean_tech_guide_tech_map.yml`, the first matching rule winning: 253 of the 634 items of the 2026 release map to a headline technology, 9 of them to more than one cell. The stage is the part of the value chain the item changes: mining and mineral extraction, fuel production sites, geothermal subsurface operations and battery recycling are Upstream; refining, fuel processing and fabrication, cells, components and equipment, and hydrogen production Midstream; power plants, complete vehicles and charging, grid operation, storage deployment and hydrogen transport and use Downstream. The Guide files oil and gas together, so their items score both. A range such as "5–6" is read as its midpoint. Items at TRL 9 are placed by the Guide's adoption stage (awaiting adoption 9, building momentum 10, commercial in many markets 11). The latest level is that of 2025, and the start level is the item's first reading among the first three years of the series. A cell's index averages its own stage's items; a stage with no item (10 of the 33 cells) takes the technology-wide value.

Input Cost Index. Upstream wages are those of mining and quarrying together with electricity, gas and water supply, or the economy-wide average where that series is not reported. The lending rate is the first available of the lending rate, euro-area rates on new and then on outstanding loans to non-financial corporations, the money-market rate, the policy-related rate and the deposit rate; mainland China, which the series does not cover, is assigned the People's Bank of China one-year loan prime rate of 3.00 per cent. A zero or negative reading in a lending-type series is a placeholder and is dropped with the averages that include it (New Zealand's lending series carries only zeros after January 2017). Wages come from the ILOSTAT copy staged on 30 September 2026. Wage and lending-rate observations more than ten years older than the newest in their panel are held out of the rankings and, for rates, of the winsorisation. Missing or held-out values are imputed within each leg from the mean of the country's World Bank region and income group, then of its income group, and finally, for wages and lending rates, from the median; on the capital side low-income economies are pooled with lower-middle-income ones, and a country with no producer-price inflation after imputation uses R_i in its place. The cost shares are those of Table 27; capital carries the larger relative weight in all but two cells, most heavily in solar and wind deployment ($0.90/0.97 \approx 0.93$ and $0.85/1.00 = 0.85$), labour the larger weight only in coal mining ($0.40/0.75 \approx 0.53$), and the two are equal in electric-vehicle manufacturing.

Equipment cost and levelised cost. The IEA relative costs are for 2023 and six regions: China, the European Union, India, Korea, other Southeast Asia and the United States. Each EU member takes the EU value and each ASEAN member on the Energy Institute's country list the value for other Southeast Asia; other economies receive the cell mean. The extract's ammonia column is assigned to downstream green hydrogen. The LCOE benchmark plants are combined-cycle gas, coal, fixed-axis photovoltaic, onshore wind and utility-scale batteries (two- and four-hour systems averaged), one asset per technology so that every market is ranked on the same asset; the reference year is 2025 in the 2026 release, and markets outside the World Bank economy list are dropped

Table 27: Labour and capital cost shares in the Input Cost Index

	Upstream		Midstream		Downstream	
Technology	α^L	α^K	α^L	α^K	α^L	α^K
Solar	0.15	0.20	0.15	0.80	0.07	0.90
Wind	0.15	0.25	0.12	0.80	0.15	0.85
Geothermal	0.15	0.68	0.12	0.80	0.22	0.70
Nuclear	0.18	0.50	0.15	0.80	0.18	0.70
Gas	0.18	0.50	0.10	0.60	0.07	0.20
Coal	0.40	0.35	0.15	0.68	0.15	0.30
Oil	0.18	0.55	0.10	0.60	0.15	0.25
Green Hydrogen	0.06	0.30	0.10	0.55	0.08	0.18
Electric Grid	0.20	0.28	0.25	0.68	0.38	0.53
Electric Vehicles	0.20	0.45	0.20	0.20	0.10	0.20
Batteries	0.22	0.45	0.08	0.15	0.08	0.30

Notes: α^L and α^K are the labour and capital shares of costs assigned to each technology and stage in `cost_competitiveness_labor_weights()` and `cost_competitiveness_capital_weights()` (`R/categories/economic opportunity/cost_competitiveness.R`). They need not sum to one, since materials and other inputs make up the remainder; the Input Cost Index uses only the relative weights $w^L_{ts} = \alpha^L_{ts}/(\alpha^L_{ts} + \alpha^K_{ts})$ and $w^K_{ts} = 1 - w^L_{ts}$ of equation (44).

before ranking.

Foreign dependency: electric vehicles. The IEA regional data report five regions that partition the world: China, the European Union, North America, other Asia-Pacific and the rest of the world. Regional values are copied to China, to the EU members, to the economies of the Energy Institute's North America sub-region, to the rest of its Asia-Pacific region, and to every other economy, European economies outside the EU included, as the rest of the world. The net-export ratio $(P_R - S_R)/S_R$ is higher for a net exporter, which is less import-dependent. The row mean `Overall Production` leaves the category score unchanged whenever all four EV indices are observed. Because the regions are coarse, the members of a region are not ranked apart on these variables, and the same holds for the EU members on the 2035 manufacturing-capacity shares.

A.2 Inputs of the policy, alignment, coalition and partnership layers

Table 28 lists the inputs of the layers built on the two pillars. Three features of the final construction deserve note. First, the Policy index has two components, IEA PAMS (required) and the Climate Action Tracker rating (added where it rates the country); the New Industrial Policy Observatory enters only a robustness variant, and a dual-use score built for earlier drafts is excluded because it takes one value for every economy of a pair. Second, the coalition is derived by rule from the U.S.–China alignment axis, GDP and two membership routes (agreements and major-supplier status), each recorded per economy, so every member's route can be traced. Third, Friendshore has

seven components: the conduit rule removes positions with sixteen financial centres from the IMF outward-investment measure, and in four Midstream cells the investment tie and the Chinese-parent exposure come from the CM-FDI project census.

Table 28: Inputs of the Policy index, the Strategic Index, alignment, the coalition rule and the partnership indices

Variable	Definition	Source	Construction and use
Policy index POL_{its} *(weights renormalised over the components present)*			
IEA PAMS component	Policies matched to technology and stage by whole-word rules; strength of a policy = product of weights for its type, status, jurisdiction and technology and stage match; summed per cell	IEA Policies and Measures, export of 22 Jan. 2026	ϕ_γ of the log of summed strength; weight 3; required (215 economies)
CAT component	Country rating $c_i \in [0,1]$ (EU members without a rating take the bloc's) combined with the technology weight g_t: $c_i g_t + (1-c_i)(1-g_t)$	Climate Action Tracker, 39 countries and the EU	Weight 2, where rated (65 economies: 39 own ratings, 26 via the EU)
NIPO component (robustness only)	Strength of industrial-policy interventions in force by jurisdiction, technology and stage	GTA New Industrial Policy Observatory, October 2026 release (in force on 5 Oct. 2026)	Not in the composite; the robustness variant adds it with weight 4 (1,070 rows, 77 jurisdictions, EU members individually)
Dual-use score (excluded)	Judgement of dual-use relevance per technology and stage, no country dimension	Project-authored	Excluded: constant within each pair, so it cannot rank economies
Fixed terms of the Strategic Index (Section 9)			
pol_{its}	Within-country S-curve of POL_{its} over the economy's in-sample sectors	Policy index	Weight 0.2
λ_s	Stage weight: Upstream 0.5, Midstream 0.75, Downstream 0.25	Project choice	Weight 0.1
g_t	Technology climate weight: min-max of minus lifecycle GHG intensity (nuclear 0.999, gas 0.42, coal 0.02, oil 0)	IPCC AR5 WGIII Annex III medians; oil from SRREN Annex II	Weight 0.1; also in the CAT component and the climate penalty of the bilateral indices
Resource base	Technology-level Reserves or Production > 0	Reserves and production themes	Eligibility of an Upstream cell for a member's allied pick; SI values unchanged
Geopolitical alignment (Section 8)			
θ_i	Mean all-votes UNGA ideal point, sessions 74–78 (2019–2023)	Bailey, Strezhnev and Voeten, Dataverse V39	193 economies with own votes
s_i	$(\theta_i - \theta_{\mathrm{CHN}})/(\theta_{\mathrm{USA}} - \theta_{\mathrm{CHN}})$, clamped to $[0,1]$	As above	U.S.–China axis
Bloc, p_i	U.S.-leaning ($s_i \geq 0.511$, $p_i = 1$), China-leaning ($s_i \leq 0.142$, $p_i = 0$), nonaligned ($p_i = 0.5$); quartile cuts	As above; territory map (project-authored)	44 territories inherit their sovereign's p_i; Taiwan, the West Bank and Gaza and Kosovo 0.5
A_{ij}	$p_i p_j + (1-p_i)(1-p_j)$	As above	Friendshore component `geo_alignment`, weight 2
Coalition rule (Section 8)			
GDP	Latest GDP, current US dollars (2025)	WDI `NY.GDP.MKTP.CD`	Floor of USD 100 billion for core members and agreement partners

Continued on next page

Table 28 (continued)

Variable	Definition	Source	Construction and use
Agreement memberships	Members of U.S. free-trade agreements, IPEF, the Minerals Security Partnership, the G7 and NATO; MSP Forum (sensitivity only)	Official lists, accessed 8 Oct. 2026 (project-authored file)	Partner route through agreements
Supply shares	World shares of production, refining, manufacturing capacity or HS6-basket exports in an Upstream or Midstream stage of a headline technology	IEA Critical Minerals Dataset; EI; Red Book; IEA ETP; Atlas	Supplier route, ≥ 5 percent, not China-leaning, no GDP floor
Reserves shares	World shares of reserves of a headline mineral, fuel or uranium	USGS MCS 2026 (end-2025); EI (end-2020); Red Book (<USD 260/kgU)	Supplier route, ≥ 10 percent
EU stage weights	Member shares within the EU node per stage: production if it covers at least half of the EU total, else exports, else GDP	EI; Red Book; Atlas; WDI	Network node `EUU` only
Partnership indices (Section 10)			
Bilateral trade	Reconciled flows $F_{ij,ts}$, importer first, else exporter $\times 1.06$ (equation (8)), 2020 and 2024	UN Comtrade, all-reporter dyads	S-curve of last-year level and growth (2:1) across trading dyads, then within the reporter; FS imports (weight 2), PO exports (weight 3)
Need and capability	Reporter's $1 - \text{ES}$ (FS, weight 2); partner's EO (FS, weight 2); reporter's EO (PO, weight 3); partner's $1 - \text{ES}$ (PO, weight 1)	Pillars	As built
Outbound investment	Outward direct-investment positions in debt instruments, mean of 2020–2024; positions with conduit centres unobserved	IMF Direct Investment Positions	FS, weight 2; replaced by the CM-FDI tie in Batteries, EV, Solar and Wind Midstream
Conduit centres	Netherlands, Luxembourg, Hong Kong, British Virgin Islands, Bermuda, Singapore, Cayman Islands, Switzerland, Ireland, Mauritius; Bahamas, Barbados, Turks and Caicos Islands, Isle of Man, Guernsey, Cyprus	Damgaard et al. (2024) (first ten); offshore centres	Excluded as reporters and counterparts of the IMF measure
CM-FDI investment tie	Attributed count and disclosed value of manufacturing projects of reporter-headquartered parents in the partner (joint ventures 0.5)	CM-FDI, edition of 5 Oct. 2026 (905 of 1,042 projects)	Mean of S-curves of count and value within the cell, then S-curve within the reporter
Chinese-parent exposure E_{jt}	Share of the partner's foreign-invested manufacturing in t owned by China-, Hong Kong- or Macao-headquartered parents; missing below two projects	CM-FDI	FS component scored $1 - E_{jt}$, weight 1
Climate penalty	$(1 - g_t)$ times the CAT component, scaled by a penalty weight	CAT; IPCC	Subtracted from FS and PO

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Table 28 (continued)

Variable	Definition	Source	Construction and use
Development components	WDI development and investment-environment indices, Doing Business score (2019), partner's ES and EO, supply-chain aid (absolute and per GDP), energy use	WDI; World Bank; OECD CRS	Development Potential, non-high-income partners; weights 1, 1, 1, 1, 1.5, 1.5, 2, 1

Appendix B Data sources

Table 29 lists every source the indices read, grouped as in Table 4. For each it gives the dataset and the staged file, the publisher, the coverage of the staged release, the release used by the final run and the themes and categories it feeds. Releases of pattern-resolved inputs are those selected by the rule of equation (OA.1) (Online Appendix OA.1) and recorded in the run's vintage stamp (`data/raw/raw_inputs_resolved.yml`); single staged files carry the access date recorded in the input manifest. Theme names are those of the processed tables (`data/processed/<theme>_tbl.rds`); ES and EO denote the Energy Security and Economic Opportunity pillars, FS, PO and DP the Friendshore, Partnership Opportunity and Development Potential indices. Licensed inputs are marked † (subscription) and ‡ (proprietary). Where a dataset has no bibliographic entry, the publisher's landing page recorded in the manifest is given.

Table 29: Data sources, coverage, releases used and categories fed

Dataset (staged file)	Publisher	Coverage of the staged release	Release used	Themes → categories fed
Energy statistics				
Statistical Review of World Energy, long format (`ei_stat_review_world_energy.csv`)	Energy Institute, with Ember, Kearney and KPMG (Energy Institute, 2026)	90 annual variables, 1965–2025; 142 country and region names (regional totals and the USSR dropped)	2026 (75th) edition; latest year 2025	energy_access → Energy Access (ES, EO); energy_consumption → Consumption, Energy Access (ES, EO); import_dependence → Energy Imports (ES); production_depth_momentum → Production (EO); coalition supplier test; EU node weights
Statistical Review of World Energy, workbook (`ei_stat_review_world_energy_wide.xlsx`)	Energy Institute	Proved reserves of oil, gas and coal (end-2020); mine production of cobalt, lithium, graphite, rare earths, copper, manganese, nickel, zinc and PGMs	2026 edition	reserves → Reserves (ES, fossil fuels); critical_minerals_production → Production (ES); coalition supplier and reserves tests
International Energy Statistics, bulk file (`eia_intl_fossil_production.csv`)	U.S. Energy Information Administration (U.S. Energy Information Administration, 2026)	Fossil-fuel production and consumption by country; oil production to 2025, gas and coal to 2024	INTL.zip of 3 Oct. 2026, accessed 11 Oct. 2026	Oil, gas and coal production where the Statistical Review has no row, as observations: import_dependence (ES), production_depth_momentum (EO)
Reserves and uranium				
Mineral Commodity Summaries 2026, world data (`MCS2026_World_Data.csv`)	U.S. Geological Survey (U.S. Geological Survey, 2026), doi: `10.5066/P1WKQ63T`	Country reserves and mine production of the nine critical minerals; holders without a figure (NA, W, s) counted as zero in roll-ups	MCS 2026, reserves at end-2025; v1.3 erratum (Brazil rare earths 11 Mt) applied	reserves → Reserves (ES, minerals); coalition reserves test

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Table 29 (continued)

Dataset (staged file)	Publisher	Coverage of the staged release	Release used	Themes → categories fed
Uranium Resources, Production and Demand, Table 1.2a (`uranium_resources.csv`)	OECD Nuclear Energy Agency and IAEA (OECD Nuclear Energy Agency and International Atomic Energy Agency, 2026)	Identified recoverable resources by cost category; 59 economies below USD 260/kgU	*Uranium 2026* (NEA No. 7770), resources at 1 Jan. 2025	reserves → Reserves (ES, Nuclear Upstream); coalition reserves test
Uranium Resources, Production and Demand, Table 1.18 (`uranium_production.csv`)	OECD Nuclear Energy Agency and IAEA (OECD Nuclear Energy Agency and International Atomic Energy Agency, 2026)	Mine production by economy, 2019–2024; 18 producers in 2024	*Uranium 2026* for 2022–2024, earlier editions for 2019–2021	critical_minerals_production → Production (ES); production_depth_momentum → Production (EO); coalition supplier test
Trade				
UN Comtrade, all-reporter reporter–partner dyads in the crosswalk's products (`comtrade_energy_dyads_all.csv`)	UN Statistics Division (United Nations Statistics Division, 2026)	173 reporters in either year (168 in 2020; 142 exporting and 143 importing in 2024); 208 economies as reporter or partner; partner S19 mapped to Taiwan	2020 and 2024; main pull 7–8 Oct. 2026, supplements 9 and 12 Oct. 2026	Reconciled flow matrix (importer first, else exporter ×1.06) → FS imports, PO exports, network edges and observed export concentration
UN Comtrade, energy-technology HS6 exports and imports with the world (`comtrade_energy_trade.csv`)	UN Statistics Division (United Nations Statistics Division, 2026)	World rows of the dyad pull: 142 exporting and 143 importing reporters; every current basket code (the 13 absent codes are legacy codes)	2024; derived 9 and 12 Oct. 2026	trade_concentration → Trade (ES); export_feasibility → Trade (EO): export size, Balassa RCA, export concentration (Hong Kong excluded as a re-export hub)
UN Comtrade, total exports (`comtrade_total_export.csv`)	UN Statistics Division (United Nations Statistics Division, 2026)	All-product exports by reporter	2024; pulled 9 Oct. 2026	Denominator of RCA

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Table 29 (continued)

Dataset (staged file)	Publisher	Coverage of the staged release	Release used	Themes → categories fed
Atlas of Economic Complexity, HS92 six-digit country–product–year (`hs92_country_product_year_6.csv`)	Growth Lab at Harvard University (The Growth Lab at Harvard University, 2019; Hausmann et al., 2014)	1995–2023; exporter and importer reports reconciled	Latest year in the panel (2023)	World market share of the basket (ES, EO); net basket imports relative to GDP (ES); coalition supplier test; EU node weights; the vintage build
Atlas of Economic Complexity, HS92 four-digit (`hs92_country_product_year_4.csv`)	Growth Lab at Harvard University (The Growth Lab at Harvard University, 2019)	1995–2023; product-space distance by economy and product	Latest year with distance (2023)	Feasibility → Trade (EO)
Critical minerals				
Critical Minerals Dataset (`IEA Critical Minerals Dataset 2026.xlsx`)	International Energy Agency (International Energy Agency, 2026a)	Supply by economy for copper, cobalt, lithium, nickel, graphite and magnet rare earths (mining, refining), base year and projections; demand by technology, Stated Policies	2026 edition	Mineral weights by technology; foreign_dependency, critical_minerals_processing → Foreign Dependency (ES); production_depth_momentum → Production (EO); coalition supplier test
UN Comtrade, critical-minerals imports, exports and total exports (`critmin_import_2024.csv`, `critmin_export_2024.csv`, `critmin_total_export_2024.csv`) and partner-reported imports from non-reporters (`critmin_mirror_import.csv`)	UN Statistics Division (United Nations Statistics Division, 2026)	130, 128 and 130 reporters; 110 HS codes	2024; the 2025 pulls (3 reporters) fall below the 50-reporter floor	critical_minerals_trade → Minerals Trade (ES)
Manufacturing capacity				

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Table 29 (continued)

Dataset (staged file)	Publisher	Coverage of the staged release	Release used	Themes → categories fed
Clean-technology midstream manufacturing capacity (`iea_cleantech_Midstream.csv`)	International Energy Agency, Energy Technology Perspectives 2024 (International Energy Agency, 2024)	Shares of current and announced capacity for solar PV, wind, batteries and electrolysers; China, Viet Nam, India, EU, United States and "Other"; Europe reported as a region	ETP 2024	foreign_dependency → Foreign Dependency (ES); market_share_manufacturing → Foreign Dependency (EO); coalition supplier test
Electric-vehicle production and sales by region (`ev_Midstream_capacity.csv`)	International Energy Agency (Global EV Data Explorer)	China, EU, North America, other Asia-Pacific, rest of world; 2021–2024	Staged extract	foreign_dependency (ES) and market_share_manufacturing (EO), Electric Vehicles Midstream
Demand and technology				
World Energy Outlook Annex A free dataset, World (`WEO2025_AnnexA_Free_Dataset_World.csv`)	International Energy Agency (International Energy Agency, 2025b)	World totals by fuel and flow; historical and Stated Policies series	WEO 2025	future_demand → Technology Demand (EO)
Global EV Outlook country data (`EV data by country 2026.csv`)	International Energy Agency (International Energy Agency, 2026d)	73 economies and regions, 2010–2035; Stated Policies used	Global EV Outlook 2026	future_demand → Technology Demand (EO)
New Energy Outlook (`2024-10-29 - New Energy Outlook 2024.csv`)†	BloombergNEF (BloombergNEF, 2024)	Regional projections of demand, capacity and generation; Economic Transition Scenario	NEO 2024 (data file of 29 Oct. 2024)	energy_consumption → Consumption (ES, EO); future_demand → Technology Demand (EO)
Energy Transition Supply Chains (`BNEF_Energy Transition Supply Chains 2025.xlsx`)†	BloombergNEF, `https://about.bnef.com/`	Ratio of manufacturing capacity to demand by component; batteries, solar, wind, hydrogen	2025	overcapacity_premium → Technology Demand (EO)
Market Size for Technology and Supply Chain (`Market Size for Technology and Supply Chain.xlsx`)‡	Boston Consulting Group (consultancy extract)	Serviceable addressable market, United States, by technology and stage	Undated extract	future_demand → Technology Demand (EO)

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Table 29 (continued)

Dataset (staged file)	Publisher	Coverage of the staged release	Release used	Themes → categories fed
ETP Clean Energy Technology Guide, public dataset (`IEACleanTechGuide publicdataset.csv`)	International Energy Agency (International Energy Agency, 2026c)	Readiness levels 2020–2025 for 634 designs and components; no country dimension	2026 release	technological_readiness → Technological Readiness (EO), by stage
Costs and prices				
LCOE Data Viewer (`2026-08-11 - LCOE Data.xlsb`)†	BloombergNEF, `https://about.bnef.com/`	LCOE by market and technology; reference year and 2050; Mid scenario	Release of 11 Aug. 2026; reference year 2025	lcoe_competitiveness → Cost Competitiveness (EO)
Relative technology costs (`Relative_Costs_IEA.csv`)	International Energy Agency, Energy Technology Perspectives 2024 (International Energy Agency, 2024)	Equipment costs relative to China for six regions, 2023	2023	cost_competitiveness → Cost Competitiveness (EO)
Average monthly earnings of employees by economic activity (ILOSTAT `EAR_EMTA_SEX_ECO_CUR_NB_A`; `ilo_earnings_by_activity.csv`)	International Labour Organization	Economies, annual	Staged copy downloaded 30 Sep. 2026	cost_competitiveness (labour cost) → Cost Competitiveness (EO)
Lending rates, monetary and financial statistics (`imf_lending_rates.csv`)	International Monetary Fund, `https://data.imf.org/en/Data-Explorer`	Economies, monthly to Oct. 2025; placeholder zero readings dropped	Staged extract	cost_competitiveness (capital cost) → Cost Competitiveness (EO)
Producer price indices (`imf_ppi.csv`)	International Monetary Fund, `https://data.imf.org/en/Data-Explorer`	Economies, 2021 to Sep. 2025	Staged extract	cost_competitiveness (input prices) → Cost Competitiveness (EO)

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Table 29 (continued)

Dataset (staged file)	Publisher	Coverage of the staged release	Release used	Themes → categories fed
Primary Commodity Price System (`imf_commodity_prices.csv`)	International Monetary Fund (International Monetary Fund, 2026b)	World prices, monthly, to Dec. 2025; units from the series metadata	API, staged copy as fallback	energy_prices → Energy Prices (ES, EO)
Resource potential and investment				
Global Solar Atlas, country potential (`solar_potential_clean.csv`)	World Bank Group (ESMAP) with Solargis, `https://globalsolaratlas.info`	241 territories; photovoltaic output potential	No time dimension	solar_pv_potential → Reserves (ES)
Global Wind Atlas, country potential (`wb_wind_country.csv`)	Technical University of Denmark and World Bank Group (ESMAP), `https://globalwindatlas.info`	225 territories; wind power density	No time dimension	wind_potential → Reserves (ES)
Enhanced geothermal technical potential and LCOE (`geothermal_lcoe_mw.csv`)	Franzmann, Heinrichs and Stolten (2025), Mendeley Data, version 2, doi: `10.17632/sbs8k66bwf.2`	175 economies; LCOE for 500 MW and 5 GW resources; technical potential	Version 2	geothermal_potential → Reserves (ES)
Clean Investment Monitor, investment and capacity (`GCIM_Investment_Capacity_aggregated.xlsx`)	Rhodium Group and MIT Center for Energy and Environmental Policy Research (Rhodium Group and MIT Center for Energy and Environmental Policy Research, 2026)	Annual investment 2018–2025 by technology and segment; capacity pipeline	2026 first-quarter release	investment_momentum → Investment (ES, EO)
Policy				

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Table 29 (continued)

Dataset (staged file)	Publisher	Coverage of the staged release	Release used	Themes → categories fed
Policies and Measures database export (`IEA_PAMS_Export.csv`)	International Energy Agency (International Energy Agency, 2026e)	Policies by country, jurisdiction, status, type and sector	Export of 22 Jan. 2026	IEA PAMS component → Policy index (required; 215 economies)
Country ratings (`CAT_country ratings data.csv`)	Climate Action Tracker (Climate Analytics and NewClimate Institute) (Climate Action Tracker, 2026)	40 ratings (39 countries and the EU), transcribed from the country pages	Ratings last updated May 2022–June 2025	CAT component → Policy index where rated (EU members without a rating take the bloc's; 65 economies); climate penalty of FS and PO
Lifecycle greenhouse-gas intensities (`ipcc_ghg_intensity.csv`)	IPCC Fifth Assessment Report, WGIII, Annex III, Table A.III.2 (Schlömer et al., 2014)	12 generating technologies, g CO_2-eq/kWh; oil from the SRREN Annex II	AR5 (2014) medians	Technology weight g_t of the Strategic Index; CAT component; climate penalty of FS and PO
New Industrial Policy Observatory (`GTA NIPO - October 2026.xlsx`)[†]	Global Trade Alert, St. Gallen Endowment for Prosperity through Trade (Global Trade Alert, 2026)	Interventions by implementing jurisdiction, with dates and affected products; in force on 5 Oct. 2026	October 2026 run; 1,070 component rows, 77 jurisdictions	Robustness only: the "NIPO added back" variant of the Policy index (not in the composite)
Bilateral, alignment and coalition				
Direct Investment Positions by counterpart economy (`imf_dip.csv`)	International Monetary Fund, `https://data.imf.org/en/Data-Explorer`	Outward positions in debt instruments, gross assets, by reporter and counterpart	Mean of reported years 2020–2024; 16 conduit centres excluded	FS outbound investment (cells without CM-FDI ties)
Cleantech Manufacturing FDI Dataset (`cm_fdi_projects.csv`)	Net Zero Industrial Policy Lab, Johns Hopkins University (Bandara and Net Zero Industrial Policy Lab, 2026); CC BY 4.0	1,042 cross-border solar, wind, battery and EV manufacturing projects; 905 used	Edition 2026.10.05, downloaded 11 Oct. 2026	FS investment tie and Chinese-parent exposure, Batteries, EV, Solar and Wind Midstream

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Table 29 (continued)

Dataset (staged file)	Publisher	Coverage of the staged release	Release used	Themes → categories fed
UN General Assembly ideal points, all-votes series (`un_ga_ideal_points_all.csv`)	Bailey et al. (2017); Harvard Dataverse (Voeten, 2009)	Sessions 1–78 (1946–2023)	Dataverse V39.0; sessions 74–78 averaged	Alignment axis, blocs, A_{ij} (FS); coalition core and China-leaning screen
UN General Assembly ideal points, final-passage series (`un_ga_ideal_points.csv`)	As above	Sessions 1–80 (1946–2025)	Dataverse V39.0	Sensitivity of the blocs and the coalition only
Economic-security agreement memberships (`economic_security_agreements.csv`)	Project-authored from the official lists (USTR, IPEF, MSP, G7, NATO)	U.S. FTA partners, IPEF, Minerals Security Partnership and MSP Forum, G7, NATO	Accessed 8 Oct. 2026	Coalition partner route (MSP Forum as sensitivity)
Territory and non-voting economy map (`territory_sovereign_map.csv`)	Project-authored	Economies without a UN vote and the alignment each takes	Version of 8 Oct. 2026	Territories inherit their sovereign's bloc; Taiwan, the West Bank and Gaza and Kosovo neutral
World Development Indicators, development panel (`wb_wdi.csv`)	World Bank (World Bank, 2026)	Economies; 2022–2024 values used	API	DP (enabling environment, energy use)
Doing Business indicators, historical (`wb_doingbusiness.csv`)	World Bank (World Bank, 2020); discontinued (World Bank Group, 2021)	190 economies; ease-of-doing-business score, 2019	Doing Business 2020 release	DP (business environment)
Creditor Reporting System, aid flows (`oecd_crs_api.csv`)	OECD, `https://data-explorer.oecd.org/`	ODA disbursements by donor, recipient and sector, 2013–2024	Staged extract; summed over years	DP (aid by stage)
World Development Indicators, GDP in current US dollars, `NY.GDP.MKTP.CD` (`wdi_gdp.csv`)	World Bank (World Bank, 2026)	2007–2025	Pulled at ingestion	Net imports relative to GDP (ES Trade); coalition GDP floor; scaling of investment and aid in the bilateral indices; demand weights and portfolio caps of the network

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Table 29 (continued)

Dataset (staged file)	Publisher	Coverage of the staged release	Release used	Themes → categories fed
WDI country reference table (`wdi_country_info.csv`)	World Bank (World Bank, 2026)	217 economies and 79 aggregate codes; Taiwan added by the pipeline	Pulled at ingestion	Canonical economy names; comparison sets of the trade, cost and LCOE themes (Section 3.4)
State System Membership List	Correlates of War Project (Correlates of War Project, 2024)	195 states in 2024 (UN members, Kosovo, Taiwan)	v2024, accessed 7 Oct. 2026	Sovereignty condition of the rankability rule
Project-authored crosswalks and lists				
Energy HS6 master catalogue (`data/reference/energy_hs6_master.csv`) and its generated views	Project-authored	259 HS6 codes in 415 basket rows (17 legacy bridge codes); Upstream and Midstream; excluded rows kept with their reasons	Repository version after the October 2026 review	Trade baskets of both trade themes, the bilateral trade components and the network (Section 3.2)
Conduit financial centres (`config/partnership.yml`)	Project-authored after Damgaard et al. (2024)	16 economies: 10 pass-through economies and 6 offshore centres	Repository version	Exclusion from the IMF outward-investment measure of FS

Notes to Table 29. Coverage describes the staged release files, before harmonisation; the number of economies scored after harmonisation is reported in Figure 3. Two bookkeeping files complete the manifest: the Comtrade retrieval stamp and the vintage stamp of resolved releases. Inputs that the final indices do not read are not listed: the dual-use scores, which are excluded from the Policy index because they take one value for every economy of a technology–stage pair; the February, June and July 2026 NIPO extracts (the October 2026 run is used for the robustness variant); the earlier direct Comtrade extract of energy-technology trade and the 22-reporter allied dyad pull, both replaced by the all-reporter pull; a USGS Mineral Commodity Summaries 2025 file; and staged releases of pattern-resolved inputs that the rule does not select. The IPCC values are those of the Fifth Assessment Report, except the oil-fired value, which is taken from the Annex II of the Special Report on Renewable Energy Sources.

Appendix C Network design algorithms

This appendix states the algorithms behind the allied network design of Section 11, proves the properties used there, and documents the calibration of the diversification weight. Notation follows Section 11: $\mathcal{V}$ is the set of 37 network nodes, a_p the producer score, d_c the demand weight, W_{pc} the edge weight, δ_p the development potential, $\mathcal{X}_{ts}$ the nodes barred from stage (t, s) by the portfolio caps and $\mathcal{E}_{ts}$ the eligible producers. The functions of `R/indices/allied_network_design.R` read and write no files; all input and output is in `scripts/30_build_allied_network_design.R`. Table 30 lists the production settings.

C.1 Overview and network nodes

Algorithm 1 gives the top level. The stage programs are solved one at a time and independently; the only link between stages is the portfolio loop, which changes the exclusion sets and re-solves the stages it affects.

Algorithm 1 Allied network design (top level)

Require: cell indices EO, ES; directed partner indices FS, PO; development table; GDP; members and their nodes; EU stage weights; 2024 basket trade among members; parameters of Table 30

collapse every input to one record per key; **assert** uniqueness
aggregate the members of the EU node (rules below); every other member is its own node
for each stage, S-curve EO and ES across the nodes; compute a_p, d_c, W_{pc}
$\kappa_i \leftarrow$ eq. (80) for every node; $\mathcal{X}_{ts} \leftarrow \emptyset$
for all stages (t, s) **do**
    $R_{ts} \leftarrow \mathit{SolveStage}(t, s, \mathcal{X}_{ts})$ ▷ Algorithm 2
end for
$(R, \mathcal{X}) \leftarrow \mathit{PortfolioCaps}(R, \mathcal{X}, \kappa)$ ▷ Algorithm 4
split each EU share to the members by stage weight; rank build candidates by eq. (OA.5)
return designs, diagnostics, caps and counts, member shares, build list, input checks

Table 30: Parameters of the allied network design (production configuration)

Symbol	Code argument	Value	Role
Nodes, demand and edges			
–	`network_members`	57 + 6, 37 nodes	Coalition members and EU-only members; the EU-27 as node `EUU`
–	`supply_chains`	Up, Mid	The 22 stages solved
γ	–	0.5	S-curve exponent of node components
$\omega_{\mathrm{EO}}, \omega_{\mathrm{ES}}$	`node_weights`	0.5, 0.2	Producer score, eq. (74)
–	`node_policy`	`FALSE`	Policy term off
–	`demand_mode`	`mixed` (0.5, 0.5)	Need and size, eq. (75)
$\beta_{\mathrm{FS}}, \beta_{\mathrm{PO}}$	`edge_weights`	0.5, 0.5	Edge weight, eq. (76)
Stage program			
w_N, w_E, w_D	`w_node`, `w_edge`, `w_dev`	1, 0.5, 0	Objective weights, eq. (78a)
λ	`diversity_lambda`	0.78	Diversification weight (calibrated)
h	`diversity_breakpoint_step`	0.01	Share grid of the tangent cuts
$\bar{s}, \underline{s}$	`max_share`, `min_share`	0.40, 0.05	Share bounds
m_P, m_S	`min_producers`, `min_suppliers_per_consumer`	3, 2	Producers per stage, suppliers per consumer
ε	`epsilon_supplier_share`	0.10	Smallest share of a consumer's demand on an active link
Solver and portfolio caps			
–	`auto_milp_max_nodes`	60	Largest stage sent to the MILP
T	`milp_stage_time_limit_sec`	300	Time limit per stage solve (s)
K	`portfolio_top_k`	5	Top-K rule, eq. (81)
$\kappa_{\min}, \kappa_{\max}$	`portfolio_min_cap`, `portfolio_max_cap`	2, 10	Caps, eq. (80), log-GDP scale
$I_{\max}$	`portfolio_max_iters`	30	Largest number of cap passes

Notes: Values set in `scripts/30_build_allied_network_design.R` or, where the script passes none, the defaults of `allied_network_design()`. Component weights are renormalised over the components observed; objective weights are used as given.

The EU node. The design is solved over nodes, and the members of the EU node are aggregated before anything is ranked, so that the node is one point in each within-stage S-curve. The member weights $\mu_{m,ts}$ are the stage weights of Section 8 (members' production shares for Coal, Gas and Oil Upstream, their basket-export shares where the stage has a basket, their GDP shares otherwise), renormalised over the members. A producer input of the node (EO, ES, development potential) is the μ-weighted mean over the members that have a value, the weights renormalised over them. Demand uses the members' summed GDP for size, and for need the GDP-weighted mean of the members' raw security, which is then S-curved across nodes like any other node's. An edge component (FS or PO) between the node and a partner is the mean of the member edges, weighted by the members' 2024 basket trade on that edge (the members' imports from the producer when the EU consumes, their exports to the consumer when it produces) or, without trade, by the stage weights; components are averaged before the missing-component zero is applied. Edges between two members are dropped, and the node's self-supply edge follows eq. (77). The portfolio cap uses the summed GDP. The node's designed share is split back to the members in proportion to $\mu_{m,ts}$; these member shares are derived, not designed.

C.2 The stage MILP with the diversification term

The quadratic penalty of eq. (78a) is carried into the mixed-integer linear program by an outer approximation. Let $\mathcal{B} = \{0, h, 2h, \dots, \bar{s}\}$ be the breakpoints of the share grid ($h = 0.01$, 41 points). Each producer gets an auxiliary variable $\tau_p \geq 0$ and the cuts

$$\tau_p \;\geq\; 2b\,x_p - b^2\,y_p \qquad \forall b \in \mathcal{B}, \tag{83}$$

and the objective subtracts $\lambda \sum_p \tau_p$ instead of $\lambda \sum_p x_p^2$. With $y_p = 1$ the cuts are the tangents of x^2 at the breakpoints; with $y_p = 0$, so that $x_p = 0$, they reduce to $\tau_p \geq 0$. Because the program maximises, τ_p settles on the upper envelope of the tangents.

Approximation error. For $x \in [b, b+h]$ the envelope is $\max\{2bx - b^2,\, 2(b+h)x - (b+h)^2\}$, and x^2 minus the envelope is largest where the two tangents meet, at $x = b + h/2$, where it equals $(h/2)^2$. The envelope therefore under-states x_p^2 by at most $h^2/4 = 2.5 \times 10^{-5}$ per producer, and by at most 5×10^{-4} of the Herfindahl index even with the largest feasible number of producers (20); unselected nodes contribute no error. The diagnostics record the realised error; in the production design it equals the bound in every stage, because an interior optimal share sits on a kink of the envelope, at a midpoint $b + h/2$ of the grid, unless a bound or a linking constraint binds. Designed shares such as 0.325, 0.275 or 0.135 are of this kind, which is why ties are frequent and the top-K rule uses a strict rank.

Perspective form. Written with y_p multiplying b^2, the cuts are the perspective of the tangents. They leave the integer solutions and their objective values unchanged but tighten the linear relaxation: with plain tangents $\tau_p \geq 2bx_p - b^2$, a fractional y_p could hold a share below $\underline{s}$ at almost no penalty.

On a Nuclear Upstream stage with near-tied candidates the perspective form cut GLPK's solution time from 875 to 51 seconds with the same solution.

Size. With n nodes the program has n^2 flows, n selection and n^2 link binaries, and $2n$ share and auxiliary variables: $2n^2 + 3n$ variables, $n^2 + n$ of them binary. It has n demand constraints, $2n$ share bounds, one producer-count constraint, $3n^2$ linking constraints, n redundancy constraints, n share definitions and $|\mathcal{B}|\, n$ tangent cuts, plus one equality per barred node. For $n = 37$ this is 2,849 variables (1,406 binary) and 5,810 constraints.

Solution and fallback. Algorithm 2 solves one stage. A stage is sent to the MILP when the solver packages are installed and the stage has at most 60 nodes, which every stage satisfies. GLPK receives its own time limit of $T = 300$ seconds, an R-level limit guards the model build, and only a solution with optimal status is accepted; a time-limited or infeasible run falls back to the greedy heuristic, and the diagnostics record the method and the solver's message.

Algorithm 2 Solving one stage (*SolveStage*)

Require: stage (t, s); exclusion set $\mathcal{X}_{ts}$; node and edge data of the stage
$C \leftarrow$ the nodes of the stage; **if** $|C| < 3$ or no a_p is defined **then return** skipped
complete W: a missing component is 0; $W_{pp} \leftarrow$ eq. (77)
$\mathcal{E} \leftarrow C \setminus \mathcal{X}_{ts}$, restricted to nodes with a defined a_p if at least $\max\{m_P, \lceil 1/\bar{s} \rceil, m_S\}$ remain
if the MILP packages are installed **and** $|C| \le 60$ **then**
    build (78) with the cuts (83) and $y_p = 0$ for $p \notin \mathcal{E}$; solve with GLPK under the limit T
    if the build or solve fails, or the status is not optimal **then**
        record the message; $R \leftarrow$ *Greedy*$(C, \mathcal{E})$ ▷ Algorithm 3
    else
        $R \leftarrow$ the optimal (y, f) with $x_p = \sum_c f_{pc}$
    end if
else
    $R \leftarrow$ *Greedy*$(C, \mathcal{E})$
end if
record the method, producers, J with the exact penalty $\lambda \sum_p x_p^2$, HHI, $\max_p x_p$, realised redundancy and the tangent error
return R

C.3 Properties of the stage program

Throughout, self-supply is allowed, $d_c \ge 0$ and $\sum_c d_c = 1$.

Feasibility. At the production values the program is feasible if and only if $|\mathcal{E}_{ts}| \ge 3$. *Proof.* Summing the demand constraints gives $\sum_p x_p = 1$, and $x_p \le \bar{s}$ requires at least $\lceil 1/\bar{s} \rceil = 3$ selected nodes, all eligible. Conversely, select any three eligible nodes, link each to every consumer and set $f_{pc} = d_c/3$: shares of $1/3$ lie in $[\underline{s}, \bar{s}]$, each link carries $d_c/3 \ge \varepsilon d_c$, and every consumer has three suppliers. The penalty does not affect feasibility. □

Concentration bounds. Any feasible design satisfies eq. (79). *Proof.* With k producers, each with $x_p \geq \underline{s}$, $k \leq \lfloor 1/\underline{s} \rfloor$ and $k \leq |\mathcal{E}|$, and $\sum_p x_p^2 \geq 1/k$ by the Cauchy–Schwarz inequality. The Herfindahl index is convex, so its maximum over $\{x : \sum_p x_p = 1,\ 0 \leq x_p \leq \bar{s}\}$ is attained at a vertex, where $\lfloor 1/\bar{s} \rfloor$ shares equal $\bar{s}$ and one equals the remainder, here $(0.4, 0.4, 0.2)$ and 0.36. □

Redundancy. Every consumer with $d_c > 0$ has between $m_S = 2$ and $\lfloor 1/\varepsilon \rfloor = 10$ suppliers, and at least εd_c of its demand comes from another node, because an active link carries at least εd_c and at least two links are active.

Interior shares. Fix the producer set S and write v_p for the marginal value of a unit of p's share (its node score plus the edge value of the marginal flow). Maximising $\sum_{p \in S} v_p x_p - \lambda \sum_{p \in S} x_p^2$ subject to $\sum_p x_p = 1$ and $\underline{s} \leq x_p \leq \bar{s}$ gives, by the Karush–Kuhn–Tucker conditions,

$$x_p = \min\left\{\bar{s},\ \max\left\{\underline{s},\ \frac{v_p - \mu}{2\lambda}\right\}\right\}, \tag{84}$$

with the level μ set so that the shares sum to one. Two interior shares therefore differ by $(v_p - v_q)/(2\lambda)$, and as $\lambda \to 0$ the shares move to the bounds, the corner solutions of the linear design. Equation (84) is the water-filling rule of the greedy fallback.

Monotonicity. Holding the eligible set, scores, demand and edges fixed, the optimal value is weakly decreasing in the stringency of each resilience parameter (lower $\bar{s}$, higher $\underline{s}$, m_P, m_S or ε), because each parameter enters one family of constraints and a more stringent value only removes feasible points. It is also weakly decreasing in λ. The portfolio loop can produce different exclusion sets at different parameter values, so the property holds stage by stage for given exclusions, not for the design as a whole; the non-monotone median Herfindahl index of the calibration grid arises in this way.

C.4 Greedy fallback

The heuristic (Algorithm 3) replaces the joint choice of producers, shares and flows by three steps. It ranks the eligible nodes on $g_p = w_N a_p + w_E \bar{W}_p + w_D \delta_p$, where $\bar{W}_p = \sum_c W_{pc} d_c$ is the demand-weighted connectivity, with ties broken by a_p, then $\bar{W}_p$, then the code, so that the selection never depends on row order; nodes with no positive edge to a consumer with demand go last. For each $k \geq m_P$ it water-fills the first k nodes with eq. (84), using $v_p = g_p$ (the edge term as if p served every consumer in proportion to demand) and finding μ by bisection, and keeps the k with the highest penalised value. For candidates in value order and without a development bonus this is the exact optimum of the node part. Flows are then fitted to the target shares and to demand by biproportional (RAS) scaling of the edge matrix, $f_{pc} = r_p W_{pc} k_c$, which splits each consumer's demand in proportion to edge quality and minimises the Kullback–Leibler divergence from the edge matrix among the matrices with the required margins. Links below ε of a consumer's demand are

then pruned one at a time and the fit repeated, but only where the consumer keeps m_S suppliers and the refit still meets the share targets: the share cap is a hard constraint, redundancy a soft one.

Algorithm 3 Greedy fallback with water-filling (*Greedy*)

Require: nodes C, eligible set $\mathcal{E}$; a, d, δ; complete edge matrix W; $\lambda > 0$

$\bar{W}_p \leftarrow \sum_c W_{pc} d_c$; $g_p \leftarrow w_N a_p + w_E \bar{W}_p + w_D \delta_p$ ($a_p = 0$ if undefined)

order $\mathcal{E}$ by g_p, a_p, $\bar{W}_p$ (decreasing) and code; move nodes that reach no consumer with demand to the end

for $k = m_P, \ldots, |\mathcal{E}|$ **do**

    $x^{(k)} \leftarrow$ water-filling (84) of the first k nodes with $v_p = g_p$ (bisection on μ)

    $V_k \leftarrow \sum_p g_p x_p^{(k)} + w_D \sum_p \delta_p - \lambda \sum_p (x_p^{(k)})^2$

end for

$k^* \leftarrow \arg\max_k V_k$; $S \leftarrow$ the first k^* nodes; $\sigma \leftarrow x^{(k^*)}$

$f \leftarrow$ RAS fit of $W_{S \times C}$ to row sums σ and column sums d (at most 2,000 iterations, tolerance 10^{-12})

prune links with $f_{pc} < \varepsilon d_c$ one at a time where the consumer keeps m_S suppliers; refit; reject a refit that misses σ or d by more than 10^{-6}

return S, $x = \sum_c f_{pc}$, f and J evaluated with eq. (78a)

A greedy design that satisfies every constraint of the MILP is feasible for it, so an optimal MILP solution scores at least as high. In the production configuration without caps the MILP objective is at least the greedy one in all 22 stages, by 2.4% in the median stage and 4.0% at most, and the greedy design is more concentrated (median index 0.154 against 0.139); with the caps on, solving every stage greedily gives a median index of 0.160, above the calibration target, and a summed objective 2.5% lower. The fallback is a safeguard, not an equivalent; it is not triggered in the production design.

C.5 Portfolio caps

Algorithm 4 enforces the caps of eq. (80). The caps are computed once: GDP (latest finite WDI year, the coalition median if none) is transformed to $\log(1 + \text{GDP})$, scaled to $[0, 1]$ over the nodes, mapped linearly to $[2, 10]$, rounded and clamped, and the largest economy is set to ten. An appearance requires a positive share and a strict rank of at most K, ties broken by producer score and then by code; with shares on the midpoints of the tangent grid a dense rank would count every tied producer and let one stage contribute more than K appearances.

Exclusion sets only grow and a barred node remains a consumer, so at most $(1 + I_{\max})$ times the number of stages are solved. Within a pass the offenders' exclusions are chosen from one snapshot and do not depend on the order in which offenders are visited; across passes the outcome is path-dependent, because barring a node can promote another into a stage's top K. Compliance requires $\sum_i \kappa_i \geq 3S^*$ for S^* solved stages, since every stage has at least three producers within its top K; here $\sum_i \kappa_i = 218 \geq 66$. In the production design the loop converges after seven passes (62 stage solves) with no violation, and every run of the calibration grid converges, in four to eight passes. A joint program with the linear constraints $\sum_{(t,s)} y_{i,ts} \leq \kappa_i$ would optimise the stages

Algorithm 4 Portfolio caps across stages (*PortfolioCaps*)

Require: stage results R; exclusion sets $\mathcal{X}$; caps κ; $K = 5$; $I_{\max} = 30$

for $\iota = 1, \ldots, I_{\max}$ **do**
  in every stage, $\rho_{i,ts} \leftarrow$ strict rank of $x_{i,ts}$ (decreasing; ties by a, then code)
  $A_i \leftarrow |\{(t,s) : x_{i,ts} > 0,\ \rho_{i,ts} \le K\}|$ ▷ eq. (81)
  $O \leftarrow \{i : A_i > \kappa_i\}$; **if** $O = \emptyset$ **then** converged; **break**
  for all $i \in O$ **do** ▷ all offenders use the same snapshot of R
    bar i from the $A_i - \kappa_i$ counted stages with the smallest $x_{i,ts}\, a_{i,ts}$: $\mathcal{X}_{ts} \leftarrow \mathcal{X}_{ts} \cup \{i\}$
  end for
  re-solve every stage whose exclusion set changed ▷ Algorithm 2
end for
return R, $\mathcal{X}$, $(\kappa_i, A_i)_i$, convergence status, passes, violations left

together, at the cost of a program 22 times as large; it is left for future work.

C.6 Calibration of the diversification weight

Target. For each stage with an HS6 basket, the nodes' observed export concentration is $\mathrm{HHI}^{\mathrm{N}}_{ts} = \sum_{j \in \mathcal{V}} \left(X_{jts} / \sum_{k \in \mathcal{V}} X_{kts}\right)^2$, where X_{jts} are node j's 2024 basket exports in the reconciled Comtrade matrix (importer reports, else the exporter's mirror declaration $\times 1.06$; unobserved cells count as zero). A single-country node counts its exports to every other economy; the EU node counts only exports to importers outside the EU, because trade among its members is internal to the node, as a country's domestic sales are, and would otherwise make the EU the leading exporter of nearly every stage. The target is the median over the 18 basket stages, 0.1556; the stage values range from 0.095 (Batteries Midstream) to 0.355.

Procedure. For each λ on the grid, the full production configuration is re-solved: 37 nodes, 22 stages, caps on, MILP with the 300-second limit, up to 30 cap passes, and the inputs assembled exactly as `scripts/30` assembles them. The rule selects the smallest grid value whose median $\mathrm{HHI}^{\mathrm{D}}_{ts}$ over the 22 stages is at most the target. The run at the selected value reproduces the stored production design to 5×10^{-16} in every share, and the production script warns if the designed median exceeds the observed one on its own inputs, so a change of inputs that breaks the rule cannot pass unnoticed. Table 31 lists the 36 grid points.

Results. Without the penalty the median stage has three producers, two at the cap, and an index of 0.345; at $\lambda = 0.5$ it is still 0.216. Between 0.70 and 0.77 the median lies within 0.009 of the target and above it, and the rule selects 0.78 (0.1520, a margin of 0.0036). The median rises in eight of the 35 grid steps (at 0.72–0.73, 0.79–0.80, 0.84–0.85, 0.86–0.87, 0.90–0.91, 0.93–0.94, 0.96–0.97 and 0.99–1.00), by at most 0.0017, and is flat between 0.81 and 0.82; no point above the pick returns above the target. Every run up to $\lambda = 0.91$ solves all 22 stages by the MILP in every pass. The score cost of diversification grows smoothly, from 5.8% at the pick to 7.6% at $\lambda = 1$ and 10.6% at 1.5, while the number of producer roles rises from 201 to 280. Without caps, the same λ gives a

Table 31: The calibration grid of the diversification weight

λ	$\widetilde{\text{HHI}}$	Eff.	Max.	Roles	Floor	Δ	M	λ	$\widetilde{\text{HHI}}$	Eff.	Max.	Roles	Floor	Δ	M
0	.3450	2.90	.400	79	6	0.00	22	0.86	.1480	6.76	.305	209	56	−6.39	22
0.5	.2156	4.64	.400	149	35	−2.71	22	0.87	.1493	6.70	.305	209	54	−6.44	22
0.70	.1640	6.10	.355	190	54	−5.02	22	0.88	.1469	6.81	.305	211	54	−6.54	22
0.71	.1630	6.14	.345	192	54	−5.17	22	0.89	.1414	7.07	.302	212	54	−6.62	22
0.72	.1622	6.16	.335	193	51	−5.27	22	0.90	.1411	7.09	.302	214	59	−6.70	22
0.73	.1630	6.14	.335	194	53	−5.36	22	0.91	.1423	7.03	.295	213	55	−6.71	22
0.74	.1620	6.17	.335	197	57	−5.50	22	0.92	.1347	7.43	.310	216	58	−6.86	21
0.75	.1593	6.28	.335	197	58	−5.55	22	0.93	.1323	7.56	.309	217	61	−6.99	21
0.76	.1583	6.32	.331	199	62	−5.66	22	0.94	.1340	7.46	.285	221	64	−7.07	22
0.77	.1558	6.42	.325	200	61	−5.72	22	0.95	.1331	7.51	.305	221	62	−7.18	21
0.78	**.1520**	6.58	.325	201	61	−5.80	22	0.96	.1326	7.54	.285	225	70	−7.28	22
0.79	.1485	6.74	.325	203	63	−5.90	22	0.97	.1343	7.45	.282	225	66	−7.39	22
0.80	.1502	6.66	.320	205	63	−6.03	22	0.98	.1310	7.63	.276	227	65	−7.44	22
0.81	.1500	6.66	.315	205	54	−6.09	22	0.99	.1288	7.77	.300	226	65	−7.53	21
0.82	.1500	6.66	.315	206	53	−6.14	22	1.00	.1291	7.74	.275	230	70	−7.58	22
0.83	.1492	6.70	.315	205	55	−6.16	22	1.1	.1196	8.36	.257	238	63	−8.17	22
0.84	.1487	6.73	.305	208	58	−6.26	22	1.2	.1126	8.88	.245	251	77	−8.96	22
0.85	.1501	6.66	.305	207	56	−6.29	22	1.5	.0945	10.58	.215	280	92	−10.64	22

Without portfolio caps: $\lambda = 0.78$: $\widetilde{\text{HHI}} = .1393$, Eff. 7.18, Max. .312, Roles 214, Floor 57, $\Delta = -5.14$, M 22.

Notes: Each row re-solves the production configuration (37 nodes, 22 Upstream and Midstream stages, portfolio caps on, MILP with a 300-second stage limit) at one value of λ. $\widetilde{\text{HHI}}$: median over the 22 stages of the Herfindahl index of designed shares; the rule selects the smallest λ with $\widetilde{\text{HHI}} \leq 0.1556$, the median observed export Herfindahl index of the nodes over the 18 basket stages (bold), and every row from 0.78 on meets it. Eff.: median effective number of producers. Max.: largest designed share. Roles: producer roles summed over stages. Floor: shares at $\underline{s} = 0.05$. Δ: summed node and edge terms of the 22 stage objectives (without the penalty), in percent of their value at $\lambda = 0$ with caps. M: stages solved by the MILP in the final pass; in the four runs with 21 (0.92, 0.93, 0.95, 0.99), Nuclear Upstream reached the time limit and was solved by the greedy fallback. The portfolio loop converged in every run (four to eight passes). Source: lambda study on the inputs of the final rerun of 12 October 2026.

median of 0.139, so the calibrated value is specific to the capped configuration, as it is to the edge weight $w_E = 0.5$ (Section 12).

C.7 Implementation, verification and options not used

The stage MILP is written with `ompr` (Schumacher, 2023) and passed through `ompr.roi` and the ROI plug-in to GLPK (Makhorin, 2020). The production script writes the tables `allied_network_*.csv` (specialization, flows, diagnostics, input checks, nodes, edges, member shares, observed concentration, portfolio and build candidates); in the flow and edge tables the reporter is always the producer and the partner the consumer, and the friendshore column holds FS_{cp}, the consumer's rating of the producer. The six test files of the design (`tests/testthat/test-allied-network-*.R`), on synthetic fixtures only, run 43 tests with 443 passing expectations and no failure. They check the uniqueness and orientation of nodes and edges, the sums of shares, demand and flows, the share bounds and redundancy rules, the tangent approximation and its error bound, the water-filling fallback, the aggregation of the EU node and the split of its shares, the strict top-K count, the portfolio loop and the deterministic treatment of ties.

Options not used in production. Two options are retained in the code for other callers and are not used for any result in the paper. The first is the three-component producer score with the Policy index (node weights 0.5, 0.3 and 0.2 on EO, policy and ES), in which a node without a policy value receives the stage median of the observed values before the S-curve,

$$\mathrm{POL}_{pts} \;\leftarrow\; \mathrm{median}\{\mathrm{POL}_{qts} :\; q \in \mathcal{V},\; \mathrm{POL}_{qts} \text{ observed}\} \quad \text{if } \mathrm{POL}_{pts} \text{ is missing;} \tag{85}$$

the production design switches the policy term off, so nothing is imputed. The second is a capacity-denominated variant, in which flows are in US dollars against a stage target, producers have existing and maximum capacities and capacity builds carry a cost that development potential can lower. It needs stage targets and capacity data that do not exist at the resolution of the indices, and its solvers lack the deterministic tie rule and the input checks of the share-based ones.

Appendix D Corrections relative to the shipped release

The indices were first released in September 2026. That *shipped release* combined artefacts built at different dates: the Energy Security and Economic Opportunity pillars and the Strategic Index of 3 September 2026 rested on a policy component table of 6 February 2026, the partnership tables dated from 24 March 2026 and the allied network outputs from 23 February 2026. Before the results of this paper were produced, the pipeline was audited end to end, from the staged raw inputs to the published tables, and every defect found was corrected. A set of data and method changes followed, decided while the paper was written. All results in Sections 5–12 come from one complete run of the corrected pipeline, the final run of 12 October 2026, built from a single revision of the code.

This appendix lists every correction and change and compares the shipped release with the final one, so that a reader who holds numbers from the shipped release can tell which changed and why.

D.1 Scope and kinds of change

Table 32 gives one line per item. The items coded A to N are defects. They fall into three kinds. *Input defects* change what a variable measures: a join on mismatched keys, a wrong supply-chain stage, a ranking taken over the wrong population. *Aggregation defects* change how correctly measured variables are combined: an unobserved category scored as zero, a row counted several times. *Reporting defects* leave every value unchanged but misstate a diagnostic, a label or the documentation. The distinction matters for reconciliation: reporting defects change no number, aggregation defects mostly shift the levels of whole cells, and input defects can reorder countries within a cell. The uncertainty and sensitivity analyses recommended for composite indicators vary weights, normalisation and aggregation for given inputs (Saisana et al., 2005; Nardo et al., 2008); they can reveal the second kind of defect but not the first, which is why the audit traced each variable from its source to the index.

The items coded D (data) and M (method) are not defects. They are changes of source, basket or design adopted after the audit, most of them by explicit decision, and they are described in the main text where they apply. For each defect the audit re-ran the affected builders on the inputs saved with the shipped release and compared the outputs. These *pre-rerun effects*, reported in Online Appendix OA.11 with the symptom and the correction of each item, isolate one correction on inputs that still embed the others; they are not results of the paper. Each correction comes with unit tests that encode the corrected behaviour, and the code in `R/`, `scripts/` and `config/` is its specification.

Table 32: Index of corrections and changes relative to the shipped release

Item	Correction or change	Commit	Outputs affected	Status
Aggregation				
A1	Available-case aggregation: an unobserved category no longer counts as zero	`7003d11`	ES, EO, PSI	R
A2	Imputation only in groups with an observed country	`7003d11`	ES, EO	L
A3	Imputed flag for groups with several components	`7003d11`	ES, EO diagnostics	D
A4	Energy Prices scored; guards for unproduced score variables	`7003d11`, `cd3b6cc`	ES, EO Upstream	R
A5	Orphan EO Reserves weight and 88 phantom EO cells removed	`7003d11`	EO	R
A6	Contribution tables reconcile to the index	`7003d11`	decompositions	D
A7	Overall scores no longer collapse into one row	`7003d11`	ES, EO	L
Country universe				

continued on next page

Table 32, continued

Item	Correction or change	Commit	Outputs affected	Status
C1	One canonical name per economy	`9a5a72a`, `c840032`	all	R
C2	Aggregates, residual and historical entities dropped	`9a5a72a`	all	R
C3	Comtrade vintages need at least 50 reporters	`9a5a72a`	ES Minerals Trade	L
Energy Security and production themes				
E1	Harmonised mineral names (graphite, rare earths)	`c840032`	mineral categories	R
E2	Refining assigned to the Midstream	`c840032`	ES Foreign Dependency	R
E3	Zero supply for non-suppliers, ranked below every supplier	`c840032`	ES Foreign Dependency	R
E4	Reserves roll-up over all of a technology's minerals	`c840032`	ES Reserves	R
E5	Wind consumption, within-technology growth, one regional value per country	`c840032`	ES, EO Consumption	R
E6	Minerals Trade scores market share and concentration	`c840032`, `cd3b6cc`	ES Minerals Trade	R
E7	Minerals investment split by demand; no manufactured zero cells	`c840032`	ES, EO Investment; SI	R
E8	Electrolyser capacity mapped to Green Hydrogen	`c840032`	EO Foreign Dependency	R
E9	Mineral production depth as a demand-weighted share of world mined supply	`c939ad6`	EO Production	R
E10	EV Midstream keeps the non-EU European economies	`c939ad6`	ES, EO EV Midstream	R
E11	Self-sufficiency and net-export ratios named for what they measure; ties	`c939ad6`	ES Energy Imports	R
E12	Absent EI production no longer read as zero (EIA fill, same-year pairing)	`cce8985`, `7f0190c`	ES Energy Imports, EO Production	R
Trade and Economic Opportunity themes				
T1	Separate ES and EO trade themes; ES deficit term computed	`fe605b6`, `cd3b6cc`	ES, EO Trade	R
T2	Export size and feasibility ranked across economies	`fe605b6`	EO Trade	R
T3	Market share as the share of the basket	`fe605b6`	ES, EO Trade	R
T4	HS6 crosswalk: raw materials, logs, duplicate keys	`fe605b6`	ES, EO Trade	R
T5	Legacy HS codes bridge the HS92 Atlas	`fe605b6`	ES, EO Trade	R
T6	Levelised cost counted once in Cost Competitiveness	`fe605b6`, `cd3b6cc`	EO Cost Competitiveness	R

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Table 32, continued

Item	Correction or change	Commit	Outputs affected	Status
T7	Technological readiness broadcast to economies	`fe605b6`	EO, SI	R
T8	Technology Demand counts each source once	`fe605b6`, `cd3b6cc`	EO Technology Demand	R
T9	Export-concentration index ranked over headline cells only	`23f20da`	ES, EO Trade	R
T10	Energy-trade file completed (60 basket codes, 13 reporters, Qatar's LNG)	`9778e34`	ES, EO Trade	R
T11	Unique demand keys; IEA 2035 shares over the world total incl. "Other"	`c939ad6`	EO Technology Demand	D
T12	Zero lending-rate stubs dropped (New Zealand)	`c939ad6`	EO Cost Competitiveness	R
T13	IMF price units and source labels	`c939ad6`	labels	D
Policy and Strategic indices				
P1	Separate jurisdiction weights for IEA PAMS and NIPO	`81ede9d`	Policy, SI	R
P2	Whole-word stage tagging of IEA policies	`81ede9d`	Policy, SI	R
P3	Policy coverage of Korea, the UAE, France, Italy and Denmark	`81ede9d`	Policy, SI, network	R
P4	Policy component table rebuilt with provenance	`81ede9d`	Policy, SI	R
P5	NIPO as-of date capped at the extract's last announcement	`81ede9d`	NIPO (robustness only)	R
P6	Security and security need stored under their own names in the SI	`81ede9d`	SI tables, charts	D
P7	No imputation of pillar or policy values in the SI	`81ede9d`	SI	R
P8	Sectors with a constant pillar detected and excluded	`81ede9d`	SI, allied pick	R
P9	Oil lifecycle emissions in the technology weight (SRREN median)	`6ef4831`	SI	R
Bilateral indices				
B1	Duplicated Friendshore rows and self-dyads removed	`45efcf2`	FS, PSI, network	R
B2	Outbound investment: three measures, name matching, missing vs zero	`45efcf2`	FS, PSI, network	R
B3	Development potential at partner level	`45efcf2`	DP, PSI, network	R
B4	Available-case PSI; partner identity by ISO3 code	`45efcf2`	PSI	R
B5	Direction conventions; headline technologies only	`45efcf2`	FS, PO, PSI, network	R
Allied network				
N1	Development join fan-out	`2eeb111`	network	R

continued on next page

Table 32, continued

Item	Correction or change	Commit	Outputs affected	Status
N2	Duplicate edges and split country records	`2eeb111`	network	R
N3	Friendshore half of the edge oriented consumer to producer	`2eeb111`	network	R
N4	Selection and portfolio counts over distinct countries	`2eeb111`	network	R
N5	Coalition code of the United Arab Emirates	`2eeb111`, `81ede9d`	network, SI	R
N6	MILP threshold, solver reporting, greedy flows	`2eeb111`	network	R
N7	Missing node components and stages without signal	`2eeb111`	network	R
Data changes				
D1	Nuclear Upstream on Red Book uranium resources and production	`6d874a3`	ES, EO Nuclear Upstream	X
D2	Technological readiness mapped by stage (IEA Clean Technology Guide)	`fefbc37`	EO, SI	X
D3	All-reporter Comtrade dyads, importer first, else mirror $\times 1.06$; Taiwan	`c8ca2d5`	trade, FS, PO, network	X
D4	HS6 basket review; electricity out of Electric Grid Upstream	`3f52711`	ES, EO Trade	X
D5	Copper and aluminium basket for Electric Grid Upstream	`0bbcf79`	ES, EO Trade	X
D6	Critical-mineral reserves from USGS MCS 2026	`8ef7ab0`	ES Reserves, coalition	X
D7	HS 851762 out of Grid Midstream; Hong Kong out of export measures	`23f20da`	ES, EO Trade	X
D8	ILO earnings from a staged copy	`9776f9f`	EO Cost Competitiveness	X
D9	Upstream trade baskets for EV, Wind, Nuclear, Solar, Semiconductors	`0fc5b73`	ES, EO Trade	X
D10	CM-FDI investment ties and Chinese-parent exposure in Friendshore	`78be8b5`	FS, PSI, network	X
Method changes				
M1	Rankability rule: sovereign, at most half the weight imputed	`5f60417`	ES, EO, SI, PSI	X
M2	Coalition derived from UN voting, in two tiers; EU as one node	`07e1783`, `19bf99e`	SI picks, PSI, network	X
M3	Policy index: IEA PAMS required, CAT where rated; NIPO (robustness only, October 2026 run) and dual use out	`0aaf68d`, `aefd3ef`	Policy, SI	X
M4	Endowment screen for the allied pick	`0aaf68d`	allied pick	X
M5	Alignment in Friendshore; conduit centres out; core-view PSI	`6ef4831`	FS, PSI	X

continued on next page

Table 32, continued

Item	Correction or change	Commit	Outputs affected	Status
M6	Network on Upstream and Midstream, diversification term, caps, no policy term	`5afe34c`	network	X
M7	Calibration rule for λ; $\lambda = 0.78$ on the final inputs	`3122cea`	network	X

Notes: Status R: the defect changed values in the shipped release. L: latent; it did not change the shipped release but would have changed a rerun. D: diagnostics, labels or documentation only. X: change of data or method adopted after the audit, not a defect. Commits are on branch `fix/allied-base-paper`; the audit corrections (`7003d11` to `cd3b6cc`) were applied to revision `c0cd1ce`. Item detail: Online Appendix OA.11.

D.2 The main corrections

Four defects account for most of the difference between the releases, and each is stated here formally. Write c_{itsk} for the score of category k in cell (i, t, s), w_k for its weight and R_{its} for the categories with a row for the cell, observed or not.

Unobserved categories as zeros (A1). The shipped builder kept an unobserved category in the denominator,

$$I_{its}^{\text{shipped}} = \frac{\sum_{k \in R_{its}} w_k \, c_{itsk} \, \mathbf{1}[c_{itsk} \text{ observed}]}{\sum_{k \in R_{its}} w_k}, \tag{86}$$

so it counted as zero and a cell with no observed category scored zero. The corrected pillar is the available-case mean,

$$I_{its} = \frac{\sum_{k \in R_{its}} w_k \, c_{itsk} \, \mathbf{1}[c_{itsk} \text{ observed}]}{\sum_{k \in R_{its}} w_k \, \mathbf{1}[c_{itsk} \text{ observed}]}, \tag{87}$$

and a cell without an observed category is missing (Section 4). Combined with the zero cells that the investment theme manufactured (E7), the shipped rule made Energy Security in Nuclear Upstream zero for every economy, so its security need was maximal everywhere and it was the top allied pick of 8 of the 18 shipped allies.

Market share of a basket (T3). The shipped trade theme averaged a country's product-level world shares with its own exports as weights, which measures dominance in the products it ships rather than weight in the basket:

$$\widetilde{ms}_{its} = \frac{\sum_{h \in B_{ts}} X_{ih} \, (X_{ih}/W_h)}{\sum_{h \in B_{ts}} X_{ih}}, \qquad ms_{its} = \frac{\sum_{h \in B_{ts}} X_{ih}}{\sum_{h \in B_{ts}} W_h}, \tag{88}$$

with X_{ih} exports of product h, W_h world exports and B_{ts} the basket. The corrected measure is ms_{its}. In the shipped Batteries Midstream basket the Democratic Republic of the Congo had $\widetilde{ms} = 0.862$ against a basket share of 0.0074. Together with the ranking of export size across economies rather than within each reporter (T2), the correction changes the ten leading EO Trade scores in Batteries Midstream, on the shipped inputs, from ore exporters (Chile, Peru, Zimbabwe) to cell and material manufacturers (China, Germany, the United States). The ES and EO trade themes had, moreover,

been identical tables (T1).

The PSI denominator (B4). Development potential is defined only for partners that are not high-income, but for a high-income partner it kept its weight and contributed nothing,

$$\mathrm{PSI}_j^{\mathrm{shipped}} = 0.4\,C_j^{\mathrm{FS}} + 0.4\,C_j^{\mathrm{PO}} \;\le\; 0.8, \tag{89}$$

so high-income partners sat at the bottom: the United States ranked 185th of 218. With the duplication of Friendshore rows (B1) and the outbound-investment defect (B2), the shipped PSI ordering does not survive the corrections.

The network fan-out (N1). The node table joined the dyadic development table without aggregating it, and the greedy solver took the first rows of the ranked candidates rather than distinct countries, so in the affected stages one country was selected with k^2 rows of share $1/k$:

$$n_{\mathrm{producers}} = k^2, \qquad \mathrm{HHI} = \sum_{r=1}^{k^2} \left(\frac{1}{k}\right)^2 = 1. \tag{90}$$

The shipped specialisation table had 606,866 rows, of which 2,339 were distinct. With the transposed Friendshore half of the edge (N3), counts over rows (N4) and the invalid code for the United Arab Emirates (N5), the shipped network outputs carry no usable information.

The remaining defects are smaller but not negligible. The country standardiser left 30 economies under two or three spellings, each taking its own rank in every S-curve (C1); graphite and the rare earths fell out of every mineral join (E1); refining was pooled with mining (E2); and the policy block had not written its table since February 2026 (P4), with substring stage rules that tagged 37 percent of IEA policies Upstream (P2). Three latent defects (A2, A7, C3) would have changed a rerun; A7 alone would have removed every category built from an overall score. Item E12 is a defect found late: an economy without its own production row in the Energy Institute's tables (Energy Institute, 2026) was read as producing nothing, so New Zealand counted as importing all its gas. Its production now comes from the EIA (U.S. Energy Information Administration, 2026).

D.3 Data and method changes

The data changes replace or extend sources. Nuclear Upstream, which the shipped release scored on a manufactured zero and the audited code on the world uranium price alone, is scored on the identified resources and mine production of the OECD-NEA/IAEA Red Book (OECD Nuclear Energy Agency and International Atomic Energy Agency, 2026) (D1). Critical-mineral reserves come from the USGS Mineral Commodity Summaries 2026 (U.S. Geological Survey, 2026) rather than the older USGS figures repeated in the Energy Institute's sheets (D6). Technological readiness is mapped by stage from the IEA Clean Technology Guide (International Energy Agency, 2026c) (D2). Bilateral trade comes from all-reporter Comtrade dyads, importer report first, else the exporter's mirror declaration times 1.06 (D3). The HS6 baskets were reviewed against a dominant-use rule, which removed 92 basket rows; electricity left Electric Grid Upstream, which received a copper and

aluminium input basket, telecommunications switching equipment left Electric Grid Midstream, and five technologies received Upstream baskets (D4, D5, D7, D9). Hong Kong left the Comtrade export measures as a re-export hub (D7). Friendshore gained investment ties and a Chinese-parent exposure component from the CM-FDI project data (Bandara and Net Zero Industrial Policy Lab, 2026) (D10). One limitation of the sources remains: the IEA's 2035 announced manufacturing-capacity shares are regional for Europe, so every EU member carries the EU's share in the variables built on them (Section 13.2).

The method changes are the rankability rule (M1); the coalition derived from UN-voting alignment in two tiers, with the EU-27 as one network node (M2), which replaced the shipped hand-picked list of 23 economies; a Policy index built on IEA PAMS, combined with CAT where CAT rates the country, which dropped the GTA NIPO component and the dual-use score of the shipped four-component index (M3; NIPO survives only as a robustness variant, rebuilt from the October 2026 extract for 77 jurisdictions); the endowment screen of the allied pick (M4); the alignment component, the exclusion of conduit financial centres and the core-coalition view of the PSI (M5); and a network restricted to the Upstream and Midstream stages, with a concave diversification term, GDP-log portfolio caps and no policy term (M6). The diversification weight follows a rule rather than a number (M7): the smallest λ whose median designed Herfindahl index does not exceed the nodes' median observed export Herfindahl index. Applied to successive input vintages, the rule gave 1.5 on the hand-picked list, then 0.94 on 34 nodes, 0.92 on 36, 0.95 and 0.94 on 37 nodes with 14 basket stages, and 0.78 on the final 37-node inputs with 18 basket stages; only the last is used in this paper.

D.4 Reconciliation with the shipped release

Matching and coverage. The comparison uses the headline tables of the two releases. Every shipped country name is mapped to its canonical name (C1), units without an ISO3 code are dropped (C2), and where two spellings give the same key the record under the canonical spelling is kept. Harmonisation alone reduces the shipped Energy Security table from 270 strings to 237 economies, merging 162 rows of 29 economies that appeared under two or three spellings. Table 33 reports the counts. Of the harmonised ES cells 5,522 are in both releases; 710 are new in the final one, mostly economies reached by the rebuilt trade themes and the Green Hydrogen Midstream cells the shipped ES did not score. For EO, 6,598 cells are in both and 533 are new, mostly imputed (median imputed weight share 0.68) and only 11 rankable.

Three rules apply to the pillar statistics. A technology–stage cell in which a pillar is constant in either release is excluded: this removes ES in Nuclear Upstream, zero for every economy in the shipped release, and the shipped Geothermal Upstream and Midstream cells. Rank statistics are computed over the cells rankable in the final release (Section 4), with statistics over all matched cells alongside. Correlations are Spearman unless labelled otherwise. Table 34 summarises the agreement.

Table 33: Coverage of the headline tables: shipped and final release

	Shipped release	Final release
Energy Security		
Rows (country–technology–stage cells)	5,746	6,232
rankable	–	4,163
Distinct country strings	270	219
non-country units dropped (rows)	17	–
merged spellings (rows)	162	–
Economies after harmonisation	237	219
Technology–stage pairs scored	31	30
Cells in both releases	5,522	5,522
rankable in the final release	–	3,731
Cells in one release only	45	710
Economic Opportunity		
Rows (country–technology–stage cells)	6,923	7,131
rankable	–	4,301
Distinct country strings	272	219
non-country units dropped (rows)	56	–
merged spellings (rows)	234	–
Economies after harmonisation	237	219
Technology–stage pairs scored	33	33
Cells in both releases	6,598	6,598
rankable in the final release	–	4,290
Cells in one release only	35	533
Strategic Index		
Scored cells	571	3,154
Countries	18	174
Sectors	32	30
Cells in both releases	477	477

Notes: Headline tables of the eleven technologies. Shipped release: `index_outputs.rds` of 3 September 2026; final release: the same file of the final 12 October 2026 run. Shipped country names are mapped to canonical names with `standardize_country_names()` (item C1); units without an ISO3 code ("Global", "USSR", "Northern Cyprus") are dropped (item C2) and, where two spellings give the same cell, the record under the canonical spelling is kept. "Rankable": the economy is sovereign and at most half of the cell's index weight is imputed (`config/rankability.yml`); the shipped release had no such rule. "Cells in one release only" counts shipped cells absent from the final release (first column) and final cells absent from the shipped release (second column). A Strategic Index cell of the final release requires a Policy index (IEA PAMS, combined with CAT where CAT rates the country) and a rankable EO and ES value; a shipped cell required the four-component Policy index of February 2026 and imputed missing pillar values (item P7).

Table 34: Rank agreement between the shipped and the final release

		Pooled Spearman		Median within-cell Spearman			
Index	Units in both	Rankable	All	All	Up	Mid	Down
Energy Security (ES)	3,537 cells	0.81	0.84	0.83	0.84	0.70	0.90
Economic Opportunity (EO)	4,290 cells	0.67	0.71	0.77	0.61	0.80	0.88
Strategic Index (SI)	477 cells	0.74	–	0.74	–	–	–
PSI, world view	187 partners	−0.09	–	–	–	–	–
PSI, core view	191 partners	−0.33	–	–	–	–	–

Notes: Units present in both releases after harmonising shipped country names (C1) and dropping non-country units (C2). ES and EO: rankable matched cells of 194 and 178 economies, technology–stage cells constant in either release excluded; "All" pools every matched cell (5,305 ES, 6,598 EO). Within-cell correlations are across economies within a technology–stage cell (28 ES and 33 EO cells; medians by stage); for the SI they are across sectors within each of the 18 economies of the shipped list scored in both releases (range 0.42 to 0.92). The shipped PSI is reconstructed from the March 2026 partner tables with the shipped code's logic (B4) and compared with the final world view (like for like) and with the headline core view. Source: `paper/results/reconcile.json`.

The pillars. Over the rankable matched cells the Spearman correlation between the releases is 0.81 for ES and 0.67 for EO (Figure 46). Agreement differs by stage in opposite directions: for ES it is lowest in the Midstream (pooled 0.55), where the Trade category was rebuilt from all-reporter dyads and reviewed baskets; for EO it is lowest in the Upstream (0.50), where categories enter the cell and Trade and Investment were rebuilt. Levels moved as well as ranks: the mean change is +0.021 in ES but +0.081 in its Upstream, partly because 89 Upstream cells of 21 economies scored exactly zero under rule (86). A change of more than 0.1 affects 31 percent of rankable ES cells and 17 percent of EO cells.

A pooled correlation mixes level shifts of whole cells with reorderings within them. Figure 47 separates the two. The median within-cell correlation is 0.83 for ES and 0.77 for EO; 45 percent of ES values and 56 percent of EO values move by more than ten places in their cell (median moves of 9 and 13 places). A two-factor Shapley decomposition of each cell's change, with an unobserved shipped category counted as zero at full weight, splits it into score effects of categories present in both releases and a composition effect from categories that enter or leave. Composition accounts for 25 percent of the summed absolute change in ES and 40 percent in EO. When the entering or leaving category is constant within a cell (the zero Investment cells, Energy Prices, readiness) it moves the cell's level and leaves the ordering unchanged; the reorderings come from rebuilt inputs, chiefly Foreign Dependency in the ES Upstream, Trade in the ES Midstream and Consumption or Investment in the ES Downstream. The extreme case is EO Green Hydrogen Midstream, with a within-cell correlation of −0.55: both releases rank its economies mostly on imputed values, and neither ranking should be read as a measurement. ES Nuclear Midstream combines the largest level shift (a mean gain of 0.24, from the removal of the zero Investment category) with a within-cell correlation of 0.59 (from the rebuilt Trade category).

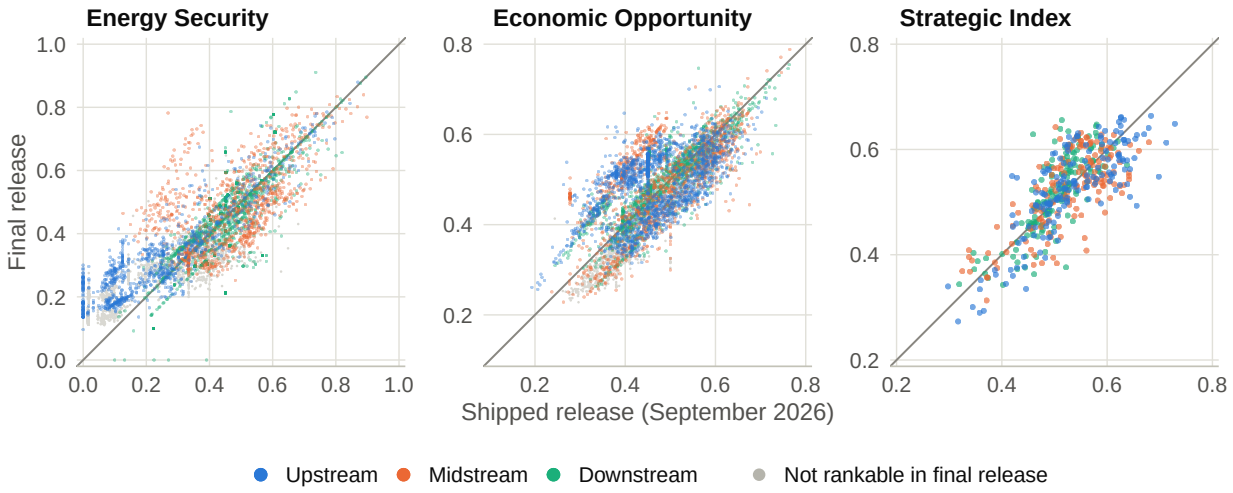


Figure 46: Shipped against final release, cell by cell. Each point is a country–technology–stage cell (Energy Security, Economic Opportunity) or a country–sector cell (Strategic Index) present in both releases, with the shipped value (3 September 2026) on the horizontal axis and the final value (12 October 2026) on the vertical axis. Colour marks the stage of cells rankable in the final release (sovereign economy, at most half of the index weight imputed); grey points are matched cells that are not rankable. Energy Security: 5,305 matched cells, 3,537 rankable; Economic Opportunity: 6,598 and 4,290; Strategic Index: the 477 cells of the 18 economies of the shipped hand-picked list scored in both releases. Technology–stage pairs in which a pillar is constant in either release are excluded. Spearman correlations over rankable cells: 0.81, 0.67 and 0.74. Shipped names are mapped to canonical names before matching (item C1). Sources: `index_outputs.rds` of the two releases; `scripts/paper/fig_reconcile.R`.

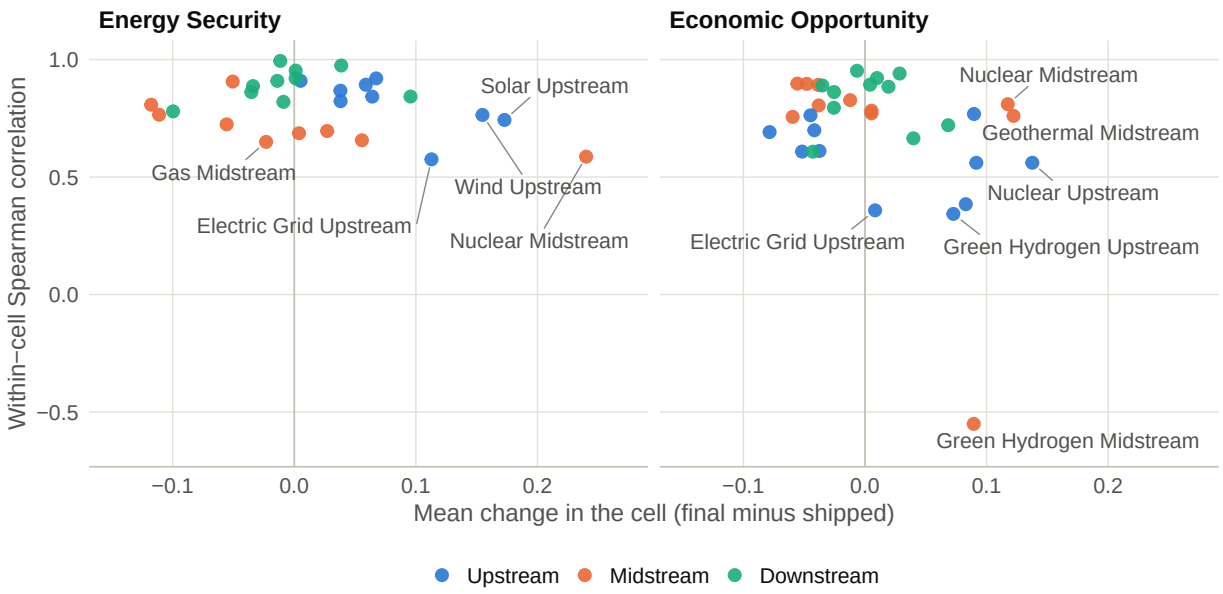


Figure 47: Level shift and rank agreement by technology–stage cell. Each point is one cell. Horizontal axis: mean change of the pillar (final minus shipped) across the economies scored in the cell in both releases and rankable in the final release. Vertical axis: Spearman correlation of the shipped and final values across the same economies. Colour marks the stage; labels name the three cells with the lowest correlation and the three with the largest absolute mean change in each panel. Energy Security: 28 cells (Nuclear Upstream, constant in the shipped release, is excluded; Geothermal Upstream and Midstream and Electric Grid Downstream have no final score; Green Hydrogen Midstream had no shipped one); Economic Opportunity: all 33 cells. A point far from zero horizontally but high vertically is a level shift that left the ordering largely intact. Sources: as in Figure 46.

Coalition members. Among the 57 members the rankable matched cells number 1,468 (ES) and 1,716 (EO), with correlations of 0.82 and 0.68, close to the full sample. Figure 48 and Table 35 show the twelve largest changes in each pillar, at most two per cell. The largest are Canada and Namibia in Nuclear Upstream, from zero to 0.74 and 0.72 in ES: the shipped cell held only the zero Investment category, and the Red Book places both among the leading holders and producers of uranium. The others trace the rebuilt themes: consumption (E5), the zero-filled Investment cells (E7), the new Upstream trade baskets, mirror trade for Gabon, which did not report imports in 2024, and the reviewed copper and aluminium basket for the United States in Electric Grid Upstream. Eight of the twelve ES changes and seven of the twelve EO changes are mainly composition effects, so they matter most for comparisons across one country's cells, which is what the Strategic Index makes.

Table 35: Largest pillar changes among coalition members and their attribution

Member	Cell	Shipped	Final	Composition effect	Largest score effect	Attributed cause
Energy Security						
Canada	Nuclear Upstream	0.000	0.741	+0.741	–	Zero Investment cell removed (E7); uranium Reserves and Production added (Red Book); Energy Prices added (A4); Trade added (T1–T5; all-reporter dyads)
Namibia	Nuclear Upstream	0.000	0.716	+0.716	–	Zero Investment cell removed (E7); uranium Reserves and Production added (Red Book); Energy Prices added (A4); Trade added (T1–T5; all-reporter dyads)
Germany	Nuclear Midstream	0.345	0.729	+0.286	Trade +0.099	Zero Investment cell removed (E7)
Croatia	Nuclear Midstream	0.288	0.665	+0.251	Trade +0.126	Zero Investment cell removed (E7) + Trade rebuilt (T1–T5; all-reporter dyads)
Turkiye	Oil Downstream	0.468	0.787	+0.000	Consumption +0.320	Consumption rebuilt (E5)
Thailand	Wind Upstream	0.112	0.415	+0.127	Production +0.123	Zero Investment cell removed (E7); Energy Prices added (A4); Trade added (T1–T5; all-reporter dyads) + Production rebuilt (E1, E3)
Ghana	Wind Upstream	0.018	0.296	+0.105	Minerals Trade +0.088	Zero Investment cell removed (E7); Energy Prices added (A4); Trade added (T1–T5; all-reporter dyads) + Minerals Trade observed, was imputed (E6)
Turkiye	Batteries Downstream	0.580	0.312	+0.000	Investment −0.165	Investment zero-filled (E7) + Consumption rebuilt (E5)
Gabon	Solar Upstream	0.000	0.263	+0.105	Foreign Dependency +0.086	Zero Investment cell removed (E7); Energy Prices added (A4); Trade added (T1–T5; all-reporter dyads, mirror imports) + Foreign Dependency rebuilt (E2–E3)
Oman	Solar Upstream	0.000	0.262	+0.090	Foreign Dependency +0.086	Zero Investment cell removed (E7); Energy Prices added (A4); Trade added (T1–T5; all-reporter dyads) + Foreign Dependency rebuilt (E2–E3)
Czech Republic	Batteries Downstream	0.580	0.331	+0.000	Investment −0.165	Investment zero-filled (E7) + Consumption rebuilt (E5)
New Zealand	Green Hydrogen Downstream	0.451	0.212	+0.000	Consumption −0.238	Consumption rebuilt (E5)

Table 35: continued

Member	Cell	Shipped	Final	Composition effect	Largest score effect	Attributed cause
Economic Opportunity						
Canada	Nuclear Upstream	0.398	0.647	+0.266	Technology Demand −0.015	Zero Investment cell removed (E7); Energy Prices added (A4); Production added (E1, E3; uranium, Red Book); Trade added (T1–T5; all-reporter dyads)
Namibia	Nuclear Upstream	0.379	0.616	+0.257	Cost Competitiveness −0.016	Zero Investment cell removed (E7); Energy Prices added (A4); Production added (E1, E3; uranium, Red Book); Trade added (T1–T5; all-reporter dyads)
Slovak Republic	Electric Vehicles Midstream	0.358	0.591	+0.000	Foreign Dependency +0.148	Spellings merged (C1); Foreign Dependency observed + Investment observed
Germany	Green Hydrogen Midstream	0.278	0.503	+0.066	Cost Competitiveness +0.098	Cost Competitiveness rebuilt (T6) + Foreign Dependency added (E8)
Portugal	Green Hydrogen Midstream	0.278	0.479	+0.070	Cost Competitiveness +0.118	Cost Competitiveness rebuilt (T6) + Foreign Dependency added (E8)
United States	Wind Upstream	0.397	0.584	+0.164	Production +0.064	Zero Investment cell removed (E7); Energy Prices added (A4); Trade added (T1–T5; all-reporter dyads)
United States	Electric Grid Upstream	0.427	0.608	+0.069	Trade +0.079	Trade rebuilt (T1–T5; all-reporter dyads, basket review) + Production rebuilt (E1, E3)
Viet Nam	Coal Downstream	0.465	0.640	+0.000	Production +0.052	Spellings merged (C1); Production observed + Consumption rescored
France	Nuclear Midstream	0.474	0.647	+0.124	Trade +0.034	Zero Investment cell removed (E7)
Colombia	Nuclear Midstream	0.340	0.511	+0.094	Trade +0.044	Zero Investment cell removed (E7)
Norway	Geothermal Midstream	0.434	0.602	+0.148	Cost Competitiveness +0.038	Zero Investment cell removed (E7)
Colombia	Geothermal Midstream	0.389	0.557	+0.134	Cost Competitiveness +0.050	Zero Investment cell removed (E7)

Notes: The 12 largest absolute changes, final minus shipped release, of ES_{its} and of EO_{its} among the 57 members of the final coalition, at most 2 per technology–stage cell, over the headline cells present in both releases that the final release ranks (sovereign economy, at most half of the index weight imputed); cells that were constant across countries in the shipped release are included. The change is split by a two-factor Shapley decomposition of the weighted category mean: the score effect of a category present in both releases is $\frac{1}{2}(\omega_k^{\text{shipped}} + \omega_k^{\text{final}})(c_k^{\text{final}} - c_k^{\text{shipped}})$, with ω_k the normalised category weight and an unobserved shipped category counted as $c_k = 0$ (item A1); the composition effect is the remainder, due to categories that enter or leave and to the renormalisation of the weights. The cause names the largest effect, and the second when it is at least half as large; item codes refer to Table 32. Data changes: "uranium, Red Book" = identified recoverable resources and mine production of the OECD-NEA/IAEA Red Book; "USGS MCS 2026" = critical-mineral reserves of the USGS Mineral Commodity Summaries 2026; "all-reporter dyads" = trade rebuilt from all-reporter Comtrade dyads (importer report first, else the exporter's declaration ×1.06) and the energy-trade file rebuilt from the same pull; "basket review" = a stage whose HS6 basket the October 2026 review changed; "mirror imports" = an economy that did not report 2024 imports, whose imports come from partners' declarations; "TRL by stage" = technology readiness mapped by stage from the IEA Clean Technology Guide. Source: `category_contributions` of the two `index_outputs.rds`.

Strategic Index. The shipped index scored 571 cells of 18 economies of its hand-picked list; the final one scores 3,154 cells of 174 countries in 30 sectors. On the 477 cells scored in both the correlation is 0.74, and across sectors within a country the median is 0.74 (Argentina 0.42 to Kenya

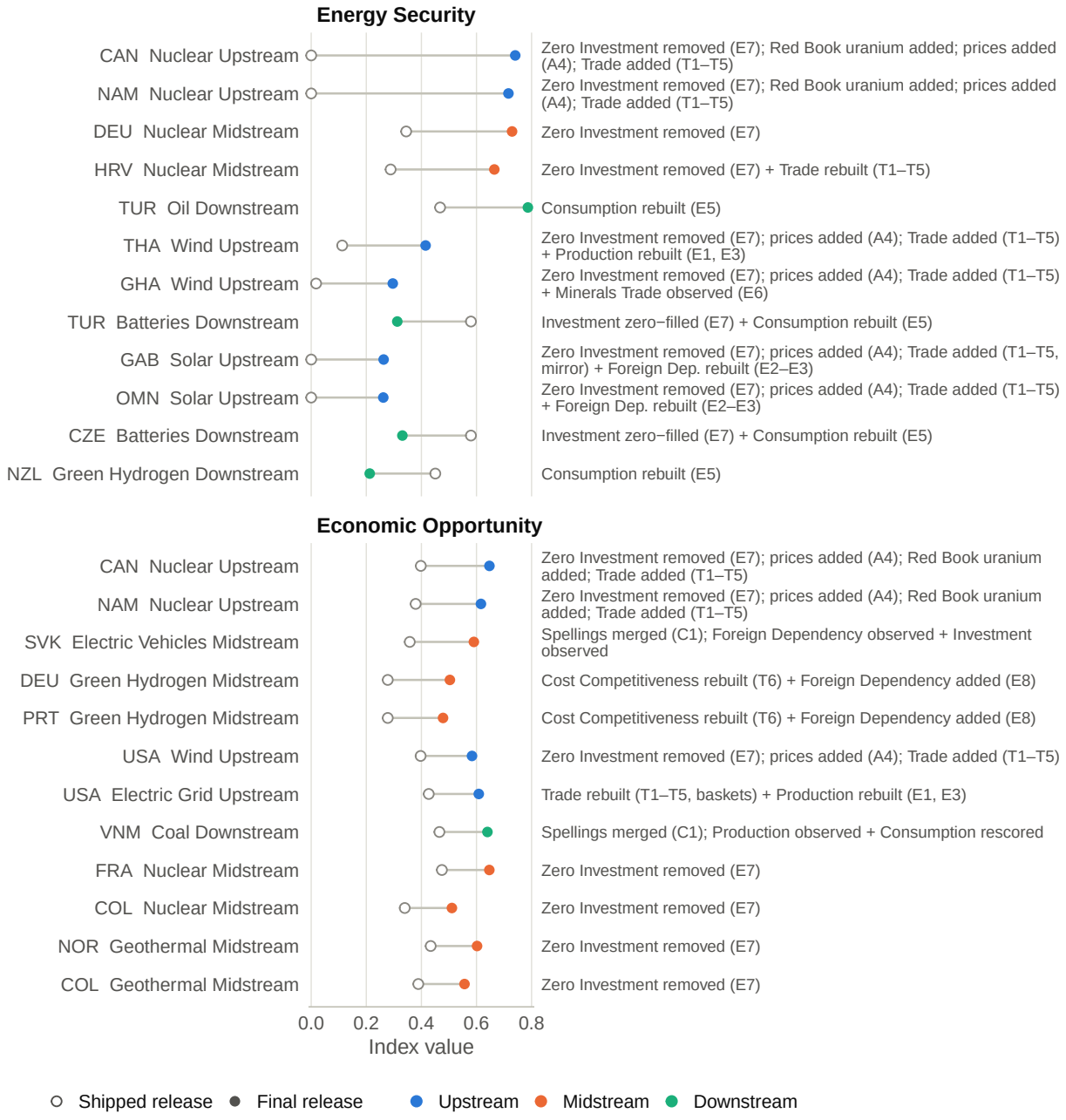


Figure 48: The twelve largest changes among coalition members, by pillar, at most two per technology–stage cell. Each row is one member's cell (ISO3 code). Hollow circles mark the shipped value and filled circles the final value, coloured by stage. The text at the right names the cause: the largest effect in a two-factor Shapley decomposition of the change into category score effects and a composition effect (categories entering or leaving and the renormalisation of weights), and the second when it is at least half as large, with the data change behind it ("Red Book uranium": OECD-NEA/IAEA resources and production; "mirror": trade measured from partners' declarations). Item codes refer to Table 32; the full decomposition is in Table 35. Sample: the 57 members of the final coalition, cells present in both releases and rankable in the final release (Energy Security in Nuclear Upstream, constant in the shipped release, is included as a mover). Sources: `category_contributions` of the two `index_outputs.rds`.

0.92). The policy term changed most. The shipped Policy index combined IEA PAMS, CAT, the February 2026 NIPO component and the dual-use score; the final one combines IEA PAMS and CAT, and all 18 economies have both. The within-country policy term correlates only 0.61 between releases and accounts for 40 percent of the summed absolute change, against 35 percent for security need, 22 percent for EO and 3 percent for the fixed weights, which differ only through the technology weights of oil (0.12 shipped, 0 final) and coal (0 and 0.02; item P9). In the final run 215 countries are policy-eligible (65 on IEA PAMS and CAT, 150 on IEA PAMS alone); all 23 economies of the shipped hand-picked list are among them, as are all 57 members of the derived coalition. Figure 49 splits the twelve largest changes into their terms. For the final release Geothermal Midstream is excluded by rule, as before; no sector is constant in either pillar, because Nuclear Upstream now varies with the Red Book data; and Electric Grid Downstream and Geothermal Upstream have no ES value, so the index covers 30 sectors.

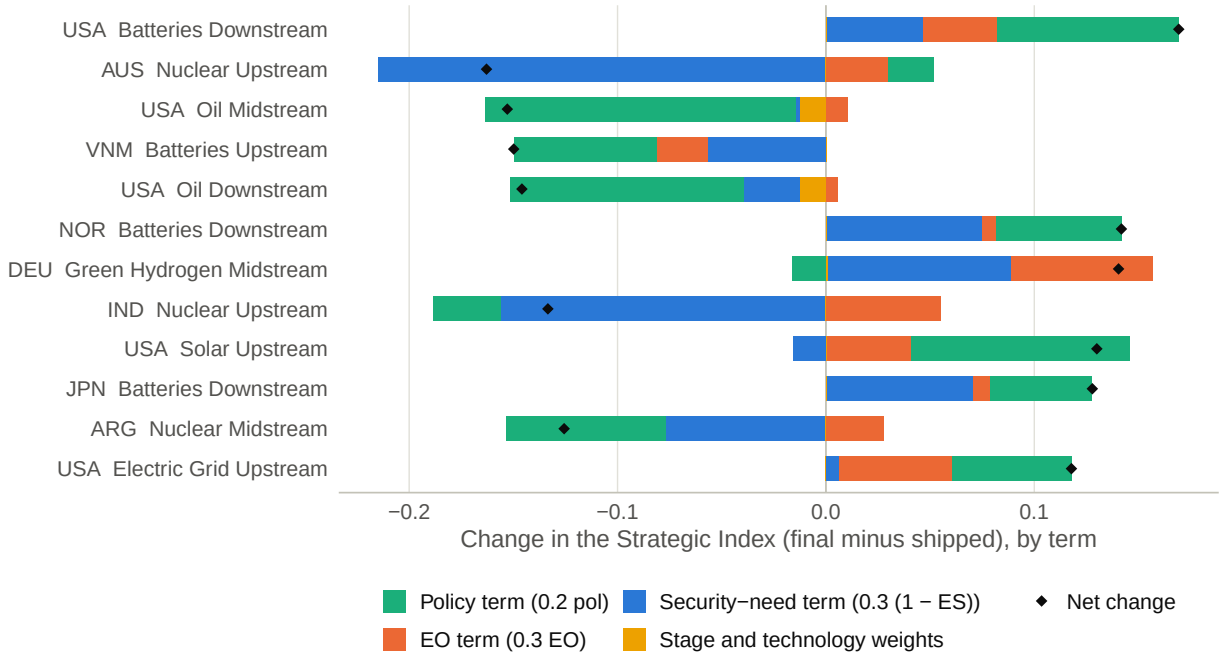


Figure 49: The twelve largest changes in the Strategic Index among the economies scored in both releases, split into the weighted terms of $\mathrm{SI} = 0.2\,\mathrm{POL} + 0.3\,\mathrm{EO} + 0.3\,(1 - \mathrm{ES}) + 0.1\,\lambda_s + 0.1\,g_t$. Each bar is one country–sector cell; the segments are the changes of the policy term, the EO term, the security-need term and the two fixed weights combined, and the diamond is the net change, which the segments sum to exactly. The shipped policy term is the within-country S-curve of the four-component Policy index of February 2026 (IEA PAMS, CAT, NIPO, dual use); the final one is that of IEA PAMS and CAT. The policy term is the largest contributor in seven of the twelve, security need in five. Sample: the 477 cells of the 18 economies of the shipped hand-picked list scored in both releases. Sources: `strategic_index` of the two `index_outputs.rds`; `scripts/paper/fig_reconcile.R`.

The allied picks. The shipped picks were made over a hand-picked list of 23 economies, the final ones over the 57 members of the derived coalition, so the comparison covers the 21 they

share (Table 36). Kenya and the United Arab Emirates are not members of the final coalition, and Denmark, France, Italy and South Korea had no shipped pick because they lacked a Policy index (P3). Of the 17 members with a pick in both releases, 16 change it; only Chile keeps Nuclear Upstream. Six changes (Australia, Brazil, Canada, India, Mexico and the United States) are reorderings by the security-need term: all six picked Nuclear Upstream where the shipped ES was zero for everyone, and the Red Book now records uranium for all six. Four (Argentina, Indonesia, South Africa, Viet Nam) are reorderings by the policy term, four (Germany, Japan, Morocco, Norway) are barred by the endowment screen, and two (Saudi Arabia, the United Kingdom) are no longer scored because the EO cell of the shipped pick is not rankable. On these members the picks move from Nuclear Upstream (8) and Batteries Upstream (5) to Batteries Midstream (13).

Network and partnership strength. The shipped network outputs are not comparable with the final ones (Table 37). The shipped diagnostics record a greedy solution in all 33 stages over 22 effective members, and in 14 stages a producer count of k^2 with a Herfindahl index of one, the signature of (90); the portfolio loop stopped at its limit with 12 members above their caps. The final design solves 22 stages over 37 nodes by MILP, with 7 to 10 producers per stage and a median designed Herfindahl index of 0.152, and the loop converges in seven passes. The shipped release did not write the PSI to disk; reconstructed from the March 2026 partner tables with the shipped logic, it correlates -0.09 with the final world view over 187 partners and -0.33 with the core view over 191. The denominator defect (B4) placed high-income partners below the others, so correcting it moves that group up as a block: the United States moves from 168th to 4th of the 187 in the world view and to 1st in the core view, and Germany from 135th to 5th. China moves from 22nd to 1st in the world view but to 44th in the core view, whose Friendshore scores rate China from the core members' viewpoints, alignment included.

Using numbers from the shipped release. Within a technology–stage cell, shipped pillar rankings are a rough guide in the Downstream (median within-cell correlations 0.90 for ES and 0.88 for EO) and in the EO Midstream (0.80), a poor guide in the ES Midstream (0.70) and the EO Upstream (0.61), and no guide for EO Green Hydrogen Midstream. Comparisons across cells, and with them the shipped Strategic Index and its picks, inherit the level shifts and a different Policy index; the shipped picks should be replaced by those of Section 9. The shipped network outputs and any PSI computed from the shipped partner tables should not be used. As the composite-indicator literature warns, the ordering a composite produces can depend on its construction as much as on its data (Saisana et al., 2005; Saltelli, 2007); here, however, the large changes come from defects and new data rather than from defensible alternative choices, so they are corrections, not a measure of uncertainty.

Table 36: Allied pick per member of the shipped list: shipped and final release

	Shipped release		Final release		
Member	Pick	SI	Pick	SI	Status or cause of change
Argentina	Electric Vehicles, Midstream	0.624	Batteries, Upstream	0.551	reordered by the policy term
Australia	Nuclear, Upstream	0.656	Batteries, Midstream	0.610	reordered by the security-need term
Brazil	Nuclear, Upstream	0.613	Batteries, Midstream	0.606	reordered by the security-need term
Canada	Nuclear, Upstream	0.638	Electric Grid, Midstream	0.615	reordered by the security-need term
Chile	Nuclear, Upstream	0.626	Nuclear, Upstream	0.645	unchanged
Denmark	–	–	Batteries, Midstream	0.636	new
France	–	–	Batteries, Midstream	0.642	new
Germany	Batteries, Upstream	0.712	Batteries, Midstream	0.623	shipped pick screened: no resource base
India	Nuclear, Upstream	0.643	Batteries, Midstream	0.628	reordered by the security-need term
Indonesia	Electric Vehicles, Midstream	0.642	Nuclear, Midstream	0.568	reordered by the policy term
Italy	–	–	Batteries, Midstream	0.639	new
Japan	Batteries, Upstream	0.657	Nuclear, Upstream	0.620	shipped pick screened: no resource base
Kenya	Batteries, Upstream	0.701	*not a member*	–	not in final coalition
Mexico	Nuclear, Upstream	0.624	Batteries, Midstream	0.564	reordered by the security-need term
Morocco	Solar, Upstream	0.683	Electric Grid, Midstream	0.609	shipped pick screened: no resource base
Norway	Nuclear, Upstream	0.729	Batteries, Upstream	0.618	shipped pick screened: no resource base
Saudi Arabia	Batteries, Upstream	0.688	Nuclear, Upstream	0.567	shipped pick not scored: EO not rankable
South Africa	Electric Grid, Midstream	0.679	Batteries, Midstream	0.620	reordered by the policy term
South Korea	–	–	Batteries, Midstream	0.635	new
United Arab Emirates	–	–	*not a member*	–	not in final coalition
United Kingdom	Batteries, Upstream	0.690	Batteries, Midstream	0.617	shipped pick not scored: EO not rankable
United States	Nuclear, Upstream	0.613	Batteries, Midstream	0.632	reordered by the security-need term
Viet Nam	Batteries, Upstream	0.698	Batteries, Midstream	0.606	reordered by the policy term

Notes: The pick is the member's Upstream or Midstream sector with the highest Strategic Index; Geothermal and Green Hydrogen are excluded by rule of the selection. Shipped release: the Strategic Index of 3 September 2026 over its hand-picked list of 23 economies, of which 18 had a pick. Final release: `data/processed/charts/allied_strategic.csv`, over the derived coalition of 57 members, with the endowment screen (an Upstream extraction stage is eligible only where the member has recorded reserves or production of its resources). The comparison is on the 21 members the two lists share; 2 economies of the shipped list are not in the final coalition. "new": a pick only in the final release. Cause of a change: the shipped pick is no longer scored (its exclusion), is barred by the endowment screen, or is outscored, in which case the term of the index that contributes most to the reversal of the two sectors' order is named (an exact split of the reversal into the five weighted terms). The picks of the other 36 final members are reported in Section 9.

Table 37: Allied network diagnostics: shipped and final release

	Shipped release	Final release
Network nodes	22	37
Stages designed	33 (all three stages)	22 (Upstream, Midstream)
Solver used (stages)	greedy (33)	MILP (22)
Stages with $n_{\text{producers}} = k^2$ and HHI $= 1$	14	0
Reported producers per stage	3–193,600	7–10 (median 9)
HHI of designed shares	0.333–1.000	0.126–0.209 (median 0.1520)
Diversification weight λ	–	0.78
Policy term in node scores	yes	no
Portfolio loop converged	no	yes
Portfolio passes	10 (limit)	7
Members above their portfolio cap at exit	12	0

Notes: Shipped release: `allied_network_diagnostics.csv` written on 23 February 2026, the only shipped network artefact read here; it designed every stage of the hand-picked list's 22 effective members (the United Arab Emirates was coded "UAE", item N5). In 14 shipped stages a single country was selected with k^2 rows of share $1/k$, so the producer count was k^2 (with k from 16 to 440) and the Herfindahl index summed over the rows was exactly one (item N1, equation (90)). Final release: the same file of the final run of 12 October 2026, 37 nodes (the EU-27 as one node), Upstream and Midstream stages only, MILP with the concave diversification term ($\lambda = 0.78$, the smallest value on the calibration grid whose median designed HHI does not exceed the observed one) and portfolio caps, and no policy term in the node scores; the median designed HHI compares with a median observed export HHI of the nodes of 0.1556 over the 18 stages with an HS6 basket. The portfolio loop runs over all stages jointly, so it has one outcome per release; the last row counts the members that, when the loop stops, are a top producer in more stages than their cap allows.

Online Appendix

Opportunity, Security, and the Allied Industrial Base

Further detail, derivations and diagnostics

OA.1 Data: releases, harmonisation and years

The release rule. Inputs that are published in successive releases are staged under a file-name pattern, and the run selects one release by rule rather than by hand. Let F_P be the staged files matching the pattern of input P, $k(f)$ the vintage key parsed from a file's name (a date or a year), $m(f)$ its modification time and $R(f)$ the number of distinct reporters in a tabular extract. The release used is

$$f_P^* = \operatorname*{lex\,max}_{f \in F_P:\ R(f) \geq \underline{R}_P} \big(k(f),\, m(f)\big), \tag{OA.1}$$

the newest declared vintage, with the modification time breaking ties and ordering unversioned names, among the files that meet the coverage floor $\underline{R}_P$. The floor is 50 reporters for the Comtrade critical-minerals extracts and absent elsewhere. Comtrade publishes a year's data as reporters file, so the newest file can be a stub: the 2025 critical-minerals extracts have three reporters each and are skipped in favour of the 2024 extracts (130 import, 128 export and 130 total-export reporters). Every run writes the releases it resolved, with their reporter counts, to a vintage stamp (`data/raw/raw_inputs_resolved.yml`); Appendix B reports them. Inputs that are not pattern-resolved (the Comtrade dyads, the Red Book and USGS tables, the EIA bulk file, CM-FDI, the ideal points and the project-authored lists) are single staged files whose access dates are recorded in the input manifest.

Harmonisation of economies. Every source is put on one canonical name per economy before anything is ranked: the World Development Indicators name of the economy's ISO3 code, matched through the `countrycode` package and a curated table of spellings. Taiwan, which the World Bank's country table does not list, is added under its own code, including to the rosters of the trade themes; Kosovo is carried as XKX. World, regional, income-group and residual rows ("Other Asia Pacific", "Rest of world") and historical entities are dropped from every theme table, so that no aggregate is imputed or averaged as if it were an economy; theme builders also remove aggregates from their comparison sets before ranking, while keeping them in world denominators where the source defines the world total. Keys that two spellings of one economy share are collapsed, with a warning if both carry different non-missing values. Sovereignty, which the rankability rule uses, follows the Correlates of War state-system list (Correlates of War Project, 2024): 195 states in 2024, the 193 UN members plus Kosovo and Taiwan.

Fossil production from EIA. The Statistical Review lists fewer producers than consumers and counts the rest in regional "Other" residuals. Where it has no production row for an economy and fuel, the value is taken from the EIA International Energy Statistics (U.S. Energy Information Administration, 2026) (bulk file of 3 October 2026) in the Review's units: oil as crude oil including lease condensate plus natural-gas plant liquids, in thousand barrels a day; gas as dry gas in billion cubic metres times 0.036 EJ; coal as terajoules (gross calorific value) times 0.95×10^{-6} EJ. On economies both sources report, the median ratio of the Review's value to the converted EIA value is 1.006 to 1.011 for oil, 0.97 to 0.99 for gas and 0.99 to 1.05 for coal. An EIA value is not used when EIA's consumption of the fuel covers less than a tenth of the Review's, since the Review then counts under the fuel something EIA excludes; in 2015–2025 this affects only Estonian coal (oil shale), which stays missing. Where EIA's latest production year precedes the Review's consumption year (2025 gas and coal from EIA 2024), self-sufficiency is computed with consumption of the same year and carried forward. EIA values are observations, not imputations.

Reserves and uranium. The USGS Mineral Commodity Summaries 2026 data release (U.S. Geological Survey, 2026) gives country reserves at end-2025; the report's version 1.3 erratum puts Brazil's rare-earth reserves at 11 Mt (the Statistical Review's mineral sheet, an older USGS copy, gives 21 Mt). A reported zero scores 0; a holder listed without a figure (NA, W or s) is missing in its mineral variable, scored 0 and counted as imputed, and counts as zero in the technology roll-ups, so world shares are measured on quantified holdings. "Other countries" and world-total rows are never holders. The Red Book (OECD Nuclear Energy Agency and International Atomic Energy Agency, 2026) supplies identified recoverable resources (reasonably assured plus inferred) below USD 260/kgU at 1 January 2025 (Table 1.2a of *Uranium 2026*) and mine production (Table 1.18; *Uranium 2026* for 2022–2024, earlier editions for 2019–2021), completed with observed zeros over the economies of the Statistical Review and the World Development Indicators.

Bilateral trade. The all-reporter dyad file was pulled on 7–8 October 2026 for the pre-review product list, with supplements on 9 October (the copper and aluminium inputs of the grid) and 12 October (the new upstream codes). Comtrade's partner S19 ("Other Asia, n.e.s.") is mapped to Taiwan. Liberia, Mali and São Tomé and Príncipe filed 2024 data after the pull and enter through their partners' declarations. The country-level energy-trade file is derived from the pull's World rows; an earlier direct extract, with 129–130 reporters, lacked 60 basket codes and Qatar's exports of liquefied natural gas. Critical-minerals exports of non-reporting economies are measured from partners' imports, moved from CIF to FOB with an HS-chapter ratio estimated on importers that file both values.

Regional and staged inputs. The IEA's 2035 announced manufacturing-capacity shares and its electric-vehicle production and sales are regional for Europe, so EU members carry the EU's value; economies outside the IEA's named regions take its "Rest of world" values. ILOSTAT earnings are read from a copy staged on 30 September 2026, since the ILO's interface returns empty responses

Table OA.1: Years of the inputs in the final run

Source	Years used
EI Statistical Review 2026	Consumption and production to 2025 (growth 2020–2025); fossil reserves end-2020
EIA International Energy Statistics	Oil production to 2025; gas and coal to 2024
USGS Mineral Commodity Summaries 2026	Reserves end-2025
Red Book (Uranium 2026)	Resources at 1 January 2025; production 2019–2024
IEA Critical Minerals Dataset 2026	Base year and 2035 (Stated Policies)
IEA ETP 2024 capacity; IEA EV regional data	Announced capacity to 2035; EV production and sales 2024
UN Comtrade dyads; energy-trade file	2020 and 2024; 2024
Comtrade critical-minerals extracts	2024 (2025 below the floor)
Atlas of Economic Complexity (HS92)	2023
IEA WEO 2025; Global EV Outlook 2026	2024 and 2035; 2022, 2025 and 2035
BNEF New Energy Outlook 2024; Supply Chains 2025	2024 and 2035; 2025
IEA Clean Energy Technology Guide 2026	Readiness 2020–2025
BNEF LCOE; IEA relative costs	2025 and 2050; 2023
ILOSTAT; IMF lending rates, PPI, PCPS	Latest within ten years; to October, September and December 2025
Clean Investment Monitor	Investment 2018–2025; capacity 2025
IEA PAMS; Climate Action Tracker	Export of 22 January 2026; ratings of May 2022–June 2025
IMF Direct Investment Positions; CM-FDI	Mean of 2020–2024; edition of 5 October 2026
WDI; Doing Business; OECD CRS	2022–2024 (GDP to 2025); 2019; 2013–2024
UNGA ideal points (Dataverse V39)	Sessions 74–78 (2019–2023)

Notes: Each component enters at the latest year its source reports (equation (13)). Appendix B gives the releases.

to scripted requests. In the IMF lending-rate series a zero reading in a lending-type rate is a placeholder and is dropped with the averages that include it: New Zealand's lending series carries only zeros after January 2017, so its last genuine reading (4.76 percent) is used. Commodity prices are in the units of the IMF series metadata.

Index vintages. The headline indices are a single snapshot. A separate build re-cuts each raw input to a requested year and reruns the same theme and pillar builders, so that the pillars can be compared across years (Section 12). Only sources with a genuine time dimension move: the Statistical Review, the Atlas trade data, IMF prices and rates, ILOSTAT earnings and Clean Investment Monitor investment. Resource potential, reserves, the IEA critical minerals release, BloombergNEF demand and costs, overcapacity and technology readiness are snapshots or projections and are held fixed; a 2025 vintage uses Atlas 2023. The build records, for each vintage, the share of each pillar's scored weight that re-derives and the year to which each source resolved.

Algorithm OA.1 Construction of a pillar index $P \in \{\mathrm{ES}, \mathrm{EO}\}$ from its theme tables

Require: theme tables $h \in \mathcal{H}_P$; category weights w_k (`config/weights.yml`); missing-data rules $m_h \in \{\text{zero}, \text{global average}\}$ (`config/missing_data.yml`); for each category its score variable and components $\mathcal{V}_k$ (`config/index_definition.yml`)

Bind the theme tables; average rows that differ only by sub-sector; keep rows whose data type is *index*
For each (i, t, s, k, v, h) keep the observation with the latest year ▷ eq. (13)
for each technology–stage pair (t, s) **do**
    Complete the grid of economies × component groups
    for each group $g = (t, s, k, v, h)$ **do**
        if at least one economy is observed in g and theme h has a rule **then**
            Fill the missing economies of g by rule m_h from the observed values ▷ eq. (14)
        end if
    end for
end for
Average each component over themes; score each category by the mean of its present components ▷ eq. (15)
Aggregate the observed categories of each cell with renormalised weights ▷ eq. (16)
Record category and variable contributions, ι_{its} and the pair status; flag rankable cells ▷ eqs. (19)–(22)

Ensure: $P_{its} \in [0, 1]$, or missing when no category of the cell is observed

OA.2 Index construction: derivations and further diagnostics

This appendix collects the construction algorithm, the properties of the S-curve that Section 4 uses without proof, the conventions of the theme and aggregation layers, and the diagnostics of the final build that the main text summarises.

Theme tables and orientation. Every theme table has one row per economy, technology, stage, sub-sector, category, variable, data type and year, with its value, source and a short explanation. Rows of type *raw* keep a metric in natural units for tracing; only rows of type *index*, on $[0, 1]$, enter the pillars. Rows that differ only by sub-sector are averaged with equal weights before any other step. Each variable is oriented in its theme: concentration enters through $-$HHI, a trade deficit relative to GDP, a levelised cost and price volatility through their negatives. By the symmetry of ϕ_γ, the S-curve of $-x$ is one minus the S-curve of x for any comparison set with at least two observations, so orientation and transformation commute. The opportunity builder verifies that every category score lies in $[0, 1]$.

The S-curve. Differentiating equation (10),

$$\phi'_\gamma(r) = \frac{\gamma\,[r(1-r)]^{\gamma-1}}{\left[r^\gamma + (1-r)^\gamma\right]^2}, \qquad \phi'_\gamma(1/2) = \gamma, \tag{OA.2}$$

which for $\gamma < 1$ diverges as $r \to 0$ or $r \to 1$. With $\gamma = 0.5$ the quartiles map to 0.366 and 0.634, so near the median a step in rank moves the index by half as much as it moves the percent rank, while gaps between adjacent units are widened in the tails. In levels, $\phi_\gamma(r) > r$ below the median and $\phi_\gamma(r) < r$ above it. With n untied observations the index takes exactly the values $\phi_\gamma(k/(n-1))$, $k = 0, \ldots, n-1$, whatever the raw distribution, so every untied S-curved variable has mean $1/2$ and a dispersion that depends only on n and γ: the population standard deviation is 0.260 for $n = 10$, 0.224 for $n = 20$ and 0.206 for $n = 40$, and tends to 0.19, against 0.29 for the percent rank and 0.38 for $\gamma = 2$. Components ranked over comparison sets of a given size therefore enter a category on a common scale; smaller sets have larger spreads, and ties, the zero anchor and imputed values move a component away from the grid. The smallest step above zero, $\phi_\gamma(1/(n-1))$, is coarse in small sets (0.261 for $n = 10$, 0.140 for $n = 40$). A single observation scores 0, two score 0 and 1, and a fully tied set scores $1/2$. In the simulated lognormal sample of Figure 4, min-max rescaling puts 29 of 40 observations below 0.1 and spreads the top four over 0.79 of the scale; the percent rank gives each group of four a spread of 0.077; the S-curve gives the bottom and top four 0.22 each and the middle four 0.04. Rank-based normalisation is a standard option for composite indicators (Nardo et al., 2008); it is adopted because the inputs differ in units and skewness by orders of magnitude and the questions the indices answer are comparative. Min-max or z-score normalisation would let a few very large values compress the rest of the distribution.

The zero anchor. Equation (12) is equation (10) applied to the positive values with a single zero prepended to the comparison set and then removed. An observed zero and an unrecorded economy both score 0, and the smallest positive value scores $\phi_\gamma(1/m) > 0$: for $(0, 2, 5, 9)$ the positive entries score 0.414, 0.586 and 1. A set in which every value is zero scores 0 throughout.

Comparison sets and cardinal exceptions. Most themes rank the economies observed in a pair, sometimes split by sub-sector, year or mineral before averaging to the pair. World-level series are ranked across the units over which they vary: commodity-price volatility across technologies, world export concentration across the cells of the eleven headline technologies, and global demand across fuels. A theme that builds a composite of several S-curved terms averages them and may pass the average through the S-curve again. Index values are positions in a cross-section: adding or removing economies changes the scores of the others, and a change over time records a change in standing against the field. The three cardinal EO inputs are the readiness score (a raised-cosine bell in the readiness level, zero at levels 2 and 11 and peaking at 6, blended 0.7:0.3 with a capped concave score of its recent increase), the overcapacity term $2^{-\rho}$ in BloombergNEF's ratio ρ of capacity to forecast demand, and the capital-cost term of the input-cost index, a lending rate winsorised at its 5th and 95th cross-country percentiles and rescaled by min-max. They keep interval information and do not share the fixed distribution of an S-curved variable; the first two are technology-level values.

Latest year. The selection of equation (13) is made on the year field, with undated rows at year 0, separately for each economy and variable. A row recorded for a later year with a missing value therefore supersedes an earlier observed value. In ES the mixed years lie between 2023 and 2025, apart from the 2035 horizon of the announced manufacturing-capacity shares. In EO part of the mixing reflects the projection horizons of the outlook series; counting only years up to 2026, 91 percent of EO cells still combine more than one vintage, with at most four years and a median spread of three among the cells that mix. The extreme case is the input-cost index of Papua New Guinea, whose latest observation is for 1980 and which enters all 33 of its EO cells through Cost Competitiveness.

The grid and the rules. Because a pair's grid is the union of the economies of all the pillar's themes for the pair, one broadly distributed theme sets its extent: the commodity-price series is copied to every economy of the World Bank list at the upstream stage, so every such economy enters the upstream grid of each technology with a price series and is scored there on its other components, observed or filled. Rows of categories a pillar does not score cannot add economies to its grid. Every one of the 219 economies in the upstream grids has a country-specific upstream component in at least one upstream pair of each pillar. When a single economy is observed in a group, the global-average rule copies its value to every other economy: this is how the Boston Consulting Group estimate of the U.S. serviceable addressable market, one value per pair, enters EO Technology Demand, flagged as imputed for every economy but the United States. Because the average is taken after the latest-year selection, it can combine observations of different years. The zero rule equates "not recorded" with "none" and so understates the position of an economy an incomplete register omits.

Overall variables. A component reported by several themes (energy consumption per capita, for example) is averaged across them with equal weights before the category mean. Some component lists follow a one-vote-per-source principle: where a source enters through a composite of its own terms, the composite is one component. Two checks guard the mapping from configuration to data: the builder warns when none of a category's listed components occurs in the inputs, and it discards any theme row that carries the name of a configured overall variable, so the configured definition always governs. A configuration test requires the weighted categories of each pillar to coincide with its configured categories.

Why not zero-fill. Replacing each unobserved category by the cell's available-case mean and averaging over all categories with the nominal weights returns ES_{its} unchanged. Scoring an unobserved category as zero while keeping its weight in the denominator gives instead

$$\mathrm{ES}^{\mathrm{zero}}_{its} = \mathrm{ES}_{its} \cdot \frac{\sum_{k \in K_{its}} w_k}{\sum_{k \in \mathcal{K}^P} w_k}, \tag{OA.3}$$

which multiplies the index by the observed share of the nominal weight. An earlier version of the code behaved in this way for categories present as rows but unobserved; Appendix D documents the correction. Since no cell observes all ten categories, listwise deletion at the category level would leave every cell unscored.

Category sets. Figure OA.1 shows the categories that enter each pillar in each headline pair. The 30 ES pairs use 16 distinct category sets and the 33 EO pairs 18; no pair observes all ten categories of either pillar. The variation is structural: only Trade, Foreign Dependency and Investment exist at the midstream stage of ES, and only Consumption, Energy Access, Investment and, for the three renewables with a resource-potential series, Reserves downstream. In seven ES pairs the pillar is a single category (Trade in the coal, gas, nuclear and oil midstream pairs; Foreign Dependency in Green Hydrogen midstream; Investment in Electric Vehicles downstream; Consumption in Green Hydrogen downstream). Nuclear upstream is scored on Production, Reserves, Trade and Energy Prices; its Trade category, like those of the Electric Vehicles, Solar and Wind upstream pairs in both pillars, comes from the new upstream baskets. The narrowest EO pair, Electric Grid downstream, has only Cost Competitiveness and Technological Readiness. Energy Prices is constant within the ten upstream pairs with a price series, Technological Readiness within all 33 EO pairs, and Technology Demand within five EO pairs (Batteries upstream and midstream, Geothermal at all three stages).

Effective weights and constant inputs. Averaging effective weights over cells instead of pairs changes the stage means by at most 0.032, because the category set is common within a pair; the largest gap between effective and nominal share is 0.49. Upstream ES weight is spread more evenly, led by Reserves (0.20), Trade (0.18) and Production (0.18); Energy Prices, nominally 0.048, carries 0.10. In EO the redistribution favours categories observed at every stage: Technology Demand (0.20 to 0.25, nominal 0.133), Cost Competitiveness (0.19 to 0.24, nominal 0.100) and Technological Readiness (0.12 to 0.16, nominal 0.067). Categories constant within the pair take a mean of 0.34 of the weight of an upstream EO cell, 0.22 at the midstream stage and 0.18 downstream; adding the constant components inside varying categories (world export concentration in Trade; the addressable-market, global-demand and overcapacity terms of Technology Demand) raises these shares to 0.46, 0.39 and 0.27. The corresponding component-level shares in ES are 0.16, 0.20 and 0, the midstream figure being the world export-concentration term of Trade. Energy Prices is copied to every economy before ranking, so the economies of a technology form one block of tied ranks: with T technologies the technology in position p receives a percent rank close to $(p - \frac{1}{2})/T$ rather than $(p-1)/(T-1)$, and no technology scores exactly 0 or 1.

Imputation burden by category. Table OA.2 gives the values behind Figure 6. Upstream, Reserves (0.84) alone accounts for 0.17 of the ES stage mean of 0.41; Energy Imports (0.55), Foreign Dependency (0.50) and Production (0.43) also cover only part of the grid, and the upstream Trade category is imputed at 0.19. In EO, Technology Demand is imputed at 0.73 to 0.86 and contributes

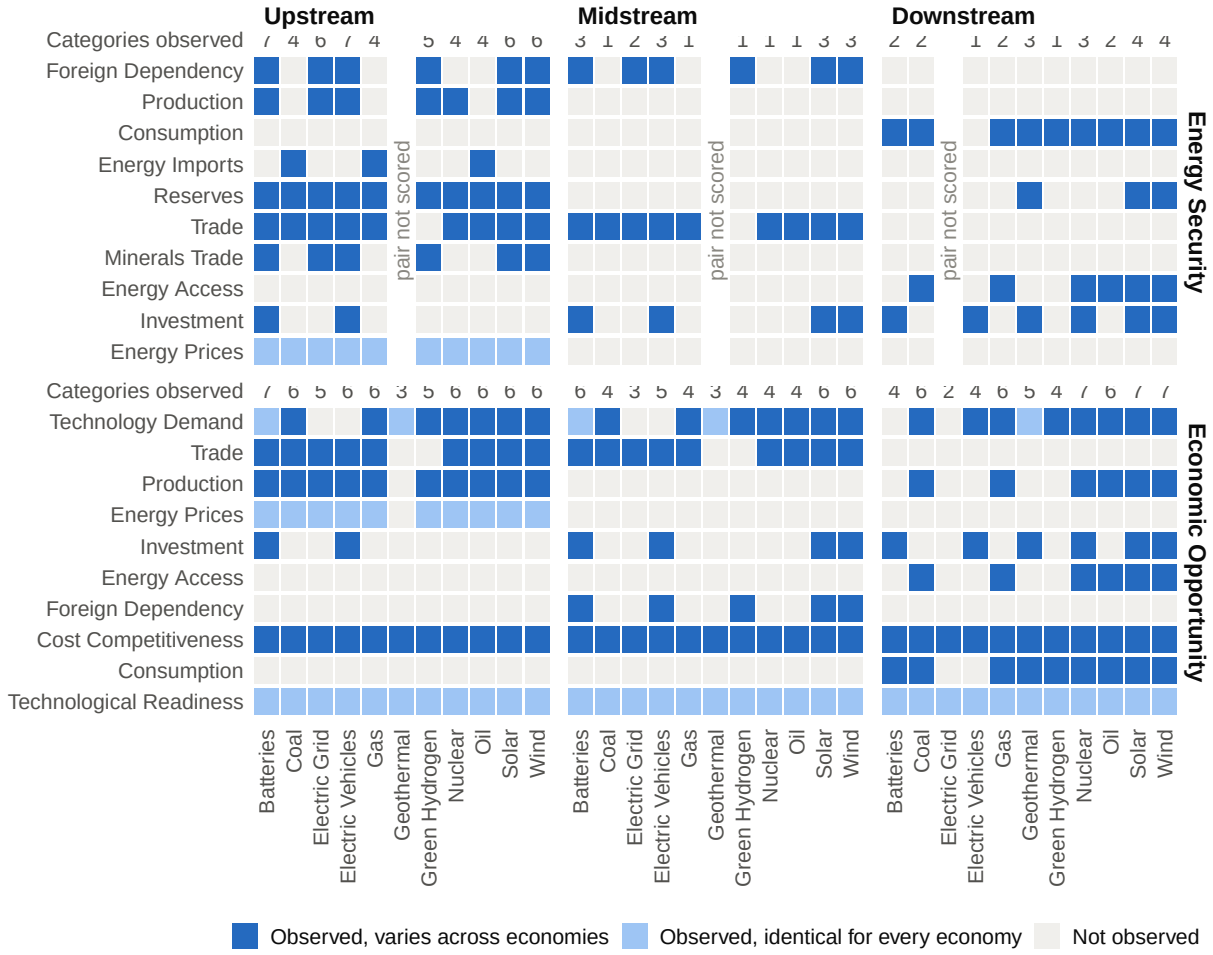


Figure OA.1: Category composition of the pillars by technology–stage pair. Each column is one of the 33 headline technology–stage pairs, grouped by stage. Each row is a category of the Energy Security pillar (upper block) or the Economic Opportunity pillar (lower block), ordered by nominal weight as in Table 7. Dark blue: the category is observed in the pair and its score varies across economies. Light blue: the category is observed but takes the same score for every economy of the pair (a world- or technology-level input). Gray: the category is not observed in the pair. The number above each column is the number of observed categories, which is the same for every economy of the pair. "Pair not scored" marks the three pairs without an ES value. Sample: the country–technology–stage cells of the eleven headline technologies that have a pillar value (6,232 for ES, 7,131 for EO). Source: category contributions in `data/processed/outputs/index_outputs.rds`; `scripts/paper/fig_method.R`.

Table OA.2: Mean imputed share by category, stage and pillar, headline technologies

	Energy Security			Economic Opportunity		
Category	Up	Mid	Down	Up	Mid	Down
Consumption	–	–	0.03	–	–	0.05
Cost Competitiveness	n.a.	n.a.	n.a.	0.34	0.42	0.36
Energy Access	–	–	0.65	–	–	0.66
Energy Imports	0.55	–	–	n.a.	n.a.	n.a.
Energy Prices	0.01	–	–	0.01	–	–
Foreign Dependency	0.50	0.54	–	–	0.79	–
Investment	0.83	0.78	0.64	0.83	0.78	0.66
Minerals Trade	0.02	–	–	n.a.	n.a.	n.a.
Production	0.43	–	–	0.68	–	0.55
Reserves	0.84	–	0.10	n.a.	n.a.	n.a.
Technological Readiness	n.a.	n.a.	n.a.	0.04	0.02	0.03
Technology Demand	n.a.	n.a.	n.a.	0.86	0.73	0.83
Trade	0.19	0.01	–	0.26	0.15	–
Pillar, $\bar{\iota}$	0.41	0.23	0.27	0.41	0.44	0.43
Cells	2,181	2,002	2,049	2,394	2,346	2,391

Notes: Category rows give the mean, over the country-technology-stage cells of the eleven headline technologies in which the category is observed, of the share of the category score carried by imputed components, $\sum_v \lambda_{vkits} \mathbf{1}[\tilde{x}_{itsv}$ imputed$]$. The pillar row gives the mean over all scored cells of the stage of the imputed weight share ι_{its} of equation (21). –: category not observed at that stage; n.a.: category not part of the pillar. Source: `data/processed/outputs/index_outputs.rds` (variable contributions and the pillar diagnostics); `scripts/paper/fig_method.R`.

0.16 to 0.19 to each stage mean: the addressable-market term counts as imputed for every economy but the United States, the regional outlook is observed for a median of 12 economies per pair, and the world-level global-demand series is recorded for 108 economies and filled with the same value for the rest. Investment is imputed at 0.64 to 0.83 in both pillars. World-level series show almost no imputation (Energy Prices below 0.01, Technological Readiness at most 0.04), which is why ι_{its} records how a value was obtained rather than whether it is country-specific, and why the rankability rule can admit a cell whose observed weight is largely world-level. A component averaged across two themes counts as imputed if either value was, so ι_{its} is an upper bound there. At the other extreme, 2,055 ES cells (33 percent) and 407 EO cells (6 percent) contain no imputed component. Rebuilding ι_{its} from the variable table reproduces the builders' value to within 1.1×10^{-16}.

Mass points and pair status. Economies without country-specific data share one value within their pair: in five EO pairs (Electric Grid downstream, Green Hydrogen upstream and midstream, Geothermal upstream and midstream) 65 to 70 economies sit at a single interior value, and none of these cells is rankable. Observed zeros create mass points that the rule does not remove: in ES, 98 of the 109 economies scored in Green Hydrogen midstream share the pair's minimum (0.256), 95 of them rankable, and in Nuclear upstream 127 of 218 share one value (0.172), 105 of them rankable. Such a tie records a real absence of production or resources but becomes a uniform, sector-wide

Table OA.3: Rankable cells by pillar and stage, headline technologies

	Scored cells	Rankable cells	Not rankable		Share rankable	Economies ranked
			Territory	$\iota_{its} > 0.5$		
Energy Security						
Upstream	2,181	1,124	231	826	0.52	195
Midstream	2,002	1,397	169	436	0.70	192
Downstream	2,049	1,642	207	200	0.80	193
All stages	6,232	4,163	607	1,462	0.67	195
Economic Opportunity						
Upstream	2,394	1,681	250	463	0.70	178
Midstream	2,346	1,307	221	818	0.56	166
Downstream	2,391	1,313	247	831	0.55	169
All stages	7,131	4,301	718	2,112	0.60	178

Notes: Country–technology–stage cells of the eleven headline technologies with a pillar value. A cell is rankable when the economy is a sovereign state (a Correlates of War state-system code) and at most $\tau = 0.5$ of its index weight rests on imputed components, $\iota_{its} \leq \tau$ (`config/rankability.yml`). Reasons are exclusive: a territory is counted as a territory whatever its imputed share; the next column counts sovereign cells over the threshold. Economies ranked: economies with at least one rankable cell. Source: `data/processed/outputs/index_outputs.rds`; `scripts/paper/fig_method.R`.

need under the security-need reading of the Strategic Index. A pair is *unscored* if no economy has an observed category, *constant* if at least two are scored and their range is at most 10^{-12}, and *ok* otherwise. Over every technology, including those outside the headline set, all 45 ES pairs and all 40 EO pairs are ok; Nuclear upstream, constant before the uranium data were added, now varies. Electric Grid downstream and Geothermal upstream and midstream have no observed ES category.

Rankability in detail. Table OA.3 gives the counts behind Figure 7. Counted over sovereign cells only, the rule keeps 74 percent of ES and 67 percent of EO cells. Rankable sovereign cells score higher on average than sovereign cells over the threshold in every pillar and stage, most markedly in midstream ES (0.46 against 0.30) and midstream EO (0.49 against 0.37). The narrowest pairs keep a usable comparison set: Green Hydrogen midstream keeps 40 rankable EO cells (19 percent of those scored), and Nuclear downstream, the narrowest ES pair, keeps 68. The 17 sovereign states without a rankable EO cell are Liechtenstein, Monaco, Djibouti, Dominica, Eritrea, Haiti, Kiribati, the Marshall Islands, Micronesia, Nauru, North Korea, Palau, São Tomé and Príncipe, Somalia, St Kitts and Nevis, St Vincent and the Grenadines and Tuvalu; their only rankable EO cells had been upstream, and the upstream Trade category, whose export-size and comparative-advantage terms are imputed for economies absent from the Comtrade reporter file, lifted their imputed share just over the threshold. Taiwan keeps 19 rankable ES and 18 rankable EO cells. In the Strategic Index the 29 policy-scored economies that lose every cell are nine territories and special administrative regions (among them Hong Kong, Macau and the West Bank and Gaza) and 20 sovereign states: the 17 above, Andorra, the Central African Republic and Grenada. The pillars are built before the rule is applied, so excluding a cell does not change the ranking of the cells that remain.

OA.3 Energy Security: further results

Rankability. The rankability rule keeps 4,163 of the 6,232 scored cells. It removes the 607 cells of the 24 territories and 1,462 cells of sovereign economies whose score rests mainly on imputation. Among the rankable cells the mean imputed weight share falls to 0.18 upstream, 0.08 midstream and 0.22 downstream. A pair holds between 68 rankable economies (Nuclear downstream) and 195 (Nuclear upstream). Within each pair every scored economy observes the same set of categories. The upstream trade baskets give Trade categories to Electric Vehicles, Wind, Solar and Nuclear upstream, so nine of the ten upstream pairs carry one; Green Hydrogen upstream is the exception.

Distribution by pair. In Electric Grid, Solar and Green Hydrogen upstream the interquartile range is narrower than 0.05. Upstream pair medians run from 0.17 in nuclear and 0.20 in batteries and electric vehicles to 0.47 in oil. Midstream and downstream the pillar rests mainly on median-centred S-curves; the four midstream pairs scored on Trade alone (gas, coal, oil and nuclear) have medians between 0.48 and 0.56. Green Hydrogen midstream, scored on Foreign Dependency alone, gives 90 per cent of its 106 rankable economies the value 0.26; Green Hydrogen downstream, scored on Consumption alone, is bimodal (interquartile range 0.10 to 0.72). Within stages, technology-stage effects explain 33, 50 and 22 per cent of the upstream, midstream and downstream variance, and country effects 51, 65 and 29 per cent of the within-pair remainder. Subtracting the contributions of Energy Prices and of the world-concentration terms of Trade and Minerals Trade leaves the pair-effect R^2 at 48 per cent.

Drivers. Investment, defined midstream only for batteries, electric vehicles, solar and wind, supplies 1.4 per cent of the midstream score against a mean effective weight of 7.0 per cent, because the investment monitor records neither manufacturing investment nor capacity for most economies (72 per cent of rankable economies score zero in the median of these pairs). Downstream, resource potential supplies 10 per cent and Investment 9.8 per cent against a weight of 19 per cent. The concentration term of Minerals Trade is constant for all economies except Taiwan and Réunion, which lie outside the World Bank list. Energy Prices scores range from 0.19 for gas to 0.81 for the electric grid, raising every economy's upstream score by between 0.025 (batteries) and 0.088 (nuclear) without changing any within-pair rank.

Correlation among categories. The median within-pair Spearman correlation between two categories is 0.42 upstream (one of 78 pair-specific correlations is negative, -0.01), 0.33 midstream and 0.11 downstream. Reserves and Production correlate at a median of 0.84 over the seven pairs that carry both, and at 0.87 among the economies with a positive score on both (a median of 16 per pair). Foreign Dependency correlates with Production at 0.76 and with Reserves at 0.69. Midstream, Foreign Dependency and Trade correlate at only 0.28 (range -0.23 to 0.48): economies with large shares of refined supply and manufacturing capacity are not always those with large export shares and small net imports. Downstream, Energy Access and Consumption correlate at

0.12 and resource potential correlates negatively with Consumption (−0.11). Within a pair the pillar's ranking follows Energy Imports (median Spearman correlation with ES_{its} of 0.80), Minerals Trade (0.79) and Reserves (0.79) upstream, Trade (0.94) midstream, and Investment (0.76) and Consumption (0.74) downstream.

Uranium. An economy with neither resources nor production scores the Energy Prices contribution, $0.704/8 = 0.088$, plus its Trade category. For the 105 economies whose uranium trade is not observed, Trade is imputed and the score is 0.172 (shared rank 67); the other 32 economies without uranium score between 0.145 (Trinidad and Tobago) and 0.201 (Norway). Their imputed weight shares lie between one quarter and one half, so all remain rankable. Producers dominate the top, though not all of it: Türkiye, which holds resources but reports no mine output, ranks 18th (0.34), and the lowest-ranked producer is 26th. Nine of the 15 highest-ranked economies are coalition members. Among the 26 members without identified resources are France, the United Kingdom, South Korea, the Netherlands, Norway and Poland.

Leaders. China's eleven first places are Batteries, Electric Vehicles and Wind upstream, where its shares of world mineral supply, production and trade dominate; Batteries, Electric Grid, Electric Vehicles, Solar and Wind midstream; and Batteries, Nuclear and Wind downstream. South Africa leads green hydrogen upstream, whose mineral weights are dominated by the platinum-group metals, and Chile the grid and solar upstream, in which copper carries most of the demand weight; in Electric Grid upstream, whose basket holds copper and aluminium inputs, the Democratic Republic of the Congo and Peru follow Chile. The midstream pairs scored on Trade alone are led by Poland (coal), South Korea (gas and oil) and Russia (nuclear, whose basket holds enriched uranium, reactor components and steam-cycle equipment). Indonesia leads Green Hydrogen midstream, in which 95 of 106 rankable economies share one value. Downstream, Qatar leads oil, Kuwait gas, Indonesia coal, Saudi Arabia solar, the United States geothermal and green hydrogen, and Türkiye electric vehicles, a pair that rests on Investment alone.

In Oil upstream (Figure 11) Saudi Arabia leads (0.81); the United States is 12th (0.71) and China 34th (0.55). The cell-constant inputs give every economy 0.16; of the gap between the top-ten mean and the pair mean, Reserves accounts for 58 per cent, Energy Imports for 28 and the country-varying part of Trade for 13. China's Reserves and Trade scores (0.64 and 0.79) are close to those of the United States (0.69 and 0.87), but its Energy Imports score is 0.25 against 0.68. In Batteries and Solar midstream China leads with 0.77 (United States 0.56) and 0.82 (India 0.67), and coalition members fill the other nine top-ten places of each, among them Indonesia and Malaysia in batteries and Thailand, Malaysia, Viet Nam and Indonesia in solar. Investment accounts for 62 and 58 per cent of the top-ten gap in these pairs (mean scores 0.13 and 0.09, top ten 0.61 and 0.58); China's lead over the runner-up comes from Foreign Dependency (0.37 against 0.21 in batteries, 0.40 against 0.28 in solar), while on Investment the United States scores higher than China in batteries (0.86 against 0.80).

The coalition's tiers. The partners, several of which enter as major suppliers, have the higher upstream mean percentile (0.63 against 0.56 for the core), while the core is ahead midstream (0.78 against 0.64) and downstream (0.64 against 0.59). The 21 EU members of the coalition are weaker upstream than the other members (0.50 against 0.64) and stronger midstream (0.79 against 0.67). By mean within-pair percentile the United States (0.95), Canada (0.89), Australia (0.82), India (0.79) and Poland (0.77) lead, and Qatar (0.44), Ghana (0.45), Gabon (0.46), Namibia (0.46) and Luxembourg (0.47) trail. Midstream, Japan has the highest mean percentile of any member (0.95), ahead of Germany, the United States, India and South Korea. Of the 1,710 member cells in the 30 scored pairs, 1,606 are rankable, 96 scored but not rankable and 8 unscored; the non-rankable cells concentrate in Namibia, Costa Rica, the Dominican Republic, Guatemala and Ghana (rankable in 13 to 15 of their 28 or 29 scored pairs) and Canada (23 of 29).

The United States, China and Taiwan. The United States leads two pairs (Geothermal and Green Hydrogen downstream); its weakest positions are 30th in Nuclear midstream, 23rd in Oil downstream, 22nd in Electric Grid midstream and 17th in Nuclear downstream. China is in the top ten of every midstream pair. Its weak positions are those of a fossil-fuel importer (33rd in gas and 34th in oil upstream, 24th in gas downstream) and two downstream pairs in which deployment data place it low: 63rd in green hydrogen and last of 186 in oil, where Consumption carries five-ninths of the weight and no rankable economy has lower projected growth of oil-fired capacity. The United States ranks above China in 8 of the 30 pairs, all upstream or downstream fossil-fuel and resource pairs and none midstream. Taiwan, scored in 28 cells and rankable in 19, has a midstream mean percentile of 0.75 (ranks 27th to 61st on Trade) and 0.01 across its seven rankable upstream pairs, with no place in any top ten.

Ties. In Green Hydrogen midstream the United States' rank of 12 is shared with 94 other economies, and in Green Hydrogen downstream 49 economies share third place at 0.72.

OA.4 Economic Opportunity: construction details and further results

OA.4.1 Construction details

Trade. Revealed comparative advantage complements market share: a small economy can be highly specialised in a basket without a large share of the world market, and specialisation is what the complexity literature reads as revealed capability. Basket export value is a second measure of scale from a source with different coverage and year. Feasibility is the forward-looking element: a country with no exports of a basket can score well if it exports products that require similar capabilities. An economy that does not report to Comtrade is scored on its two Atlas components and the world-market term. The net-import term of the ES category is not used in EO.

Production. Absolute growth rewards large producers whose output expands by the most units, relative growth rewards fast growth from a small base, and the composite balances the two; level and change each enter the category twice, directly and through the composite. In the mineral block, tonnes of different minerals cannot be added, so each mineral enters as a share of world supply weighted by the technology's share of the mineral's demand; the horizon level is divided by base-year world supply, so the change is a change in quantity, not in market share. A country the IEA supply table does not name for a mineral counts as zero there. Uranium mine output from the Red Book is scored like oil, gas and coal.

Technology Demand. Growth in the World Energy Outlook composite is absolute because a relative measure ranked across fuels would put hydrogen's near-zero base at the top; the value is a property of the technology, the same at every stage and for every economy. The BCG addressable market is ranked across the 26 cells it sizes and copied to every economy by the global-average rule. Coverage is uneven: the electric grid has no input in any source and so no Technology Demand category, electric vehicles have it only downstream, downstream batteries lack it because BCG does not size that cell, and geothermal is scored on the BCG entry alone.

Technological Readiness. The bell rises over four steps and falls over five, so an item at TRL 9 keeps a level score of $\cos(3\pi/10) \approx 0.59$ and one at TRL 3 scores $\cos(3\pi/8) \approx 0.38$; the concave momentum term gives the first step of progress the largest increment. Of the 253 mapped items, 9 map to more than one cell; 98 carry a post-TRL-9 adoption stage, 43 per cent sit at TRL 9 or above, and the 26 items that are commercial in many markets score zero on level. Of the items with a 2020 reading, 54 per cent did not move. Ten of the 33 headline cells take their technology's value because no item maps to the stage; the index differs across the three stages of ten of the eleven technologies (wind keeps one value), with a median range across a technology's stages of 0.16. The parameters (bounds 2 and 11, peak 6, momentum weight 0.3, cap of three steps, exponent 1/2) are the theme builder's defaults.

Overcapacity. Of the four midstream technologies the BloombergNEF supply-chain data cover, solar ($\rho = 2.3$, index 0.20), batteries and wind ($\rho = 3.0$, 0.13 each) sit on the steep part of the curve, and green hydrogen ($\rho = 18.2$) is effectively at zero.

Cost Competitiveness. Wages are converted at market exchange rates rather than purchasing-power parities because the price-level difference that PPP conversion removes is the cost advantage being measured. The lending rate is winsorised at its 5th and 95th cross-country percentiles before min-max rescaling; producer-price inflation enters as a rate because each country's price index has its own base year. Stale observations and gaps are filled from regional and income-group peers. The Input Cost Index carries the category wherever the LCOE and equipment cost components are absent and half of it where one is present; counting each source once avoids giving BNEF three of four votes wherever it covers a market.

OA.4.2 Further results

Rankability. Of the 2,830 cells the rule sets aside, 2,112 carry more than half of their weight on imputed components, 74 belong to territories and 644 fail on both counts. The rule bites hardest where country-specific sources are thinnest: 70.2 per cent of upstream cells are rankable, against 55.7 per cent midstream and 54.9 per cent downstream. In Batteries midstream 42 of the 54 rankable economies have an observed Foreign Dependency or Investment input, in Green Hydrogen midstream 30 of 40. The mean imputed weight share of rankable cells is 0.29. Ranked among all scored economies of their pair, cells excluded for imputation alone have a mean percentile of 0.43 (63.4 per cent below the median) and territory cells 0.42, against 0.55 for rankable cells. Both decompositions reconcile exactly (largest gaps 0 and 1.1×10^{-16}); 45.3 per cent of category scores contain an imputed component.

Distribution. Over all scored cells the midstream median is 0.454 and its lower quartile 0.328, because the zero rule places economies without recorded capacity or investment at the bottom of the clean-technology midstream pairs, and the rule removes most of them. Four of the five clean-technology midstream pairs (wind, electric vehicles, solar, batteries) have the lowest midstream medians (0.351 to 0.426). The widest boxes owe their width to one category each: in EV midstream (interquartile range 0.144) Foreign Dependency, observed for 97 of 145 rankable economies, accounts for 46 per cent of the variance (0.053 without it); in Green Hydrogen downstream (0.122) Consumption accounts for 94 per cent; in Wind downstream (0.117) Investment accounts for 46 per cent.

Variance and persistence. Technology effects alone explain 27.8 per cent of the variance and stage effects almost nothing (0.01 per cent), so a pair's level is set by its composition rather than its stage. Over all scored cells the pair share is 71.5 per cent. Across the 159 economies rankable at all three stages, the Spearman correlation of mean within-pair percentiles is 0.86 between upstream and midstream, 0.70 between upstream and downstream and 0.79 between midstream and downstream. Part of this common factor is the Input Cost Index, a country-level index whose stage variation comes only from the wage activity and the factor shares: across the 178 economies with a rankable cell, mean EO percentile and mean Cost Competitiveness percentile correlate at 0.77 (0.82 across the 57 coalition members).

Drivers. Level shares: upstream Cost Competitiveness 21.9 per cent, Technology Demand 17.3, Production 16.5 and Energy Prices 16.2; midstream Cost Competitiveness 29.2, Technology Demand 24.7 and Trade 24.6; downstream Cost Competitiveness 27.5, Technology Demand 19.7 and Technological Readiness 18.0. Within-pair variance shares: upstream Trade 26.9 and Production 22.4; midstream Trade 25.5 and Foreign Dependency 22.2; downstream Consumption 23.8 and Investment 22.2. Investment contributes 0.3, 1.2 and 4.2 per cent of the level by stage but 8.3, 12.7 and 22.2 per cent of the within-pair variance, because under the zero rule the few economies with recorded

investment stand far above the majority without it (in 9 of the 12 pairs that carry the category). The upstream constants are led by price volatility (15.9 per cent of effective weight), the TRL Index (12.5) and the addressable market (10.1). Before the rankability rule the observed country-specific share is 32.3 per cent and the imputed share 30.1 per cent.

Leaders. Chad combines low recorded wages with a 2017 lending rate of 2.45 per cent, the joint fifth lowest of the 144 rates in the cost theme; the runner-up in its three pairs is Cameroon, with the same rate (2018), and Chad's margins are 0.0020, 0.0005 and 0.0011. Canada leads nuclear upstream (0.65) and China electric-grid upstream (0.71, ahead of Chile at 0.65), the pair built on the copper and aluminium basket. The leader's margin is below 0.01 in seven pairs; the United States has the largest non-Chinese margin (0.153 over Indonesia in geothermal downstream). Coalition members hold 69.1 per cent of the 330 top-ten places; China is in the top ten of 30 pairs and the United States of 20. The United States ranks in the bottom half of one pair only, 109th of 141 in electric-grid downstream, a cost-only pair; in the mineral-based upstream pairs it ranks between third (wind) and 62nd (green hydrogen). China's one low rank, 98th of 104 in oil downstream, comes from Consumption and the growth component of Production.

The coalition. Among core members the United States (0.89), Portugal (0.84), Poland (0.84), the Czech Republic (0.81) and Spain (0.80) rank highest and the United Kingdom (0.57), Romania (0.52), Australia (0.46), New Zealand (0.34) and Ukraine (0.11) lowest; among partners, Thailand (0.82), Indonesia (0.80), Malaysia (0.79), India (0.78) and Saudi Arabia (0.66) rank highest and the Dominican Republic (0.26), Brazil (0.23), Argentina (0.19), the Democratic Republic of the Congo (0.13) and Ghana (0.10) lowest. Türkiye, Brazil, Argentina and Ghana record nominal lending rates of 39.0, 46.0, 59.7 and 27.0 per cent, at the winsorisation ceiling; with the Cost Competitiveness contribution set to its pair mean their mean percentiles would be 0.73, 0.61, 0.45 and 0.39 rather than 0.30, 0.23, 0.19 and 0.10, and Ukraine's (rate 18.9 per cent) 0.50 rather than 0.11; six members gain more than 0.2. At the top the same change would lower Portugal from 0.84 to 0.62, Thailand from 0.82 to 0.64 and Ireland from 0.71 to 0.51, while the United States barely moves. The BNEF regional composite of Technology Demand is observed for at most twelve economies and the global-average rule gives the others their mean, so a modelled economy below that mean ranks below many unmodelled ones: Australia's value is below the mean in all 18 pairs in which it is observed, the United Kingdom's in 14 of 18 and France's in 15 of 21; with Technology Demand at its pair mean their mean percentiles would be 0.52, 0.65 and 0.80 rather than 0.46, 0.57 and 0.76.

Of the core members' 1,023 scored cells 1,002 are rankable, against 716 of the partners' 858. Gabon has only 9 rankable cells, Oman 21, Saudi Arabia and Singapore 22 each, and Morocco, Namibia and Costa Rica 23. The mean imputed weight share of rankable cells is 0.20 for the core, 0.27 for the partners, 0.33 for other economies and zero for the United States, partly because the BCG addressable market is recorded for the United States alone. Within pairs the median Spearman correlation between EO_{its} and the imputed share is -0.28.

Illustrative pairs. Oil upstream is led by the United States (0.659), China (0.644), Canada (0.637), Saudi Arabia (0.622) and Norway (0.607); Saudi Arabia, Iraq, Iran, Kuwait and the United Arab Emirates share the smallest Cost Competitiveness contribution (0.095, an imputed value) and owe their places to production and trade. In Batteries midstream China (0.771) leads the United States (0.633) on Foreign Dependency (0.167 against 0.106) and Cost Competitiveness (0.142 against 0.073), while the United States' Investment contribution (0.143) exceeds China's (0.134); nine of the ten are coalition members. In Solar midstream China (0.788) is followed by India (0.640) and the United States (0.635), then Viet Nam, Thailand, Indonesia, Malaysia and the Philippines, with France and Germany eighth and ninth. In EV downstream emerging economies take eight of the ten places: China (0.755), Viet Nam (0.731), Thailand (0.705) and Türkiye (0.702) lead, because the EV-market composite, half of whose indices are growth rates (equation (82)), rewards markets growing fast from a small base; Türkiye, Viet Nam and the Philippines receive larger Technology Demand contributions than China.

OA.5 The opportunity-security space: further results

Sample. Of the 3,075 cells scored by both pillars but not rankable on both, 607 belong to territories and 2,468 to sovereign economies whose imputed weight share exceeds one half in Energy Security only (721), in Economic Opportunity only (1,006) or in both (741). The number of rankable economies per pair ranges from 40 (Green Hydrogen midstream) to 175 (Nuclear upstream). In this sample technology-stage fixed effects account for 41.9 per cent of the variance of ES_{its} and 53.4 per cent of that of EO_{its}. The six EU members that enter the allied network only through the EU node are not counted as coalition members.

Co-movement by technology. The within-pair correlation exceeds 0.5 in 19 of the 30 pairs and is negative only in Green Hydrogen midstream. Pooled on levels it is 0.48 and on pooled within-pair ranks 0.57. The stage gradient holds within wind (0.37, 0.72 and 0.89 by stage), nuclear (0.16, 0.49, 0.89), coal (0.35, 0.56, 0.79), gas (0.40, 0.54, 0.78) and oil (0.50, 0.55, 0.75). Solar (0.48, 0.78, 0.58) and electric vehicles (0.60, 0.94, 0.80) peak midstream; in batteries the correlation falls along the chain (0.63, 0.61, 0.40); in green hydrogen the midstream value (-0.21) lies below the upstream one (0.03). Medians by technology run from 0.03 (green hydrogen) to 0.80 (electric vehicles). Upstream the correlation is positive in all ten pairs. In the five mineral pairs with a trade basket on both sides (batteries 0.63, grid 0.60, electric vehicles 0.60, solar 0.48, wind 0.37), opportunity's Trade and security's Minerals Trade measure exports of overlapping materials and correlate at 0.70 to 0.85. The fossil-fuel pairs follow (oil 0.50, gas 0.40, coal 0.35); nuclear (0.16) has half of its economies at one security value, and green hydrogen (0.03), the only upstream pair without a trade basket, rests on platinum-group endowments on the security side and on cost, demand, readiness and production on the opportunity side.

Shared components. A component counts as shared when the other pillar contains a component of the same category with identical observed values: every component of Consumption, Energy Access, Investment and Energy Prices, two of the three security and two of the five opportunity components of Trade, and five of the six components of Foreign Dependency in each pillar; no Production component. In seven downstream pairs (batteries, coal, electric vehicles, gas, green hydrogen, nuclear, oil) the security score is a reweighting of part of the opportunity score, which is why Green Hydrogen (0.92) and Nuclear (0.89) downstream show some of the highest correlations. The non-shared median correlation has an interquartile range of 0.20 to 0.40, a 45 per cent reduction from the full pillars. The level covariance is positive in all 30 pairs, including Green Hydrogen midstream. Of the pooled within-pair covariance, the two cross terms of equation (49) account for 24 per cent and the non-shared term for 16 per cent (58 per cent upstream, nothing downstream); in the median pair the shared term accounts for 19 per cent. The cross terms are not definitional but run through shared inputs.

The non-shared rank correlation is 0.37 midstream and 0.30 upstream. The pillars' distinctive trade components (net imports and, upstream, minerals trade for security; revealed comparative advantage, export size and feasibility for opportunity) all reward exporters of the stage's basket; removing opportunity's trade components as well lowers the medians to 0.15 midstream and 0.02 upstream. In the three downstream pairs where it can be computed (geothermal, solar, wind) it is -0.04.

Category correlations. Consumption, Energy Access and Investment correlate perfectly with their counterparts; Trade, sharing two components, 0.62; Production, from different sources and transformations of the same physical quantity, 0.27; and Foreign Dependency, whose shared components are missing for many economies and imputed under different rules, 0.04. The median correlation between categories with different names is 0.14 over 47 category pairs. Technology Demand and Cost Competitiveness have absolute correlations of at most 0.12 and 0.17 with every security category (the most negative is Cost Competitiveness with Reserves, -0.17). Within security, Production and Reserves average 0.79 across seven upstream pairs (0.72 to 0.91 in the six critical-mineral pairs, where 62 to 85 per cent of rankable economies record zero on both); Nuclear upstream has the lowest value (0.61), because many holders of uranium resources do not mine them.

Regions. Laggards outnumber leaders because a tied cell at a median counts as low and ties are common where a pillar rests on a zero-anchored component. Across pairs, the observed share of opportunity without security tracks the copula-implied one (correlation 0.90). Off-diagonal shares are 19.5 and 17.2 per cent upstream, 13.8 per cent (opportunity without security) midstream, and 12.4 and 10.2 per cent downstream; by technology the leapfrog share runs from 9.5 per cent (electric vehicles) to 20.2 per cent (green hydrogen and nuclear). In Green Hydrogen midstream 34 of 40 economies share the lowest security value, which is also the median; in Nuclear upstream 87 of 175 share 0.172, the median, and 37 of the 49 cells in the leapfrog region are such economies. Counting ties as high empties the region in Green Hydrogen midstream and lowers it to 6.9 per cent in

Nuclear upstream. The next pairs, Green Hydrogen upstream (24 per cent), Coal upstream (22) and Batteries downstream (21), have few ties and shares near the copula value.

The coalition. The partners, many of them major suppliers, have 21.4 per cent of cells with security but not opportunity. The median core member has 16.7 per cent of its pairs in the leapfrog region, the median partner 6.9 per cent. Austria, Portugal, Luxembourg (11 of 30 each), Ireland (10) and Croatia (9) lead; without the two tie pairs their counts are 9, 10, 10, 8 and 7. Among partners the largest counts are Thailand and the Philippines (5 of 29), then Malaysia, Viet Nam and Qatar (4). Thirteen members have no leapfrog pair: Canada, Japan, the United Kingdom, Australia and Ukraine in the core, and Türkiye, Brazil, Argentina, the Democratic Republic of the Congo, the Dominican Republic, Gabon, Namibia and Ghana among partners. Brazil, a partner on its reserves alone, is the clearest case of security without opportunity. The pairs most often in the leapfrog region across members are Gas and Oil upstream and Green Hydrogen midstream (14 members each), Coal upstream (13), Solar downstream (12), Nuclear upstream (11) and Grid upstream (10); without the two tie pairs the coalition share falls from 12.8 to 11.8 per cent. Members' rankable pairs number 22 to 30 in the core and 9 to 29 among partners, so the shares of Costa Rica, Gabon, Namibia (9 pairs), Guatemala (10), the Dominican Republic and Ghana (11) rest on few cells.

Within countries. Of the 174 countries with a Strategic Index, seven with fewer than five cells are left out; the remaining 167 have 5 to 30 cells (median 22; 50 have fewer than ten). The correlation is positive for 164 countries; the three exceptions (Afghanistan, Bhutan, the Maldives) are not members. Among members it averages 0.52, from 0.27 (New Zealand) to 0.77 (Viet Nam). Security varies more than opportunity across a country's sectors (mean standard deviations 0.15 and 0.07); the covariance share ranges from 16.5 per cent (Ghana) to 66 per cent (India) among members, and the standard deviation of Δ_{its} from 0.068 (Lao PDR) to 0.26 (the Gambia). The pairs' mean pillar values are only weakly related across pairs (0.33); after subtracting pair means the within-country correlation rises to 0.67 and the standard deviation of the gap falls to 0.08, so part of the gap's spread is common to every country.

The Strategic Index and the regions. Every Strategic Index cell (3,154 for 174 countries) lies in this sample. A country's top sector, before eligibility rules, lies in the leapfrog region for 47 of 174 countries (27.0 per cent; 15 members), against 15.2 per cent of cells. Nuclear upstream is the top sector of 62 countries, 43 of them at the common security value, which gives a large need whichever side of the opportunity median they fall. After the endowment screen, 169 countries keep an eligible cell and for 35 of them (20.7 per cent) the top one lies in the region. Of the 57 allied picks (43 midstream), the eight in the region are Austria, Belgium, Italy, the Netherlands, Romania and Viet Nam in Batteries midstream and Indonesia and Slovakia in Nuclear midstream; 21 are leaders, 14 have security without opportunity and 14 are laggards, while 61.0 per cent of eligible sectors are leaders. None of the eight Nuclear upstream picks is in the region: all eight members hold or mine uranium. Costa Rica, the Dominican Republic and Guatemala have four

eligible sectors each (nuclear, oil, gas and coal midstream) and Namibia five (the same and Nuclear upstream). Midstream nuclear security is built from Trade alone and compressed around its median (quartiles 0.435, 0.482, 0.530), so their security ranks of 0.50 to 0.71 lie at or above it, and the technology weight (nuclear 0.999, gas 0.422, coal 0.024, oil zero) favours nuclear.

Caveats. Percentile ranks discard distances, and the median split turns small differences into differences of region; where many economies tie at a median the convention decides the region of the whole group (counting ties as high lowers the upstream leapfrog share from 19.5 to 15.5 per cent and the coalition's from 12.8 to 10.8). Regions are relative to the rankable economies of each pair, a set that leans towards economies with good statistical coverage. The rank-based measure of mechanical overlap covers 23 pairs; the covariance decomposition covers all 30 but concerns levels and is dominated by the pairs with the largest covariances. Within-country statistics compare category bundles that differ across pairs. The endowment screen is a rule for the allied picks, not part of either pillar or of the index. More generally, shared components make the implied trade-offs of a composite hard to see in the aggregate (Ravallion, 2012); inspecting the correlation structure among indicators (Nardo et al., 2008) is what reveals them here.

OA.6 Coalition rule: data and further results

Supply and reserves measures of the supplier route. An economy passes the supplier test if, in at least one Upstream or Midstream stage of a headline technology, it holds at least 5 percent of the world total of one of the following supply measures, or at least 10 percent of world reserves of a resource the stage extracts. Supply: exports of the stage's HS6 basket (2023; The Growth Lab at Harvard University, 2019); oil, gas and coal production (2025; Energy Institute, 2026); uranium mine production (2024, OECD Nuclear Energy Agency and International Atomic Energy Agency, 2026); mine production and refining of the minerals a technology uses (2025; International Energy Agency, 2026a, with manganese, zinc and platinum-group metals from the Energy Institute); and clean-technology manufacturing capacity (2024). Reserves: the critical-mineral reserves of U.S. Geological Survey (2026) at the end of 2025 (cobalt, lithium, graphite, rare earths, copper, manganese, nickel, zinc and platinum-group metals; the report's erratum puts Brazil's rare earths at 11 Mt), proved oil, gas and coal reserves at the end of 2020 (Energy Institute, 2026) and identified uranium resources recoverable below USD 260/kgU (OECD Nuclear Energy Agency and International Atomic Energy Agency, 2026). Holders whose reserves USGS does not quantify (entries NA, W or s) count as zero, so world shares are measured on quantified holdings. The route reads the same reserves tables as the Reserves category of the Energy Security pillar (Section 5). The 10 percent bar binds for a small group: six economies that are not China-leaning hold between 5 and 10 percent of a world reserve without otherwise qualifying (Madagascar 8.7 percent of graphite, Mozambique 8.1 percent, Kuwait 5.9 percent of oil, Tanzania 5.8 percent, Niger 5.7 percent of uranium, New Caledonia 5.1 percent of nickel), as do Iraq, Turkmenistan and the United Arab Emirates, which are China-leaning.

GDP floor. The floor of USD 100 billion (current dollars, latest year, 2025 for every member) excludes 18 U.S.-leaning economies, among them six EU members and small European and Pacific states, and 16 members of the five agreements, the largest Lithuania (USD 95 billion), which enters through the EU node, and Panama (USD 91 billion). Puerto Rico inherits the U.S. position and exceeds the floor but has no vote of its own and is not a member.

Supply cover of the coalition. Across the 18 Upstream and Midstream stages with an HS6 basket, the 63 economies of the network account for a median 70.7 percent of world exports, from 45.2 percent in Wind Upstream to 88.4 percent in Batteries Upstream, and China for a median 17.6 percent, with a maximum of 40.9 percent in Batteries Midstream. The coalition's share exceeds China's in all 18 stages, but in two Upstream stages the excluded China-leaning suppliers other than China hold large shares: 31.3 percent of Wind Upstream exports (coalition 45.2) and 47.6 percent of Nuclear Upstream exports (coalition 50.0). Coalition economies mine 81.9 percent of the world's cobalt, 79.4 percent of its nickel, 60.0 percent of its lithium and 54.5 percent of its copper, but 3.6 percent of its graphite and 14.3 percent of its rare earths; they refine 17.4 percent of cobalt, 23.2 percent of copper, 24.6 percent of lithium, 5.5 percent of graphite and 12.6 percent of rare earths, against China's 75.6, 46.8, 73.7, 94.4 and 83.9 percent; nickel refining is 55.7 percent in the coalition. Coalition economies hold 75.8 percent of quantified world cobalt reserves, 77.3 percent of lithium, 85.5 percent of manganese, 78.8 percent of nickel and 83.9 percent of platinum-group metals, but 33.3 percent of rare earths (China 58.4) and 35.7 percent of graphite (China 32.3). In oil and gas the excluded China-leaning suppliers other than China hold about as much as the coalition or more (35.3 against 35.7 percent of oil reserves, 43.4 against 30.8 percent of gas).

EU node weights. For each of the 22 Upstream and Midstream cells the weights are the members' shares in the first usable basis: production for Coal, Gas and Oil Upstream, exports of the stage's HS6 basket for 15 cells, and GDP for Geothermal and Green Hydrogen, Upstream and Midstream. A basis is usable only if it lists the members individually and covers at least half of the EU total, and only if the EU holds at least 0.1 percent of the world total; the IEA manufacturing and mineral data, which report the EU as a bloc or list only leading producers, do not qualify. Germany carries the largest weight in 11 of the 22 cells (23.8 percent in the GDP-weighted cells, 38.9 percent of the Electric Vehicles Midstream basket exports), France and Poland in four each, and Finland (Electric Vehicles Upstream, 27.4 percent), Italy (Oil Upstream, 38.5 percent) and Romania (Gas Upstream, 29.5 percent) in one each. France accounts for 97.9 percent of the EU's exports of uranium ores and concentrates and 46.5 percent of its silicon-metal exports, and Estonia, below the GDP floor, for 25.5 percent of the Wind Upstream basket of rare-earth compounds. The Herfindahl index of the weights ranges from 0.081 to 0.959 across cells. The production weights rest on small totals: the EU produces 1.6, 0.8 and 0.3 percent of world coal, gas and oil, and the listed members cover 95, 89 and 64 percent of the EU's own totals. The node's GDP, USD 21.2 trillion, is the second largest after that of the United States.

Dyadic alignment among members. Counting EU members individually, the 1,596 pairs of members split 29.3, 64.8 and 5.8 percent across $A_{ij} = 1$, $\frac{1}{2}$ and 0. The three China-leaning partners have $A_{ij} = 0$ with every core member and $A_{ij} = 1$ with China and Russia.

Sensitivity of the blocs. With session 78 alone (192 economies; Spearman 0.966 with the headline, 89.1 percent in the same bloc), New Zealand falls just below the U.S.-leaning cut ($s = 0.583$ against 0.584) and becomes a partner through IPEF, Saudi Arabia becomes China-leaning and loses its supplier route, Myanmar becomes nonaligned and enters as a supplier, and India becomes China-leaning but remains a partner through IPEF and the MSP. With the final-passage series (sessions 76–80, 193 economies, 89.6 percent in the same bloc) the cuts move to 0.446 and 0.055: China-leaning economies cluster more tightly around China in final-passage votes, and Russia ($s = 0.117$), Kazakhstan (0.056) and Myanmar (0.163) would enter the coalition as major suppliers; Malaysia becomes China-leaning but stays through IPEF.

The earlier hand-picked list. Of the 23 members fixed by hand in an earlier version, the United States, Canada, Japan, Australia, Korea, the United Kingdom, Germany, France, Italy, Norway and Denmark are core members, and India, Mexico, Brazil, Saudi Arabia, South Africa, Indonesia, Viet Nam, Argentina, Morocco and Chile are partners. The United Arab Emirates (largest supply share 4.2 percent, oil reserves 5.6 percent) and Kenya leave. The 16 new partners are Colombia, Costa Rica, the Democratic Republic of the Congo, the Dominican Republic, Gabon, Ghana, Guatemala, Malaysia, Namibia, Oman, Peru, the Philippines, Qatar, Singapore, Thailand and Türkiye.

OA.7 The Policy index and the Strategic Index: further detail

Tagging and strength of IEA policies. Each policy of the IEA Policies and Measures database (International Energy Agency, 2026e) is assigned to technologies first from its own technology list and, failing that, from keyword rules applied to its title and text; a policy matched to no technology is cross-cutting and counted in every technology. Stages are assigned by keyword rules on the text: Upstream on whole words for mining, mines, exploration, extraction, ores, leases, concessions and drilling; Midstream on processing, refining, smelting, manufacturing, factories, plants, components, supply chains and domestic content; Downstream on deployment, installation, consumers, end use, adoption, efficiency, demand, charging, grid connection, power plants and generation. A policy that matches no stage is cross-cutting across the three. Its strength is $\sigma_h = w_{\text{type}}\, w_{\text{status}}\, w_{\text{juris}}\, w_{\text{tech}}\, w_{\text{stage}}$ with type weights 0.85 (financial incentives), 0.70 (public procurement), 0.50 (regulations and standards; research, development and demonstration), 0.20 (information and labelling; other) and 0.10 (voluntary and coordination measures); status weights 1 (in force), 0.60 (planned), 0.50 (proposed), 0.20 (expired or unknown) and 0 (ended); jurisdiction weights 1 (national), 0.66 (international), 0.50 (subnational, state, provincial or regional), 0.30 (unknown) and 0.25 (local); a technology discount of 0.70 for a cross-cutting policy and 0.30 for a technology found only in the text; and a stage discount of 0.75 for a cross-cutting stage. A policy counted in several cells carries

its full strength in each. Upstream words are matched as whole words, because as substrings “ore” and “mine” match “more”, “Korea” and “determine”.

The Climate Action Tracker component. The Tracker rates 65 economies in the final build: 39 in their own right and 26 EU members through the Union’s rating. Forty are rated insufficient, nine critically insufficient, six highly insufficient and ten almost sufficient; among the 46 rated coalition members the counts are 31, 7, 4 and 4. Because the component varies across a country’s cells only through g_t, it tilts the within-country policy ranking. Within the Strategic Index sample the median within-country Spearman correlation between POL^{*}_{its} and g_t is 0.82 for the 40 countries rated insufficient and 0.86 for those rated almost sufficient, -0.92 for the nine rated critically insufficient, and 0.12 for the 103 countries scored on the IEA component alone. The IEA component itself is uncorrelated with g_t within a country (median -0.07 for the 65 countries rated by the Tracker). The mean Policy index is 0.64 for the economies scored on both components and 0.39 for those scored on the IEA component alone; the IEA component distinguishes a median of 31 values among the 33 cells of the first group and 11 among those of the second. The index enters the Strategic Index only through within-country ranks, so these level differences do not move one country against another.

Technology weight. The lifecycle medians are 820 $\mathrm{gCO_2e/kWh}$ for pulverised coal, 490 for gas combined cycle, 48 and 41 for utility and rooftop photovoltaics, 27 for concentrated solar, 38 for geothermal, 12 for nuclear and 11 and 12 for onshore and offshore wind (Schlömer et al., 2014); oil-fired generation takes the SRREN median of 840 (Moomaw et al., 2011) in place of the 720 of the input file, because AR5 reports no oil row. Technologies without a row take proxies: batteries the solar value, the grid the mean of solar and wind, green hydrogen the mean of gas and solar, and electric vehicles the mean of six generation technologies. The resulting g_t is 0 (oil), 0.024 (coal), 0.422 (gas), 0.569 (electric vehicles), 0.694 (green hydrogen), 0.967 (solar, batteries and geothermal), 0.983 (grid), 0.999 (nuclear and wind). The sector prior $0.1\lambda_s + 0.1g_t$ spans 0.025 to 0.175; the gap between oil and wind in the technology term (0.100) is twice the gap between adjacent stages (0.025 to 0.05).

Sample. Of the 6,918 country–sector rows of the output, 442 belong to countries without a Policy index and 6,476 to policy-eligible countries. Among the latter, 1,000 fail because the Economic Opportunity cell is not rankable, 721 because the Energy Security cell is not rankable, 979 because neither is, and 622 because the Energy Security cell is missing; 3,154 are scored. Before the Upstream trade baskets were added on 11 October 2026 the index covered 192 countries; the new Trade category of the Upstream cells is imputed for many small economies and pushes their cells over the imputed-share threshold, which leaves 174. Sector fixed effects explain 72.6 percent of the variance, technology fixed effects 67.5 percent, stage fixed effects 4.2 percent and country fixed effects 4.4 percent. The highest sector means are Solar Upstream (0.634), Batteries Upstream (0.631), Nuclear

Upstream (0.620) and Batteries Midstream (0.618), the lowest the coal and oil stages (0.410–0.441). Upstream cells have the highest mean need (0.677, against 0.513 Midstream and 0.552 Downstream).

The endowment screen. An Upstream cell carries a resource base if the technology-level variable of the Reserves or of the Production category of the Energy Security pillar (for instance the uranium resources and mine production behind Nuclear Upstream) is positive for the country; every technology with such a variable is screened. Unquantified USGS holdings count as zero in these technology roll-ups. Cells without a resource base carry `resource_base` FALSE and the pick exclusion "no resource base" in the output; their index values are unchanged. Of the 41 picks the screen changes, the forgone index has median 0.026 (interquartile range 0.016–0.040); 16 picks are unchanged. Every Nuclear Upstream pick under the screen lies in a member with identified uranium resources (Chile, the Democratic Republic of the Congo, Finland, Gabon, Greece, Japan, Saudi Arabia and Sweden).

Variants of the Strategic Index. Each variant recomputes the index over the headline cells with the endowment screen kept. Within-member Spearman correlation with the headline (median), identical picks and members whose headline pick is in the variant's top three: security as strength 0.36, 5 and 12 of 57 (core 0.38, partners 0.24, negative for ten partners); strength without the fixed weights 0.01, 1 and 9; need without the fixed weights 0.90, 31 and 49; opportunity and need ranked within the country 0.89, 20 and 49; no policy term 0.88, 30 and 55; policy from the IEA component alone (the 46 members scored on both) 0.89, 34 and 42; the October 2026 NIPO component added back (50 members) 0.96, 27 and 44. The NIPO variant uses a GTA NIPO extract of October 2026 (measures in force on 5 October 2026, refreshed on 13 October with the debiased strength measure), built exactly as the pipeline builds the NIPO Policy Index rows: 77 jurisdictions with EU members individually, 939 rows in the headline technologies, all of them economies that already have a Policy index; NIPO enters with weight 4 beside IEA PAMS (3) and CAT (2), renormalised over the components present. The component stored with the February 2026 data covered only 23 economies. Requiring both policy components (`OPSI_POLICY_REQUIRE_ALL=1`) leaves 65 eligible countries and 1,666 Strategic Index cells, identical to the headline in value; 46 members keep their pick and the 11 scored on the IEA component alone have none.

OA.8 Bilateral partnership indices: further detail

Trade components. The reconciled flow matrix holds, for each exporter, importer and technology–stage cell, the flow in 2020 and 2024: the importer's declaration where the importer reports the cell in that year, and otherwise the exporter's declaration times 1.06, the conventional CIF/FOB margin. An economy that reports a cell but no flow with a partner records a zero; a pair in which neither side reports is unobserved. The growth term is defined only where both years are observed and the 2020 flow is positive; a dyad that trades only in 2024 is scored on its level. The first-stage ranking runs over dyads with a positive flow in either year, and the second ranks them within the reporter,

with recorded zeros as one tie block below every trading partner (so they take the block's mid-rank). A dyad without an observed 2024 flow takes the mean of the observed values in its cell and counts as imputed in the rankability measure; in the 15 cells without an HS6 basket (every Downstream cell and the Upstream and Midstream stages of Geothermal and Green Hydrogen) every dyad carries the constant 0.5, which ranks nothing and is left out of the imputed-share measure. The Comtrade dyads have 168 reporters in 2020 and 143 in 2024, and 208 economies appear as reporter or partner. Core members' imports are almost all self-declared (0.3 percent of their observed import components are mirrors; 5.2 percent of their components are imputed). Dropping the mirror fill and re-running every later step leaves the headline PSI_j almost unchanged (Spearman 0.997, median rank change 1, largest 35, the same top fifteen) but moves C_j^{PO} (Spearman 0.816), because a third of the core exporters' observed export flows are mirrors of their markets' own import declarations; the United Arab Emirates (91st to 126th) and Russia (105th to 117th) would lose most.

Outward investment. The IMF indicator is outward direct investment, assets (gross), debt instruments, all entities, reported official data. A position is a stock, so the reported years are averaged; an absent record is missing, while a reported zero is a real zero and joins a tie block below every positive position before the within-reporter ranking. The 16 conduit centres are the ten economies that host more than 85 percent of phantom FDI according to Damgaard et al. (2024) (the Netherlands, Luxembourg, Hong Kong, the British Virgin Islands, Bermuda, Singapore, the Cayman Islands, Switzerland, Ireland and Mauritius) and six offshore centres excluded before (the Bahamas, Barbados, the Turks and Caicos Islands, the Isle of Man, Guernsey and Cyprus); positions in which a centre is reporter or counterpart are unobserved. Before the list was extended the Netherlands and Singapore were the second and third Friendshore partners of the United States on the strength of pass-through positions. The United States' largest outward counterparts are now the United Kingdom, Australia, Taiwan, Malaysia and Canada. In the four CM-FDI cells the tie is built from the projects used by the index measures (Bandara and Net Zero Industrial Policy Lab, 2026): manufacturing projects that are announced, under construction, completed or operating (cancelled, paused and closed plants are dropped), with a home economy other than the destination; a joint venture attributes half of the project to the listed parent. Attributed counts and disclosed capital are ranked within the cell over the dyads with a project and averaged, then ranked within the reporter with partners without a project as one tie block below; a reporter with no project in the cell has the component missing. Because the ties are sparse, 7.2 percent of the dyad cells in these four cells carry an investment component, against 19.5 percent elsewhere (18.1 percent overall); 29 economies invest (18 of them core members). Reporter need is missing in 12.5 percent of Friendshore cells, among them every cell of Electric Grid Downstream and the two Geothermal stages.

Chinese-parent exposure by supplier. Among the heavily exposed suppliers are Türkiye (batteries 1.00, solar 0.97, electric vehicles 0.82), Thailand (batteries 0.97, electric vehicles 0.88), Malaysia (electric vehicles 0.90), Viet Nam (electric vehicles 1.00), Indonesia (solar 0.85), Saudi

Arabia and Oman (solar 1.00), and among the core Korea (batteries 0.91, wind 1.00), Hungary (electric vehicles 0.64, batteries 0.55) and Slovakia (electric vehicles 0.61). The United States (at most 0.27, in solar), Canada and Mexico (zero in batteries) are among the least exposed. Of the 94 supplier–technology cells outside China, Hong Kong and Macao, 48 are at 0.5 or above and 19 at zero. Where defined, the component contributes 0.051 to the composite on average (8.1 percent). Suppliers with a share of at least one half lose 0.063 of their core-view Friendshore rating on average, those below 0.2 gain 0.043. Without the component Hungary would rise from 20th to 16th in the headline PSI_j, and Denmark and Finland would fall three places.

Climate penalty. The Tracker rates 65 economies (39 national ratings and the EU's rating for 26 members). Forty of them are rated insufficient, so the median rating given to every unrated economy is 0.75, and the penalty is a common value, $0.15(1 - g_t)$ for Friendshore and $0.0375(1 - g_t)$ for Opportunity, for every dyad of unrated or insufficient parties; the within-cell ranking removes it. The largest penalties, for parties rated almost sufficient, are 0.200 and 0.050 for oil, 0.195 and 0.049 for coal, 0.116 and 0.029 for gas, 0.086 and 0.022 for electric vehicles and 0.061 and 0.015 for green hydrogen; for the clean technologies they are below 0.01. Among the ten leading suppliers of the United States and Japan the penalty averages 0.018 and 0.033.

Development Potential inputs. The development index $d^{(1)}$ averages, over the series observed for each economy, the S-curves across economies of 2022–2024 means of a 40-series WDI basket (World Bank, 2026) (Table OA.4), with 13 series inverted, and ranks the mean. Coverage is uneven: the ten B-READY series are observed for 50 economies, central-government debt for 51 and electrical outages for nine, against more than 200 for GDP per capita or the governance percentiles; the human capital index has no observation in the window. The business environment uses the Doing Business score of 2019 (World Bank, 2020), the last release with broad coverage, although the series was discontinued (World Bank Group, 2021). Aid is the sum over donors and the years 2013–2023 of constant-price ODA disbursements from the OECD Creditor Reporting System (OECD, 2026) (44 official donors),

$$A_{js} = \max\Big(0,\ \sum_{d}\sum_{y}\Big[\sum_{\sigma\in\Sigma_s^{+}} a^{\sigma}_{dj,y} - \sum_{\sigma\in\Sigma_s^{-}} a^{\sigma}_{dj,y}\Big]\Big), \tag{OA.4}$$

with Σ_s^{+} the stage's sector (industry for Midstream, energy for Downstream, mineral resources and mining for Upstream) and Σ_s^{-} the coal and upstream oil-and-gas sub-rows for Upstream; $d^{(6)}$ and $d^{(7)}$ rank A_{js} and A_{js}/Y_j within each stage. Stage aid is observed for 121 (Upstream), 131 (Midstream) and 131 (Downstream) of the 132 partners. Several orientations are choices: high bank capital and liquidity ratios are read as under-intermediation; ODA received is inverted inside $d^{(1)}$ (aid dependence) but enters positively in $d^{(6)}$ and $d^{(7)}$ (existing engagement); energy use per capita is read as the depth of the energy system. The donor is not matched to the reporter, so DP is not reporter-specific.

Table OA.4: WDI series in the development basket and their orientation

Orientation	Series (WDI code)
Higher is better (27)	Access to electricity (`EG.ELC.ACCS.ZS`); ten B-READY pillar scores: business entry, business insolvency, business location, dispute resolution, financial services, international trade, labor, market competition, taxation, utility services (`IC.BRE.*.OS`); control of corruption, government effectiveness, political stability, rule of law (percentile ranks, `CC/GE/PV/RL.PER.RNK`); domestic credit to the private sector, total and by banks (`FS.AST.PRVT.GD.ZS`, `FD.AST.PRVT.GD.ZS`); external balance on goods and services (`NE.RSB.GNFS.ZS`); FDI net inflows (`BX.KLT.DINV.WD.GD.ZS`); GDP growth (`NY.GDP.MKTP.KD.ZG`); GDP per capita (`NY.GDP.PCAP.KD`); gross capital formation (`NE.GDI.TOTL.ZS`); high-technology exports (`TX.VAL.TECH.MF.ZS`); human capital index (`HD.HCI.OVRL`; no observation in 2022–2024); industry value-added growth (`NV.IND.TOTL.KD.ZG`); tax revenue (`GC.TAX.TOTL.GD.ZS`); trade openness (`NE.TRD.GNFS.ZS`).
Inverted (13)	Bank liquid reserves to assets (`FD.RES.LIQU.AS.ZS`); bank capital to assets (`FB.BNK.CAPA.ZS`); deposit interest rate (`FR.INR.DPST`); net energy imports (`EG.IMP.CONS.ZS`); external debt stocks (`DT.DOD.DECT.GN.ZS`); total debt service (`DT.TDS.DECT.GN.ZS`); public and publicly guaranteed debt service (`DT.TDS.DPPF.XP.ZS`); central government debt (`GC.DOD.TOTL.GD.ZS`); firms experiencing electrical outages (`IC.ELC.OUTG.ZS`); net bilateral aid from the United States (`DC.DAC.USAL.CD`); net ODA received per capita (`DT.ODA.ODAT.PC.ZS`); applied tariff rates, all and manufactured products (`TM.TAX.MRCH.WM.AR.ZS`, `TM.TAX.MANF.WM.AR.ZS`).
Excluded (4)	Consumer price index (`FP.CPI.TOTL`); net foreign assets in local currency (`FM.AST.NFRG.CN`); net trade in goods, current US$ (`BN.GSR.MRCH.CD`); energy use per capita (`EG.USE.PCAP.KG.OE`, used as $d^{(8)}$).

Notes: Series as in `data/raw/wb_wdi.csv`; orientation and exclusions as in `partnership_strength_build_development_indicators()`. Series are averaged over 2022–2024 and S-curved across economies before inversion and averaging.

Rankability of partners. A cell's imputed weight share is the share of its partner- and pair-specific component weight resting on imputed or missing inputs: a trade component imputed at the cell mean, a missing investment position, an alignment assigned rather than read from both countries' own votes, a missing development indicator, and, for components read off the partner's pillar cells, those cells' own imputed share. The reporter's own attributes, the trade component of cells without a basket and the penalties are left out. A dyad's share is the mean over the five cells of its best-five summary, and the PSI averages and weights the shares as it does the scores. The median Friendshore dyad share is 0.42 over all reporters and 0.26 over the core members. In the core view the ranked partners' shares have median 0.20 (maximum 0.42); the highest-scoring unranked partners are territories such as Guam.

Further results. The market score C_j^{PO} follows the core members' observed exports to the market (Spearman 0.82) and correlates negatively with the market's own security need (-0.54), although need enters it with a positive weight, because the economies most in need of supply are mostly small importers. The strongest supplier–weak market contrasts are small European states and some low-income suppliers (San Marino, Andorra, Albania, Chad, Niger); the reverse are markets such as the United Arab Emirates (17th on C_j^{PO}, 132nd on C_j^{FS}), Kazakhstan, Liberia and Uganda. The Spearman correlations among the category scores are 0.59 (C^{FS} and C^{PO}, 192 partners), 0.31 (C^{FS} and C^{DP}) and 0.29 (C^{PO} and C^{DP}, 130 partners); Opportunity accounts for 14.5 percent of the variance of PSI_j among the 62 ranked partners without Development. High-income partners average 0.827 and the other partners on the World Bank list 0.672. Without the alignment component, 19 rather than 24 core members would be in the top 25, and China, Saudi Arabia and Indonesia would replace Sweden, Korea and Ireland in the top fifteen; Viet Nam would rise from 64th to 20th,

the United Arab Emirates from 91st to 21st and Oman from 140th to 59th. The world view's top fifteen add China, India, Indonesia, Viet Nam and Malaysia and drop Belgium, Poland, Czechia, Sweden and Ireland; in the full-coalition view China is 9th and India 15th. Comparing supplier and market views, the pair means agree at 0.733 over 3,192 ordered coalition pairs, and the mean within-cell partial correlation given the shared capability term is 0.44; for the 2,070 pairs with a nonaligned member the two agree on average (0.016). The pairs the market view favours most are in opposite blocs, such as Luxembourg supplying Oman (FS = 0.37 against PO = 0.71) and Indonesia supplying Canada (0.42 against 0.75); those the supplier view favours most have Ukraine as supplier, to Greece (0.70 against 0.23) and Sweden. After Canada's rating of the United States, the highest dyadic PSI_{ij} are Spain's of Portugal, the United States' of the United Kingdom, and Austria's and Czechia's of Germany (0.970–0.976). The Development leaders after China are Serbia, Rwanda, Viet Nam, North Macedonia and Mauritania (0.872), with best cells 40 percent Upstream, 26 percent Midstream and 34 percent Downstream.

OA.9 The allied network design: further results

Inputs. The design reads five collapsed inputs restricted to the 63 economies of the network: 2,079 EO and 1,881 ES records, 726 development-potential records, and 128,898 friendshore and 116,622 opportunity dyads. For each, the number of rows read equals the number of keys kept, with no duplicated or multivalued key, so the collapse rules (one record per member, technology and stage; one record per ordered pair and stage; the latest year and the mean of the distinct finite values otherwise) act as guards only. All 37 nodes have an EO record, and hence a producer score, in every one of the 22 stages. In Geothermal Upstream and Midstream all 37 scores rest on EO alone, and in Green Hydrogen Midstream the scores of Costa Rica, the Dominican Republic, Ghana, Guatemala and Namibia do. Without ES, demand rests on size alone, so Namibia, the smallest node, has zero demand in these three stages; it remains a potential producer.

Producers. Eight nodes produce only Upstream, all of them resource economies (Australia, Chile, Saudi Arabia, Peru, Argentina, Oman, Namibia and the Democratic Republic of the Congo), while Japan takes ten of its twelve roles and Korea nine of its ten in Midstream. Canada (16 roles, nine of them Upstream), the EU node (15), the United States and Indonesia (13 each) and India and Japan (12 each) are followed by Thailand and Mexico (11 each). The largest share of a stage is at most 0.30 in 18 of the 22 stages, and 30% of all shares sit at the floor of 0.05. The partner tier carries 58% of designed Upstream and 45% of Midstream production. Indonesia leads Batteries, Coal, Electric Vehicles, Green Hydrogen, Solar and Wind Upstream with shares of 0.225 to 0.325, and Oil Midstream.

The EU node. The EU node produces in 15 of the 22 stages (six Upstream, nine Midstream), with shares from 0.05 to 0.215 and a median of 0.105. It is the second-largest producer in Green Hydrogen Midstream (0.215) and Nuclear Midstream (0.205), and its 0.205 in Coal Midstream

equals Japan's leading share, but it is the sole largest producer in no stage. It is a top-5 producer in ten stages, exactly its cap, and the portfolio loop bars it from Electric Grid Upstream, Electric Vehicles Midstream, Nuclear Upstream, Solar Midstream and Wind Upstream, so Electric Vehicles and Solar are the two Midstream stages without EU production. By its extra-EU exports the EU is the largest exporting node in nine of the 18 basket stages, all Midstream. The design sees it differently in two ways: its producer score averages strong and weak members, and its demand is rank-compressed, with a median weight of 0.035 across stages (0.029 to 0.046) against 24.6% of the nodes' GDP, so that it consumes about as much as the United States (35.6% of the nodes' GDP, median weight 0.033). Table OA.5 splits the EU's designed shares back to its members in proportion to the stage weights that aggregate their inputs; summed over stages, Germany (0.41), France (0.24), Poland (0.19) and Italy (0.16) carry the largest derived shares. These figures are derived, not designed: the program never sees the members separately.

Table OA.5: The EU node's designed production shares and their split to member states

Technology	Stage	EU share	Weights	Three largest members (derived share)	Top-3 weight
Green Hydrogen	Upstream	.125	GDP	DEU .030, FRA .020, ITA .015	.52
Solar	Upstream	.115	exports	FRA .053, DEU .020, NLD .012	.74
Geothermal	Upstream	.085	GDP	DEU .020, FRA .013, ITA .010	.52
Batteries	Upstream	.055	exports	DEU .016, FRA .006, FIN .005	.49
Coal	Upstream	.050	production	POL .025, DEU .014, CZE .006	.91
Electric Vehicles	Upstream	.050	exports	FIN .014, DEU .007, NLD .007	.54
Green Hydrogen	Midstream	.215	GDP	DEU .051, FRA .034, ITA .026	.52
Nuclear	Midstream	.205	exports	FRA .058, DEU .054, NLD .028	.68
Coal	Midstream	.205	exports	POL .082, DEU .032, ITA .022	.66
Wind	Midstream	.155	exports	DEU .045, ITA .016, POL .012	.47
Gas	Midstream	.135	exports	DEU .038, ITA .025, FRA .010	.54
Batteries	Midstream	.105	exports	POL .023, HUN .022, DEU .020	.62
Electric Grid	Midstream	.095	exports	DEU .031, ITA .008, FRA .006	.47
Geothermal	Midstream	.070	GDP	DEU .017, FRA .011, ITA .008	.52
Oil	Midstream	.055	exports	DEU .016, ITA .008, FRA .004	.51

Notes: Stages in which the EU-27 node produces in the production design (15 of 22). EU share: the node's designed production share. The design solves the EU as one node; its share is split back to the 27 member states in proportion to the stage weights that aggregate their inputs (Weights: the members' production or export shares in the stage, else their GDP shares), so the member figures are derived, not designed. Top-3 weight: the combined weight of the three largest members within the node. ISO3 codes. Source: `allied_network_member_shares.csv`.

Flows and edges. In Batteries Midstream the United States carries 0.295 of production, India and Malaysia 0.135 each, Canada 0.114 and the EU 0.105, with four more nodes near the floor; in Solar Midstream India (0.275), the United States (0.205), Viet Nam (0.135) and Thailand (0.125) lead; in Wind Midstream India (0.275), the United States (0.250) and the EU (0.155). Of the 811 consumer–stage pairs with positive demand, 793 draw on exactly two suppliers and 18 on three. Self-supply is positive in two stages only, 0.8% of demand in Electric Grid Upstream and 0.2% in Electric Vehicles Midstream, because a node's own edge is worth the median edge while the edges to its best partners lie above it. The flow-weighted mean edge weight over flows between distinct nodes ranges from 0.385 to 0.910 across stages. The low end is Geothermal, whose edges lack the

opportunity component and so enter at no more than half weight (flow-weighted 0.385 to 0.387), against 0.73 to 0.75 in the two Green Hydrogen stages and 0.85 to 0.91 elsewhere; the flow-weighted mean exceeds the unweighted mean in all 22 stages. The EU node's edges are weighted by its members' 2024 basket trade with the partner where it exists (76% of the EU edges) and by the stage weights otherwise.

Concentration and current exporters. The design is less concentrated than the nodes' exports in 10 of the 18 basket stages and more in eight; the spread of concentration is compressed, from 0.095–0.355 in the nodes' exports to 0.126–0.209 in the design. Where the nodes' exports are most concentrated, in Nuclear Upstream (Canada 54%), Coal Upstream (Australia 41%) and Nuclear Midstream (the EU 48%), the design spreads production; where they are dispersed, as in Batteries Midstream ($HHI^N = 0.095$), it concentrates it on the highest-scoring producers. The largest designed share is below the largest node's export share in 13 of the 18 stages, and the designed and observed leaders coincide in four. Against world exports the design is more concentrated in 13 of the 18 stages, but so are the nodes' own exports in 16. China leads eight of the nine Midstream baskets (Nuclear Midstream, led by France, is the exception) and the Solar and Wind Upstream baskets (silicon metal, 38%, and magnet rare earths, 30%); Kazakhstan leads Nuclear Upstream (44%). The number of exporters behind each index differs (34 to 139 world reporters, at most 37 nodes, seven to ten designed producers), so part of any difference reflects the number of units. All three largest exporting nodes are designed producers in eight of the 18 stages and at least one is in every stage. The lowest correlations between designed and observed shares, in Batteries and Electric Vehicles Upstream (0.27 and 0.29), illustrate the combinatorial logic: in Batteries Upstream Indonesia (0.235) is placed level with Australia, the largest exporter, and Canada and Chile receive 0.12 each, while South Africa and the Philippines, the second and third exporters, receive 0.05 and 0.06.

Endowment. Applying the endowment screen of the allied picks after the fact (technology-level reserves or production above zero; the EU node counts as endowed when any member is), the 21 roles held by nodes without a recorded resource base carry 5% of Batteries Upstream (the United Kingdom), 10% of Nuclear Upstream (Israel and Thailand at the floor), 12.5% of Coal Upstream (Peru and the Philippines), 23.5% of Wind Upstream (Japan, Israel, Korea and Singapore), 27.5% of Electric Grid Upstream (five nodes at or near the floor), 28% of Green Hydrogen Upstream (Thailand 0.145, Israel and the United Kingdom) and 36% of Solar Upstream (the United Kingdom, Norway, South Africa and Singapore), and none of Electric Vehicles, Gas and Oil Upstream. These nodes score on the stage through its trade basket, readiness and the other categories of EO and ES.

Portfolio caps. Eleven nodes have no top-5 appearance. The final pass bars Thailand and the United States from nine stages each and Indonesia and Malaysia from eight. Without caps, six nodes would exceed their cap: the United States (17 appearances against a cap of ten), Indonesia (12 against seven), India (ten against eight), Thailand and Malaysia (ten against six) and Canada (eight against seven). The caps pass these appearances chiefly to Japan and Mexico (from three to

six each), the Philippines (none to four), South Africa (one to four) and Israel (two to four); 17 stages lose objective value. Four Upstream stages (Coal, Green Hydrogen, Nuclear and Solar) have a tie at fifth place, which the strict rank resolves by producer score.

Build candidates. The build priority is

$$b_{p,ts} = \delta_{p,ts}\,(0.6\,a_{p,ts} + 0.4\,\bar{W}_{p,ts}), \qquad \bar{W}_{p,ts} = \textstyle\sum_c W_{pc}\,d_c, \tag{OA.5}$$

development potential times a weighted mean of the producer score and the demand-weighted connectivity. Restricted to the 22 nodes with an observed development potential, the ten leading candidates are India in Solar Midstream (0.76), Wind Midstream (0.75), Gas Midstream (0.72) and Nuclear Midstream (0.69); Indonesia in Electric Vehicles Upstream and Coal Upstream (0.74 each), Batteries Upstream (0.71), Solar Upstream (0.69) and Wind Upstream (0.68); and Viet Nam in Solar Midstream (0.67). All ten are already designed producers, as the construction implies: the priority weights the quantities on which the program itself selects producers, so the list shows where the design already relies on an emerging economy rather than where new capacity would add a producer.

OA.10 Robustness and validation: further detail

Reproduction. The weighting experiments use a re-implementation of the available-case mean of equation (16) on the stored category scores, which reproduces the stored indices at the configured weights to within 10^{-15}; the curvature, imputation and data-choice experiments rebuild both pillars from the theme tables with the pipeline's builders and configuration; the Strategic Index and the Policy index are rebuilt with the pipeline's own functions. Each reproduces the stored values exactly at the baseline settings. Table OA.6 collects the within-pair correlations of the pillars.

Vintages. A separate build re-derives both pillars as of 2020 and 2025 by slicing the time-varying sources (the EI Statistical Review, the Atlas of Economic Complexity, GCIM investment, and the IMF and ILO price, rate and earnings series) to each year, holding the other themes (resource potential, reserves, the IEA critical-mineral and clean-technology tables, the mineral production sheets, BNEF demand, levelised costs, overcapacity and technology readiness) identical, and measuring Trade on the Atlas basis. Categories carrying 52 per cent of the nominal ES weight and 70 per cent of the EO weight are time-varying by classification, those whose scores differ in at least one per cent of their cells 40 and 50 per cent, and, after renormalisation over observed categories, 34 per cent of the average ES cell's weight and 36 per cent of the average EO cell's re-derive. In ES (Figure OA.2) the mean absolute change is 0.015, 31 per cent of cells do not change, and the mean percentile shift is 3.8 points; 77 per cent of the squared Upstream changes is a common shift of whole pairs from the Energy Prices category, which is constant within a pair. In EO the mean absolute change is 0.030, no cell is unchanged and the mean percentile shift is 10.8

Table OA.6: Robustness of within-pair country rankings: Spearman correlations with the baseline

	Energy Security (ES)			Economic Opportunity (EO)		
Variant	Pooled	Median	Minimum	Pooled	Median	Minimum
Category weights						
Dirichlet draws	0.99	1.00	0.98	0.98	0.99	0.97
Equal weights	0.97	1.00	0.98	0.98	1.00	1.00
S-curve curvature (baseline $\gamma = 0.5$)						
$\gamma = 0.25$	0.98	1.00	0.97	0.98	0.99	0.98
$\gamma = 1$ (percent rank)	0.98	0.99	0.90	0.98	0.99	0.97
$\gamma = 2$	0.93	0.96	0.74	0.94	0.97	0.90
Global-average imputation rules						
Removed (available case)	0.99	1.00	0.62	0.85	0.97	0.29
Replaced by zero	0.87	0.98	0.58	0.62	0.77	0.42
Vintage						
2020 against 2025	0.97	0.99	0.68	0.89	0.87	0.49
2025 vintage against headline	0.99	1.00	0.87	0.99	1.00	0.79

Notes: Spearman rank correlations between the baseline pillar and each variant, over the baseline's rankable cells (sovereign economies with at most half of the index weight imputed). "Pooled" is computed over all rankable cells of the eleven headline technologies; "Median" and "Minimum" are taken across technology–stage pairs of the correlation computed within each pair. For the Dirichlet draws the pooled entry is the median over 1,000 draws and the within-pair entries summarise, across pairs, each pair's median over the draws. The vintage rows compare the 2020 and 2025 builds of `scripts/40_build_index_vintages.R`, in which only the time-varying sources differ (4,139 ES and 4,301 EO rankable cells observed in both vintages), and the 2025 build with the headline index. Sample: 30 ES pairs (4,163 rankable cells) and 33 EO pairs (4,301 rankable cells). Entries are rounded to two decimals. Source: `scripts/paper/fig_robustness.R`; `paper/results/robustness.json`.

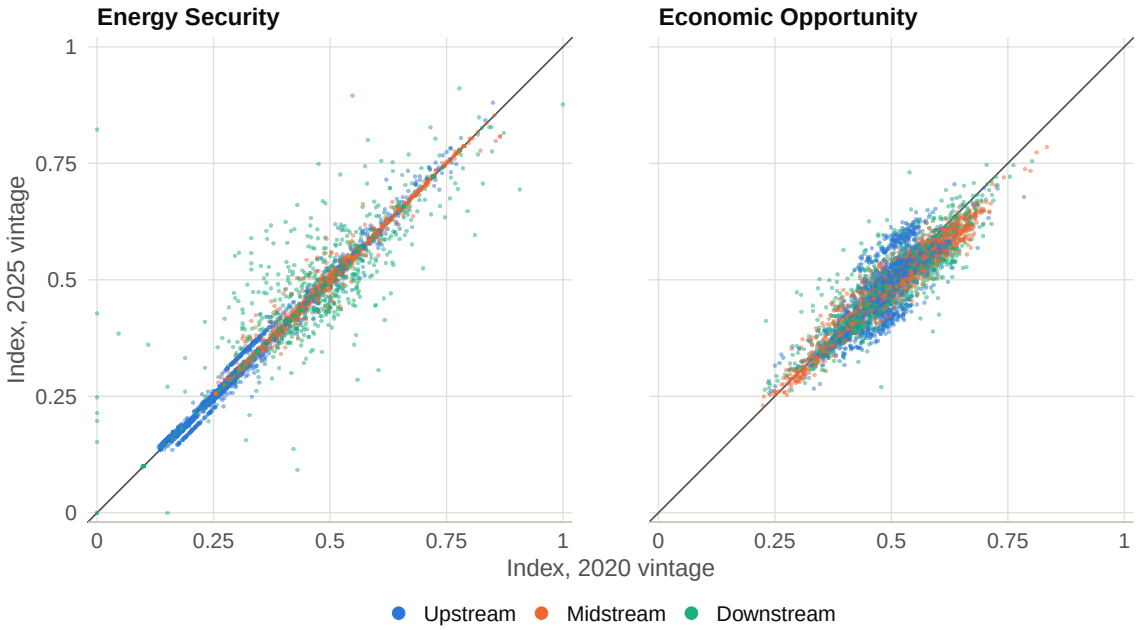


Figure OA.2: The 2020 and 2025 vintages of the pillars. Each point is a rankable country cell of the eleven headline technologies observed in both vintages (4,139 in ES, 4,301 in EO), coloured by stage; the line is the 45-degree line. Both vintages are built by `scripts/40_build_index_vintages.R` with the headline builders and configuration; only the time-varying sources are sliced to the vintage year, and Trade is measured on the Atlas basis. Source: `data/processed/vintages`; `scripts/paper/fig_robustness.R`.

points, because Cost Competitiveness re-derives in every pair from annual earnings, lending-rate and producer-price series. Oil Downstream has the lowest correlation in both pillars (0.68 and 0.49). The largest single moves occur where a pair rests on one flow variable: in Electric Vehicles Downstream ES, built from annual GCIM flows alone, Viet Nam moves from 94th of 101 to 3rd. The 2025 vintage coincides with the headline except in Trade (median within-pair correlation 1.00 in both pillars; minima 0.87 in ES Nuclear Midstream and 0.79 in EO Nuclear Upstream, whose baskets the final revision changed). Because only about a third of the weight re-derives, the comparison likely understates instability over time; the vintage years are nominal for two sources (Atlas 2023 in the 2025 vintage, IMF producer prices 2021 in the 2020 vintage).

Category weights. The weight vectors are drawn as $w \sim \mathrm{Dir}(c\,\bar{w})$ around the configured shares $\bar{w}$, so that $E[w_k] = \bar{w}_k$ and the coefficient of variation of w_k is $\sqrt{(1-\bar{w}_k)/[\bar{w}_k(c+1)]}$; $c = 99$ for the ten categories of a pillar gives 0.3 for a category of average weight (Decancq and Lugo, 2013). In seven ES pairs and three EO pairs a single category varies, and there the ranking does not depend on the weights at all; on average 3 per cent of the observed weight of an ES pair and 25 per cent of an EO pair sits on categories constant within it. Restricted to pairs with two or more varying categories (23 in ES, 30 in EO), the median correlation over pair–draws is 0.996 and 0.987, but the minimum falls to 0.77 and 0.30. For each coalition member and pair, the 5th to 95th percentile band of its percentile position over the draws has a median width of 2.8 points in ES (1,606 member cells)

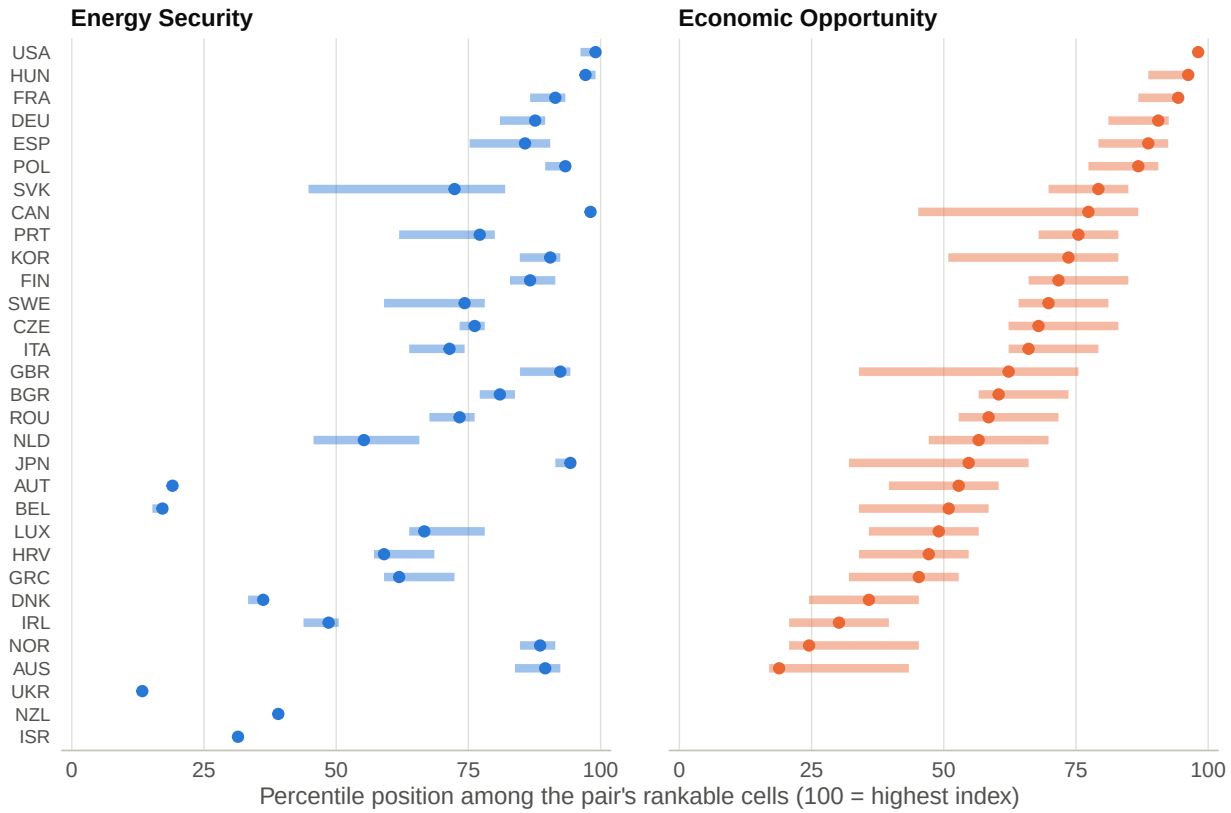


Figure OA.3: Rank uncertainty of the core members under perturbed category weights, Batteries Midstream. For each core member with a rankable cell (31 in ES, 28 in EO) the dot is its percentile position at the configured weights and the bar spans the 5th to 95th percentiles over 1,000 Dirichlet draws (coefficient of variation 0.3 for a category of average weight). Source: category scores in `index_outputs.rds`; `scripts/paper/fig_robustness.R`.

and 9.6 in EO (1,718), with a 90th percentile of 18 and 31 points. Wide bands arise among large blocks of tied scores: in ES Batteries Downstream South Korea ranks 69th of 186 at the configured weights with a band from 15th to 186th; in EO the widest bands are Upstream, such as Brazil in Nuclear Upstream (163rd of 175, band 21st to 174th). Figure OA.3 shows one pair.

Curvature. Because $\operatorname{logit}\phi_\gamma(r) = \gamma \operatorname{logit} r$, a component at curvature γ' follows from the stored value as $\phi_{\gamma'}(r) = \operatorname{logistic}\left((\gamma'/0.5)\operatorname{logit}\phi_{0.5}(r)\right)$, which also maps anchored zeros to zero. The map is exact for a component that is a single S-curve of a rank; 46 of the 57 components that enter a scored category pass this test, carrying 84 per cent of the average ES cell's weight and 63 per cent of the EO cell's. For the others (the input- and levelised-cost terms, readiness, overcapacity, the production surplus of Energy Imports, the mineral-supply and production shares of Foreign Dependency, the minerals-trade terms and the geothermal and wind potential indices) it re-curves a mean and is an approximation. Levels move more than ranks: the mean absolute change of ES is 0.03, 0.04 and 0.08 at $\gamma' = 0.25$, 1 and 2.

Table OA.7: The rankability threshold: cells ranked, robustness of their rankings, and the Strategic Index

	ES			EO			Strategic Index			PSI
τ	Cells	ρ_{none}	ρ_{zero}	Cells	ρ_{none}	ρ_{zero}	Cells	Countries	Picks	Ranked
0.3	3,147	1.00	1.00	1,967	0.99	0.94	1,577	107	29/48	159/102
0.5*	4,163	1.00	0.98	4,301	0.97	0.77	3,154	174	57/57	192/188
0.7	5,350	0.99	0.98	5,974	0.86	0.68	5,156	194	42/57	194/194

Notes: A cell is rankable when the economy is sovereign and at most a share τ of its index weight rests on imputed components (* the headline, `config/rankability.yml`). Cells: rankable headline cells (of 6,232 scored ES and 7,131 scored EO cells). ρ_{none} and ρ_{zero}: median across technology–stage pairs of the within-pair Spearman correlation, over the cells rankable at τ, between the baseline pillar and the pillar rebuilt with every global-average rule removed (available case) or replaced by zero. Strategic Index: cells and countries in the sample when the pillars' rankable flags are taken at τ (the Policy index is unchanged), and the number of coalition members whose allied pick is the baseline one, over the members with a pick at τ. PSI: partners ranked in the core-coalition (headline) and world views. Within a technology–stage pair the order of two cells rankable under two thresholds is the same, because the rule filters cells and does not change values. Source: `scripts/paper/fig_robustness.R` (`rankability` in `paper/results/robustness.json`).

Imputation. Nine ES themes and twelve EO themes fill uncovered countries with the observed mean of their component group (equation (14)). Such components carry on average 7 per cent of an ES cell's weight, mainly in Energy Access and Foreign Dependency, and 25 per cent of an EO cell's, two thirds of it in Technology Demand and most of the rest in Production and Cost Competitiveness. The lowest within-pair correlations are Nuclear Upstream (0.62) without imputation and Coal Downstream (0.58) with zeros in ES, and Coal Midstream (0.29) without imputation in EO. Zero-filling places every uncovered economy at or below every covered one on the component; removing the rule scores it on fewer measurements than its neighbours; the global average compresses the differences the component would make. None is neutral.

Rankability. Table OA.7 varies τ. Because the rule filters cells without changing values, it changes the sample, not the order of cells rankable under both thresholds. At $\tau = 0.3$ the rankable cells are 56 per cent of the sovereign scored cells in ES and 31 per cent in EO (74 and 67 per cent at the headline), at $\tau = 0.7$ 95 and 93 per cent. The median within-pair correlation of EO with its zero-filled variant is 0.94 at $\tau = 0.3$, 0.77 at the headline and 0.68 at $\tau = 0.7$. The top ten of 27 of the 30 ES pairs are identical at $\tau = 0.3$ but only 15 of the 33 EO top-ten lists. At $\tau = 0.3$, 48 of the 57 members keep a pick and 29 of them the baseline one; at $\tau = 0.7$, 42 of 57 keep the baseline pick. At $\tau = 0.3$ the world view of the PSI ranks 102 partners and the core view 159.

Leave one category out and effective importance. Figure OA.4 drops one category at a time. In ES the ranking depends most on Consumption (median within-pair correlation 0.74 over eight pairs) and Minerals Trade (0.79 over six); Trade has the widest range (minimum 0.40 in Electric Grid Midstream, whose basket the completed trade file filled) and is the only category of the four Midstream fossil and nuclear pairs. Production and Foreign Dependency, which share the largest

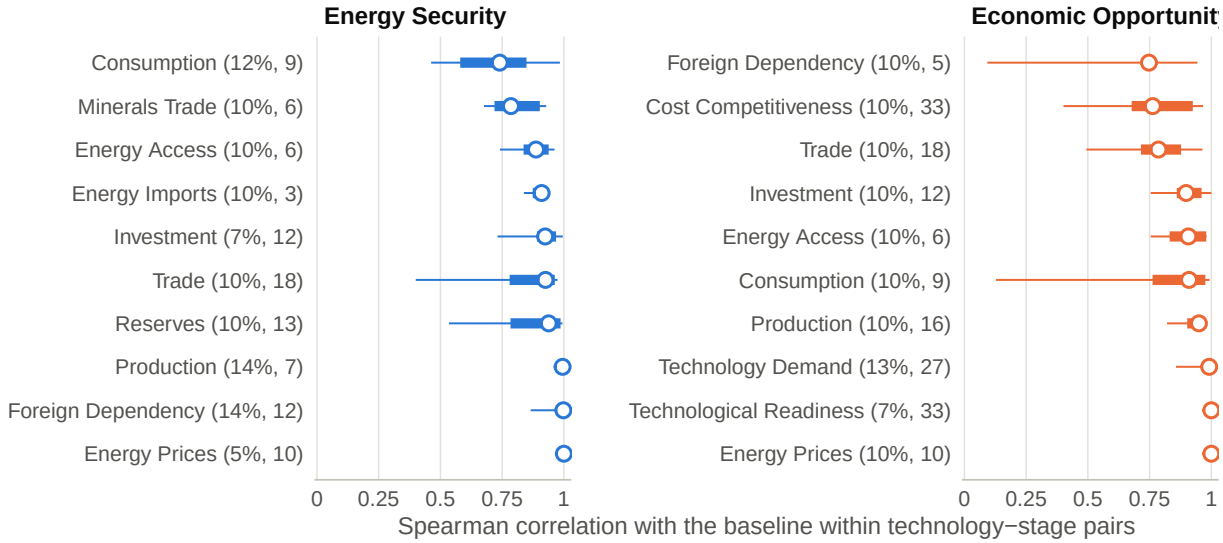


Figure OA.4: Leave-one-category-out sensitivity of within-pair rankings: Spearman correlation between the baseline and the pillar without the category (remaining categories renormalised), over the pairs in which the category is observed and the baseline's rankable cells; open circle, median; thick bar, interquartile range; thin line, range. Labels give the nominal weight share and the number of pairs observed. Pairs in which the dropped category is the only varying one are omitted. Source: `index_outputs.rds`; `scripts/paper/fig_robustness.R`.

nominal ES weight, change almost nothing (0.995 and 0.998). In EO, Foreign Dependency (0.75), Cost Competitiveness (0.76) and Trade (0.79) carry most of the ranking, while Technology Demand changes it little (0.99) and Technological Readiness and Energy Prices, constant within pairs, not at all. A complementary, exact measure is the share of a pillar's cross-country variance within a pair for which each category accounts. Because the pillar is the sum of its category contributions C_{kits} (equation (19)),

$$S_{kts} = \frac{\mathrm{Cov}_i(C_{kits},\, \mathrm{ES}_{its})}{\mathrm{Var}_i(\mathrm{ES}_{its})}, \qquad \sum_k S_{kts} = 1, \tag{OA.6}$$

and likewise for EO, a linear counterpart of the main effects of Paruolo et al. (2013). Table OA.8 compares nominal weights, mean effective weights and mean variance shares. The share of weight that would have to move for nominal and variance shares to coincide is 0.32 in ES and 0.37 in EO. In ES the gap is mainly coverage: Trade, observed in 18 of the 30 pairs, has 9.5 per cent of the nominal weight but 24.9 per cent of the effective weight and 21.5 per cent of the variance. In EO it arises within pairs: Technology Demand and Technological Readiness hold 21.8 and 14.6 per cent of the effective weight but 3.6 and 0.0 per cent of the variance, while Cost Competitiveness holds 21.9 and 32.8 per cent.

Strategic Index weights and policy variants. The Dirichlet draws of the five weights of equation (55) use concentration 43.4, giving coefficients of variation of 0.30 for the policy weight, 0.23 for the two pillar weights and 0.45 for the two priors. Indonesia keeps its pick in 15 per

Table OA.8: Nominal weight, effective weight and variance share of each category within technology–stage pairs

Category	Nominal share (%)	Effective weight (%)	Variance share (%)	Pairs	Leave-one-out ρ
*Panel A. Energy Security (*ES*), 30 pairs*					
Trade	9.5	24.9	21.5	18	0.93
Investment	7.1	11.3	18.1	12	0.92
Reserves	9.5	9.5	17.4	13	0.94
Consumption	11.9	15.8	13.2	9	0.74
Foreign Dependency	14.3	16.1	10.1	12	1.00
Production	14.3	5.8	10.0	7	0.99
Energy Access	9.5	7.2	5.0	6	0.89
Energy Imports	9.5	2.9	2.8	3	0.91
Minerals Trade	9.5	3.1	1.8	6	0.79
Energy Prices	4.8	3.4	0.0	10	1.00
*Panel B. Economic Opportunity (*EO*), 33 pairs*					
Cost Competitiveness	10.0	21.9	32.8	33	0.76
Trade	10.0	11.0	17.5	18	0.79
Investment	10.0	6.6	14.4	12	0.90
Production	10.0	8.1	12.5	16	0.95
Consumption	10.0	5.0	7.9	9	0.91
Foreign Dependency	10.0	2.9	7.4	5	0.75
Energy Access	10.0	2.8	3.8	6	0.91
Technology Demand	13.3	21.8	3.6	27	0.99
Energy Prices	10.0	5.3	0.0	10	1.00
Technological Readiness	6.7	14.6	0.0	33	1.00

Notes: Nominal share: $w_k / \sum_{k'} w_{k'}$ from `config/weights.yml`. Effective weight: mean over the rankable headline pairs of the weight $w_k / \sum_{l \text{ observed}} w_l$ a category receives after renormalisation over the observed categories; zero where the category is not observed. Variance share: mean over the same pairs of S_{kts} (equation (OA.6)), the covariance of the category's contribution with the pillar across the rankable cells of the pair divided by the pillar's variance; zero where the category is not observed. Each of the three columns sums to 100 within a panel up to rounding. Pairs: number of pairs in which the category is observed. Leave-one-out ρ: median, over the pairs in which the category is observed and the reduced pillar still ranks countries, of the within-pair Spearman correlation between the baseline pillar and the pillar without the category (Figure OA.4). Rows are ordered by the variance share. Sample: rankable cells of the eleven headline technologies. Source: category scores in `index_outputs.rds`; `scripts/paper/fig_robustness.R` (`weights.<pillar>.variance_importance` in `paper/results/robustness.json`).

cent of the draws, its two leading sectors differing by 0.0007. Under security in place of need, 28 members pick Nuclear Midstream and 10 Gas Midstream; without the priors, ten picks move to Oil Midstream, the sector the technology prior penalises most; without the policy term, 20 members pick Nuclear Upstream instead of 8. Requiring both Policy components leaves 1,666 Strategic Index cells; dropping CAT for every country gives a median within-country Strategic Index correlation of 1.00 (interquartile range 0.91 to 1.00, minimum 0.30). The October 2026 NIPO component (GTA NIPO as of 5 October 2026, debiased strength; 1,070 rows), added with weights 4, 3 and 2 on NIPO, IEA PAMS and CAT and IEA PAMS still required, is observed in Upstream and Midstream cells only, in 841 Strategic Index cells (27 per cent); for its jurisdictions the within-country Spearman correlation between the policy ranks with and without it has a median of 0.88 (range 0.00 to 1.00), the pooled Strategic Index correlation is 0.97, and 27 of the 50 members with a NIPO score keep their pick.

Data choices of the final build. Table OA.9 and Figure OA.5 report the data choices of the last revision (11–12 October 2026). The headline carries 807 EIA-sourced rows in the import-dependence theme and 656 in the production theme; setting them to zero gives within-pair correlations of 0.97 to 0.98 in ES and 0.95 to 0.96 in EO in Oil, Gas and Coal Upstream. Removing Trade from the four Upstream cells with new baskets gives median within-pair correlations of 0.94 in ES and 0.73 in EO (minimum 0.58, Solar Upstream EO) and returns the Strategic Index to 192 countries, because the newly imputed Trade category had pushed small economies over the rankability threshold; three of the four changed picks move to Nuclear Upstream. The superseded energy-trade file is not kept, but interim builds record the picks: on 8 October, 45 of the 56 members then in the coalition had a pick, 17 Electric Grid Midstream and 11 Batteries Midstream; on 10 October, 4 and 26 of 57. Without Trade, Electric Grid Midstream picks rise to 10 and Batteries Midstream falls to 13; without Technological Readiness 32 of 57 picks survive (Electric Grid Midstream would take 27), without Technology Demand 41 and without Foreign Dependency 44. The CM-FDI comparison is not re-run on the final outputs, because the dyad tables exceed the memory budget of the replication; in the integration study, switching off the Chinese-parent component or the CM-FDI investment ties alone gives core-PSI correlations of 0.9999.

Vertical complementarities. The opportunity test regresses the five-year change in a stage's S-curved world market share, and entry into revealed comparative advantage, on the adjacent stage's market-share index, with country–year and technology–year fixed effects and standard errors clustered by country. The cross-sectional association is positive and precisely estimated (0.046, standard error 0.007, for Midstream on Upstream; 0.102, standard error 0.015, for the reverse) but concentrated in fossil fuels and nuclear (0.062 and 0.151) and insignificant for the clean-technology chains (0.012 and 0.028); with country–technology fixed effects it is −0.001 (0.005) and 0.004 (0.015); lags of the stage's own capability absorb about a third of it; and it does not improve out-of-sample prediction (within-cell AUC for Midstream entry 0.800 without the term and 0.794 with it; 0.739 and 0.741 for Upstream). The 2023 co-location lift, the ratio of the joint to the independent probability

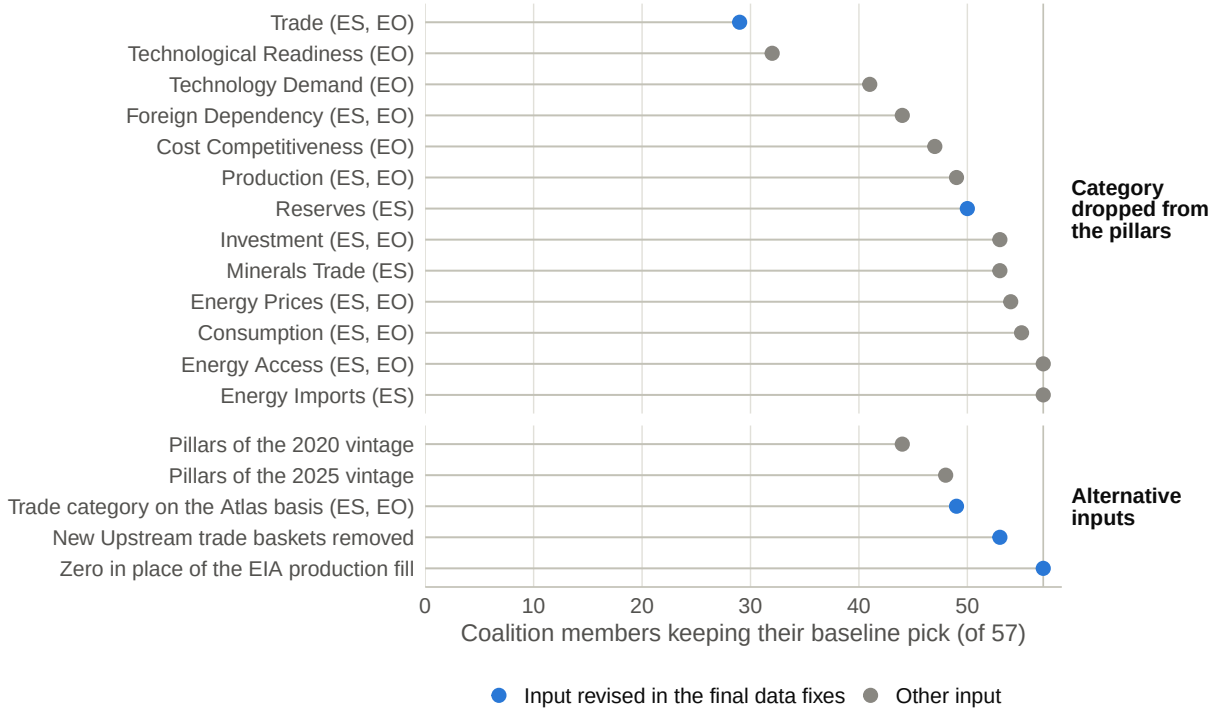


Figure OA.5: Stability of the allied picks under the data inputs: number of members, of 57, whose pick is the baseline one when the Strategic Index takes pillars rebuilt under a variant. Upper panel: one category dropped (without Trade three members have no eligible rankable sector, so 29 of the 54 with a pick keep it). Lower panel: pillars from the 2020 or 2025 vintage, Trade on the Atlas basis, Trade removed from the four Upstream cells whose baskets were added on 11 October 2026, and EIA-filled fossil production set to zero. Blue marks inputs revised in the final data fixes. The Policy index and the endowment screen are the headline ones. Source: `index_outputs.rds`, theme tables, vintages; `scripts/paper/fig_robustness.R`.

Table OA.9: Data and policy choices of the final build: rankable cells, rankings and allied picks

	Rankable cells		Median ρ, touched pairs		Strategic Index		
Variant	ES	EO	ES	EO	Cells	Countries	Same pick
Baseline	4,163	4,301	–	–	3,154	174	57/57
Data							
Zero in place of the EIA fill	4,164	4,301	0.98	0.96	3,155	174	57/57
New Upstream baskets removed	4,161	4,307	0.94	0.73	3,144	192	53/57
Policy index							
Both components required	=	=	–	–	1,666	65	46/46
IEA PAMS alone	=	=	–	–	3,154	174	45/57
October 2026 NIPO added back	=	=	–	–	3,154	174	34/57
Allied pick rule							
Endowment screen off	=	=	–	–	3,154	174	16/57

Notes: Rankable cells: headline cells of sovereign economies with at most half of the index weight imputed, after the variant (= unchanged). Median ρ: median across the touched technology–stage pairs (Oil, Gas and Coal Upstream for the EIA fill; Electric Vehicles, Wind, Nuclear and Solar Upstream for the baskets) of the within-pair Spearman correlation between the baseline and the variant over the baseline's rankable cells. Strategic Index: scored cells and countries, and the coalition members whose allied pick is the baseline one over the members with a pick. Zero in place of the EIA fill: the oil, gas and coal production that the Energy Institute does not report (807 EIA-sourced rows of the import-dependence theme and 656 of the production theme in the headline) set to zero, the rule until 11 October 2026, both themes rebuilt with the pipeline's builders. New Upstream baskets removed: the Trade category of the four Upstream cells whose trade baskets were added on 11 October 2026 dropped and both pillars rebuilt; the uranium codes moved out of Nuclear Midstream and the rare-earth codes moved out of EV and Wind Midstream are not restored. Policy rows as in Table 22. The partnership side is not rebuilt: in the CM-FDI integration study (inputs of 11 October 2026), switching off both CM-FDI parts of Friendshore gives a core-view PSI with a Spearman correlation of 0.9998 with the CM-FDI build over 192 ranked partners, a largest rank change of 5 places and 15 of the top 15 in common. Source: `scripts/paper/fig_robustness.R` (`data_choices` and `policy` in `paper/results/robustness.json`).

Table OA.10: Vertical complementarities: sensitivity of the pillars and the allied picks to an adjacent-stage term

Term	Weight	Median within-pair ρ	Minimum ρ	Picks changed
EO linkage, within-country deviation	Mid 0.07, Up 0.23	0.999	0.987	1
Placebo: scores permuted within cell	same	0.999 [0.999, 0.999]	–	0 [0, 1]
ES input security, sourcing-weighted	15% share	0.917	0.556	10
Placebo: constant 0.5	same	1.000	1.000	3
Placebo: scores permuted within cell	same	0.902 [0.891, 0.909]	–	11 [8, 16]

Notes: EO linkage: a category added to the Midstream and Upstream EO cells, the S-curved world market share of the adjacent stage (Atlas of Economic Complexity, HS92 baskets, 2023) minus the economy's mean of that measure over the 9 basketed headline technologies, S-curved within the cell (Midstream 1,890; Upstream 1,890 cells). Weights in EO weight points (the nominal category weights are 2 to 4): the evidence weights of the vertical-complementarity study, 0.6 times the ratio of the adjacent-stage to the feasibility coefficient in the five-year market-share regression, upper bounds because about a third of the coefficient is absorbed by lags of own capability. ES input security: the domestic supply share times the economy's own supply security plus the import-share-weighted security of its suppliers (half the supplier's ES in the input stage, half the dyadic alignment), 2024 sourcing, S-curved within the cell and given 15 per cent of the Midstream and Downstream cells' weight (Downstream 1,626; Midstream 1,893 cells). Median and minimum ρ: within-pair Spearman correlation with the baseline pillar over the baseline's rankable cells of the receiving stages; for the permutation placebo, the median and the 5th and 95th percentiles over 200 draws of the per-draw median. Picks changed: coalition members (of 57) whose allied pick differs from the baseline when the Strategic Index takes the modified pillar (for the placebo, median and 5th–95th percentiles over the draws). Neither term enters the headline indices. Source: `scripts/paper/fig_robustness.R` (`vertical` in `paper/results/robustness.json`), from the study's inputs.

of comparative advantage in both stages, is 4.4 for nuclear and 2.7 for coal but 0.97 for wind and below one for gas, electric vehicles and the grid. The security test builds the domestic share times the economy's own supply security plus the import-share-weighted security of its suppliers (half the supplier's ES in the input stage, half dyadic alignment). Its 2020 value does not predict import shortfalls in 2021–22 (0.10, standard error 0.31) or Midstream export growth after 2020 (0.02, standard error 0.19), whereas sourcing concentration does (negative, $p = 0.035$); the post-2022 shift of Upstream input trade towards aligned suppliers, 0.58 log points (0.11), falls to 0.07 (0.06) without Russia. Table OA.10 reports what the two terms would do at the evidence weights. The EO term, the adjacent stage's S-curved Atlas share minus the economy's mean of that measure over the nine basketed technologies, S-curved within the cell, enters with 0.07 weight points in Midstream and 0.23 in Upstream (0.6 and 1.5 per cent of a cell's mean observed weight): median within-pair correlation 0.999 (minimum 0.987), one pick changed (Luxembourg's, the smallest margin in the coalition), against 0 to 1 for a placebo with scores permuted within cells. The ES term at a 15 per cent share gives median correlations of 0.92 in Midstream and 0.91 in Downstream (minimum 0.56, Batteries Downstream) and changes 10 picks, against 11 (8 to 16) for the permuted placebo and 3 for a constant placebo that only rescales the other categories.

OA.11 Corrections to the shipped release: item detail

This appendix gives, for each item of Table 32, the defect, its symptom in the shipped release of September 2026, the correction and, where the audit measured it, the *pre-rerun effect*: the change the correction makes when applied alone to the theme tables and inputs saved with the shipped release. Pre-rerun effects isolate one correction on inputs that still embed the other defects; they are not results of the paper.

Aggregation (A1–A7). A1: an unobserved category kept its weight in the denominator and counted as zero, and a cell with no observed category scored zero; 1,809 of 2,773 headline ES Reserves rows were unobserved. The available-case rule alone changes 1,682 Upstream cells and raises mean Upstream ES by 0.139 (Oil), 0.112 (Gas) and 0.104 (Coal). Because the same commit gives Reserves a zero rule (the reserve tables list every material holder, so absence means none), the two changes offset on the shipped tables; the rule matters for categories without a missing-data rule and for the PSI (B4). A2: a zero or global-average rule filled groups in which no country was observed; it now fills only groups with an observed country and reports the rest. A3: `any(isTRUE(x))` is `FALSE` for vectors longer than one, so the diagnostics reported 11.9 and 29.5 percent of ES and EO category scores as imputed against 37.0 and 69.2 percent in fact; the final run has 38.4 and 45.0 percent. A4: Energy Prices was configured on a variable no theme produced and dropped from both pillars without warning, because the guard compared the join keys with themselves; it re-enters ten Upstream cells as a world price series that shifts cell levels and leaves within-cell order unchanged (Spearman 1.000 in every varying cell). A second guard now warns when a theme's own composite is scored in place of a configured overall. A5: an EO Reserves weight of 2 for a category EO does not have let resource-potential rows create 88 fully imputed EO cells (the Holy See, the Pitcairn Islands); EO falls from 6,923 to 6,835 headline cells. A6: the variable-contribution table divided by the number of components including missing ones; both decompositions now reconcile to the index (largest error 1.1×10^{-16}). A7: a development commit made `bind_rows()` read a named list of tables as one row; a rerun from that revision would have removed every overall-based category.

Country universe (C1–C3). C1: the standardiser attached ISO3 codes but did not rename rows, so the shipped ES, EO, Strategic and Policy tables carried 272 strings for 237 economies, with 30 codes under two or three spellings (Viet Nam and Vietnam, Korea and South Korea, the two Congos). Each spelling took its own rank in every S-curve with a partly observed, often imputed profile. Every variant now maps to the WDI name of its code (World Bank, 2026); territories with their own code stay separate. C2: the EI "USSR" series, a CIS aggregate, was coded RUS, "Northern Cyprus" CYP, and a pseudo-country "Global" carried readiness values (128 rows); aggregates, residuals and historical entities are now dropped, not merged into successors. C3: the 2025 critical-minerals Comtrade pulls hold three reporters against about 130 in 2024; a vintage needs 50.

Energy Security themes (E1–E12). E1: graphite and rare earths fell out of every join because the IEA and EI name them differently; with harmonised names the rare earths carry about 79 percent of Wind's Mineral Supply weight. E2: the stage recode mapped refining to the Upstream under both branches; refining now scores the Midstream. E3: listed non-producers (Germany, Japan) tied at 0.441 just below the Congo at 0.473 in Batteries Upstream while unlisted economies scored zero; zero supply now maps to zero for every economy and suppliers rank above it. E4: the reserves roll-up averaged only over listed minerals, so Chile scored 1.000 on Wind Reserves on copper alone. E5: BNEF onshore and offshore wind were never recoded, so Wind had no Consumption category; growth was ranked across technologies; Viet Nam's rows fall from 160 to 63 once each country receives one regional value. E6: Minerals Trade kept only the market-share index; it is now the mean of a demand-weighted market share and a concentration index $\phi_\gamma(-\mathrm{HHI}_m)$. E7: critical-minerals investment was copied in full to Batteries and EV Upstream (79.8 each; now 9.2 and 70.6), and stage completion manufactured zero Investment cells in Nuclear Upstream and Midstream and Geothermal Upstream and Midstream, which made those cells constant across countries and gave Nuclear Upstream the maximum security need for every country. E8: electrolyser capacity was filed outside the headline technologies. E9: mineral Upstream production depth was an unweighted mean of different minerals' kilotonnes; it is now a demand-weighted share of world mined supply, with growth as an absolute change (EO moves by up to 0.067 in six cells, within-cell Spearman 0.955–0.992). E10: a placeholder region dropped the eight non-EU European EI economies from EV Midstream. E11: the EI "Import share" (a self-sufficiency ratio) and the EV "import_share" (a net-export ratio) are renamed for what they measure, and non-producers now tie exactly in Energy Imports. E12: an economy without its own EI production row (EI counts it in a regional residual) was read as producing nothing, so New Zealand counted as importing all its gas; production then comes from EIA International Energy Statistics (U.S. Energy Information Administration, 2026), flagged and counted as observed, paired with consumption of the same year (Estonia coal stays missing because EI counts oil shale). On the inputs of 10 October 2026 this moves 605 ES Upstream fossil cells and no allied pick.

Trade and Economic Opportunity (T1–T13). T1: the ES and EO trade themes were identical tables and the ES deficit term was never computed; ES Trade is now concentration, basket share and net basket imports over GDP, EO Trade share, RCA (Balassa, 1965), export size, feasibility and concentration (earlier–corrected Spearman 0.54–0.91 across cells). T2: export size and feasibility were ranked within reporter, so Moldova and Samoa ranked beside China in Solar Midstream. T3: market share was the export-weighted mean of product-level world shares; the Congo had 0.862 against a basket share of 0.0074 in Batteries Midstream. T4: ores, slag and scrap were filed under Batteries Midstream, untreated softwood logs under three grid and renewable baskets, and 16 keys appeared twice. T5: 73 crosswalk codes had no HS92 counterpart, so the Atlas EV Midstream basket contained no cars; 19 predecessor codes bridge them. T6: the levelised cost entered Cost Competitiveness four times (0.78 of the category's weight where quoted). T7: readiness

was written for a pseudo-country and reached real economies only by imputation. T8: Technology Demand listed composites beside their own sub-indices and ranked a zero addressable market as the smallest. T9: the export-concentration index was ranked across every basket, so unscored baskets (Semiconductors) moved scored cells; it is now ranked over the headline cells. T10: the Comtrade energy-trade file lacked 60 basket codes and 13 reporters, among them Qatar's LNG (USD 56.9bn in 2024); it is rebuilt from the World rows of the all-reporter dyad pull. T11–T13: duplicate WEO and BNEF demand keys, IEA 2035 manufacturing shares taken without the "Other" row (China's 2035 solar share 0.90 rather than 0.80), zero lending-rate stubs (New Zealand 0.0 percent, now 4.76) and mislabelled IMF price units (International Monetary Fund, 2026b); only the lending rates move an index value.

Policy and Strategic indices (P1–P9). P1: the NIPO module's jurisdiction weights overwrote the IEA module's, so international IEA policies were weighted 0.20 instead of 0.66. P2: Upstream stage tags matched substrings ("ore" in "more", "Korea", "Singapore"), tagging 36.9 percent of IEA policies Upstream; whole-word rules tag 12.5 percent. P3: name mismatches left South Korea, the United Arab Emirates (coded "UAE"), France, Italy and Denmark without a Policy index; EU members without a national CAT rating now take the EU's (Climate Action Tracker, 2026). P4: the policy block had stopped before writing its table since 6 February 2026, so the September Strategic Index combined September pillars with February policy components. P5: NIPO's default as-of date was the latest announcement or implementation date (1 October 2028 in every 2026 extract), counting scheduled measures as in force; it is now the extract's last announcement, capped at the release month and run date. NIPO is no longer in the composite; the robustness variant uses the October 2026 extract as of 5 October 2026. P6: the strategic table overwrote the security column with $1 - \text{ES}$; it now carries security and need under their own names (no value changes). P7: missing pillar or policy values were filled with country means (51 of 571 shipped cells); a cell now needs all three. P8: sectors with a constant pillar were not detected; Nuclear Upstream, constant at zero, was the top pick of 8 of 18 shipped allies. P9: the staged oil lifecycle emissions (720 g CO_2eq/kWh) appear in no IPCC table; the SRREN median of 840 (Moomaw et al., 2011) replaces it next to the AR5 values of the other technologies (Schlömer et al., 2014), so the oil weight falls from 0.12 to 0 and coal's rises from 0 to 0.02.

Bilateral indices (B1–B5). B1: grouped `transmute()` calls, name keys and a self-crossed backbone duplicated the Friendshore table (17.5 million rows for 1.44 million cells), so a dyad's best-five mean weighted its top cells 2, 2 and 1. B2: `mean(a:b)` returned `a`, so outward investment was the rank of the level only; 99 of 259 IMF counterpart names did not match, so China and the Netherlands scored zero from every reporter. B3: development potential, a partner-level construct, was stored per reporter with donor fan-out up to 38 rows, with 56 aggregates and mis-oriented WDI series. B4: a high-income partner's undefined development score kept its weight, capping its PSI at 0.8 and placing the United States 185th of 218. B5: an opportunity column named for security held the partner's need, and the tables covered 18 to 20 pseudo-technologies rather than the eleven

headline ones.

Network (N1–N7). N1: the dyadic development table joined the node table without aggregation, so each node was copied per reporter; the greedy solver then filled every producer slot with copies of one country (606,866 specialisation rows, 2,339 distinct). N2: duplicated edges doubled the edge term (connectivity up to 1.53, above the largest edge weight). N3: both halves of the edge were read with the reporter as producer, transposing Friendshore; with the correct orientation the two halves correlate 0.54 instead of 0.01. N4: caps and producer counts were computed over rows (portfolio counts of 10^4–10^5 against caps of 2–10). N5: the code "UAE" matched no table, so the shipped coalition had 22 effective members. N6: the MILP was used only up to 18 nodes, fewer than the coalition, while the parameters reported "milp". N7: a node missing any component had no score, and stages without signal were filled alphabetically.